SCIENCE

The Biotic Associations of Cockroaches

Louis M. Roth and Edwin R. Willis

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"EVERY MAN IS A VALUABLE MEMBER OF SOCIETY WHO, BY HIS OBSERVATIONS, RESEARCHES, AND EXPERIMENTS, PROCURES KNOWLEDGE FOR MEN"--JAMES SMITHSON

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LEONARD CARMICHAEL, _Secretary, Smithsonian Institution_.

SMITHSONIAN MISCELLANEOUS COLLECTIONS VOLUME 141 (WHOLE VOLUME)

THE BIOTIC ASSOCIATIONS OF COCKROACHES

(WITH 37 PLATES)

BY LOUIS M. ROTH AND EDWIN R. WILLIS

Pioneering Research Division, United States Army Quartermaster Research and Engineering Center Natick, Mass.

[Illustration: Smithsonian Institution Logo]

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[Illustration: Plate 1 _Blaberus craniifer_, c. × 2. 1. (Photograph by Jack Salmon, Philadelphia Quartermaster Depot.)]

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FOREWORD

People having only casual interest in insects usually express amazement when they learn how much is known about this most numerous group of animals. However, while entomologists have good reason to take pride in the accomplishments of their contemporaries and predecessors, they are more likely to be appalled by how much remains to be learned. We are indeed ignorant of even the identity of fully half and probably much more than half the total number of insect species. Of those that have been described, we have reasonably complete information about the behavior and basic environmental relationships for only a comparative few. The great majority of the remainder are known only from specimens found in museum collections. Such information as we have about these species usually amounts to no more than date and locality of collection.

This is true of the cockroaches, which now include approximately 3,500 described species. Conservative estimates based on partially studied museum collections and the percent of new species found in recent acquisitions, particularly from tropical and subtropical countries, indicate that at least 4,000 species remain unnamed. Although the group is well known in general terms to nearly all entomologists, there is an almost complete void of information about all except the few domestic species and, to a progressively diminishing degree, some 400 others. Many details about the lives of even those that share man's habitations are not fully understood. This then is a rough measure of how little is known about cockroaches.

With the exception of mosquitoes and a few other comparatively small groups of insects on which work has been concentrated, it is doubtful if any other comparable segment of the world's insect fauna is better known. Already an estimated 800,000 kinds of insects have been described, and since this figure is generally regarded as less than half the actual total, think what this means in terms of knowledge yet to be assembled. No wonder entomology is a growing science with a promising future, but the magnitude of the task also presents a serious obstacle to progress. Progress can continue only if the scattered literature resulting from the diversified labors of hundreds of contributors is brought together and summarized in thorough and well-organized compilations that can serve as a solid basis for future research.

The present work is such a compilation, for it assembles what has been gleaned from approximately 1,700 sources, including correspondence with a large number of other workers. Original observations during some eight years of concentrated effort in U. S. Army Quartermaster research laboratories are a valuable supplement to what others have done, and with this background of experience the authors are especially well qualified to appraise previous work. Seldom has a compilation been done so thoroughly or a single large group of insects been the subject of such uninterrupted effort.

The contents gives the categories of subject matter treated and the introduction discusses the value of this assembled information and offers suggestions for future study. No longer are cockroaches regarded only as disagreeable pests; many species appear to be important, actually or potentially, as carriers of disease. Recognition of this importance has grown considerably, even in the period since World War II. Consequently, anything that increases our knowledge of the basic bionomics of cockroaches will be consulted widely for factual information and for clues to new approaches.

In spite of this extensive compilation, the limitations of present information about cockroach bionomics must be kept in mind. The cited observations of many writers were fragmentary, or their conclusions disagreed. But it is fundamental to scientific inquiry that we should know and attempt to evaluate the results of previous study, and that is what Drs. Roth and Willis have done. Fortunately, their review is readily available. Sometimes, a piece of work fails to be of maximum value because the results are not generally accessible to later students. For this reason I am especially glad that the Smithsonian Institution, by disseminating the results of the authors' labors, has this opportunity to exercise one of its traditional functions--that of diffusing knowledge.

Throughout the period of research by Drs. Roth and Willis at Natick, I was in frequent correspondence with them, and I admire their many accomplishments. Our warmest commendations should go not only to them personally but also to those in administration who encouraged their fundamental research and who aided in the financial support of this publication.

ASHLEY B. GURNEY _Entomology Research Division_ _United States Department of Agriculture_

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THE BIOTIC ASSOCIATIONS OF COCKROACHES[1]

BY LOUIS M. ROTH AND EDWIN R. WILLIS[2]

Pioneering Research Division, United States Army Quartermaster Research and Engineering Center Natick, Mass.

(With 37 Plates)

With most of us collectors the life history of an insect begins in the net and ends in the bottle.

HANITSCH (1928)

I. INTRODUCTION

Recently we brought together much of the literature linking cockroaches with the transmission of certain organisms that cause disease in man and other vertebrates. In that paper (1957a) we concluded that cockroaches, being potential vectors of pathogenic agents, should not be regarded simply as minor annoyances. Obviously the associations of cockroaches with agents of vertebrate diseases are of more immediate importance than their relations with pathogens of lower animals or with nonpathogens. On the other hand, cockroaches are of general economic as well as medical importance, and their control is sought by many who are unaware of their medical significance. That the control of domiciliary cockroaches is far from satisfactory may be inferred from current entomological and pest-control journals in which new insecticides are continually advocated to replace others found to be inadequate. Possibly new approaches to the control of cockroaches are needed. Whether these lie in the direction of increased use of parasites and predators for the biological control of these insects remains to be seen. In any event, the more we know about any insect, especially its ecology, the greater the likelihood of achieving satisfactory control. In order to advance knowledge in any field of science, new research should proceed from the results of prior investigations when these exist. We hope that the observations and experiments cited herein may suggest areas for future research and exploitation.

To the best of our knowledge no previous publication has brought together the vast literature on the parasites, predators, commensals, and other symbiotic associates of the Blattaria. For this reason, we have tried to assemble observations on all such known associations. Undoubtedly we have overlooked some records, as, for example, those buried in papers dealing with other phases of cockroach biology. We hope that

such inadvertent omissions will not seriously impair the usefulness of this compilation. Whatever its defects, this review should be a unified source of information for all who are interested in the biotic associates of cockroaches.

In addition to previously published information, this monograph also contains original records and observations on the associations of cockroaches that are reported here for the first time. Although some of the observations were made by us, others were made by colleagues who have graciously made their knowledge available to us in private communications.

HISTORICAL

Chopard (1938) in his book _La Biologie des Orthoptères_ reviewed much of the literature on cockroaches, but of the many biotic associations that exist he discussed only the commensal cockroaches, gregariousness, and familial associations. Asano (1937), who reviewed the natural enemies of cockroaches, mentioned about 10 groups of animals that attack cockroaches. Thompson (1951) in his _Parasite Host Catalogue_, which was based mainly on papers abstracted or noted in the Review of Applied Entomology, listed only 19 insect parasites of cockroaches. Eighteen of these were Hymenoptera which attack only cockroach eggs; the single dipteron listed (_Sarcophaga lambens_ Wiedemann, supposedly parasitic on _Pycnoscelus surinamensis_) is not a parasite in this case, but deposits its eggs on the dead insects (see p. 229). Cameron (1955) listed as parasites and predators of the cockroach 24 species of hymenopterous egg parasites, 7 species of _Ampulex_ which hunt nymphs and adults, 17 Protozoa, 13 nematodes, 5 bacteria, 2 mites, and a few other miscellaneous predators. In his classified list of the protozoan parasites of the Orthoptera of the world, Semans (1943) listed about 26 species from cockroaches. Linstow (1878, 1889) recorded 14 species of helminths from cockroaches. Van Zwaluwenburg (1928) listed 33 names of roundworms which are commensals or secondary parasites of cockroaches, but some of these names are synonyms. La Rivers (1949) extended this list with 13 additional species. Chitwood (1932) recognized 24 species of nematodes which are primary parasites (probably commensals) of blattids. Steinhaus (1946) gave many instances of biological relationships between cockroaches and bacteria, fungi, and yeasts, but the cockroaches were not discussed as an entity and the information is scattered throughout the book.

In surveying the literature on this subject we have collected a far more extensive list of animals and plants associated with cockroaches than one might have expected from an examination of any one of the previous papers on this subject. In our review of the medically important organisms associated with the Blattaria, we pointed out that in addition to many experimental associations cockroaches have been found to harbor, naturally, 4 strains of poliomyelitis virus, about 40 species of pathogenic bacteria, the eggs of 7 species of pathogenic helminths, and to serve as intermediate hosts of 12 other species of helminths pathogenic for vertebrates; cockroaches have also been found to carry, on occasion, 3 species of Protozoa that are pathogenic to man and 2 species of fungi which are sometimes found associated with pathological

conditions.

In addition to the above organisms of medical importance, we have compiled records of other organisms, nonpathogenic to vertebrates, which are naturally associated in some way with cockroaches. None of the following numbers can be considered absolute because some names may be synonyms. However, we believe that these figures are very close to the actual numbers of species that have been isolated because we have attempted to refer all obvious synonyms to the currently accepted name for each organism. On this basis there are about 45 species of bacteria, 40 fungi, 6 yeasts, 90 Protozoa, and 45 helminths that have been found associated naturally with cockroaches. Of the arthropods there are about 2 species of scorpions, 4 spiders, 15 mites, 4 centipedes, and 90 insects. Of vertebrates there are 4 species of fish, 16 amphibians, 12 reptiles, 20 birds, and 27 mammals. Besides these there are many records of experimental associations that have been contrived in the laboratory.

Some idea of the increase in our knowledge of the biotic associations of cockroaches, during the last 70 years, may be gathered from a comparison of the above figures with those of Miall and Denny (1886) who presented "...a long list of parasites which infest the Cockroach." This list included 2 bacteria, 6 Protozoa (some of the names are synonyms), 7 nematodes (some of these names are also synonyms), 1 mite, 1 wasp, and 1 beetle. In addition, they mentioned as other foes of the cockroach: monkeys, hedgehogs, polecats, cats, rats, birds, chameleons, and frogs.

METHODS

We have listed the organisms known to be associated with cockroaches systematically by phylum, class, order, and family. Within each family the organisms are listed alphabetically by genus and species. Under each organism the associated cockroaches are listed as natural or experimental hosts, vectors, or prey. Identified cockroaches are listed by the currently accepted name. Unidentified cockroaches are indicated by the word "Cockroaches." The name of each cockroach is followed by the country in which the observation was made, the authority for the record, and with a few exceptions[3] pertinent biological information, where this is known. Question marks following the names of organisms or countries indicate tentative or questionable identifications.

Records of predators capturing and feeding on cockroaches in zoos and on shipboard we consider natural, even though it is very likely that these particular predators would not normally have access to this prey in nature.

Experimental prey are cockroaches that were fed to predators in the laboratory. Although these predators may have little, if any, access to these cockroaches in nature, we have included such records to indicate the relative acceptability of cockroaches as food by a wide variety of animals.

Records of presumed or known cockroach associates that give no information about an associated cockroach are not included in this review, even though certain of

these (e.g., species of _Ampulex_, _Evania_, _Podium_) probably prey upon or parasitize cockroaches exclusively.

The validity of a host-parasite or predator-prey record is dependent upon the accuracy and knowledge of the observer. In assembling these records we have had to accept, in most instances, the identifications of species made by the original authors. However, as a result of our studies on the biology of various species of cockroaches, including some work on their hymenopterous parasites, we have questioned certain records in the literature. Other dubious records which have been perpetuated from one publication to the next, but which apparently were not based on fact, have also been questioned or have been clarified with the aid of specialists in particular groups.

Because the records cover a period of many years, the names of many of the organisms as well as the names of some of the cockroaches have been changed. Although it would have been comparatively simple to list the names as they appeared in the original references, this would have resulted in misleading redundancy with the same organism being catalogued under several synonyms. We have attempted to list each organism by its currently accepted name. However, no attempt was made to prepare complete taxonomic synonymies; the only synonyms given are those that identify the organisms by the names used by the authors of the papers cited. The synonyms under which the cockroaches may have been cited originally are listed in section II. The synonyms of associated organisms are listed with each organism. Although authorities for the name changes of the cockroaches are given, these workers are not necessarily those who were initially responsible for the synonymies. Various sections have been checked by specialists in the particular groups. Although we have accepted name changes suggested by these reviewers, we assume full responsibility for the names.

FUTURE WORK

After having examined thousands of references on cockroaches, we are impressed by how little is known about the biology of most species. As a conservative estimate there are 3,500 described species of Blattaria (J. W. H. Rehn, 1951). In our literature survey we found records of biotic associations for about 400 species. Unfortunately, many of these records contain only a sentence or two of biological information. Our detailed knowledge of cockroaches is based on studies of the few domiciliary pests that man attempts to eradicate. Comparable studies of the bionomics of the less-well-known species should add much valuable information to our knowledge of this ancient group.

Our understanding of most predator-prey and parasite-host relationships has barely progressed beyond the taxonomic stage. The total effect of predators and parasites in limiting natural populations of cockroaches remains to be determined. It is still not known how, for example, predatory or parasitic wasps select specific cockroaches from among all other insects. Secretions produced by certain cockroaches (e.g., 2-hexenal by _Eurycotis floridana_) will ward off certain predators. The identities and biological activities of most cockroach secretions are unknown, but

the use of protective chemicals against predators may be widespread among cockroaches. If so, how effective are these repellents in protecting the individual or the species? It is not known whether cockroaches are protected by apparent mimetic resemblances to other arthropods. There is no experimental proof that insect parasites can successfully attack the eggs of cockroaches that incubate their eggs while they are being carried by the female.

It is conceivable that biological control of cockroaches might be achieved in limited areas such as man-made structures or sewers, but this possibility has not been thoroughly explored. It would be informative to know what effects, if any, organisms such as bacteria, Protozoa (e.g., gregarines), intestinal nematodes, or other helminths have on cockroaches. Possibly pathogenic microorganisms can be used for biological control of cockroaches; this approach seems to have been little explored.

Associations of colonial cockroaches (e.g., _Cryptocercus_ spp.) may be truly familial or they may merely result from gregariousness. Newly hatched nymphs of species that carry their oöthecae until the eggs hatch cluster near the mother. This may be a response to the mother as such, a search for shelter beneath the nearest object (thigmotaxis? or negative phototaxis?), or there may well be yet another explanation. Tepper in 1893 stated that the native cockroaches of Australia are almost wholly carnivorous; little supporting evidence for this claim has been brought forward since that time. The apparent supersedure of one species of domiciliary cockroach by another may result from antagonism between different species, or it may result from more rapid breeding and more effective utilization of available food and space; but which? Several species of cockroaches are frequently found associated with certain plants (e.g., bromeliads and bananas); the ecological relations in these associations remain to be determined. Many of the obscure associations between cockroaches and other insects, spiders, birds, and burrowing animals have never been adequately defined. The factors influencing cannibalism have never been thoroughly investigated experimentally. These are only a few ideas for future work that have occurred to us during the preparation of this review. We hope that these suggestions as well as other questions that may occur to readers will stimulate further research in areas where it is obviously needed.

ILLUSTRATIONS

Unless otherwise credited, the illustrations were prepared from photographs taken by the authors. Except where otherwise stated, all photographs were taken of unposed living specimens.

II. SPECIES OF COCKROACHES

The cockroaches referred to in this paper are listed below. The currently accepted name for each species is given alphabetically by genus and species irrespective of its taxonomic affinities. Synonyms used by certain authors whose work we have quoted are given in brackets under the respective species; the synonymy is supported by the reference citation that follows each synonym. References to illustrations of certain

species (e.g., _Blaberus craniifer_) that appear in the paper follow the names of the describers.

Agis orientalis Chopard

Aglaopteryx absimilis Gurney _diaphana_ (Fabricius) [_Ceratinoptera diaphana_ Fabricius; Rehn and Hebard (1927)] _facies_ (Walker) [_Aglaopteryx devia_ Rehn; Princis (1929). _A. diaphana_ (Fabricius) in records from Puerto Rico only; Rehn (1932b); Gurney (1937)] _gemma_ Hebard [In Florida records = _Ceratinoptera diaphana_ R. and H.; Hebard (1917)] _vegeta_ Rehn _ypsilon_ Princis

Allacta similis (Saussure) [_Phyllodromia obtusata_ Brunner; Zimmerman (1948)]

Alluaudellina cavernicola (Shelford) [_Alluaudella cavernicola_ Shelford; Chopard (1932)]

Amazonina emarginata Princis

Anaplecta asema Hebard _azteca_ Saussure _decipiens_ Saussure and Zehntner _fallax_ Saussure _hemiscotia_ Hebard _lateralis_ Burmeister _mexicana_ Saussure

Aneurina tahuata Hebard _viridis_ Hebard

Apotrogia angolensis Kirby [_Acanthogyna deplanata_ Chopard; Princis (1957)]

Aptera fusca (Thunberg) [_Aptera cingulata_ (Burmeister); Gurney (personal communication, 1957)]

Apteroblatta perplexa Shelford

Archiblatta hoevenii Vollenhoven

Archimandrita marmorata (Stoll) _tessellata_ Rehn

Arenivaga apacha (Saussure) _bolliana_ (Saussure) _erratica_ (Rehn) _floridensis_ Caudell _grata_ Hebard _roseni_ (Brancsik) [_Heterogamodes roseni_; Bei-Bienko (1950). _"Polygamia" roseni_ is undoubtedly an erroneous citation of _Polyphaga roseni_, as there is no genus _Polygamia_ (Gurney, personal communication, 1957)]. _tonkawa_ Hebard

Aristiger histrio (Burmeister) [_Plumiger histrio_ (Burm.); Bruijning (1948). _Hemithyrsocera histrio_ Burm.; Hebard (1929)]

Aspiduchus boriquen J. W. H. Rehn [In Puerto Rico records = _Aspiduchus deplanatus_ R. and H.; Rehn, J. W. H. (1951a)] _cavernicola_ J. W. H. Rehn _deplanatus_ (Saussure)

Attaphila aptera Bolívar _bergi_ Bolívar _flava_ Gurney _fungicola_ Wheeler _schuppi_ Bolívar _sexdentis_ Bolívar

Atticola mortoni Bolívar

Audreia bromeliadarum Caudell _jamaicana_ Rehn and Hebard

Balta godeffroyi (Shelford) _patula_ (Walker) _platysoma_ (Walker) [_Temnopteryx platysoma_ (Walker); Hebard (1943)] _quadricaudata_ Hebard

scripta (Shelford) _torresiana_ Hebard _verticalis_ Hebard

Bantua stigmosa (Krauss) [_Derocalymma stigmosa_ Krauss; Princis (1957)]

Blaberus atropos (Stoll) [_Blabera fusca_ Brunner; Hebard (1917)] _boliviensis_ Princis _craniifer_ Burmeister (pls. 1, 2) _discoidalis_ Serville [_Blaberus cubensis_ Saussure; Hebard (1916)] _giganteus_ (Linnaeus) (pl. 3)

Blaptica dubia (Serville) [_Blaberus clarazianus_ Saussure; Rehn, J. W. H. (1951)]

Blatta orientalis Linnaeus (pl. 4) [_Periplaneta orientalis_; Hebard (1917)] (_Shelfordella_) _lateralis_ (Walker) [_Shelfordella tartara_ (Saussure); Princis (1957). _Periplaneta tartara_ Saussure; Bei-Bienko (1950)]

Blattella germanica (Linnaeus) (pls. 5, A, B; 31, F) [_Blatella germanica_; Gurney (1952). _Phyllodromia germanica_; Hebard (1917). _Ectobius germanicus_; Gurney (personal communication, 1957)] _humbertiana_ (Saussure) [_Blatta humbertiana_; _Phyllodromia humbertiana_; Hebard (1929)] _lituricollis_ (Walker) (fig. 7, A) [_Blattella bisignata_ (Brunner); Bei-Bienko (1950)] _schubotzi_ Shelford _vaga_ Hebard (pl. 5, C, D)

Buboblatta armata (Caudell) [_Latindia armata_ Caudell; Hebard (1920)]

Byrsotria cabrerae Rehn and Hebard _fumigata_ (Guérin) (pl. 6)

Cahita borero Rehn _nahua_ (Saussure)

Capucinella delicatula Hebard

Cariblatta antiguensis (Saussure and Zehntner) _cuprea_ Hebard _delicatula_ (Guérin) [_Blattella delicatula_ Guérin; _Cariblatta punctulata_ (Beauvois); Rehn and Hebard (1927)] _hylaea_ Rehn _imitans_ Hebard _insularis_ (Walker) _landalei_ Rehn and Hebard _lutea lutea_ (Saussure and Zehntner) _lutea minima_ Hebard (pl. 7, A, B) _nebulicola_ Rehn and Hebard _orestera_ Rehn and Hebard _punctipennis_ Hebard _reticulosa_ (Walker) _stenophrys_ Rehn and Hebard

Cariblattoides instigator Rehn and Hebard _suave_ Rehn and Hebard

Ceratinoptera picta Brunner

Chorisoneura barbadensis Rehn and Hebard _flavipennis_ Saussure and Zehntner _formosella_ Rehn and Hebard _parishi_ Rehn _specilliger_ Hebard _texensis_ Saussure and Zehntner [_Chorisoneura plocea_ Rehn; Rehn and Hebard (1916)] _translucida_ (Saussure)

Choristima sp.

Choristimodes sp.

Chromatonotus infuscatus (Brunner) _notatus_ (Brunner)

Compsodes schwarzi (Caudell)

Comptolampra liturata (Serville) [_Compsolampra liturata_; _Comptolampra_ is the original spelling, which is followed by Dr. K. Princis, according to Gurney (personal communication, 1959)]

Cosmozosteria lateralis (Walker)

Cryptocercus punctulatus Scudder (pl. 8, A) _relictus_ Bei-Bienko

Cutilia nitida (Brunner) _soror_ (Brunner) sp. near _sedilloti_ (Bolívar) (pl. 9) [Determined by Dr. A. B. Gurney from photographs.]

Cyrtotria capucina (Gerstaecker)

Dendroblatta sobrina Rehn

Derocalymma cruralis (Stål) [_Homalodemas cruralis_ (Stål); Gurney (personal communication, 1957)] _lampyrina_ Gerstaecker _porcellio_ Gerstaecker

Deropeltis autraniana Saussure _erythropeza_ Adelung _melanophila_ (Walker) _nigrita_ Saussure

Diploptera punctata (Eschscholtz) (pls. 10, 36) [_Diploptera dytiscoides_ (Serville); Princis (1950). _Eleutheroda dytiscoides_ (Serville); Zimmerman (1948)]

Dryadoblatta scotti (Shelford) [_Homalopteryx scotti_ Shelford; Rehn (1930)]

Ectobius africanus Saussure _albicinctus_ (Brunner) _duskei_ Adelung _lapponicus_ (Linnaeus) [_Ectobius perspicillaris_ Herbst, as used by Lucas (1920); Blair (1934)] _lucidus_ Hgb. _nicaeensis_ (Brisout) _pallidus_ (Olivier) (pls. 7, C; 29, A) [_Ectobius lividus_ (Fabricius); _Ectobius livens_ (Turton); Kevan (1952); Princis (_in_ Roth and Willis, 1957)] _panzeri_ Stephens [_Ectobius ericetorum_ (Wesmaël); Ramme (1923)] _panzeri_ var. _nigripes_ Stephens _semenovi_ Bei-Bienko _sylvester_ (Poda) [_Ectobius sylvestris_ (Poda); Ramme (1951)] _tadzihicus_ Bei-Bienko _vittiventer_ (Costa) [_Ectobius vittiventris_ (Costa); Ramme (1951)]

Ellipsidion Saussure [_Apolyta_ Brunner; Hebard (1943)] _affine_ Hebard _australe_ Saussure [_Ellipsidion pellucidum_ (Brunner); Hebard (1943)] _bicolor_ (Tepper) _simulans_ Hebard _variegatum_ (Fabricius) [_Ellipsidion aurantium_ Saussure; Hebard (1943)]

Epilampra abdomen-nigrum (De Geer) _annandalei_ Shelford _azteca_ Saussure _conferta_ Walker _conspersa_ Burmeister _grisea_ (De Geer) _maya_ Rehn _mexicana_ Saussure _mona_ Rehn and Hebard _notabilis_ Walker _sodalis_ Walker _tainana_ Rehn and Hebard _wheeleri_ Rehn sp. (fig. 7, B, C)

Eremoblatta subdiaphana (Scudder)

Ergaula capensis (Saussure) [_Dyscologamia capensis_ Saussure; _Dyscologamia wollastoni_ Kirby; Princis (1957)] _scarabaeoides_ Walker [_Dyscologamia piolosa_ (Walker); Princis (1957). _Parapolyphaga erectipilis_ Chopard; Princis (1950). _Dyscologamia chopardi_ Hanitsch; Bruijning (1948). _Miroblatta silphoides_ Chopard; Hebard (1929)].

Escala sp.

Euandroblatta palpalis Chopard

Eublaberus posticus (Erichson)

Eudromiella bicolorata Hebard _calcarata_ Bei-Bienko

Euphyllodromia angustata (Latreille) _liturifera_ [_Euphyllodromia decastigmata_ Hebard; Princis (1959)]

Eurycotis bananae Bei-Bienko _biolleyi_ Rehn [_Eurycotis carbonaria_ Biolley; Rehn (1918)] _caraibea_ (Bolívar) _decipiens_ (Kirby) _dimidiata_ (Bolívar) _ferrum-equinum_ Rehn and Hebard _floridana_ (Walker) (pl. II) [_Platyzosteria ingens_ Scudder; _Platyzosteria sabalianus_ Scudder and hence, by inference, _Eurycotis sabalianus_ (Scudder); Hebard (1917)] _galeoides_ Rehn and Hebard _improcera_ Rehn _kevani_ Princis _lixa_ Rehn _manni_ Rehn _opaca_ (Brunner)

Euthlastoblatta abortiva (Caudell)

Euthyrrhapha nigra Chopard _pacifica_ Coquebert

Geoscapheus robustus Tepper

Graptoblatta notulata (Stål) [_Blatta notulata_ Stål; Hebard (1929). _Phyllodromia hieroglyphica_ Brunner; Kirby (1904)]

Gromphadorhina laevigata S. and Z. _portentosa_ (Schaum) (pl. 12, A, B)

Gyna kazungulana Giglio-Tos _maculipennis_ (Schaum) [_Gyna vetula_ Brunner; Shelford (1909b)] _tristis_ Hanitsch

Hebardina concinna (Haan) [_Blatta concinna_ Haan; _Blattina concinna_ (Haan); Bei-Bienko (1950)]

Hemiblabera brunneri (Saussure)

Henicotyle antillarum (Brunner)

Heterogamodes krügeri (Salfi) _rugosa_ (Schulthess)

Holocompsa azteca (Saussure) _cyanea_ (Burmeister) _fulva_ (Burmeister) _metallica_ Rehn and Hebard _nitidula_ (Fabricius) _zapoteca_ Saussure

Hololampra bivittata (Brullé) _chavesi_ (Bolívar) _maculata_ (Schreber) [_Aphlebia maculata_ Schreber; Harz (1957); Gurney (personal communication, 1959)] _marginata_ (Schreber) _punctata_ (Charpentier) [_Aphlebia punctata_ Charpentier; Ramme (1951)]

Hololeptoblatta sp.

Homalopteryx laminata Brunner

Hoplosphoropyga babaulti Chopard

Hormetica apolinari Hebard _laevigata_ Burmeister _ventralis_ Burmeister

Ignabolivaria bilobata Chopard

Ischnoptera deropeltiformis (Brunner) (pl. 12A) [_Temnopteryx deropeltiformis_ Brunner; Hebard (1917)] _panamae_ Hebard _podoces_ Rehn and Hebard _rufa occidentalis_ Saussure _rufa rufa_ (De Geer) _schenklingi_ Karney

Karnyia discoidalis (Brunner)

Kuchinga hemerobina (Gerstaecker) [_Phyllodromia hemerobina_ Gerstaecker; Rehn (1932)] _remota_ Hebard

Lamproblatta albipalpus Hebard _meridionalis_ (Brunner)

Latiblattella chichimeca (Saussure and Zehntner) [_Blattella chichimeca_ S. and Z.; Hebard (1932)] _lucifrons_ Hebard _rehni_ Hebard _vitrea_ (Brunner) _zapoteca_ (Saussure)

Leucophaea maderae (Fabricius) (pl. 13) [_Rhyparobia maderae_; Hebard (1917). _Panchlora maderae_; Kirby (1904). Very probably _"Blaberus" maderae_ is a careless reference to this species; Gurney (personal communication, 1957)]

Leurolestes pallidus (Brunner)

Litopeltis biolleyi (Saussure) _bispinosa_ (Saussure) [_Audreia marginata Caudell_; Hebard (1920)] _deianira_ Rehn _musarum_ Rehn

Lobolampra subaptera Rambur

Loboptera decipiens (Germar) _thaxteri_ Hebard

Lobopterella dimidiatipes (Bolívar) [_Loboptera dimidiatipes_ (Bolívar); Princis (1957a). _Loboptera sakalava_ (Saussure); Hebard (1933a). _Loboptera extranea_ Perkins; Hebard (1922). Princis (1957a) in erecting _Lobopterella_ pointed out that only the nontypical variety of _sakalava_ is identical with _dimidiatipes_.]

Lophoblatta arawaka Hebard

Macropanesthia rhinocerus Saussure

Mareta acutiventris Chopard

Maretina uahuka Hebard

Megaloblatta blaberoides (Walker) [_Megaloblatta rufipes_ Dohrn; Hebard (1920)]

Megamareta verticalis Hebard

Melanosilpha capensis Saussure and Zehntner

Methana canae Pope _curvigera_ (Walker) _marginalis_ (Saussure)

Moluchia (?) dahli Princis

Monastria biguttata (Thunberg)

Muzoa madida Rehn

Myrmeblattina longipes (Chopard)

Myrmecoblatta rehni Mann _wheeleri_ Hebard

Namablatta bitaeniata (Stål)

Nauclidas nigra (Brunner) [_Poroblatta nigra_ Brunner; Rehn (1930)]

Nauphoeta cinerea (Olivier) (pl. 14) [_Nauphoeta bivittata_ Burmeister; Zimmerman (1948)] _flexivitta_ (Walker) [_Nauphoeta brazzae_ (Bolívar); Rehn (1937)] _punctipennis_ Chopard

Nelipophygus ramsdeni Rehn and Hebard

Neoblattella brunneriana (Saussure) [_Blattella brunneriana_; Gurney (personal communication, 1959)] _carcinus_ Rehn and Hebard _celeripes_ Rehn and Hebard _detersa_ (Walker) _dryas_ Rehn and Hebard _eurydice_ Rehn and Hebard _fratercula_ Hebard _fraterna_ (Saussure and Zehntner) _grossbecki_ Rehn and Hebard _laodamia_ Rehn and Hebard _nahua_ (Saussure) [_Blattella nahua_ Saussure and Zehntner of Caudell (1914); Hebard (1920)] _proserpina_ Rehn and Hebard _semota_ Rehn and Hebard _tridens_ Rehn and Hebard _vatia_ Rehn and Hebard

Neostylopyga rhombifolia (Stoll) (pl. 15) [_Dorylaea rhombifolia_; Rehn (personal communication, 1956)]

Nesomylacris cubensis Rehn and Hebard _relica_ Rehn and Hebard

Nocticola bolivari Chopard _caeca_ Bolívar _decaryi_ Chopard _simoni_ Bolívar _sinensis_ Silvestri _termitophila_ Silvestri

Nothoblatta wasmanni (Bolívar)

Notolampra antillarum Shelford

Nyctibora azteca Saussure and Zehntner _brunnea_ (Thunberg) _laevigata_ (Beauvois) _lutzi_ Rehn and Hebard _mexicana_ Saussure _noctivaga_ Rehn _obscura_ Saussure _sericea_ Burmeister _stygia_ Walker _tomentosa_ Serville [_Nyctibora latipennis_ Burmeister; Hebard (1917, p. 263)]

Oniscosoma granicollis (Saussure)

Opisthoplatia maculata Shiraki _orientalis_ (Burmeister)

Oulopteryx meliponarum Hebard

Oxyhaloa buprestoides (Saussure) _deusta_ (Thunberg)

Panchlora antillarum Saussure _exoleta_ Burmeister _fraterna_ Saussure and Zehntner _nivea_ (Linnaeus) (pl. 16) [_Panchlora cubensis_ Saussure; Gurney (1955). _Pycnosceloides aporus_ Hebard; Hebard (1921c)] _peruana_ Saussure _sagax_ Rehn and Hebard _virescens_ (Thunberg)

Panesthia angustipennis (Illiger) [_Panesthia javanica_ Serville; Hebard (1929)] _australis_ Brunner (pl. 8, B) _laevicollis_ Saussure _lobipennis_ Brunner _spadica_ (Shiraki)

Parahormetica bilobata (Saussure)

Parcoblatta americana (Scudder) _bolliana_ (Saussure and Zehntner) [_Kakerlac schaefferi_ Rehn; Hebard (1917)] _caudelli_ Hebard [[F][F] of _Ischnoptera insolita_ R. and H.; _Ischnoptera uhleriana fulvescens_ S. and Z. (in part); Hebard (1917)] _desertae_ (Rehn and Hebard) [[M][M] of _Ischnoptera insolita_ R. and H.; Hebard (1917)] _divisa_ (Saussure and Zehntner) [_Ischnoptera divisa_ S. and Z.; Hebard

(1917)] _fulvescens_ (Saussure and Zehntner) [_Ischnoptera uhleriana fulvescens_ S. and Z. (in part); Hebard (1917)] _lata_ (Brunner) [_Ischnoptera couloniana_ R. and H. (not Saussure); _Ischnoptera major_ R. and H. (not S. and Z.); Hebard (1917)] _notha_ Rehn and Hebard _pensylvanica_ (De Geer) (pl. 17, A) [_Ischnoptera pennsylvanica_ Saussure; Hebard (1917)] _uhleriana_ (Saussure) (pl. 18) [_Ischnoptera uhleriana_ Saussure; Hebard (1917)] _virginica_ (Brunner) (pls. 17, B; 27, A; 33, C; fig. 6) [_Ischnoptera borealis_ Brunner; Hebard (1917)] _zebra_ Hebard

Pelmatosilpha coriacea Rehn _kevani_ Princis _marginalis_ Brunner _purpurascens_ (Kirby) _rotundata_ Scudder _vagabunda_ Princis

Periplaneta americana (Linnaeus) (pls. 19, 35) [_Stylopyga americana_; _Blatta americana_ L.; Hebard (1917)] _australasiae_ (Fabricius) (pls. 20, 32) _brunnea_ Burmeister (pl. 21) _cavernicola_ Chopard _fuliginosa_ (Serville) (pl. 22) _ignota_ Shaw _lata_ (Herbst)

Perisphaerus armadillo Serville _glomeriformis_ (Lucas)

Phaetalia pallida (Brunner)

Phidon (?) dubius Princis

Phlebonotus pallens (Serville)

Pholadoblatta inusitata (Rehn)

Phorticolea boliviae Caudell _testacea_ Bolívar

"Phyllodromia" treitliana Werner

Phyllodromica brevipennis (Fischer) _graeca_ (Brunner) _irinae_ (Bei-Bienko) _maculata_ (Schreber) _megerlei_ (Fieber) _polita_ (Krauss) _pygmaea_ (Bei-Bienko) _tartara_ (Saussure) _tartara nigrescens_ Bei-Bienko

Platyzosteria analis (Saussure) [_Polyzosteria analis_ Saussure; Kirby (1904)] _armata_ Tepper _bifida_ (Saussure) _castanea_ (Brunner) _novae seelandiae_ (Brunner) (pl. 23) [_Periplaneta fortipes_ Walker; Shelford (1912); _Platyzosteria novae-zealandiae_] _scabra_ (Brunner)

Plectoptera dorsalis (Burmeister) _infulata_ (Rehn and Hebard) _lacerna_ Rehn and Hebard _perscita_ Rehn and Hebard _poeyi_ (Saussure) [_Plectoptera floridana_ Hebard; Rehn and Hebard (1927)] _porcellana_ (Saussure) _pygmaea_ (Saussure) _rhabdota_ (Rehn and Hebard) _vermiculata_ Rehn and Hebard

Polyphaga aegyptiaca (Linnaeus) [_Blatta aegyptiaca_ L.; Bei-Bienko (1950). _Heterogamia aegyptiaca_ (L.); Gurney (personal communication, 1957). _"Polygamia" aegyptiaca_; according to Gurney (p. c.), there is no genus _Polygamia_ and almost surely the reference is to _Polyphaga aegyptiaca_.] _indica_ Walker [_Polyphaga pellucida_ (Redtenbacher); Princis (1957)] _saussurei_ (Dohrn)

Polyzosteria limbata Burmeister _melanaria_ (Erichson)

Pseudoderopeltis aethiopica (Saussure) [_Blatta aethiopica_ Saussure; Gurney (personal communication, 1957)]

Pseudomops cincta (Burmeister) [_Thyrsocera cincta_ Scudder; Hebard (1917)] _laticornis_ Perty _septentrionalis_ Hebard

Pseudophoraspis nebulosa (Burmeister)

Pycnoscelus niger (Brunner) _striatus_ (Kirby) [_Leucophaea striata_ Kirby; Gurney (personal communication, 1957)] _surinamensis_ (Linnaeus) (pl. 24) [_Leucophaea surinamensis_ (L.); Hebard (1917). _Blatta melanocephala_ Stoll; Kirby (1904)]

Rhicnoda natatrix Shelford

Rhytidometopum dissimile Princis

Riatia fulgida (Saussure) [_Lissoblatta fulgida_ (Saussure); Gurney (personal communication, 1959)] _orientis_ Hebard

Robshelfordia circumducta (Walker) [_Escala circumducta_ (Walker); Gurney (personal communication, 1957)] _longiuscula_ (Walker) [_Escala longiuscula_ (Walker); Gurney (personal communication, 1957)]

Salganea morio (Burmeister)

Sibylloblatta panesthoides (Walker)

Simblerastes jamaicanus Rehn and Hebard

Spelaeoblatta gestroi Bolívar

Sphecophila polybiarum Shelford _ravana_ Fernando _termitium_ Shelford

Steleopyga (?) sinensis Walker [Dr. Gurney (personal communication, 1957) could not find a reference to this species. Walker described species named _sinensis_ in three different genera of cockroaches, and it is uncertain which one this combination represents.]

Stictolampra buqueti concinula (Walker)

Styphon bakeri Rehn

Supella hottentotta (Saussure) _supellectilium_ (Serville) (pls. 25; 30, B-E; 31, A-E) [_Phyllodromia supellectilium_ (Serv.); Bei-Bienko (1950)]

Symploce breviramis (Hanitsch) _cavernicola_ (Shelford) [_Ischnoptera cavernicola_ (Shelford); _Phyllodromia nigrocincta_ Chopard; Hebard (1929)] _curta_ Hanitsch _flagellata_ Hebard _hospes_ (Perkins) [_Symploce lita_ Hebard; Hebard (1922)] _jamaicana_ (Rehn) _kevani_ Chopard _parenthesis_ (Gerstaecker) [_Phyllodromia parenthesis_ Gerstaecker; Rehn (1932)] _remyi_ (Hanitsch) [_Ischnoptera remyi_ Hanitsch; Chopard (1938)] _ruficollis_ (Fabricius) [_Symploce bilabiata_ Rehn and Hebard; Princis (1949a)]

Tartaroblatta karatavica Bei-Bienko

Temnopteryx obliquetruncata Chopard _phalerata_ (Saussure)

Theganopteryx straminea Chopard

Therea nuptialis (Gerstaecker) [_Corydia nuptialis_ Gerstaecker; Princis (1950)]

Tivia australica Princis _brunnea_ (Chopard) _fulva_ (Burmeister) _macracantha_ Chopard _obscura_ (Chopard)

Typhloblatta caeca (Chopard) [_Spelaeoblatta caeca_ Chopard; Chopard (1924b)]

Typhloblattodes madecassus Chopard

Xestoblatta festae (Griffini) _immaculata_ Hebard

III. ECOLOGICAL RELATIONSHIPS

The ecology of extinct cockroaches is necessarily a highly speculative subject. From the coexistence of fossil cockroaches and fossil plants in the same geological stratum, one might conclude that there had been intimate associations between them during prehistoric life. Heer (1864) and Goldenberg (1877) suggested that Carboniferous cockroaches fed on the plants with which they have been found as fossils. Scudder (1879) concurred with this hypothesis. However, Bolton (1911), remarking on the noticeable associations of blattoid wings with vegetable remains, suggested that the cockroaches may have been partly carnivorous, feeding on the snails _Spirorbis pusillus_, which were attached to the leaves of _Cordaites_. Yet the proximity of fossil insects and plants in the same geological formation is hardly proof of a similar association during life. In fact, Sellards (1903), Bolton (1921), and Laurentiaux (1951) have all pointed out that the cockroach remains, particularly the more resistant wings, may have been washed into streams by heavy rains and transported with drifting plant material to places where permanent deposits were accumulating.

Some species of fossil cockroaches have long, well-developed ovipositors, very unlike present-day cockroaches whose ovipositors are small and nonprotruding. Brongniart (1889) and Zalesskii (1939, 1953) have suggested that certain Permian and Carboniferous cockroaches with long ovipositors may have inserted their eggs singly into trees and other plants, rather than protecting the eggs with an oötheca. However, Laurentiaux (1951), although conceding the possibility of egg laying in vegetable material, suggested that oviposition into the earth is more probable because of the unbending nature of the ovipositor.

Although the ecological associations of modern cockroaches should be well known from direct observation, actually most species are still little more than names on museum specimens, and our knowledge of them is fragmentary. All too frequently ecological observations have been only incidental to taxonomic or faunistic studies; yet the biological information that is contained in such papers is all that we know of many species. For this reason we have cited these observations in some detail, especially when they were brief; longer accounts of cockroach bionomics, of necessity, have been abstracted.

Very few exclusively ecological studies of insects have included cockroaches. The native woodroaches (_Parcoblatta pensylvanica_, _P. uhleriana_, and _P. virginica_) of the northern United States were included in ecological studies of the Orthoptera by Hubbell (1922), Strohecker (1937), and Cantrall (1943). Fifteen species of

cockroaches were included in an ecological study of the Orthoptera of northern Florida by Friauf (1953). The original papers should be consulted for detailed descriptions of the habitats and accounts of the associated plants and other Orthoptera.

In this chapter the cockroaches are grouped into those that have been found in man-made structures and those that occur in other habitats. Certain species may appear in several categories because they live both indoors and out. The structural pests are divided into cockroaches that occur in land-based structures, those on ships, and those in aircraft. The nonstructural cockroaches are divided into those that occur in quite specific habitats (caves, water, and deserts) and those that occur generally out of doors. Nests of various arthropods serve as microhabitats of commensal cockroaches; these latter associations are discussed on pages 310-318.

In this chapter our discussion is limited to the physical environment and specific habitats of cockroaches, and only very general references are made to associated organisms. The relationships of cockroaches to the biota are examined in detail in subsequent chapters. To show the full extent of the associations, the associates, from bacteroids to vertebrates, are arranged phyletically. These associate-centered classifications serve admirably to relate various species of cockroaches within common bounds, but fail to give an integrated account of the total biotic relationships in the ecology of each species. Although physically separated in this monograph, the many associates of each species of cockroach should all be considered in appraising the ecology of that species. To assist the reader to achieve this end, we have included a checklist (p. 290) which serves as a convenient index to certain organisms associated with particular species of cockroaches.

CAVE HABITATS

Caves, mines, and animal burrows are somewhat similar habitats that provide many species of cockroaches with shelter and frequently with food. The microclimates of these cockroach habitats have not been described in detail in the papers cited, but it seems rather obvious that natural caves, man-made caves (mines), and burrows offer relatively stable temperatures and humidities and protection from adverse climatic conditions. Although such cavernicolous animals as birds and bats periodically leave caves to search for food, cockroaches find the accumulated guano and animal and plant detritus an entirely adequate dietary (Chopard, 1938). Cockroaches in mines presumably subsist on the food and feces dropped by man and mine animals (e.g., pit ponies). Food stored in their nests by burrowing animals is undoubtedly utilized by the associated cockroaches.

Cavernicolous cockroaches show varying degrees of dependence on and adaptation to these specialized habitats. Some of the common domiciliary species (_Blatta orientalis_, _Blattella germanica_, and _Periplaneta americana_) may have accompanied man into caves and remained there after he left (Chopard, 1929a, 1936, 1938). Other species, from the paucity of records noting their occurrence in caves, are undoubtedly accidental inhabitants that may never become established. Besides

these, however, many other species of cockroaches have established large breeding colonies in caves. Although some of the latter species show very pronounced morphological adaptations to a cave life, many others resemble their noncavernicolous relatives. The possible origin of cavernicolous Orthoptera has been discussed by Chopard (1938).

Cavernicolous cockroaches have been segregated into four groups according to their ability to adapt to their environment and the degree of their specialized evolution (Chopard, 1936, 1938): (1) =Trogloxenes=: Cockroaches that occur in caves in a sporadic fashion (the domiciliary cockroaches and accidentals such as _Ectobius_ and _Heterogamodes_). (2) =Troglophiles=: Cockroaches found habitually in caves (_Symploce_, _Periplaneta cavernicola_). (3) =Guanobies=: Cockroaches that live in the guano of cavernicolous vertebrates (_Gyna_, _Acanthogyna_, _Dyscologamia_, _Pycnoscelus_). (4) =Troglobies=: Cockroaches that apparently cannot live outside of caves and which show very marked adaptive characters (_Alluaudellina_, _Nocticola_, _Spelaeoblatta_, _Typhloblatta_). For complete discussions of these groups including descriptions of the adaptive characters shown by certain genera, the original sources should be consulted.

Although we know very little of the ethology of most of the cavernicolous cockroaches, it is intriguing that three of the six known species of _Nocticola_ are cave dwellers, two are inhabitants of termite nests (p. 315), and one (_N. bolivari_) was found under stones and cement blocks (Chopard, 1950b). In the rather extensive list of cavernicolous cockroaches only two (_Arenivaga grata_ and _Parcoblatta_ sp.) were taken from caves in North America north of Yucatan. All other records are from Africa, Asia, Central America, Europe, West Indies, East Indies, and the Philippine Islands. This we find puzzling. Packard (1888) in his extensive study of the cave fauna of North America listed no cockroaches. Dearolf (1941) found only the above-mentioned _Parcoblatta_ in one of 37 caves in Pennsylvania. Kohls and Jellison (1948) listed no cockroaches among the arthropods from six bat caves in Texas. We would expect _Periplaneta americana_ to inhabit mines in North America, but we have found no such records. Have cockroaches been ignored in fauna collections from North American caves, or has our cave fauna been less extensively studied than that of other parts of the world?

The two species of cockroaches found in mines (_Blattella germanica_ and _Periplaneta americana_) are also found in caves. For this reason we have included them in the list headed Cavernicolous Cockroaches. On the other hand, the cockroaches found in animal burrows are generally different species from those found in caves, so we have grouped these together in a second list.

CAVERNICOLOUS COCKROACHES

=Alluaudellina cavernicola=

Tanganyika.--From Kulumusi caves, near Tanga. The eyes of this cockroach are reduced to a pair of slender streaks (Shelford, 1910a; Chopard, 1932a).

East Africa.--Chopard (1936).

=Apotrogia angolensis=

Belgian Congo.--A troglophile without well-marked adaptive characters. Collected in moist sand on floor of a sandstone grotto inhabited by bats (Chopard, 1927, 1950a). Taken in many caves in Bas Congo (Leleup, 1956).

=Apteroblatta perplexa=

East Africa.--Accidental inhabitant of cave (Chopard, 1936).

=Arenivaga grata=

Arizona.--"A female and many nymphs were taken by Flock in the guano in a bat cave in the Tucson Mountains" (Ball et al., 1942).

=Aspiduchus borinquen=

Puerto Rico.--In limestone cavern by thousands in grass and on walls (Rehn and Hebard, 1927; Rehn, J. W. H., 1951a).

=Aspiduchus cavernicola=

Puerto Rico.--In limestone cave, in caves inhabited by bats, and apparently seen in other caves well removed from entrance. "In this latter situation great numbers were seen on the side walls and roof" (Rehn, J. W. H., 1951a).

=Blaberus atropos=

Yucatan.--Found once, in Xmahit cave (Pearse, 1938).

=Blaberus craniifer=

Yucatan.--Collected within three caves, near the entrances (Pearse, 1938).

=Blaberus giganteus=

Panama.--Two males and several nymphs were taken under rocks in the second chamber of the Chilibrillo cave; some also were on the walls (Caudell, 1924).

=Blatta lateralis=

Turkmen S.S.R.--All stages, but more often females and nymphs, were found in the middle and back part of Bakharden cavern, which was inhabited by tens of thousands of bats (Vlasov, 1929).

=Blatta orientalis=

Turkmen S.S.R.--All stages found in front part of Bakharden bat cave. This cave was uninhabited by man but supported a variety of other animals (Vlasov, 1929).

=Blattella germanica=

South Africa.--Numerous in a gold mine on the Witwatersrand (Porter, 1930).

Tonkin.--Chopard (1929a); Colani (1952).

=Byrsotria fumigata=

Cuba.--Cueva de las Cucarachas, La Pantana, Baracoa, Oriente Province: 21 specimens, "It is evident ... that the species is also a cave inhabitant" (Rehn and Hebard, 1927).

=Deropeltis erythropeza=

East Africa.--Found at entrance of cave; not a strictly cavernicolous form according to Chopard (1936).

=Ectobius pallidus=

France.--Nymph in cave in Basses-Pyrénées, accidental inhabitant (Chopard, 1936).

=Ectobius vittiventer=

Italy.--In detritus at base of entrance shaft of Acquaviva cave in the Venezia Tridentina (Conci, 1951).

=Ectobius= sp.

Italy.--Found in the heap of saprophytic detritus at the base of the entrance shaft in the Acquaviva cave (Conci, 1951).

=Ergaula scarabaeoides=

Sumatra.--West coast (Hebard, 1929).

Malaya.--Found burrowing in bat guano among stones at entrance to caves in Selangor (Chopard, 1919, 1929).

=Euthyrrhapha nigra=

Madagascar.--Three males and six females in guano in Antsinomy grotto (Chopard, 1949a).

=Gyna kazungulana=

East Africa.--This species is especially found in caves although it shows no special adaptive characters. It is a typical guanobe (Chopard, 1936).

=Gyna maculipennis=

Belgian Congo.--Troglophile, guanophile. Found in two caves in Lualaba (Leleup, 1956).

=Gyna tristis=

Belgian Congo.--In three caves in Uele (Leleup, 1956).

=Heterogamodes krügeri=

North Africa.--An accidental inhabitant of caves (Chopard, 1938).

=Holocompsa zapoteca=

Yucatan.--Common throughout rather dry, dusty caves in southern Yucatan (Pearse, 1938).

=Hoplosphoropyga babaulti=

Stated to be a troglophile by Chopard (1938).

=Nocticola caeca=

Philippine Islands.--Bolívar (1892).

=Nocticola decaryi=

Madagascar.--A true troglobite according to Chopard (1945).

=Nocticola simoni=

Philippine Islands.--Bolívar (1892).

=Parcoblatta= sp.

Pennsylvania.--Found in Merkle cave, Berks County (Dearolf, 1941).

=Periplaneta americana=

IN CAVES

East Africa.--Its presence in the cave at Shimoni was thought to indicate that man had sought refuge there and brought the cockroaches in with baggage or provisions (Chopard, 1936).

India.--Many present in cave at Vengurla, the floor of which was covered with bird guano (Abdulali, 1942).

Madagascar.--Thought to have been introduced into the cave entrance by man (Chopard, 1945, 1949a).

IN MINES

Great Britain.--In a coal mine at Pontewydd where they had been established for some years (Lucas, 1916). In the Pentre Pit mine where they were abundant (Lucas, 1918). Abundant in a Welch mine 2,166 feet below the surface (Lucas, 1925). This species was found quite commonly in a number of South Wales coal mines; in one deep mine a white-eyed mutant form comprised about 5 percent of the cockroach population for the preceding 11 years (Jefferson, 1958).

India, western Bengal.--Very numerous in coal mines where the sole food apparently was human faeces (Chandler, 1926).

South Africa.--Numerous in four deep-level gold mines on the Witwatersrand.

Sumatra.--Numerous males and females from Sawah Lunto "'from a coal mine where they lived in great numbers on the faeces of miners'" (Hanitsch, 1929).

=Periplaneta australasiae=

Sarawak.--Found swarming on walls of caves and in soft bird guano in company with _Symploce cavernicola_ (Moulton, 1912).

Tonkin.--Chopard (1929a); Colani (1952).

=Periplaneta cavernicola=

Malaya.--Taken on walls of inner caverns, where they were particularly abundant (Chopard, 1919).

=Periplaneta lata=

Tonkin.--Chopard suggested that its presence in caves is probably linked with man (Chopard, 1929a; Colani, 1952).

=Periplaneta= sp.

Malaya.--From a cave in Jalor (Annandale et al., 1913).

=Perisphaerus= sp.

Malaya.--The wingless females and nymphs mined in bats' guano in a cavern of the Jalor caves (Annandale, 1900).

=Polyphaga aegyptiaca=

Turkmen S.S.R.--Females found in front part of Bakharden bat cave on several occasions (Vlasov, 1929).

Turkey.--At Magharadjik and Arab Dede, found in caves with various other animals (Lindberg, 1954).

=Polyphaga= sp.

Burma.--Hsin Dawng Cave, S. Shan States, 1 immature male under stone in complete darkness (Chopard, 1924b).

=Pycnoscelus niger=

Tonkin.--Apparently not an accidental inhabitant as nymphs were present (Chopard, 1929a; Colani, 1952).

=Pycnoscelus striatus=

Malaya.--Found burrowing in bats' guano at entrance to caves in Selangor, where it was very abundant 50 to 600 feet from entrance; also on walls of inner cavern (Chopard, 1919, 1929). In the absence of other evidence, the presence of _P. striatus_ in a cave indicates that bats also inhabit the cave (Chopard, 1929a).

=Pycnoscelus surinamensis=

Assam.--Found 300 to 400 feet from entrance of Siju cave in the Garo Hills (Chopard, 1924b).

South Celebes.--Hanitsch (1932).

=Spelaeoblatta gestroi=

Burma.--Chopard stated that this species shows marked characteristics of adaptation to a life in darkness (Bolívar, 1897; Annandale, 1913; Chopard, 1919).

=Symploce breviramis=

South Celebes.--Hanitsch (1932).

=Symploce cavernicola=

Sarawak, Borneo.--Swarming on walls of caves and in soft bird guano on the cave floor (Moulton, 1912). Hanitsch (1931) noted that this species was first recorded by Shelford from a cave in Sarawak and that there is a series from a cave in the Oxford University Museum, taken by Banks in 1928.

Malaya.--On the walls of the inner cavern of a cave at Biserat; the insects covered the walls in places (Chopard, 1919).

Sumatra.--From Baso cavern, on the west coast (Hebard, 1929).

=Symploce curta=

South Celebes.--Hanitsch (1932).

=Symploce remyi=

Tonkin.--This seems to be a true cavernicolous species (Chopard, 1929a; Colani, 1952).

=Tivia macracantha=

Belgian Congo.--A troglophile without well-marked adaptive characters (Chopard, 1950a). At Haut-Katanga, troglophile and guanophile (Leleup, 1956).

=Tivia= sp.

Madagascar.--Last-stage nymphs captured in guano in Antsinomy grotto (Chopard, 1949a).

=Typhloblatta caeca=

India, Assam.--An eyeless species with noticeably elongated appendages (Chopard, 1945).

=Typhloblattodes madecassus=

Madagascar.--Unpigmented integument and reduced eyes (Chopard, 1945).

=Xestoblatta immaculata=

Panama.--Found under rocks on guano-covered floor of the Chilibrillo bat caves (Caudell, 1924).

=Unidentified cockroaches=

Malaya.--The walls of a cave were covered by dense groups of a species of "_Blatta_" (Annandale, 1900).

England.--"The chief insect pests of the mines are cockroaches, which often swarm in hot mines and those with pit pony stables...." (Hardy, 1941).

COCKROACHES FROM THE BURROWS OF VERTEBRATES

=Arenivaga apacha=

Arizona.--In the nests of wood rats, _Neotoma_ sp. (Hebard, 1917).

=Arenivaga bolliana=

Texas.--In the nests of wood rats, _Neotoma_ sp. (Hebard, 1917; 1943a).

=Arenivaga erratica=

Arizona.--The wingless females were commonly found in burrows of _Dipodomys spectabilis spectabilis_ Merriam, the kangaroo rat. The winged males were never found in the burrows (Vorhies and Taylor, 1922). Found most commonly in wood-rat and ground-squirrel dens in the desert regions (Ball et al., 1942).

=Arenivaga floridensis=

Florida.--Found in a burrow of _Peromyscus polionotus rhoadsi_ (Bangs), the white-footed mouse (Young, 1949).

=Arenivaga roseni=

Turkmen S.S.R.--Occasionally found in burrows of _Rhombomys opimus_ Lichtenstein; in the burrows of the desert turtle, _Testudo horsfieldi_ Gray; and frequently in burrows of the ground squirrel, _Spermophilopsis leptodactylus_ Lichtenstein (Vlasov, 1933; Vlasov and Miram, 1937).

=Arenivaga tonkawa=

Texas.--An immature specimen was found in a prairie-dog hole (Hebard, 1943a).

=Cariblatta lutea=

Florida.--It has been taken in burrows of the pocket gopher, _Geomys_ sp. (Hubbell and Goff, 1940).

=Euthlastoblatta abortiva=

Texas.--In the nests of wood rats, _Neotoma_ sp. (Hebard, 1917).

=Parcoblatta fulvescens=

Texas.--In the nests of wood rats, _Neotoma_ sp. (Hebard, 1917).

=Polyphaga aegyptiaca=

Turkmen S.S.R.--Nymphs and adult females were often found in burrows of the sand mouse, _Rhombomys opimus_ (Vlasov, 1933).

=Polyphaga indica=

Turkmen S.S.R.--This species prefers sandy soils where it can be found in burrows of _Spermophilopsis leptodactylus_ and _Pallasiomys meridionalis pennicilliger_ Heptner (Vlasov and Miram, 1937).

=Polyphaga saussurei=

Tadzhikistan.--Found in burrows of turtles and rodents (Zmeev, 1936).

Turkmen S.S.R.--Nymphs and adult females are common in burrows of _Rhombomys opimus_ and in burrows of _Testudo horsfieldi_. Its principal habitat is rodent burrows in loess dust, where it is not infrequently found in the food stores of the host (Vlasov and Miram, 1937).

=Pycnoscelus surinamensis=

Texas.--In the nests of wood rats, _Neotoma_ sp. (Hebard, 1917).

DESERT HABITATS

There is relatively little ecological information about cockroaches that live in deserts, even though certain species, notably _Polyphaga aegyptiaca_, have long been known to inhabit arid zones. In fact, so little is known about the ecology of arid-zone insects in general that it is more a subject for research than for review (Pradhan, 1957). In their account of the cockroaches of Northern Kenya and Jubaland, Kevan and Chopard (1954) describe in some detail the vegetational areas of this arid desert or semidesert country, which averages only about 10 inches of rain per year. The other sources that are cited below contain very little more biological information than the abstracted material that is given under each species.

Nearly all the Polyphaginae are said to be marked xerophiles whose distribution coincides with that of the deserts (Bei-Bienko, 1950). With the exception of _Arenivaga floridana_, the species of Polyphaginae in the United States all occur in the Southwest, where they are (with a few exceptions) the only cockroaches that inhabit the desert regions proper (Hebard, 1917). The Polyphaginae reach their greatest diversity in the deserts of Northern Africa and Anterior and South-Central Asia (Bei-Bienko, 1950). Some of the desert-inhabiting species have also been found under nondesert conditions. This only further exemplifies the plasticity of cockroaches in adapting to different environments.

The ability of desert insects to live under what appear to be extremely unfavorable conditions has been abundantly illustrated by Pradhan (1957). Uvarov (1954) has pointed out that a desert "covers a great variety of landscapes, which provide desert animals with a wide range of habitats, some of them offering very favorable conditions for life." Pradhan (1957) stated that many desert animals avoid the extremes of desert climates by choosing suitable microclimates for diurnal resting places, that a permanent or temporary underground existence is very common among insects in arid zones, and that many nocturnal Orthoptera burrow into the soil or hide under stones where temperatures are lower. For example, the type of _Parcoblatta desertae_ was found under a boulder on the bare desert (Rehn and Hebard, 1909).

Symbiosis with burrowing animals is another solution to the problem of existence in the desert; in fact, symbiosis is a mode of life adopted by nearly half of the desert cockroaches about which we have any information. Vlasov and Miram (1937) found _Polyphaga indica_, _Polyphaga saussurei_, and _Arenivaga roseni_ in the burrows of

rodents and desert turtles. In the desert regions of Arizona, females of _Arenivaga erratica_ were found commonly in burrows of the kangaroo rat (Vorhies and Taylor, 1922) and in dens of wood rats and ground squirrels (Ball et al., 1942). _Arenivaga apacha_ and _Arenivaga bolliana_ have also been found inhabiting the nests of wood rats (Hebard, 1917; 1943a). Bei-Bienko (1950) has suggested that the adaptation of desert-inhabiting cockroaches to rodent burrows might enable these insects to survive in the severe climatic conditions of deserts in summer.

Under desert conditions in southern Arizona, the relative humidity outside of the burrows of the kangaroo rat is 1 to 15 percent during the day and 15 to 40 percent at night; but inside the burrows the relative humidity is 30 to 50 percent, and the temperature, even during the day, is below 30° C. (Schmidt-Nielsen, 1949). Thus by living in rodent burrows during the day and going outside at night, the desert cockroaches could avail themselves of the most favorable microclimates obtainable. Presumably whatever food these insects eat provides them with sufficient water to enable them to survive under desert conditions. Bodenheimer (1953) has suggested that the extent of utilization of dew, which is sometimes heavy in the desert, should be investigated; he stated that tenebrionid beetles have been seen in the early morning eating dry [dead?] herbs that were still wet with dew. It is obvious that there is a need for additional detailed information without which we can only guess about the ecology of desert cockroaches.

In the following list we have cited only those species that were stated to have been found under desert conditions. Undoubtedly, related species that have been taken in similar localities are also desert-inhabiting forms, as, for example, other species of _Arenivaga_ that were collected in Texas by Hebard (1943a). In the absence of specific information linking such other species with deserts, we have arbitrarily relegated those forms to the section on outdoor habitats. In addition to the species listed below, desert cockroaches are said to be found in the following genera: _Anisogamia_, _Mononychoblatta_, and _Nymphytria_ (Chopard, 1938).

DESERT COCKROACHES

=Agis orientalis=

Northern Kenya.--In desert-grass and thorn-bush country; scattered, dry tufts of grasses interspersed among acacia bush and scattered trees (Kevan and Chopard, 1954).

=Arenivaga apacha=

_U.S.A._--Inhabits desert regions of the Southwest, has been found in nests of wood rats (Hebard, 1917).

=Arenivaga bolliana=

_U.S.A._--On gravelly hillocks, in scattered scrub, and in the nests of wood rats in Texas. It is a desert inhabitant in the Southwest (Hebard, 1917; 1943a).

=Arenivaga erratica=

_U.S.A._--Inhabits desert regions of the Southwest (Hebard, 1917). In Arizona it has been found in rodent burrows in the desert (Vorhies and Taylor, 1922; Ball et al., 1942).

=Arenivaga roseni=

Turkmen S.S.R.--Predominantly found in burrows in sand; all stages "swim" in sand and loess dust (Vlasov and Miram, 1937).

=Blattella vaga=

Arizona.--Found in small numbers on the dry desert (Flock, 1941a).

=Compsodes schwarzi=

_U.S.A._--Occurs in the Southwest where it is confined to the desert and semidesert mountainous areas, rarely being found on the desert floor (Hebard, 1917). Taken in an ant nest in mountains of Arizona (Ball et al., 1942).

=Cyrtotria capucina=

Eastern Africa.--"Commonly met with under débris, the apterous females being most frequent." Thorn-bush country (Kevan and Chopard, 1954).

=Derocalymma lampyrina=

Northern Kenya.--Very abundant; both sexes under débris in desert-grass and thorn-bush country (Kevan and Chopard, 1954).

=Derocalymma porcellio=

Northern Kenya.--Taken in upland grassland and bush (Kevan and Chopard, 1954).

=Deropeltis autraniana=

Northern Kenya.--In thorn-bush country (Kevan and Chopard, 1954).

=Deropeltis melanophila=

Northern Kenya.--"Very commonly found at the base of tufts of grass and other débris, the apterous female particularly in the latter situation"; in upland grassland near forest; in thorn-bush country (Kevan and Chopard, 1954).

=Deropeltis nigrita=

Northern Kenya.--Taken in upland grassland and bush (Kevan and Chopard, 1954).

=Eremoblatta subdiaphana=

_U.S.A._--Apparently found in greatest abundance in the extreme desert conditions of the southwestern United States (Hebard, 1917). Two small groups of males were observed in the midst of the sandy desert north of Yuma, Ariz.; these insects alternately flew and ran over the sand in the hot sun while headed in a southwesterly direction (Wheeler, 1911).

=Euandroblatta palpalis=

Northern Kenya.--In desert-grass and thorn-bush country (Kevan and Chopard, 1954).

=Heterogamodes rugosa=

Northern Kenya.--"All from desert grass and thorn bush (on sand)." It was stated (under discussion of _Tivia fulva_) that _Heterogamodes_ females live more or less buried in the sand (Kevan and Chopard, 1954).

=Namablatta bitaeniata=

Southwestern Africa.--Limited in distribution to the more arid portions, being peculiar to extreme desert conditions (Rehn, 1937).

=Nauphoeta punctipennis=

Northern Kenya.--In desert grass and thorn bush; "probably the commonest of all the medium-sized cockroaches occurring in the area under discussion, coming very freely to light" (Kevan and Chopard, 1954).

=Parcoblatta desertae=

_U.S.A._--In the desert and semidesert mountainous areas of the Southwest; it is rarely found on the desert floor (Hebard, 1917). Found under boulder on bare desert (Rehn and Hebard, 1909).

=Polyphaga aegyptiaca=

Caucasus.--The wingless female was found buried in sand and dust (Burr, 1913).

Turkmen S.S.R.--Although this species is secondarily encountered in dwellings and courtyards, it is a very characteristic insect of the Trans-Caspian deserts; the females are encountered fairly frequently as inhabitants of sand, where they run slowly over the surface, or dig themselves into the sand to continue their forward motion not far below the surface (Fausek, 1906). Uvarov (_in_ Chopard, 1929b) indicated that females of this genus are found in various desert localities, particularly where vegetative debris occurs, but they are not strictly attached to sandy terrain.

=Polyphaga indica=

Turkmen S.S.R.--This species prefers sandy soils where the nymphs, alate males, and wingless females "swim" readily through the sand; they can also be found in the burrows of desert animals (Vlasov and Miram, 1937).

=Polyphaga saussurei=

Turkmen S.S.R.--Its principal habitats are rodent burrows in loess dust and burrows of the desert turtle (Vlasov and Miram, 1937).

=Supella hottentotta=

Northern Kenya.--Taken in bushes by dry river bed and in desert-grass and thorn-bush country at several stations (Kevan and Chopard, 1954). "...taken with light at night running on bark of a large acacia tree" (Rehn, 1947).

=Symploce kevani=

Northern Kenya.--In desert grass and thorn-bush country (Kevan and Chopard, 1954).

=Theganopteryx straminea=

Northern Kenya.--Taken at three stations in desert grass and thorn bush (Kevan and Chopard, 1954).

=Tivia brunnea=

Northern Kenya.--In open sandy, riverine bush (scanty ground cover among acacia trees and doum palms) (Kevan and Chopard, 1954).

=Tivia fulva=

Northern Kenya.--In desert grass and thorn bush; distributed in semidesert areas south of Sahara; the apterous females probably live buried in sand (Kevan and Chopard, 1954).

=Tivia obscura=

Northern Kenya.--In desert grass and thorn bush (Kevan and Chopard, 1954).

AQUATIC HABITATS

The so-called aquatic or amphibious cockroaches are all members of the subfamily Epilamprinae (Chopard, 1938). These forms are not nearly as aquatic as water beetles or aquatic Hemiptera, but in their relations to water they behave differently from nonamphibious cockroaches, which tend to avoid water except for drinking. There are apparently no special morphological characteristics that distinguish amphibious cockroaches (Shelford, 1907, 1909a; Chopard, 1938), although Takahashi (1926) listed several characters that he considered made _Opisthoplatia maculata_ adapted for an aquatic life: (1) Back of body easily wetted; (2) long hairs on underside of thorax trap air; (3) terminal abdominal spiracles open into tubes that extend rearward; (4) long hairs on ventral surfaces of cerci "protect" terminal abdominal spiracles. Annandale (1906) also suggested that the position of the posterior abdominal spiracles, at the base of tubes that project rearward from beneath the seventh tergite, are an adaptation to an aquatic life. However, as Shelford (1907) and Chopard (1938) have pointed out, this same feature may be observed in many terrestrial cockroaches. The legs of amphibious cockroaches are similar to those of nonaquatic species and are not modified for swimming (Shelford, 1909a; Takahashi, 1926).

Biological observations have been made on relatively few species, but representatives of at least six genera occur in quasi-aquatic habitats. Strictly speaking, these cockroaches live on land at the edges of streams or pools and spend relatively brief periods in the water. A few species are found in water-filled bromeliads. The behavior of the known amphibious species of cockroaches in relation to their habitats is discussed below.

AMPHIBIOUS COCKROACHES

=Audreia bromeliadarum=

Panama.--These insects when disturbed would dive into the water that had collected in the base of the bromeliad; they would disappear beneath the surface and remain submerged for some considerable time (Caudell, 1914).

=Dryadoblatta scotti=

Trinidad.--This species was taken from the leaf bases of _Tillandsia_ sp. at 3,100 feet; water had collected between the leaves and the insect was presumed to be more or less amphibious (Scott, 1912). Subaquatic in the bromeliad _Glomeropitcairnia erectiflora_: "This large and handsome species [_D. scotti_] is very common in the larger, water-filled, epiphytic bromeliads of the rain forest. Within these plants it is usually to be found, often in considerable numbers, just above the surface of the water or partly immersed in it. The cockroaches will descend rapidly into the water when alarmed and probably obtain their nourishment from the accumulated organic matter in the water. Floating material is probably taken and it seems less likely that they feed below the surface. They appear to be ovoviviparous." (Princis and Kevan, 1955.)

=Epilampra abdomen-nigrum=

Puerto Rico.--Abundant in wet "malojillo" meadows. The nymphs swim easily and remain under water for long periods, as do the adults (Seín, 1923; Wolcott, 1950).

Panama.--A swimming nymph, captured in a dipper with mosquito larvae in a lagoon of the Rio Chilibre, was kept under observations in an aquarium. If disturbed, the insect dived into the water from floating vegetation and swam rapidly below the surface for a minute or two. Finally becoming quiescent, the cockroach would then cling to submerged roots; twice it remained still for 15 minutes before climbing to the surface, where it remained for five or more minutes before emerging completely (Crowell, 1946).

=Epilampra annandalei=

Lower Burma.--One male and three nymphs were collected in the Dawna Hills by Annandale who made the following observations: "The wingless specimens were under stones in a jungle stream and behaved just as the one I obtained in Chota, Nagpur, did [Annandale, 1906]. The winged specimen was under a stone at the edge of the stream, but swam readily. It did not seem so much at home in the water, however, and apparently could not, owing to the wings, raise the tip of its abdomen above the surface." (Shelford, 1909a.)

=Epilampra= sp.

Siamese Malay States.--Wingless females rested on floating logs from which they would dive into the water upon the least disturbance; they remained under water for several minutes, then surfaced beneath the shelter of the log. In the jungle all females were taken either in the water or among matted roots on the sides of the

stream. Winged males were seen rising from the surface of the water (Annandale, 1900).

Sarawak.--All specimens were immature; they swam and dived well, but were soon drowned if prevented from rising to the surface to breathe. "When at rest the body of the cockroach is almost entirely submerged, the tip of the abdomen alone projecting above the surface of the water; the abdomen moves gently up and down and every 30-40 seconds a bubble of air issues from the prothoracic spiracle on each side." (Shelford, 1901, 1916.)

India.--A nymphal female, found in a jungle stream at Chota Nagpur, could swim with belly or back upward. When held under water it drowned in a few minutes. The tip of the abdomen was held out of water (Annandale, 1906).

Shelford (1907) has suggested that the immature stages of terrestrial species of _Epilampra_ may well be amphibious. This is an area that could profit by more field observations.

=Opisthoplatia maculata=

Formosa.--Invariably found under or between rocks near mountain streams. The wingless adult and the nymph have similar habits. Normally the cockroach lives on land, and when it goes into the water it returns to land within a few minutes. This cockroach rarely swims, but when it does, it maintains its body in a horizontal position just below the surface of the water. Ordinarily, it walks on the river bottom or on water-covered rocks. This insect feeds on decayed leaves and, according to Shikano, it will eat human feces. (Takahashi, 1926.)

This species has a large number of long hydrophobic hairs on the ventral sides of the thorax and anterior abdominal segments. When the insect submerges, air is trapped in these hairs. The thoracic and one pair of abdominal spiracles open into the bubble of trapped air. However, the insect apparently does not use this plastron of air to replenish its tracheal air supply, but, like _Rhicnoda natatrix_ (see below), it inspires air while at the surface through its posterior abdominal spiracles and expires air into the bubble under the thorax. While the insect is submerged, the air bubble increases in volume until part of it breaks away and floats to the surface. (Takahashi, 1926.)

=Opisthoplatia orientalis=

Formosa.--Lives on or in swampy ground (Takahashi, 1924).

=Rhicnoda natatrix=

Sarawak.--Immature cockroaches were found in sodden leaves at the edge of a pool, where they rested for hours at a time. Generally the fore part of the body was in the water but the tip of the abdomen was always in air. When disturbed the insects dived into the water and hid under sticks and stones on the bottom. Air is inspired through the posterior abdominal spiracles, when they projected above the water surface, and expired through the thoracic spiracles. In experiments in which the insects' abdomens were held immersed in water, with the thorax exposed, the insects

died in 6 to 12 hours or less. (Shelford, 1907.)

=Stictolampra buqueti concinula=

Westsumba.--Found under moist fallen leaves on gravelly shore of Melolo River. The nymphs distinguished themselves through their amphibious mode of life and were often good swimmers (Princis, 1957a).

=Unidentified epilamprines=

Brazil.--These cockroaches were found under stones at the side of a rocky stream at Ouro Preto. When disturbed they ran down under the surface of the water and hid under stones at the bottom. When thrown on the water surface, they were helpless, and to get beneath the water surface they had to walk down some object. When they had penetrated the surface film they could swim freely. Specimens kept in jars lived several days with only a portion of their abdomens exposed to the air. (Bristowe, 1925.)

OUTDOOR HABITATS

This category is a catchall for all cockroaches that are not limited to the more circumscribed habitats that have been previously considered. Some cockroaches in this section select specific microhabitats (e.g., _Cryptocercus_ spp., which live exclusively in rotten logs; and _Neoblattella dryas_, _N. eurydice_, and _N. grossbecki_ in bromeliads). Others are found in a wide variety of habitats (e.g., _Ischnoptera deropeltiformis_ and _Parcoblatta_ spp.). But some species are so little known that their actual habitats are barely suggested in the collection data.

Williams (1941) made an ecological study of the floor fauna of the Panama rain forest. He found Orthoptera (nearly all were unidentified nymphal cockroaches) in the litter of dead leaves, twigs, and other plant products in over 90 percent of the quadrats he examined. These insects represented about 0.25 percent of the total animal population.

Delamare Deboutteville (1948) made a quantitative study of the animal population in suspended soil that had accumulated between the roots of forest epiphytes of the lower Ivory Coast. He analyzed 2 dm.^3 samples of soil from an epiphyte located 45 meters above ground on a main branch of _Parinarium_, with these results: _Horizon A._--Superficial zone of large rootlets, 6 cm. deep: 2 cockroaches, 4 arachnids, and 4 beetles. _Horizon B._--Zone of fine rootlets, 6 cm. deep: 6 cockroaches and numerous other arthropods. _Horizon C._--Humid zone, 8 cm. deep: 7 cockroaches and numerous other arthropods. Plants, such as _Palissota_, were also living in this very original biotype.

The species of cockroaches listed below have been found in the following kinds of outdoor microhabitats: In jungle, forest, and woodlands they have been found in rotten wood; under bark of living, dead, and fallen trees; in decay cavities in trees; burrowing in living bark; on foliage of trees, shrubs, bushes, and low herbage; on vines and in bromeliads and epiphytic ferns; under signs on trees and stumps; in piles of logs and firewood; under dead leaves and debris; in and under decaying fruit on

the ground. Cockroaches have been found between the leaves and under leaf sheaths of sugarcane, corn, and other grasses; under dry fibers and fronds of coconut trees; in hollow stems and bases of tree-fern fronds; under bracts of banana blossoms and in bunches of bananas (p. 146). Cockroaches also inhabit abandoned cocoons and larval tents, wasp nests, ant nests, termite nests, bird nests, rat nests, and burrows of other rodents (pp. 23-25, 310-319). Cockroaches have been found in rock crevices and under rocks; under boards and other objects on ground; under seaweed, drift, and other debris on beaches; burrowing in soil and under clods of earth; in marshes and swamps; in dumps and rubbish heaps.

The above list does not exhaust the available outdoor microhabitats that cockroaches find suitable for their continued existence, but it is fairly representative. Although we have no measurements to substantiate this conclusion, we suggest that the microhabitats cited above have a more constant temperature and a relatively higher humidity than is provided by the surrounding macrohabitats. We would expect insects such as cockroaches, whose water balance is dependent on a continuous supply of fluid water or moist food, to seek moist environments or to avoid situations in which their transpiration might increase. Deviations, presumably brief, from this expected behavior must occur to account for the cockroaches that are found under relatively unfavorable environmental conditions. Despite the apparent preference for cryptic habitats, some cockroaches are found in hot sunlight (_Ellipsidion_ spp.; Tepper, 1893); Rehn (1945) has stated that many kinds are diurnal rather than nocturnal. Movement of cockroaches between habitats may be assumed to occur; but movement from an unfavorable environment to a more favorable one, following a shift in water balance, has not been observed in nature; however, laboratory experiments suggest that the mechanism for mediating such behavior is present in some species of cockroaches (Gunn and Cosway, 1938; Roth and Willis, 1952a). Obviously, additional research is needed on the bionomics of all species. Further conclusions based on current limited knowledge can only be speculative and possibly misleading.

COCKROACHES FROM OUTDOOR HABITATS

(Except Amphibious, Desert, and Cavernicolous Forms)

=Aglaopteryx absimilis=

Puerto Rico.--Living in rotten, wooden fence; living between leaves of _Samanea saman_ and in abandoned cocoons of _Megalopyge krugii_ on bucare trees (Wolcott, 1950).

Leeward Islands.--On coconut tree (Princis and Kevan, 1955).

=Aglaopteryx facies=

Puerto Rico.--As _diaphana_, in dead branch 10 feet above the ground on Mona Island (Hebard, 1917). In trunks of trees under bark and very often in abandoned cocoons of the "plumilla" (Seín, 1923). On rotten, wooden fence; in empty cocoons of _Megalopyge krugii_ on trunks of bucare trees, _Erythrina glauca_; on trunk of _Inga

laurina_; in larval tents of _Tetralopha scabridella_ on _Inga vera_ (Wolcott, 1936). In large numbers in nests of the gray kingbird (Wolcott, 1950).

=Aglaopteryx diaphana=

West Indies.--In Cuba, under corky bark of large tree in open; Jamaica, under loose bark of shade trees and in bracts of banana blossoms; in bromeliads and hollow bases of dead tree-fern fronds (Rehn and Hebard, 1927).

=Aglaopteryx gemma=

Florida.--On Long Key, under coquina boulder in heavy scrub; under loose, dry fibers near head of standing coconut palm (Rehn and Hebard, 1912). Climbing on roots of red mangrove, _Rhizophora mangle_, in swamp; under loose bark on trunk of _Exothea paniculata_ in dense jungle; under limestone boulder in keys scrub; under signs on oaks, sweet gum, and longleaf pines in southeastern and southern States (Hebard, 1917). Infrequent in the shrub growth of the Sandhills habitat (Friauf, 1953).

Texas.--In undergrowth of pine forest; under sign on oak near river; in _Tillandsia_ sp. (Hebard, 1917). Usually in hiding places on trees; only once found under a stone on ground (Hebard, 1943a).

=Allacta similis=

Hawaii.--Common in hollow stems and under bark (Swezey and Williams, 1932).

=Amazonina emarginata=

Trinidad.--On low herbage, on hibiscus at night, and in banana bunch (Princis and Kevan, 1955).

=Anaplecta asema=

Panama.--Under dead leaves in jungle (Hebard, 1920).

=Anaplecta decipiens=

Costa Rica.--In decayed leaves (Rehn, 1906).

=Anaplecta fallax=

Costa Rica.--Under stones on borders of Surubres River (Rehn, 1906).

=Anaplecta hemiscotia=

Panama.--Under rubbish at edge of jungle and in overgrowth of heavy vines on low bushes (Hebard, 1920).

=Anaplecta lateralis=

Panama.--Under drift on edge of coral-sand beach (Hebard, 1920).

=Arenivaga bolliana=

Texas.--In dense jungle brush of the river plain; on gravelly hillocks in scattered scrub; under debris and leaf mold under mesquite trees; in rat's nests, _Neotoma_ sp.

(Hebard, 1917). In dry earth under bush; inhabits litter on ground and nests of rats (Hebard, 1943a).

=Arenivaga floridensis=

Florida.--Male on ground under leaves of cabbage palmetto (Blatchley, 1920). Females in sand under boards and debris along lake shore (Friauf _in_ Cantrall, 1941). Infrequent on bare soil and ground under vegetation in the longleaf-pine flatwoods habitat (Friauf, 1953). In rodent burrow (Young, 1949).

=Arenivaga grata=

Texas.--Under stones in upper canyon; under rocks in pine-oak forest; from oak-manzanita forest along dry stream bed (Hebard, 1943a).

=Aristiger histrio=

Malaya.--Lives freely on bushes and flowers of _Passiflora_ sp. (Karny, 1924).

=Aspiduchus boriquen=

Puerto Rico.--"Apparently the species [as _deplanatus_] is locally numerous in suitable locations, such as caves, rock crevices and the shelter of large stones." (Rehn and Hebard, 1927).

=Audreia bromeliadarum=

Panama.--Perfectly at home in bromeliads (see p. 31) (Caudell, 1914).

=Audreia jamaicana=

Jamaica.--In bromeliads; under dead wood in dense forest (Rehn and Hebard, 1927).

=Balta godeffroyi=

Australia.--Under bark (Hebard, 1943).

=Balta quadricaudata=

Australia.--From sugarcane (Hebard, 1943).

=Balta scripta=

Australia, Queensland.--On leaves, grass, and sugarcane (Hebard, 1943).

=Balta torresiana=

Australia.--From leaves, under bark, from sugarcane (Hebard, 1943).

=Balta verticalis=

Australia.--In leaves, from tree, from sugarcane (Hebard, 1943).

=Blaberus atropus=

Trinidad.--Female in rotting log (Princis and Kevan, 1955).

=Blaberus discoidalis=

Jamaica.--Under dead coconut petioles in open spot. Gundlach found it under stones in a field in Cuba (Rehn and Hebard, 1927).

=Blaberus giganteus=

Trinidad.--Nymph in rotten palm tree (Princis and Kevan, 1955).

=Blaberus= spp.

Venezuela.--Only taken in the forests of the Orinoco near the trunks of rotten trees at night (Doumerc _in_ Blanchard, 1837).

Panama.--Among dead leaves and debris on floor of rain forest (E. C. Williams, Jr., 1941).

=Blatta lateralis=

_U.S.S.R._--Found among rocks at 2,000 or more meters elevation. It is found in cultivated areas as well as in mountainous landscapes and in semideserts (Bei-Bienko, 1950).

=Blatta orientalis=

Great Britain.--One female nymph under bark of tree 10 feet above the ground (Burr, 1900). Swarming within a rubbish heap in February (Lucas, 1912). In refuse tip under old sacks and sheets of linoleum (Hallett _in_ Lucas, 1922). Male under bark of oak far from houses (Donisthorpe, 1918). One adult female and nymph in prone dead elm 50 yards from house (Burr, 1937). An immature male at the roots of _Ballota nigra_ (Buck _in_ Gardner, 1954). Four additional records of this species outdoors away from houses (Lucas, 1920).

Southern Crimea.--Under stones, dead leaves, and detritus in small copses of _Quercus pubescens_, _Carpinus orientalis_, _Cornus mas_, _Paliurus aculeatus_, and _Dictamnus fraxinella_; 19 specimens, apparently breeding outdoors (Adelung, 1907).

North-central U.S.--Observations since 1950 indicate a marked increase in frequency and duration of infestations outdoors; observed in bare soil, vegetation, debris, alongside foundations in sodded areas, along sidewalks, and at edge of parking areas throughout the year; in some urban residential areas, the yards of whole blocks of houses were "alive" with this species on warm summer nights; in winter they have been found under stones, leaf debris, and soil near structures (Shuyler, 1956).

=Blattella germanica=

Algeria.--Under moist leaves in woods (Lucas, 1849).

California.--Under rubbish and on date palms (Herms, 1926).

Connecticut.--In city dump under loose material, very numerous (Walden, 1922). Additional infestations of dumps by this species have been reported in New York (Felt, 1926, 1928) and New Jersey (Hansens, 1949, 1950)

England.--Swarming within a rubbish heap in February (Lucas, 1912).

Formosa.--Lives among fallen leaves on the ground (Takahashi, 1924).

North-central U.S.--Reported living outdoors near buildings and in soil under basementless buildings from early summer to late fall (Shuyler, 1956).

=Blattella humbertiana=

India.--Common among decaying vegetation and on trees (Chopard and Chatterjee, 1937).

Formosa.--Normally found in sugarcane fields, pineapple fields, and grasslands where it feeds on decayed leaves and other decayed vegetable matter and dead insects. It lies concealed among and under fallen leaves and clods of earth on or close to ground and never on the upper parts of plants, except pineapple where it is found among the leaves (Takahashi, 1940).

=Blattella vaga=

Arizona.--Typically an inhabitant of irrigated fields and yards, it is found in fewer numbers on the dry desert. It is found under stones, plant debris, and clumps of earth; found in greatest numbers around decaying dates on ground (Flock, 1941a).

Texas.--Beneath duff under athel trees; rather abundant in clumps of Rhodes grass (Riherd, 1953).

=Byrsotria cabrerae=

Cuba.--In sea-coast woods: "The species [this and _Byrsotria fumigata_] are ground-dwelling, hiding under stones and other shelter" (Rehn and Hebard, 1927).

=Byrsotria fumigata=

Cuba.--Ground dwelling, hiding under stones, etc.; also a cave inhabitant (Rehn and Hebard, 1927).

=Cahita borero=

Brazil, Matto Grosso.--Beaten from tree foliage in dry scrub, from tree foliage at edge of dry riverine tangle, and from undergrowth in a dry forest area (Rehn, 1937a).

=Cahita nahua=

Honduras.--All beaten from foliage along roads or in thickets, during rainy season (Rehn, 1937a).

=Cariblatta antiguensis=

Virgin Islands, St. Croix.--Common under heaps of rubbish (Beatty, 1944).

Trinidad.--On herbage below bananas; all stages on _Hibiscus_ at night; in grass at dusk; on low herbage under old coconut (Princis and Kevan, 1955).

=Cariblatta cuprea=

Jamaica.--In leaves on leaf mold in hillside forest (Hebard, 1916a).

=Cariblatta delicatula=

West Indies.--In debris in short grass in open, Cuba. Under dead petioles of coconut palms, San Domingo. In leaves on leaf mold in hillside forest, Jamaica (Hebard, 1916a).

=Cariblatta hylaea=

Honduras.--Found at foot and on lower slopes of first ridges of the Sierra Pija, from 75 to at least 800 feet above sea level, where vegetation ranged from abandoned banana patches overgrown with _Heliconia_ and _Cecropia_ and interspersed with forest trees, at the foot of the hills, to primeval lowland forest (ceibas, figs, palms, etc.) on the slopes. In the banana patches _C. hylaea_ was found on hanging dead banana and _Cecropia_ leaves; on the slopes it was found on undergrowth foliage, hanging dead leaves, and in dead leaves on ground (Rehn, 1945a).

=Cariblatta imitans=

Panama.--Among loose leaves on leaf mold in heavy jungle (Hebard, 1916a).

=Cariblatta insularis=

Jamaica.--One of the most frequently encountered orthopterous insects in bromeliads on trees (Hebard, 1916a, 1917; Rehn and Hebard, 1927).

=Cariblatta jamaicensis=

Jamaica.--In decaying herbage (Rehn and Hebard, 1927).

=Cariblatta landalei=

Jamaica.--All specimens taken from under drying bracts of banana blossoms (Rehn and Hebard, 1927).

=Cariblatta lutea lutea=

North Carolina.--Under pine straw on ground in woods (Brimley, 1908).

Southeastern U.S.--Under dead oak leaves; under dead needles in longleaf-pine woods; in wire grass; under refuse; beaten from undergrowth in pine and oak woods (Rehn and Hebard, 1916). In undergrowth of shortleaf-pine, longleaf-pine, and oak woods; in heavy scrub in damp spot of sand dune area; from high bushes, _Ilex coriacea_ [=_lucida_] along inland swampy area (Hebard, 1916a). "The species is in large part terrestrial, being usually found among dead leaves and litter on the ground. Occasional specimens are, however, sometimes beaten from bushes. Individuals are decidedly active and are usually to be found in the greatest numbers in sandy situations" (Hebard, 1917).

Florida.--Throughout winter and spring they are frequent beneath leaves and other debris on ground, especially in dry, sandy locations (Blatchley, 1920). Friauf (1953) found this species under debris, fallen leaves, leaf mold, or decaying wood in these habitats: Dry, ruderal grassland (infrequent), scrub (frequent), sandhills (dominant), xeric hammock (infrequent), mesic hammock (dominant), pond margin (infrequent), longleaf-pine flatwoods (frequent), bayhead (occasional), low hammock

(frequent), and alluvial hammock (occasional). In the shrub stratum in these habitats: Scrub (frequent), sandhills (dominant), and xeric hammock (infrequent). In herbaceous stratum in these habitats: Sandhills (dominant), mesic hammock (dominant), and black-pine flatwoods (infrequent). On bare soil or bare sand under vegetation in these habitats: Sandhills (dominant), pond margin (infrequent), longleaf-pine flatwoods (frequent), and slash-pine flatwoods (frequent) (Friauf, 1953).

=Cariblatta lutea minima=

Florida.--Series of specimens captured on Long Key under dead petioles of coconut palm on moist ground at edges of pools of brackish water. Specimens from Key West were in dry dead grass under boards (Rehn and Hebard, 1912). Nymphs frequent under bark on decaying pine logs in pine woods; occasional in leaf mold in heavy junglelike scrub (Rehn and Hebard, 1914). In water-soaked leaves in heavy red-mangrove swamp (Hebard, 1915). Under dead petioles of coconut palm on sandy soil in grapefruit grove (Hebard, 1916a). Numerous at bases of tufts of coarse grass growing just back of sea beach (Blatchley, 1920). Friauf (1953) found this species in leaf duff, leaf mold, debris, or decaying wood in these habitats: Dry, ruderal grassland (occasional), scrub (infrequent), sandhills (infrequent), mesic hammock (infrequent), pond margin (occasional), longleaf-pine flatwoods (occasional), and low hammock (infrequent). On bare soil or bare sand under vegetation in these habitats: Longleaf-pine flatwoods (occasional) and slash-pine flatwoods (occasional). Dominant in the spartina marsh habitat in the grass stratum and duff around clumps. Frequent in the saw-grass marsh habitat in the grass stratum and, during the dry season, in decaying vegetation on the marsh floor.

=Cariblatta nebulicola=

Jamaica.--Adults in dead leaf litter alongside the trail in dense forest of tree ferns, _Podocarpus_, _Cyrilla_, and other trees; the forest was bathed in fog much of the time (Rehn and Hebard, 1927).

=Cariblatta reticulosa=

Jamaica.--In leaves on leaf mold in hillside forest (Hebard, 1916a). Moderately numerous in leaf litter in mangrove swamp; in decaying herbage (Rehn and Hebard, 1927).

=Cariblatta stenophrys=

Puerto Rico.--Between the leaves and under the leaf sheaths of corn (Sein, 1923; Wolcott, 1936).

=Cariblatta= spp.

West Indies.--The tropical species of this genus inhabit heavy forest, living among the fallen leaves resting on the leaf mold, in epiphytic bromeliads, and in dead agaves (Hebard, 1916a; Rehn and Hebard, 1927).

=Cariblattoides instigator=

Cuba.--In siftings from under sea grapes, other shrubs, and low trees (Rehn and Hebard, 1927).

=Cariblattoides suave=

Puerto Rico.--On dry limestone hills (Rehn and Hebard, 1927).

=Ceratinoptera picta=

Trinidad.--Under bark of old cacao tree (Princis and Kevan, 1955).

=Chorisoneura flavipennis=

Costa Rica.--Under stones on borders of Surubres River (Rehn, 1906).

=Chorisoneura formosella=

Jamaica.--Swept from huckleberry trees (_Vaccinium meridionale_) (Rehn and Hebard, 1927).

=Chorisoneura parishi=

Panama.--From jungle undergrowth (Hebard, 1920).

=Chorisoneura specilliger=

Panama.--In grass (Hebard, 1920).

=Chorisoneura texensis=

Florida.--"The almost impenetrable jungle on Key Largo was examined, and in its depths the two specimens of this species were secured by beating the lower branches of gumbo limbo, other trees and the lower bushes and shrubs, among which latter are to be found such tropical forms as _Ocotea catesbyana_ [=_Nectandra coriacea_] and _Citharexylum villosum_" (Rehn and Hebard, 1912). In nests of webworm and beaten from bushes of bayberry, _Myrica cerifera_, along edge of pine woods (Rehn and Hebard, 1916). Beneath dead leaves in oak woods and beaten from foliage of oak and bayberry (Blatchley, 1920). Infrequent in the tall shrub stratum of the xeric hammock habitat (Friauf, 1953).

Texas.--The great majority of specimens were beaten from foliage of bushes (Hebard, 1943a).

Southeastern and southern U.S.--In undergrowth in pine woods; beaten from shrubbery, from bayberry bushes, from lower branches of gumbo limbo and other trees, from lower bushes and shrubs in jungle, and from low oaks on hills. In Texas, beaten from tall weeds in opening in river-plain jungle scrub (Hebard, 1917).

=Chorisoneura translucida=

Panama.--In jungle vegetation, including vines covering low bushes (Hebard, 1920).

=Chromatonotus infuscatus=

Trinidad.--Males on low herbage under old cacao tree (Princis and Kevan, 1955).

=Chromatonotus notatus=

Trinidad.--Males in orchard on low herbage at night; females under refuse and in grass (Princis and Kevan, 1955).

=Comptolampra liturata=

Malaya.--Often found between dry foliage in the beakers of the epiphytic fern, _Asplenium nidus_, although the species lives mainly in bamboo bushes (Karny, 1924).

=Cryptocercus punctulatus=

North Carolina.--"They were never found except in parts of the logs [chestnut] where the decayed wood was soft, punky and wet" (Rehn and Hebard, 1910).

Oregon.--In fir logs where sap wood was soggy (Hebard, 1917).

Virginia.--In decaying chestnut and pine logs; taken six times in chestnut and once in pine (Hebard, 1917). In rotten logs in deep ravines of moist woods (Davis, 1926).

Appalachian Mountains, U.S.--In southern Virginia and eastern Tennessee, it is usually quite abundant in well-forested areas at elevations from 3,000 to 5,000 feet; "sometimes even a majority of the dead logs on a mountain side have roaches in them" (Cleveland et al., 1934). This cockroach not only lives in rotten, dead logs but also in sound logs that have been down only a few years. In Virginia it is found more often in chestnut and hemlock. "It occurs fairly often in oak, and has been found in pine, spruce, and arbor vitae.... There is little evidence that they ever leave the log and enter the ground" (Cleveland et al., 1934).

=Cryptocercus relictus=

Eastern Manchuria.--In great numbers under rotting fallen trees and in rotten dead wood (Bei-Bienko, 1950).

=Cutilia soror=

Marquesas Islands.--Males under stones and dead log (Hebard, 1933a).

Hawaii.--In soil about roots of pineapple (Illingworth, 1927). Often found about roots of grasses and weeds and other debris (Williams et al., 1931). Under stones and pineapple mulching paper (Fullaway and Krauss, 1945).

Wake Island.--Numerous, some from rotten logs. Found in bunch grass on Ocean Island (Bryan, 1926).

=Cutilia= spp.

Australia.--Frequent woods where they leave shelter soon after sunset and run actively on ground or ascend shrubs and trees in quest of prey (Tepper, 1893).

=Dendroblatta sobrina=

Panama.--Colony on tree trunk; on surface of trunk of fallen tree (Hebard, 1920).

=Diploptera punctata=

Hawaii.--"Crowds of these insects in various stages of development sometimes gather in cypress trees, in suitable chinks, in old flowerhead sheaths of palms, etc., and even more or less openly on leafy twigs, in bunch grass, and the species is at times locally abundant behind the older leaf bases of sugar cane" (Williams et al., 1931). Williams also lists the following as food plants: _Cryptomeria_, algaroba, lime trees, ripening mangoes, papayas, and oranges. However, Bianchi (personal communication, 1954) doubted that any of the above are the main dietary, because the largest populations he had seen "were found in the fairly dry litter of Star Jasmine (_Jasminum pubescens_ Willd.), well removed from any of the plants mentioned by Williams."

Raiatea, Society Islands.--Beaten out of bracken (Cheesman, 1927).

Uahuka, Marquesas Islands.--Under bark (Hebard, 1933a).

=Dryadoblatta scotti=

Trinidad.--Very common in water-filled, epiphytic bromeliads in the rain forest (see p. 31) (Princis and Kevan, 1955).

=Ectobius africanus=

Belgian Congo.--Females in forest margin and in forest undergrowth (Rehn, 1931).

=Ectobius albicinctus=

South France.--Females and young beneath stones (Blair, 1922).

=Ectobius duskei=

_U.S.S.R._--In the steppe belt, it is a very characteristic member of feather-grass steppes, where it is found in associations of typically steppe vegetation, with feather grasses at the head (_Stipa lessingiana_ and others), and on rocky slopes; it occurs frequently in cultivated fields of young crops and also in young geological strata in sections with virgin soil. The populations of this steppe cockroach average 6 to 8 individuals per square meter from the middle to the end of July. By the end of summer most individuals were observed at the bases of straw stacks with a canopy, having their south sides sheltered. This is the only species of _Ectobius_ adapted to a purely steppe biocenose. (Bei-Bienko, 1950.)

=Ectobius lapponicus=

Southeastern Europe.--Numerous under stones on Trebovic (Burr, 1898).

_U.S.S.R._--Found in wooded communities and peat bogs (in northern part of its range); males occur predominantly on herbaceous plants and bushes, but females hide under fallen leaves, moss, etc. (Bei-Bienko, 1950). It populated about 25 percent of the aspen trees in an experimental plot, feeding in galleries in the bark of young branches; there were 25 or more individuals per tree (Stark _in_ Bei-Bienko, 1950).

Germany.--Abundant in woods; in pine woods in company with _Stenobothrus vagans_ and _Tettix kraussi_. Numerous in low aspen bushes in forest. Numerous in deciduous and coniferous forests on trees and underbrush; under fallen leaves and moss; on oaks (Zacher, 1917). In foliage of young oak on top of mountain (Ramme, 1923).

Great Britain.--Under moss and dry leaves, among woodland undergrowth, and, generally, on vegetation close to the ground; occasional on bushes and trees (Lucas, 1920). Nymphs in heather in February and later; adults among rushes fringing pond in July (Lucas, 1925). Nymphs and males on rushy vegetation; unusually abundant on low herbage in dried-up swamp (Lucas, 1930).

=Ectobius nicaeensis=

France.--In dry woods, on bushes, and at the base of trees (Chopard, 1947).

=Ectobius pallidus=

Algeria.--Under stones; in moist places that are shaded and covered with plants (Lucas, 1849).

England.--Very abundant on sand dunes and among bracken in July (Buxton, 1914).

Germany.--In deciduous and coniferous forests; at edge of forest, from bare woods and bushes; numerous under leaves in oak woods and under moss (Zacher, 1917). In forest well lighted by the sun (Ramme, 1923).

Massachusetts.--Under loose lichens and bark on oak trees; under boxes, baskets, paper, etc., near houses; on Swiss chard (Flint, 1951). On roofs of houses, in shrubbery (Gurney, 1953). We have collected this species for several summers in a fairly dense, wooded area near dwellings, among fallen leaves and climbing on the erect stems and undersides of the leaves of periwinkle. Oöthecae were found on the ground under leaves and debris.

=Ectobius panzeri=

England.--Abundant on sandhills along shoreline among roots of grass (Burr, 1908). Under dead seaweed and other rubbish a few yards from shore on ground that would be washed by the sea (Lucas, 1896). Nymphs found among marram grass (Buxton, 1914). On sandhills near coast and covered with marram grass; often found on heather and low herbage; under old bark and rotten wood on posts; in decayed stump (Lucas, 1920). Swarming on _Beta maritima_ and other plants in July (Lucas, 1920a). Very common in all stages in August, being frequently found under stones (Lucas, 1925). Common on sand dunes especially under stems of dead marram grass. Viable oöthecae found buried in sand (Brown, 1952).

Germany.--In beech woods and in pine woods (Zacher, 1917).

=Ectobius semenovi=

Kazakhstan.--Along the shores of the Syr-daria it is found on and around living willows and on _Populus euphratica_; under loose bark of dying and dead trees

(Bei-Bienko, 1950).

=Ectobius sylvester=

_U.S.S.R._--In wooded steppe zones; probably only occurs in association with forests (Bei-Bienko, 1950).

=Ectobius tadzhicus=

Tadzhikistan.--Great numbers at the roots of _Eleagnus_ shrubs on the banks of reservoirs and frequently under the bark of old trees (Bei-Bienko, 1950).

=Ectobius vittiventer=

South France.--One male beneath stone (Blair, 1922).

=Ellipsidion affine=

Australia.--From leaves, from scrub (Hebard, 1943). Collected in trees (Pope, 1953a).

=Ellipsidion australe=

Australia.--On eucalyptus leaves, on wattle, under bark (Hebard, 1943). Collected in trees (Pope, 1953a).

=Ellipsidion bicolor=

Australia.--In corn and from tree (Hebard, 1943).

=Ellipsidion simulans=

Australia.--From sugarcane (Hebard, 1943).

=Ellipsidion= spp.

Australia.--All stages are diurnal moving about the foliage of shrubs and small trees in bright sunlight on hottest summer days (Tepper, 1893).

=Epilampra abdomen-nigrum=

Trinidad.--In dried-up drain; among grass; in debris under old cacao tree; under old leaves (Princis and Kevan, 1955).

Puerto Rico.--Abundant in damp lowlands (Seín, 1923). Under dead leaves in wet malojillo meadow (Wolcott, 1936).

This species is amphibious (p. 31). Shelford (1907) suggested that immature stages of other species of the genus may be aquatic, which would place them in moist situations on the shores of rivers and other bodies of water.

=Epilampra azteca=

Panama.--Very scarce, under palm trees in decaying leaf mold and litter; one found under decaying bark of a log (Hebard, 1921a).

=Epilampra mona=

Mona Island, Puerto Rico.--One specimen under bark of dead tree (Ramos, 1946).

=Epilampra tainana=

Cuba.--Under dead leaves on stream bank (Rehn and Hebard, 1927).

=Epilampra wheeleri=

Puerto Rico.--In siftings from high-altitude primeval forest (Rehn and Hebard, 1927).

=Epilampra= spp.

Australia.--By day the insects live under bark, stones, logs, dead vegetable debris, or buried in loose dust or soil. After sunset females wander in grass or ascend low objects (Tepper, 1893).

=Ergaula capensis=

Uganda.--In open bush and short grass (Princis, 1955).

=Eudromiella bicolorata=

Panama.--Under rubbish on edge of jungle (Hebard, 1920).

=Euphyllodromia liturifera=

Colombia.--In brushwood (Princis, 1946).

=Eurycotis biolleyi=

Costa Rica.--Numbers of individuals were found in the large bromeliads of the temperate localities (Picado, 1913).

=Eurycotis decipiens=

Trinidad.--In old, rotten coconut stump (Princis and Kevan, 1955).

=Eurycotis dimidiata=

Cuba.--"This species was recorded from under stones in the fields ... by Gundlach" (Rehn and Hebard, 1927).

=Eurycotis ferrum-equinum=

Cuba.--Under stones in woods (Rehn and Hebard, 1927).

=Eurycotis floridana=

Florida.--Moderately common under bark of dead pine stumps and logs; at Key West it fairly swarmed under coquina boulders in the woods (Rehn and Hebard, 1905). Many specimens under palmetto leaves on ground (Caudell, 1905). In pine woods under dry bark of dead logs; on Long Key in dry fibers at the base of the heads of coconut palms; "at Key West, a large colony was discovered among boards lying on dry grass in a field, and several were captured upon turning over coquina boulders in the dense bush" (Rehn and Hebard, 1912). Particularly numerous in tree cavities and under bark along the edge of hammock areas (Hebard, 1915). Abundant between basal leaves of _Tillandsia utriculata_; beneath loose bark of logs and stumps; in and beneath decaying palmetto trunks and leaves; under rubbish (Blatchley, 1920). On

ground in heavy tangle after dark; in decaying log of _Sabal palmetto_; in bromeliads; common under debris and bark in jungle; under signs on _Pinus caribaea_; in almost every sheltered outdoor place (Hebard, 1917). It moves about at night and hides under bark of logs and in other recesses during the day; where pines are present it almost invariably hides under bark of dead logs and stumps (Rehn and Hebard, 1914). Friauf (1953) found this species in leaf duff, leaf mold, or decaying wood in these habitats: Sandhills (infrequent), xeric hammock (dominant), mesic hammock (frequent), and low hammock (dominant); on tree trunks in sandhills habitat (infrequent) and mesic hammock (frequent); infrequent in saw-grass marsh habitat in the grass stratum and, during the dry season, in decaying vegetation on floor of marsh. Under the bark of logs and beneath logs in the woodpile habitat (Friauf, 1953).

=Eurycotis galeoides=

Cuba.--Under stones in deep woods (Rehn and Hebard, 1927).

=Eurycotis kevani=

Trinidad.--Under debris, trash, and vegetable refuse (Princis and Kevan, 1955).

=Eurycotis opaca=

Cuba.--In pine and palmetto region (Rehn and Hebard, 1927).

=Euthlastoblatta abortiva=

Texas.--Under dense tangle of bushy vegetation, palms, and vines near Rio Grande; in leaves and dry litter on ground; on dead petiole hanging from palm tree (Hebard, 1917). Under bark of dead hackberry; abundant in dead leaves, dry litter, and rats' (_Neotoma_ sp.) nests in heavy scrub (Hebard, 1943a).

=Graptoblatta notulata=

Tahiti.--On foliage in sun or concealed among dead leaves that collect between the fronds of tree ferns (Cheesman, 1927).

Hawaii.--Quite active during the day, occurring on sugarcane, etc., in the wetter districts; it is also a household insect (Williams et al., 1931).

=Hemiblabera brunneri=

Puerto Rico.--Under bark of tamarind tree (Rehn and Hebard, 1927). Under the bark on a fence post (Wolcott, 1950).

=Henicotyle antillarum=

Dominica.--From rotting wood and wood soil (Rehn and Hebard, 1927).

=Holocompsa metallica=

Dominican Republic.--Along railroad through jungle and swamp (Rehn and Hebard, 1927).

=Hololampra bivittata=

Canary Islands.--Found in numbers among pine needles; nymphs were in the majority, adults rare (Burr, 1911).

=Hololampra chavesi=

Azores.--Very common in the hedges, particularly in brambles. Contrary to most species of this genus, which live on the ground under stones, this species is exclusively dendricolous and is only captured by beating the bushes on which it abounds (Chopard, 1932).

=Hololampra maculata=

Germany.--Abundant in deciduous forest in grass and under fallen leaves; in pine forests under lichens and between fallen needles; in edge of coniferous forest; under stones (Zacher, 1917).

=Hololampra marginata=

Macedonia.--Usually found crawling on the flowers and stems of giant thistles in May; common on thistles in June (Burr, 1923).

=Hololampra= sp.

Caucasus.--Numerous beneath dry leaves in a garden (Burr, 1913).

=Hololeptoblatta= sp.

Seychelles.--Apparently only inhabits _Pandanus_ between the leaf bases (Scott, 1910, 1912).

=Homalopteryx laminata=

St. Vincent.--In decaying leaves in forest (Rehn and Hebard, 1927).

Trinidad.--In forest debris and debris under old cacao trees; it is not uncommon under dry leaves; it feigns death when disturbed (Princis and Kevan, 1955).

=Hormetica laevigata=

Brazil.--From crown of palm between leaf bases (Hancock, 1926).

=Ignabolivaria bilobata=

_U.S.S.R._--Under rocks and on the edges of woods in the lowlands in the north and in the mountains in the south (Bei-Bienko, 1950).

=Ischnoptera deropeltiformis=

North Carolina.--Under pine straw on ground in woods (Brimley, 1908).

Georgia.--Under dead oak leaves; under debris in garden; running on ground in pine and oak woods (Rehn and Hebard, 1916).

Indiana.--It is "a ground-frequenting, forest-loving insect, hiding beneath cover or about the edges of deep woodland, more frequently in damp places, and rarely taken beneath bark, signs, or at lights" (Blatchley, 1920).

Missouri.--Twenty to 30 males found resting on heads of wild oats on successive evenings (Rau, 1947).

Texas.--It preferred damp, open woodlands (Hebard, 1943a).

Eastern and southeastern U.S.--Under stone in heavy deciduous forest; under damp, dead leaves on edges of forests; under bark of pine log; in wire grass and sphagnum bordering stream thicket; in leaf mold and rubbish about pothole in pine woods, _Pinus caribaea_; under debris and leaf mold in hammock; under dead oak leaves in heavy deciduous forest (Hebard, 1917).

Florida.--"This species is distinctly geophilous and appears to prefer damp surroundings" (Rehn and Hebard, 1912). Under boards on very wet ground in everglades; in debris and leaf mold in heavy, junglelike areas of trees, bushes, and vines (Rehn and Hebard, 1914). Adults and numerous nymphs beneath weeds, grass, and other debris washed up on beach of Lake Okeechobee (Blatchley, 1920). Friauf (1953) found this species in leaf duff, leaf mold, and/or decaying wood on ground in these habitats: Dry, ruderal grassland (infrequent), scrub (infrequent), sandhills (occasional), xeric hammock (frequent), mesic hammock (dominant), shrubby, longleaf-pine flatwoods (infrequent), bayhead (dominant), and low hammock (dominant). On open bare soil or bare sand under vegetation in these habitats: Dry, ruderal grassland (infrequent), mesic hammock (dominant), moist, ruderal grassland (infrequent), pond margin (occasional), longleaf-pine flatwoods (infrequent), slash-pine flatwoods (infrequent), and low hammock (dominant). Infrequent in the herbaceous stratum of these habitats: Dry, ruderal grassland, moist, ruderal grassland, and longleaf-pine flatwoods. Infrequent in the shrub stratum of the dry, ruderal grassland habitat. (Friauf, 1953.)

Tennessee.--Taken in traps baited with cantaloupe in a parklike stand of oak, gum, hickory, and tulip trees in a creek bottom, and in a stand of oak on a dry ridge (Walker, 1957).

=Ischnoptera panamae=

Panama.--Under rubbish at edge of jungle and under drift on edge of coral-sand beach (Hebard, 1920).

=Ischnoptera podoces=

Jamaica.--In dead leaf litter along side trail through mountain forest (Rehn and Hebard, 1927).

=Ischnoptera rufa rufa=

Virgin Islands, St. Croix.--Common under rubbish and on shrubbery at night (Beatty, 1944).

Barbados.--Occasionally found in cane fields (Tucker, 1952).

West Indies.--In Puerto Rico, under stones in cultivated area, under debris on alkalie flat. In Jamaica, under dry petioles of coconut palm in grassy area; under logs, logwood on docks, and litter on limestone and near beach. In Panama, under drift on

edge of coral-sand beach; under rubbish at edge of jungle (Hebard, 1916c).

Jamaica.--Under limbs and leaf litter in mangrove swamp (Rehn and Hebard, 1927).

=Lamproblatta albipalpus=

Panama.--Under drift on edge of coral-sand beach. Several under decayed banana stem (Hebard, 1920).

=Lamproblatta meridionalis=

Trinidad.--Under debris in forest and debris under old cacao trees (Princis and Kevan, 1955).

=Latiblattella chichimeca=

Costa Rica.--Very common in the bromeliads of all Costa Rica (Picado, 1913).

=Latiblattella lucifrons=

Arizona.--"Most commonly seen feeding on pollen and dead insects on the flower stalks of _Yucca elata_ in June in the Santa Rita Mountains" (Ball et al., 1942).

=Latiblattella rehni=

Florida.--Widely distributed throughout pine woods (_Pinus caribaea_); under signs on _Pinus clausa_ and _Pinus caribaea_ (Hebard, 1917). Beneath bark of dead pine tree; beating Spanish moss; they seldom attempt flight when disturbed, but hide in crevices or drop to ground (Blatchley, 1920).

=Latiblattella zapoteca=

Costa Rica.--Under stones on borders of Surubres River (Rehn, 1906).

=Leucophaea maderae=

Barbados.--In cane fields (Tucker, 1952).

Dominica.--In vegetation of royal palms, guava, etc.; under loose bark and banana sheaths. In Jamaica, on logwood docks (Rehn and Hebard, 1927).

=Litopeltis biolleyi=

Costa Rica.--Under bark of tree in forest; in epiphytic bromeliads (Rehn, 1928).

=Litopeltis bispinosa=

Panama Canal Zone.--About 80 specimens from rotting banana stalks at bases of leaves; boring in decaying banana stem (Hebard, 1920).

=Litopeltis deianira=

Costa Rica.--In tree stump on edge of mountain forest; in dead wood on ground (Rehn, 1928).

=Litopeltis musarum=

Costa Rica.--Shaken from dead banana leaves. Footnote to specific name: "In relation to the liking of species of this genus for bananas (_Musa_) as shelter and possibly food" (Rehn, 1928).

=Lobolampra subaptera=

France.--Under stones and dead leaves, always rare (Chopard, 1947).

=Loboptera decipiens=

France.--All stages common beneath stones (Blair, 1922). Under stones and dead leaves (Chopard, 1947).

Maltese Islands.--Quite common in open country under stones (Valletta, 1955).

Dalmatia.--On seashores under rocks and seaweed cast up on shore (Bei-Bienko, 1950).

=Loboptera thaxteri=

Argentina.--Common in rubbish and leaf litter in small woodlot (Hebard, 1932).

=Lobopterella dimidiatipes=

Hawaii.--Abundant in wet districts, both in lowlands and to a considerable altitude in the forests, under trash, stones, boards, etc. (Williams et al., 1931). Often it is found with nymphs of _Periplaneta australasiae_ (Fullaway and Krauss, 1945).

=Lophoblatta arawaka=

Trinidad.--On grass, maize, and cut sugarcane fodder; under vegetable and garden refuse; under old cacao (Princis and Kevan, 1955).

=Macropanesthia rhinocerus=

Australia.--Infrequently seen during dry season from March to October. "They burrow quite deeply, about two feet below the surface of the sandy soil in stands of cypress pine (_Callitris_ sp.). They make a nest of dead leaves, grass roots, etc., frequently among the pine roots. The young nymphs rarely appear above ground, but following rain the adults burrow to the surface, especially at night.... This species is also found in the brigalow (_Acacia harpophylla_) scrub about 70 miles west of Rockhampton, Queensland, and on Fraser Island off the Coast of Queensland" (Henson _in_ Day, 1950).

=Megaloblatta blaberoides=

Panama.--Under bark on tree (Hebard, 1920).

Ecuador.--Under a dense pile of dead leaves around base of tree (Campos R., 1926).

=Megamareta verticalis=

Australia.--In sugarcane (Hebard, 1943).

=Methana canae=

Australia.--Under loose bark on dead upright tree (Pope, 1953a).

=Methana curvigera=

Australia.--Under loose bark on trees and logs; many specimens on wattle trees where in strong sunlight they hid in curled-up leaves; oöthecae attached to underside of loose bark and leaves (Pope, 1953a).

=Methana marginalis=

Australia.--Under loose bark of trees and logs (Pope, 1953a).

=Moluchia (?) dahli=

Chile.--Collected from lichens and mosses on tree trunks (Princis, 1952).

=Muzoa madida=

Costa Rica.--Under dead wood in dense second-growth forest; in thick mat of hanging dead vegetation in dense forest; under leaves in forest (Rehn, 1930).

=Nauclidas nigra=

St. Vincent.--Under rotten fruit (Rehn and Hebard, 1927).

=Nelipophygus ramsdeni=

Cuba.--Under rotten bark (Rehn and Hebard, 1927).

=Neoblattella detersa=

Jamaica.--Under dried leaves of coconut palm; in dry leaves under acacia on hillside; in debris on beach; under stones on coral rock; in leaf mold under dense brush on hillside; under bracts of banana blossoms (Rehn and Hebard, 1927).

=Neoblattella dryas=

Jamaica.--In bases of dead tree-fern fronds; numerous in bromeliads; nearly all collected specimens were taken in these plants (Rehn and Hebard, 1927).

=Neoblattella eurydice=

Jamaica.--Nearly all collected specimens taken in bromeliads (Rehn and Hebard, 1927).

=Neoblattella grossbecki=

Jamaica.--In epiphytic bromeliads and hollow bases of dead tree-fern fronds; nearly all collected specimens taken in bromeliads (Rehn and Hebard, 1927).

=Neoblattella proserpina=

Jamaica.--Under bark of huckleberry; nearly all collected specimens taken in bromeliads (Rehn and Hebard, 1927).

=Neoblattella semota=

Jamaica.--All specimens collected from under drying bracts of banana blossoms (Rehn and Hebard, 1927).

=Nesomylacris cubensis=

Cuba.--In dry region of palmettos and pines (Rehn and Hebard, 1927).

=Nesomylacris relica=

Jamaica.--Widely distributed from sea level to 5,700 feet elevation; in bromeliads in mountain forest; among dead leaves in heavy leaf mold under dense hillside scrub; under stones and in ground litter about banana trees; under bark of tree in dense ridge-type forest; in dead agave in scrub forest (Rehn and Hebard, 1927).

=Nocticola bolivari=

Ethiopia.--Always found under stones or cement blocks, but not necessarily deeply buried in the ground (Chopard, 1950b).

=Nyctibora laevigata=

Jamaica.--In cracks in dead stump of mimosa; in bromeliads (Rehn and Hebard, 1927).

=Nyctibora lutzi=

Puerto Rico.--Possibly to be found most often in rotten tree trunks in the highest mountains; found in rotten stump with termites, ants, and beetle grubs (Wolcott, 1950).

=Nyctibora obscura=

Trinidad.--Under pile of cornstalks (Princis and Kevan, 1955)

=Nyctibora stygia=

Haiti.--Under loose dead bark of mesquite tree, 52 specimens (Rehn and Hebard, 1927).

=Oniscosoma spp.=

Australia.--The females bury themselves in loose soil or dust (Tepper, 1893).

=Opisthoplatia orientalis=

Formosa.--On or in swampy ground or under rotten trees on the ground (Takahashi, 1924).

=Panchlora antillarum=

Dominican Republic.--In cultivated grounds, palms, fruits, etc. (Rehn and Hebard, 1927).

=Panchlora nivea=

Panama.--As _Pycnosceloides aporus_, in jungle under decaying banana stem in which were boring individuals of _Litopeltis bispinosa_ (Hebard, 1920).

Texas.--Lives in foliage and in the green sheaths of plants (Hebard, 1943a).

Cuba.--On cane leaves; according to Gundlach this genus lives under the loose bark of trees (Rehn and Hebard, 1927).

Puerto Rico.--In rotting trunks of coconut palms (Seín, 1923). Most specimens have been collected from the very rotten interior of coconut palms (Wolcott, 1950).

Trinidad.--On corn; under old log; flies readily to lights (Princis and Kevan, 1955).

=Panchlora sagax=

Dominica.--In decaying stump in banana patch and in rotting wood. In Puerto Rico, in rotten coconut palm (Rehn and Hebard, 1927).

=Panesthia australis=

Australia.--In burrows under the thick bark of fallen and rotting trees (Shaw, 1914). In loose detritus, beneath clods of earth, and in fissures at foot of cliffs along the seashore beyond direct action of the waves (Tepper, 1893).

=Panesthia laevicollis=

Australia.--Under decayed logs in coastal scrub. It burrows into the soft part of the log (Froggatt, 1906).

=Parcoblatta bolliana=

North Carolina.--Under pine straw on ground in pine woods (Brimley, 1908).

Texas.--Under dry cow dung in pine woods (Hebard, 1917).

Nebraska.--Under pile of old boards (Hauke, 1949).

=Parcoblatta caudelli=

North Carolina.--From under the bark of dead trees (Rehn and Hebard, 1910).

Virginia.--At night on shrubbery. In South Carolina, under sign on tree (Hebard, 1917).

Tennessee.--In traps baited with cornmeal, cantaloupe, or fish in a stand of oak on dry ridge, and in abandoned rocky field on a south-facing slope (Walker, 1957).

=Parcoblatta desertae=

Texas.--From mountains, arid, and semi-arid regions; under small boulder on desert (Hebard, 1917). On ground in dry-creek bed through scrub oak, pine, and juniper forest (Hebard, 1943a).

=Parcoblatta divisa=

Eastern and southeastern U.S.--All specimens taken from under signs on red oaks and longleaf and shortleaf pines in Georgia and Virginia (Rehn and Hebard, 1916). Trapped in molasses-baited jar in oak forest in New Jersey; under signs on red and white oaks, sweet gum, and other deciduous trees; under signs on shortleaf and longleaf pines and pine stumps (Hebard, 1917). Widespread in southeastern U.S. in habitats as diverse as dry pine lands, oak scrub, moist hammocks in northern Florida,

and deep, cool ravines along Apalachicola River (Hebard, 1943a).

=Parcoblatta fulvescens=

Eastern and southeastern U.S.--Trapped in molasses jars: in heavy, barrier-beach forest; in typical pine-barrens undergrowth; in pine barrens with heavy, grassy undergrowth; on border of pine barrens and on edge of swamp; in heavy deciduous forest; in heavy oak woods. Found under debris in dead, shortleaf-pine needles; under dead leaves on edge of oak and shortleaf-pine woods; under bark of pine log; among dead leaves under live oaks; under sign on _Pinus caribaea_ (Hebard, 1917).

Georgia.--From under bark of pine log, among dead leaves under live oaks, and under leaves on edge of oak and shortleaf-pine woods (Rehn and Hebard, 1916).

Florida.--Very common among dead leaves, under logs, beneath loose bark, and wanders about at night in pinelands, hammock, turkey oak, and sand-scrub habitats (Hubbell and Goff, 1940). Beneath drift, cow dung, leaves, boards, bark of logs, and other debris, usually in open pine woods in sandy areas; frequent at the base of thistle leaves (Blatchley, 1920). Friauf (1953) found this species in leaf duff, debris, or decaying wood in these habitats: Scrub (dominant), sandhills (dominant), xeric hammock (dominant), mesic hammock, longleaf-pine flatwoods (infrequent), low hammock (infrequent), and alluvial hammock (infrequent). In the shrub stratum in these habitats: Scrub (dominant), sandhills (dominant), xeric hammock (dominant), and longleaf-pine flatwoods (infrequent). In the herbaceous stratum of the longleaf-pine flatwoods habitat, and under bark and beneath logs in the woodpile habitat.

=Parcoblatta lata=

Southeastern and southern U.S.--Under bark of pine logs and stumps; in sweet-gum logs and stumps; moderately numerous under bark of dead shortleaf pines; under bark of longleaf-pine stumps; under signs on red oak and longleaf pines; in dead oak. In Texas, under bark of pine stumps (Hebard, 1917).

North Carolina.--All stages under loose bark of dead pines, both prostrate and upright, and stumps. "It seems to prefer the space under the bark to be rather damp" (Brimley, 1908). Under bark of dead pine trees (Rehn and Hebard, 1910).

Florida.--Infrequent in leaf duff and decayed wood of low hammock habitat (Friauf, 1953).

Indiana.--Beneath rocks on sides and tops of high hills, in limestone glades where cedar abounds (Blatchley, 1920).

Missouri.--In leaf stratum of oak-hickory forest (Dowdy, 1951). Earlier, Dowdy (1947) reported finding numerous immature Pseudomopinae [presumably _Parcoblatta_ sp.] in soil and leaf strata of oak-hickory forest.

Texas.--Captured in molasses-baited traps in low, wet, oak woods and in dry woodlot on hillside (Hebard, 1943a).

=Parcoblatta pensylvanica=

Eastern and southeastern U.S.--Trapped in molasses-baited jars; in oak and in chestnut forests, and on knoll with high deciduous trees. Found in oak and pine woods, under bark of decaying chestnut log and dead chestnut stump, and under signs on trees including oaks (Hebard, 1917).

North Carolina.--In all stages under loose bark of upright, dead pines, when the space under the bark was dry (Brimley, 1908).

Virginia, North and South Carolina.--Under signs on trees (white and red oaks); under bark of dead shortleaf-pine and sweet-gum logs and stumps (Rehn and Hebard, 1916).

Indiana.--Beneath bark of logs and stumps; empty oöthecae common beneath loose bark of logs, especially shellbark hickory (Blatchley, 1920). Under loose bark on logs in January (Blatchley, 1895).

Illinois.--In pine forest associes, in black oak forest on sand, in oak-hickory forest on clay, and in climax forest; it evidently moved into the pine associes nightly, great numbers of oöthecae were found under bark of pine logs, where, in October and November, hibernating nymphs were found (Strohecker, 1937). In nests of _Vespula maculata_ (Balduf, 1936; McClure, 1936).

Missouri.--Usually in hollow trees, under loose bark, in woodpiles, and in cracks in rural buildings (Rau, 1940).

Michigan.--Common in oak-dune and beech-maple forests, under loose bark on dead trees and fallen logs, and under debris on forest floor (Hubbell, 1922). "A characteristic inhabitant of the low shrub-terrestrial and probably the terrestrial-hypogeic stratum." It occurred throughout the upland forests; groups were found established in and under logs 100 to 200 feet from the nearest forest (Cantrall, 1943).

Ontario.--Very abundant in rocky, sparsely-wooded country, where it occurred in rotten logs and under loose bark; on tree trunk at night on rocky island in lake (Walker, 1912).

=Parcoblatta uhleriana=

North Carolina.--Under pine straw on ground in woods (Brimley, 1908). Under bark of dead trees; 92 males attracted to lights (Rehn and Hebard, 1910).

Virginia.--Resting on woods foliage; at night on road (Rehn and Hebard, 1916).

Eastern and southeastern U.S.--Trapped in molasses-baited jars: in oak and pine woods, in heavy barrier-beach forest, in both scant and typical undergrowth on pine barrens, in heavy grassy undergrowth on pine barrens, on border of pine barrens, on edge of swamp, in heavy deciduous forest, in heavy oak woods, in upland oak and chestnut forest, in chestnut forest, in forested ravine, and on ridge with heavy oak, chestnut, and maple forest. Found under damp leaves on edge of forest, under bark of decayed chestnut log, inside decaying chestnut log with _Cryptocercus punctulatus_, under palmetto roots, under bark of pine stump, and in dry leaves under live oaks (Hebard, 1917).

Tennessee.--In traps baited with cornmeal or cantaloupe in maple-gum-oak forest in a mesic valley, and in a stand of oak on a dry ridge (Walker, 1957).

Indiana.--Beneath cover on slopes of high wooded hills. "This is essentially a forest-loving species; usually occurring beneath leaves and other debris on or along the borders of heavy hardwood timber." (Blatchley, 1920.)

Illinois.--In oak-hickory forest on clay and in climax forest (Strohecker, 1937).

Michigan.--In oak-dune woods (Hubbell, 1922). Restricted to woodlands, where it inhabited piles of moist dead leaves and rotten logs in oak-hickory forest (Cantrall, 1943).

_U.S.A._--This species, _P. uhleriana_, and _P. virginica_ were attracted at night to honeydew secreted by aphids on _Pyrus_ sp. (Davis, 1918).

=Parcoblatta virginica=

New England.--Females under loose stones, boards, and other debris on ground; beneath loose bark (Morse, 1920).

North Carolina.--Under debris in dead shortleaf-pine needles (Rehn and Hebard, 1916).

Florida.--Infrequent in the shrub stratum of the scrub habitat. This was the only habitat of 25 studied in which this species was found (Friauf, 1953).

Eastern and southeastern U.S.--Trapped in molasses-baited jars: in pine and oak woods, in pine barrens, in pine woods with heavy grass undergrowth, in oak forest, in heavily forested ravine, on rocky slope with few deciduous trees, on knoll with high deciduous trees, in lofty chestnut forest, and in heavy low chestnut and oak forest on high ridge; under bark of decaying chestnut log and stump; under stones in chestnut forest; under bark of pine stumps (Hebard, 1917).

Indiana.--Frequents borders of open woods and fields; under debris, loose bark, and half-buried logs (Blatchley, 1920).

Illinois.--In black-oak forest on sand, in oak-hickory forest on clay, and in climax forest (Strohecker, 1937).

Michigan.--Common in oak-dune and beech-maple forests; under loose bark on dead trees and fallen logs and under debris on forest floor (Hubbell, 1922). Restricted to woodlands, where it inhabited piles of moist dead leaves and rotten logs in oak-hickory forest (Cantrall, 1943).

Texas.--Captured in molasses traps in moist woods of maple, oak, and pine with much undergrowth and a heavy layer of duff; in open, rather dry woodlot of Spanish oak and other trees; and in low wet woods of willow and oak along creek (Hebard, 1943a).

=Parcoblatta zebra=

Indiana.--Beneath log in cypress swamp (Blatchley, 1920).

Louisiana and Mississippi.--In decay cavity in sweet gum; under sign on shortleaf pine (Hebard, 1917).

=Parcoblatta= spp.

Alabama.--In the dry wall of a sweet-gum stump together with serropalpid and tenebrionid beetles (Snow, 1958).

Ohio.--Oöthecae under loose bark of fallen trees, where as many as 184 oöthecae were found within a few feet of each other; others found under boards and in piles of firewood (Edmunds, 1952).

=Pelmatosilpha coriacea=

Puerto Rico.--Mona Island, under bark of dead trees and under guava leaves (Ramos, 1946). Under bark of _Sideroxylon foetidissimum_ (Wolcott, 1941). Common along the coast and in mountains. "Very much at home" under the loose bark of _Sideroxylon foetidissimum_ (Wolcott, 1950).

=Pelmatosilpha kevani=

Trinidad.--Under debris in bush (Princis and Kevan, 1955).

=Pelmatosilpha purpurascens=

Dominica.--In decaying logs in forest (Rehn and Hebard, 1927).

=Periplaneta americana=

Bermuda.--Among and under decaying debris, just above high-tide line (Verrill, 1902).

Johnson Island.--Nocturnal, coming out at night in great numbers about _Tribulus_ blossoms. Under timbers on French Frigate Shoals (Bryan, 1926).

United States.--Alleyways and yards may be overrun during the summer; adults and hundreds of nymphs found in decaying maple trees along residential street (Gould and Deay, 1938, 1940). Around fumaroles where a railroad fill was burning internally (Davis, 1927). Common in palm trees along the gulf coast of Texas, where they often fly around street lights at night (Zimmern _in_ Gould and Deay, 1940).

=Periplaneta australasiae=

Bermuda.--Very abundant under stones (Rehn, 1910).

Jamaica.--Under bark of dead tree and under bases of leaves of coconut palms (Rehn and Hebard, 1927).

Virgin Islands, St. Croix.--Common in sugarcane fields and in woodlands (Beatty, 1944).

Florida.--Juveniles under bark of dead logs of _Pinus caribaea_ (Hebard, 1915). Frequently found under signs on trees near borders of towns; under bases of dead petioles of cabbage palmetto (Hebard, 1917). Beneath logs, burlap bags, and other cover in old orange orchards (Blatchley, 1920).

Marquesas Islands.--Under coconut fronds and grass (Hebard, 1935).

Nihoa Island.--Nymphs only, on _Sida_, _Pritchardia_, bunch grass, and about camp (Bryan, 1926).

=Periplaneta brunnea=

Georgia.--Under signs on oaks (Rehn and Hebard, 1916).

Florida.--Beneath bark of stump (Blatchley, 1920).

=Periplaneta fuliginosa=

Southeastern and southern U.S.--"This species is usually encountered out of doors, in or near towns. Over its range it is frequently found under signs on trees" (Hebard, 1917).

=Phidon (?) dubius=

Chile.--Collected from mosses and lichens on tree trunks (Princis, 1952).

=Phoraspis= spp.

Brazil and Guiana.--In grasslands, plantations of maize, sugarcane, and other plants on the borders of forests; the cockroaches were always found between the leaves which form the branches of the plants (Doumerc _in_ Blanchard, 1837).

=Phyllodromica brevipennis=

Asia Minor and western Europe.--On ground among grasses; under moss and brushwood in mountain meadows (Bei-Bienko, 1950).

=Phyllodromica graeca=

U.S.S.R., western Georgia.--In pine forest mixed with deciduous trees (Bei-Bienko, 1950).

=Phyllodromica irinae=

U.S.S.R., Turan Lowland.--Along margins of "tugas" under half-fallen bushes of _Salsola kali_ that overhang the ground (Bei-Bienko, 1950).

=Phyllodromica maculata=

Central Europe and western U.S.S.R.--On the edges of forests of the central-European type that are lighted by the sun; under fallen leaves; on bushes and conifers (Bei-Bienko, 1950).

=Phyllodromica meglerei=

_U.S.S.R._--Among fallen leaves under bushes; on oak branches; under mown hay (Bei-Bienko, 1950).

=Phyllodromica polita=

Caucasus.--Under fallen leaves on slopes of mountains covered by forest or brushwood (Bei-Bienko, 1950).

=Phyllodromica pygmaea=

_U.S.S.R._--In the sands of Un-dzhal-kum and Zhety-konur it is found in the dense turf of _Aristida pennata_ (Bei-Bienko, 1950).

=Phyllodromica tartara=

Central Asia.--In lowlands and in mountains up to 2,500 meters; in fruit orchards under trap rings fastened to trees to combat lesser apple worm (Bei-Bienko, 1950).

=Phyllodromica tartara nigrescens=

Southern Uzbekistan.--Under bark of _Juniperus_ sp., under stones and on flowers of _Scorzonera acanthoclada_ (Bei-Bienko, 1950).

=Platyzosteria castanea=

Australia.--Under loose wood or bark (Shaw, 1914).

=Platyzosteria novae seelandiae=

New Zealand.--Swarms under loose dry bark and logs (Walker, 1904).

=Plectoptera dominicae=

Dominica.--On moss-covered lime trees. "The species of the genus _Plectoptera_ are all foliage and flower frequenters, generally secured by beating low arborescent vegetation, or are attracted to light" (Rehn and Hebard, 1927).

=Plectoptera dorsalis=

Puerto Rico.--In caladium, grass, weeds, coffee, and bananas; in flowers of _Ipomoea tiliasea_ (Rehn and Hebard, 1927). "... living in trees between leaves, or in 'butterfly-nests' of _Tetralopha scabridella_ in leaves of _Inga vera_, or of _Pilocrocis secernalis_ in the leaves of 'capá blanco' (_Petitia domingensis_) in the mountains. Along the coast they have been found under the bracts of cotton squares or bolls, and under the leaf-sheaths of sugar cane, in curled-up leaves of grapefruit, or in the dry flower clusters of 'espino rubial' (_Zanthoxylum caribaeum_)." These observations apply also to _Plectoptera infulata_ and _P. rhabdota_ (Wolcott, 1950).

=Plectoptera floridana=

Florida.--On fringe of tall bushes at edge of mangrove swamp (Rehn and Hebard, 1914). Rehn and Hebard (1927) stated that on the Keys it frequented dry scrubby vegetation, particularly _Ilex cassine_.

=Plectoptera infulata=

Puerto Rico.--See Wolcott's (1950) comments under _Plectoptera dorsalis_ above.

=Plectoptera lacerna=

Cuba.--In grasses, sedges, etc., about a waterhole; on grass, pines and oak (Rehn and Hebard, 1927).

=Plectoptera perscita=

Dominica.--On moss-covered lime trees (Rehn and Hebard, 1927).

=Plectoptera porcellana=

Cuba.--Taken on flowers of "Júcaro" (Gundlach _in_ Rehn and Hebard, 1927).

=Plectoptera pygmaea=

Jamaica.--In relatively dense forest foliage; in shrubbery (Rehn and Hebard, 1927).

=Plectoptera rhabdota=

Puerto Rico.--In mixed vegetation; on grapefruit tree and guava (_Psidium guajava_); on bushes and shrubs (Rehn and Hebard, 1927). On coffee trees; on _Spondias_; on sugarcane; in caterpillar nests of _Tetralopha scabridella_ on _Inga vera_; in old cotton bolls; on grapefruit (Wolcott, 1936). See also Wolcott's (1950) comments under _Plectoptera dorsalis_.

=Plectoptera vermiculata=

Cuba.--On pine in palmetto region (Rehn and Hebard, 1927).

=Polyphaga aegyptiaca=

Algeria.--Nymphal females under decaying leaves at the end of November (Lucas, 1849).

Transcaucasia.--In burrows in argillaceous cliffs along ravines. Females often covered by attached clay particles, an indication, according to Bei-Bienko, that this species is ecologically connected to compact clay soils or at least does not avoid them (Bei-Bienko, 1950).

See also the section on desert habitats (p. 29).

=Polyphaga saussurei=

South-central Asia.--Occupies compact clay soils; distributed in drier regions than _P. aegyptiaca_; frequently found near dwellings, in yards, stables, and houses (Bei-Bienko, 1950).

=Polyzosteria limbata=

Australia.--Common, usually "resting among the foliage or sunning itself on a fence or stumps, seldom or never hiding under bark or logs like most of the species" (Froggatt, 1906).

=Poroblatta= spp.

Tropical America.--"The species of _Poroblatta_ apparently live as borers in stumps and logs in a manner similar to those of _Cryptocercus_ Scudder in the United States" (Gurney, 1937).

=Pseudomops septentrionalis=

Texas.--In dead-brush pile; not scarce in heavy weeds, sunflowers, etc., in openings of river-plain jungle scrub (Hebard, 1917). It lives largely in herbage

(Hebard, 1943a).

=Pycnoscelus surinamensis=

Florida.--Under stones and rubbish; very abundant under coquina boulders in woods at Key West (Rehn and Hebard, 1905). "This species is common under planks, stones, and other debris on the ground ... also found at Long Key in the dry fibres at the base of the petioles of a coconut palm" (Rehn and Hebard, 1912). At Musa Isle, found burrowing in sand (Hebard, 1915). In fallen leaves and decaying wood in xeric and mesic hammock habitats (Friauf, 1953).

Hawaii.--The soil swarmed with young of various stages during the summer (Illingworth, 1915). In soil about roots of pineapple under mulching paper; feeding on pineapple roots (Illingworth, 1927, 1929).

Fakarava, Tuamoto Archipelago.--Numerous among dead leaves in tree holes (Cheesman, 1927).

West Indies.--Under decayed stalks of sugarcane and in siftings from mangrove swamps, Cuba. Under manure, bases of leaves of coconut palm, litter, logs, and stones on coral rock and in bromeliads, Jamaica. Under wood, tiles, and boards in stable yards; immature individuals bored into the soil, Puerto Rico. (Rehn and Hebard, 1927.)

Barbados.--Frequents cane fields (Tucker, 1952).

Puerto Rico.--"Altho primarily a xerophytic species: collected among dry stones on Mona Island, under dry cow dung at Boquerón, and under boxes at Guánica, it is reasonably common in the more humid parts of Puerto Rico" (Wolcott, 1950).

Virgin Islands, St. Croix.--Common under rubbish; frequently seen feeding on chicken feces around chicken roosts (Beatty, 1944). By feeding on chicken feces it may become the vector of the chicken eyeworm, _Oxyspirura mansoni_, as described in the references cited on page 204.

Egypt.--Large numbers were found in moist soil at the site of a manure pile (Chakour, 1942).

Germany.--Under greenhouse conditions the depth to which _P. surinamensis_ penetrated the soil was determined; 21 dug down to a depth of 8 to 10 cm., 3 dug down 10 to 12 cm., but only one dug 13 cm. below the surface. Often the tubes in the soil ended in a chamber which the cockroach might not leave for several days; nymphs molted in such chambers and females bore their young there (Roeser, 1940).

=Rhytidometopum dissimile=

Trinidad.--Male on low herbage in orchard at night; under sacking; on _Hibiscus_ at night (Princis and Kevan, 1955).

=Riatia orientis=

Trinidad.--Numerous specimens of both sexes at night on roadside _Hibiscus rosa-sinensis_ or low herbage in orchard (Princis and Kevan, 1955).

=Simblerastes jamaicanus=

Jamaica.--Numerous in fragmentary debris of an abandoned termite nest on ground in the dry Liguanea Plain; a specimen was also taken under a stone in a field of short grass (Rehn and Hebard, 1927).

=Styphon bakeri=

Costa Rica.--Among humus and rubble in crevices and large cavities in rocks of the Tertiary limestone rim and the metamorphosed and igneous rocks of the interior of the islands (Baker _in_ Rehn, 1930).

=Supella supellectilium=

Virgin Islands, St. Croix.--Under rubbish heaps; in sugarcane straw (Beatty, 1944).

Africa.--"A cosmotropical species which occurs both out of doors and as a household pest in many warmer parts of the world. It is apparently endemic to non-forested areas in much of Africa north of the Equator." (Kevan and Chopard, 1954.)

=Symploce flagellata=

Puerto Rico.--Under low trees on hillside and dead leaves in thicket of sea grape (Hebard, 1916c).

=Symploce hospes=

Hawaii.--Under stones and rubbish (Illingworth, 1915).

Virgin Islands, St. Croix.--Under rubbish and on shrubbery at night (Beatty, 1944).

=Symploce jamaicana=

Jamaica.--In dead leaves under acacia and other shrubs in desert tract; under log and rubbish in open on limestone sand near beach (Hebard, 1916c). Very common in short dry grass in roadside gutter at night, often clustered together; under beach trash in stony wash of Hope River (Rehn and Hebard, 1927).

=Symploce ruficollis=

Virgin Islands, St. Croix.--Under rubbish and on shrubbery at night (Beatty, 1944).

Puerto Rico.--In siftings from sea-grape thicket on sandy soil (Rehn and Hebard, 1927). Often living under leaf-sheaths of sugarcane (Wolcott, 1950).

=Tartaroblatta karatavica=

Asia, Kara-tau Mountains.--Many hundreds of individuals found only under stones on moist earth and not where ground seemed dry; found on very stony slopes with sparse vegetation, often with undergrowth present (Bei-Bienko, 1950).

STRUCTURAL HABITATS

In this category we include all man-made structures, whether inhabited by man or not, that may become infested with cockroaches. A nonexhaustive list of such structures would include dwellings, restaurants, mess halls, barracks, groceries, markets, bakeries, dairies, drug stores, department stores, hotels, hospitals, warehouses, mills, factories, packing houses, animal houses, breweries, incinerators, privies, sewers, sewage treatment plants, ships, aircraft, etc. Although dwellings are only one of the many kinds of structures that are colonized by cockroaches, the several species that have adopted this mode of life are generally referred to as domiciliary cockroaches. This term is adequate only if we remember that these cockroaches are not restricted to domiciles but are pests in other structures as well.

Associations between man and certain species of cockroaches possibly started as casually as the short-lived association that Beebe (1953) observed when he discovered three cockroaches in the newly built couch of an orang-utan. Obviously, when man came down from the trees, his fellow travelers found his cave dwellings and other abodes particularly favorable habitats. From such primitive beginnings, domiciliary cockroaches have spread into every kind of structure that man has since devised. We predict that when man develops a suitable vehicle, cockroaches will someday accompany him into space. Yet despite the apparent predilection of certain species of cockroaches for man, man is only incidental to these associations. Only the shelter and food that man unwittingly provides for these unwelcome guests attract cockroaches to him; man's physical presence is unnecessary.

Most, if not all, of the common domiciliary cockroaches apparently originated in the Tropics or sub-Tropics from whence they have spread, through normal commercial channels, into most of the inhabited world. At least eight domiciliary cockroaches originated in Africa (Rehn, 1945): _Blatta orientalis_, _Blattella germanica_, _Leucophaea maderae_, _Nauphoeta cinerea_, _Oxyhaloa buprestoides_, _Periplaneta americana_, _P. australasiae_, and _Supella supellectilium_; and, perhaps, _Periplaneta brunnea_ as well; _Neostylopyga rhombifolia_ was probably of Indo-Malayan origin; _Pycnoscelus surinamensis_ was of oriental origin; and _Leurolestes pallidus_ was endemic in the West Indies (Rehn, 1945). Princis (1954a) rejected Africa as the original home of _Blatta orientalis_ and advanced reasons for placing its origin in Central Asia.

Several domiciliary species have become well established in temperate zones and some even in the Arctic. Bei-Bienko (1950) listed the following 10 species as sinanthropes in the Palearctic zone: _Blatta lateralis_, _B. orientalis_, _Blattella germanica_, _Leucophaea maderae_, _Periplaneta americana_, _P. australasiae_, _Polyphaga saussurei_, _Pycnoscelus surinamensis_, and _Supella supellectilium_. In the warmer parts of the temperate regions, as in their native Tropics, certain domiciliary species breed outdoors as well as indoors. In the less temperate extensions of their ranges most domiciliary species are nearly always found indoors. In regions with low winter temperatures these cockroaches do not survive in unheated structures; but in heated buildings _Blattella germanica_, for example, has been able to withstand the rigorous climate of Alaska, where it has caused severe

infestations (Chamberlin, 1949).

The limiting factors that determine whether man-made structures will provide suitable habitats for cockroaches are favorable temperature and availability of water and food. The range of temperatures that man provides for his own comfort and protection fosters the rapid increase of cockroach populations indoors. Gunn (1934, 1935) has demonstrated that the preferred temperature range (zone of indifference) of _Blatta orientalis_ is 20-29° C. The upper limit of the preferred temperature of _Blattella germanica_ and _Periplaneta americana_ is 33° C. (Gunn, 1935). The lower limits of temperature tolerance were not sharply defined in Gunn's work. However, less than optimum temperatures, if they last for only short periods, are not necessarily lethal. The 24-hour mortality for _P. americana_ that had been held for one hour at 0° C. was only 2±2 percent (Knipling and Sullivan, 1957). Gunn (1934) observed that _Blatta orientalis_ would not settle at temperatures above 33° C. and would react violently against higher temperatures (e.g., 39° C.) by running away; thus the thermotactic behavior of cockroaches might be presumed to bring them into favorable environments within structures. Thermal death points have been determined for the above three species by Gunn and Notley (1936).

It is common knowledge among those who rear cockroaches experimentally that, unless the water content of the food is high, fluid water is essential in the insects' dietary. Ten species of domiciliary cockroaches have been shown to be unable to survive as long on dry food alone as they could on food and water at 36-40 percent relative humidity (Willis and Lewis, 1957). _Blatta orientalis_, when in a state of normal water balance, usually spent more time in the drier part of a humidity gradient; but desiccated insects tended to become hygropositive (Gunn and Cosway, 1938). We presume that other domiciliary species behave similarly. If water is available nearby, it may be presumed that partially desiccated cockroaches could locate a source through the mediation of a humidity sense. Hygroreceptors have been demonstrated on the antennae of _Blattella germanica_ (Roth and Willis, 1952a) and suggested for _Blatta orientalis_ (Gunn and Cosway, 1938).

Drinking water is available to cockroaches in the traps of sinks, wash basins, tubs, and toilet bowls; in flush tanks; as condensation on cold pipes, flush tanks, and windows; around leaking pipes and faucets; as spillage; in miscellaneous water-filled containers, such as pet drinking dishes, aquaria, vases; empty beverage bottles; and drainage from ice boxes. Soft, juicy fruits and vegetables can provide both moisture and food. There seems to be a tendency for certain species (_Blatta orientalis_ and _Blattella germanica_) to become established in the more humid parts of structures, such as basements, around sinks, and in bathrooms. Whether this is a reaction to a preferred humidity or merely a fortuitous aggregation near sources of drinking water and food has never been clearly demonstrated. The rather widespread dissemination of these species into zones of low as well as high humidity suggests that detailed studies of the microclimatic conditions of structural microhabitats will be needed before meaningful conclusions can be drawn about the stratification of cockroaches within structures according to species.

In nearly all structures infested by cockroaches, food of some kind is available, either in the structure itself or nearby. This may be the food stored by man for his own use or the use of kept animals; it may be crumbs, food spillage, garbage, or excreta; glues and pastes on cartons, boxes, stamps, envelopes, labels, and wall paper; sizing on cloth and book covers; various dried animal and plant products; dead insects; living plants; etc. In fact, it is almost impossible, despite good housekeeping, to keep any structure used by man free of all food suitable for cockroaches.

That the requisite temperature, water, and food are provided, more or less adequately, by a variety of structures is attested by the innumerable infestations of cockroaches that develop when control measures are relaxed. Within structures the accessibility of certain harborages to cockroaches probably depends on the habits of the species and to some extent on their size. Similar types of harborages in different structures may be used by the same species, although there seems to be some overlapping by different species into the same kinds of daytime shelters. The comparative ecology of domiciliary cockroaches has not been thoroughly investigated, so any interpretation of observational data is necessarily speculative and inconclusive at this time. Our discussion on pages 324 to 343 is also pertinent to this section.

LAND-BASED STRUCTURES

Dwellings provide a variety of microhabitats that are acceptable to cockroaches. It has been stated that old houses, or houses that have many cracks and crevices, or have basement kitchens that are not kept clean and in good repair are particularly liable to invasion by cockroaches (Laing, 1946; British Museum [Nat. Hist.], 1951). Although this statement is undoubtedly true, it has been our personal experience, as well as the experience of others, that new, clean, and well-planned houses and apartments are also easily and sometimes quickly invaded by cockroaches. Mallis (1954) has cited the following places that are frequently infested by cockroaches in homes. In the kitchen, cockroaches are found in and around sinks, in cupboards above and below sinks, under tables and chairs, in stoves, around breadboards, in utility cabinets, in kitchen closets, under linoleum, behind, under, and inside refrigerators and iceboxes. In living rooms cockroaches are found in furniture, studio couches, sewing machines, closets, and bookshelves; behind picture frames, pennants, calendars, and other wall ornaments. In bathrooms cockroaches are found in and behind utility cabinets and toilets; they may be found in wicker clothes hampers, in brooms and mops, and in door hinges. Ordinarily, cockroaches are not found in bedrooms unless they are abundant elsewhere in the dwelling. Additional harborages are cited under specific cockroaches in the list below.

In markets DeLong (1948) found the German cockroach in bags of potatoes and onions, in crates of citrus fruits, in pads and shredded papers in banana boxes, and in cases of bottled beverages. The insects were attracted by coffee and crawled into the folds of coffee bags. They were found in cartons of canned goods; in bread and baked goods; in cartons of packaged cookies, cakes, and crackers. Packaged cereals were attractive, and cockroaches were sometimes found in packages of cigarettes. The

insects occurred in scales (by the hundreds) and in cash registers. Rather heavy infestations were found under stainless-steel cappings that covered wooden arms on the fish cleaning stand. The insects were numerous in display cases where they were warm and sheltered. They were also found behind mirrors above produce racks, in electrical switch boxes and conduits, and in telephone boxes, as well as generally in cracks and behind loose moldings or loose wall boards. Enclosed boxlike tables were frequently heavily infested.

In restaurants cockroaches may be found in the following places: Crevices in wood, plaster, concrete, and metal; in the bar; in the kitchen and in the associated equipment; in cupboards, lavatories, and garbage storage areas; and on the undersides of chairs and tables (Mallis, 1954).

In drug stores Frings (1948) found cockroaches behind the mirror and between the sink and the cooler. Thousands were found in hollow ornamental shelf edging. The hollow bases of malted-milk dispensers and drink mixers were cockroach havens.

In a hospital Frings (1948) found cockroaches in decorative trim around doorways, by the thousands in wicker laundry baskets, and in incubators for premature babies. In military hospitals we have seen cockroaches (_Blattella germanica_) in kitchens and dining halls in the usual hiding places mentioned above and on the undersides of stainless-steel serving tables.

In department stores cockroaches have been found in food departments, beauty salons, rest rooms, dressing rooms, linen departments, and stationery departments (Anonymous, 1952). The infestation in the linen department was traced to clean towels which, when returned from the laundry, contained at least 500 cockroaches per bundle. The insects were carried into the rest rooms and beauty salon when the towels were distributed.

The microhabitats of cockroaches in privies and sewers have not been studied. These habitats are particularly important in view of the demonstrated migrations of cockroaches from sewers into dwellings and the possible dissemination of pathogenic microorganisms from feces to food. The reader is referred to our 1957(a) paper for a summary of the known information on cockroach dispersal from sewers.

COCKROACHES ASSOCIATED WITH LAND-BASED STRUCTURES

Most of the cockroaches listed below are either known domiciliary species or they have been found one or more times in houses or other man-made structures. The known structural pests breed within the building. Certain other species, which have been observed only infrequently in structures and are not known to breed there, may possibly be incipient pests; these latter species may attain future economic importance if they establish breeding colonies within a structure. A few species have undoubtedly wandered indoors by accident. It is difficult to decide whether a particular species was an accidental invader or whether it was attracted indoors in response to some stimulus. Only additional information will provide the desired

answers.

=Aglaopteryx ypsilon=

Trinidad.--Male found indoors (Princis and Kevan, 1955).

=Allacta similis=

Hawaii.--Found only indoors at Nauhi. Otherwise this is apparently an outdoor species (Swezey and Williams, 1932).

=Blaberus craniifer=

Cuba.--Household pest (Deschapelles, 1939). Particularly abundant in houses in Santiago and Havana (Rehn and Hebard, 1927).

Florida.--Under boards in woodshed (Rehn and Hebard, 1912, 1914).

=Blaberus discoidalis=

Ecuador.--In eating places (Campos R., 1926).

Hispaniola.--In houses (Rehn and Hebard, 1927).

New Jersey.--In greenhouse (Weiss, 1917).

Puerto Rico.--In homes (Seín, 1923). In fruit stores (Wolcott, 1950).

=Blatta lateralis=

Central Asia.--Household pest, often found in homes with clay floors (Bei-Bienko, 1950).

Turkmen S.S.R.--Males and females occurred in dwellings (Vlasov, 1929).

=Blatta orientalis=

This species is a cosmopolitan domiciliary pest (Hebard, 1917; Rehn, 1945). It is reported to occur particularly in basements and crawl spaces under basementless houses (Mallis, 1954). In damp basements where food is available large colonies are not unusual, but it also may infest offices and apartments several floors off the ground (Gould and Deay, 1940). The number encountered on upper floors is seldom large, but the frequency of occurrence may reach 30 percent of the observations (Spear et al., _in_ Shuyler, 1956). In supermarkets this species hides during the day inside concrete blocks or cracks in the foundation, under furniture, or behind cartons; it is conspicuous on the floors of the markets at night (De Long, 1948). In Great Britain the kitchen is preferred by this pest (and by _Blattella germanica_); they shelter beneath steam radiators and gas stoves, behind hot-water pipes, underneath furniture and floor coverings, sinks and baths; basements and underground kitchens are especially likely to be infested (Laing, 1946; British Museum [Nat. Hist.], 1951). Goodliffe (1958) noted that _B. orientalis_ may travel long distances to find food.

=Blattella germanica=

This species is a cosmopolitan domiciliary pest (Hebard, 1917; Rehn, 1945). It is one of the commonest insects in homes and restaurants (Gould and Deay, 1940). It is found in kitchens, larders, bathrooms, furnace rooms, and storage rooms of bakeries, breweries, hospitals, barracks, as well as dwellings, where, during daylight, it hides behind cupboards, furniture, hanging pictures, panels and skirting boards, in cracks around drains, water pipes, electric wires, and hot-water and steam heating units (Wille, 1920). The German cockroach may be found in cracks around baseboards, pipes, conduits, sinks, and drawers; behind cabinets; inside switch boxes and refrigerators; on under surfaces of tables, chairs, and shelves; between stacks of stored goods, and in almost every place that is not readily observed (Kruse, 1948). We have also seen this species packed in electric-clock cases and loud-speaker baffles, in cash registers, and clinging to the undersurface of stainless-steel steam tables. The infestation of markets by this species has been described above. Very narrow cracks provide refuges for the German cockroach. Wille (1920) found first-instar nymphs in cracks 0.5 mm. wide and adult males and females without oöthecae in cracks 1.6 mm. wide.

Shuyler (1956) has observed extensions into relatively new structural habitats by _Blattella germanica_ in the north-central area of the United States. A few German cockroaches are now being encountered in living rooms, bedrooms, clothes closets, bedroom furniture, lobbies, entrance halls, checkrooms, nonfood storerooms, nonfood warehouses, and coin-vending machine repair shops. In these situations this species is behaving much like the brown-banded cockroach, _Supella supellectilium_.

=Blattella schubotzi=

Cameroon.--Five specimens in a house (Princis, 1955).

=Blattella vaga=

_U.S.A._--Although this is mostly an outdoor species, during dry seasons it may temporarily enter houses in great numbers, occasionally breeding indoors in Arizona (Flock, 1941a). Two adults were collected indoors in Texas (Riherd, 1953).

=Chromatonotus notatus=

Trinidad.--A male was found indoors (Princis and Kevan, 1955).

=Cutilia soror=

Hawaii.--Almost as common in houses as _Neostylopyga rhombifolia_ (Hebard, 1922).

=Ectobius duskei=

_U.S.S.R._--Frequently occurs in living apartments in farming localities as an accidental inhabitant (Bei-Bienko, 1950).

=Ectobius pallidus=

Massachusetts.--A summertime pest in houses along coast (Gurney, 1953; E. R. Willis, personal observation). Generally an outdoor species.

=Epilampra abdomen-nigrum=

Trinidad.--Male, indoors (Princis and Kevan, 1955). An outdoor species generally.

=Ergaula capensis=

Cameroon.--A male taken in a house (Princis, 1955).

=Eublaberus posticus=

Trinidad.--Indoors, feeding on bat feces (Princis and Kevan, 1955).

=Eurycotis floridana=

Florida.--Occasionally found in homes (Creighton, 1954; Roth and Willis, 1954a).

=Euthyrrhapha pacifica=

Hawaii.--Found outdoors and indoors where it breeds in neglected cupboards and in rubbish (Fullaway and Krauss, 1945).

=Holocompsa azteca=

Mexico.--Household pest (Ball et al., 1942).

=Holocompsa cyanea=

Costa Rica.--One specimen in house (Rehn, 1906).

=Holocompsa nitidula=

Apparently domiciliary in American Tropics (Hebard, 1917). In houses under chests, etc., Cuba (Gundlach, 1890-1891); Puerto Rico (Gundlach, 1887). In folds of burlap bag, Florida (Rehn and Hebard, 1914).

=Ischnoptera rufa occidentalis=

Panama.--Thrives about human habitations under litter, though not domiciliary (Hebard, 1920).

=Ischnoptera rufa rufa=

Jamaica.--In hotel. "While hardly a domiciliary form it would seem to frequent environments where man has considerably disturbed natural conditions, as under debris, docks, under logs and stones in cultivated areas" (Rehn and Hebard, 1927).

=Leucophaea maderae=

West Indies.--In habitations, warehouses, and other structures; "At times it is a very abundant and serious pest" (Rehn, 1945). In Puerto Rico it was also found in fruit stores, markets, and inns (Seín, 1923; Wolcott, 1950).

Reported as a domiciliary pest in Madeira (Heer, 1864); Windward Islands (Marshall, 1878); Tropics and sub-Tropics (Rehn, 1937); Philippine Islands (Uichanco, 1953); New York City (Anonymous, 1953; Gurney, 1953); Trinidad (Princis and Kevan, 1955). This species is also established in coastal Brazil, Central America, all the Greater Antilles, several other tropical islands, and tropical Africa, where it probably originated (Rehn, 1945).

=Leurolestes circumvagans=

Hispaniola, Grenada.--Largely domiciliary (Rehn and Hebard, 1927).

=Leurolestes pallidus=

Cuba.--All over island, in houses, under lockers, etc. (Rehn, 1945; Gundlach, 1890-1891).

Florida.--Rehn and Hebard (1914).

This species has been recorded from various islands in the West Indies, from Mexico, Guatemala, and Brazil (Rehn, 1945).

=Methana marginalis=

Australia.--Reported entering houses (Pope, 1953a).

=Nauphoeta cinerea=

Australia.--In hospital (Mackerras and Mackerras, 1948). In dwellings, grain stores, and fowl-feeding pens (Pope, 1953).

Sudan.--Domiciliary in huts of the Shilluk natives; fairly widely distributed in eastern Africa (Rehn, 1945).

Hawaii.--In feed rooms of poultry plants (Illingworth, 1942).

Florida.--Established in feed mills around Tampa (Gresham, 1952; Gurney, 1953).

The wide distribution of this species from East Africa, where it originated, to the Orient and the New World was undoubtedly mediated by shipping (Rehn, 1945).

=Neoblattella= sp.

Puerto Rico.--Observed [as _Blatta caraibea_, Rehn and Hebard (1927)] in houses (Gundlach, 1887).

=Neostylopyga rhombifolia=

Domiciliary in Indo-Malaya and New World Tropics (Rehn, 1945); Philippine Islands (Uichanco, 1953); and Hawaii (Hebard, 1922).

=Oxyhaloa buprestoides=

Presumably to some extent domiciliary, as it evidently spread from Africa to the New World via slave ships (Rehn and Hebard, 1927; Rehn, 1945).

=Panchlora nivea=

Colombia.--A male and a female taken in a dwelling (Princis, 1946). This is primarily an outdoor species which is frequently taken indoors as an adventive on bananas (see p. 150 for references).

=Parcoblatta fulvescens=

Florida.--Males found in laboratory and dormitory buildings, ostensibly attracted by lights (Friauf, 1953).

=Parcoblatta lata=

Connecticut.--Domiciliary pest (Moore, 1957). Generally an outdoor species.

=Parcoblatta notha=

Arizona.--It may occasionally be a nuisance in houses (Ball et al., 1942).

=Parcoblatta pensylvanica=

_U.S.A._--Country houses often badly infested, Indiana (Blatchley, 1920). Frequently taken in houses in wooded areas, Michigan (Hubbell, 1922). Infestation by males, females, and nymphs on fourth floor of building, South Dakota (Severin, 1952). Houses in wooded areas infested by nymphs and occasionally by adults (Gould and Deay, 1940).

Canada.--Pest in summer cottages in Ontario (Walker, 1912).

=Periplaneta americana=

This is a cosmopolitan domiciliary pest (Hebard, 1917; Rehn, 1945). It is common in restaurants, grocery stores, bakeries, and where food is prepared or stored; it was trapped regularly in the basement and upper floors of store buildings, and it was also found in all heated parts of an old meat-packing plant (Gould and Deay, 1938). _P. americana_ was numerous in latrines in Iran (Bei-Bienko, 1950) and in privies in Texas (Dow, 1955) and Georgia (Haines and Palmer, 1955). Large numbers of this species also occur in sewers adjacent to human habitations (Roth and Willis, 1957a).

=Periplaneta australasiae=

Generally domiciliary, but also occurs outdoors in the West Indies (Rehn and Hebard, 1927). It is a very abundant domiciliary pest in tropical Africa and tropical America (Rehn, 1945); Ecuador (Campos R., 1926); Puerto Rico (Sein, 1923); Philippine Islands (Uichanco, 1953); Australia (Pope, 1953).

Also occurs as a pest in greenhouses in Pennsylvania (Thilow and Riley, 1891); France (Giard, 1900); Italy (Boettger, 1930); Great Britain (Laing, 1946; British Museum [Nat. Hist.], 1951).

=Periplaneta brunnea=

Circumtropical domiciliary pest which is apparently more nearly peculiar to the Tropics and adjacent regions than _P. americana_ (Hebard, 1917). This species has been trapped in significant numbers in privies and dwellings in Georgia (Haines and Palmer, 1955). It is the species of _Periplaneta_ present in homes in San Antonio, Texas (Mallis, 1954).

=Periplaneta fuliginosa=

_U.S.A._--Frequently encountered out of doors, but has been reported common after dark about a hotel in Alabama and was captured in a house in Louisiana; it was also extremely abundant on the wharves at night in Jacksonville, Florida (Hebard,

1917). As a domiciliary pest it was, next to _Blattella germanica_, the most common cockroach inside homes in southwest Georgia, where it was also the most common cockroach in privies (Haines and Palmer, 1955). This species has become a very common domiciliary pest in Texas (Eads, personal communication, 1955). It infested a greenhouse for five years in Indiana (Gould and Deay, 1940).

=Periplaneta ignota=

Australia.--It occurs in dwellings occasionally (Pope, 1953).

=Phaetalia pallida=

Colombia.--Three specimens from three dwellings (Princis, 1946).

Trinidad.--Male indoors; male and female at light (Princis and Kevan, 1955).

=Plectoptera dorsalis=

Puerto Rico.--According to Gundlach (1887) it enters houses at night attracted by light (Rehn and Hebard, 1927).

=Polyphaga aegyptiaca=

Iraq.--Common in houses (Weber, 1954).

Caucasus.--Winged male in kitchen (Burr, 1913).

_U.S.S.R._--Listed as a sinanthrope (Bei-Bienko, 1950).

=Polyphaga saussurei=

South-central Asia.--One of the commonest domiciliary species (Bei-Bienko, 1950).

=Pseudophoraspis nebulosa=

East Indies.--This species is sometimes difficult indoors (Karny, 1925).

=Pycnoscelus surinamensis=

A household pest in the East Indies (Karny, 1925); Philippine Islands (Uichanco, 1953); Tanganyika (Smith, 1955); Trinidad, eight records indoors (Princis and Kevan, 1955). It is also a greenhouse pest (Hebard, 1917; Zappe, 1918; Doucette and Smith, 1926; Saupe, 1928; Roeser, 1940). Common in or around chicken batteries and yards in Hawaii (Schwabe, 1949).

=Supella supellectilium=

Domiciliary wherever distributed (Rehn, 1945), this species is especially difficult to control because of its apparently nonselective dispersal throughout dwellings. For example, Mallis (1954) observed in Texas that it was widely distributed throughout the apartment and was probably the most common cockroach seen in the bedroom; its favorite harborages were beneath and behind corner braces on kitchen chairs, underneath tables, behind pictures and other objects on walls, and in shower stalls; its oöthecae were commonly fastened on walls and ceilings throughout the house. Gould and Deay (1940) reported that this species prefers high locations, such as

shelves in closets, behind pictures, and picture molding; oöthecae were found about kitchen sink, desks, tables, and other furniture, and even in bedding. Hafez and Afifi (1956) stated that the adult wanders in nearly all rooms of the house and only visits the kitchen when searching for food; it hides in cupboards, pantries, closets, bookshelves, drawers, and behind picture frames; the nymphs normally hide in the corners of drawers, behind frames, and in similar situations.

=Symploce bicolor=

Puerto Rico.--In houses, Sardinera Beach, Mona Island (Ramos, 1946).

=Symploce hospes=

North American Tropics.--Domiciliary, but not exclusively so, and apparently widely distributed (Hebard, 1917). In Florida, as _Ischnoptera rufescens_, found in a greasy cupboard (Rehn and Hebard, 1914).

Hawaii.--Illingworth (1915).

SHIPS

Sailing ships have long been notorious for their unwelcome hordes of cockroaches, and it was by ship that at least 11 domiciliary species migrated from their centers of origin to other parts of the world (Rehn, 1945). Over 40 nondomiciliary species have been carried by ship from the American Tropics to other parts of the world in cargoes of bananas (p. 146). In addition to these, other adventive cockroaches appear from time to time in ports to which they have been carried by ships. Yet by far the most numerous cockroaches on shipboard are the breeding populations of a few common domiciliary pests. Except in the most rigorously disinsectized ships, this commerce in cockroaches has continued to the present day.

Cockroaches undoubtedly infested the first ships that sailed the Mediterranean; of these we have no records. The earliest recognizable record of cockroaches on shipboard is Moffett's (1634) statement that when Drake captured the ship _Philip_, he found it overrun with cockroaches [_Blattarum alatarum_]. Bligh (1792) described disinfesting H.M.S. _Bounty_ with boiling water to kill cockroaches. Chamisso (1829) reported that he had seen ships casks, in which rice or grain had been stored, that were found to be filled with _Blattella germanica_ when opened. During a voyage from England to Van Diemen's Land, Lewis (1836) was greatly annoyed by hundreds of cockroaches flying about his cabin at night; the most numerous resembled _Periplaneta americana_ and another was similar to _Ectobius lapponicus_. This latter was undoubtedly _B. germanica_, which is the only ship-infesting species that resembles the feral _E. lapponicus_. Lewis continued, _P. americana_ "were in immense profusion, and had communication with every part of the ship, between the timbers or skin. The ravages they committed on everything edible were very extensive; not a biscuit but was more or less polluted by them, and amongst the cargo 300 cases of cheeses, which had holes in them to prevent their sweating, were considerably damaged, some of them being half devoured and not one without some marks of their residence."

Kingsley (1870), Kellogg (1908), Gates (1912), Heiser (1936), and Bronson (1943) have all reported that cockroaches were so numerous on ships that they gnawed the skin and nails of the men on board. These are all independent observations of what may well have been a common occurrence on ships. We have discussed in detail the subject of cockroaches biting man in our 1957(a) paper.

Mosely (1892) reported, "At the time that England was left the ship [H.M.S. _Challenger_] seemed nearly free of animals, other than men, dogs, and livestock required for food. The first cockroaches apparently came on board at St. Vincent, Cape Verdes.... Cockroaches soon became plentiful on board, and showed themselves whenever the ship was in a warm climate.

"At one period of the voyage, a number of these insects established themselves in my cabin, and devoured parts of my boots, nibbling off all the margins of leather projecting beyond the seams on the upper leathers."

Sir Edmund Freemantle (1904) recalled some of his experiences in the British Navy. "Cockroaches in the tropics were also terrible scourges. One saw little of them in fine, dry weather, but in damp, wet weather they seemed to come from every hidden corner ... our remedy in the 'Spartan' was to make the boys catch them--on pain of being caned.... One brig, the 'Lily,' was so overrun by cockroaches that the officers' clothes smelt of them."

The quotation from Sonan (1924) on page 348 describes similar conditions in the Japanese Navy.

On modern cargo ships cockroaches are reported to be extremely numerous in the galley, the crew's quarters, and sometimes in the holds; they dwell in hot, humid environments such as the casing around steam pipes (Monro, 1951). Williams (1931) reported that _Blattella germanica_ was the most important cockroach pest on ships seen at New York. Although often numerous in the holds, the cockroaches as a rule congregated in living quarters. They were also frequently found between tarpaulins covering the hatches. It was not unusual to kill 20,000 to 50,000 in the forecastle, and more than 20,000 have been taken from a single stateroom. Simanton (1946) inspected the S.S. _William Kieth_ when it berthed at San Francisco from a 10-month voyage to the South Pacific. The holds were infested with thousands of _Periplaneta americana_, but in the crew's quarters, mess halls, and storerooms _B. germanica_ predominated. After insecticidal treatment about 2,000 _P. americana_ were seen in each hold and as many as 24 _B. germanica_ in each cabin. Richardson (1947) reported that in Army transports inspected between 1943 and 1946 at New York, _P. americana_ was found in the galleys and messes, and occasionally heavy infestations were found deep in the holds; _B. germanica_ was found in the galleys and messes; _Blatta orientalis_ was found only in the hold.

Additional references indicating the presence of _Blattella germanica_ on ships may be found in the account of its parasite _Ripidius pectinicornis_ (p. 232). Although Rice (1925) and Williams (1931) cite _B. germanica_ as the most numerous cockroach on ships, Brooke (1920) stated that the great majority of ship cockroaches were _Periplaneta americana_. In addition to citing cockroach infestations on ships, the

following authors reported various methods for disinfecting ships: Canalis (1916), Pryor (1918), Brooke (1920), Rice (1925), Williams (1931), Simanton (1946), Richardson (1947), and Anonymous (1951, 1954).

COCKROACHES ASSOCIATED WITH SHIPS

In the following list we include some previously unpublished data on cockroaches that were recovered from ships at the Miami, Fla., Quarantine Station for the periods November 1945 through May 1946; May, June, August, and September 1950; and 17 July 1957 (Porter, personal communication, 1958). These data were lumped, without breakdown to species, under the entry Orthoptera in Porter (1958).

Certain of the species listed below occur only accidentally on shipboard and will probably never establish breeding colonies on ships or become pests on shipboard or elsewhere; some were merely passengers between one land-based colony and another. Others, the truly domiciliary pests, are as likely to be pests on shipboard as they are in land-based structures.

=Blaberus discoidalis=

Hispaniola.--On board ship (Rehn and Hebard, 1927).

=Blatta orientalis=

_U.S.A._--At Port of New York (Richardson, 1947).

=Blattella germanica=

At sea?--In ships casks (Chamisso, 1829).

_U.S.A._--Port of New York (Williams, 1931; Richardson, 1947). San Francisco (Simanton, 1946). At Miami, 7,852 live specimens recovered from ships (Porter, personal communication, 1958). Most numerous species on ships (Rice, 1925).

=Epilampra maya=

At sea.--One male and one female found dead on S.S. _Tenadores_ (Hebard, 1917).

=Epilampra= sp.

Florida.--One dead specimen, Miami (Porter, personal communication, 1958).

=Ischnoptera= sp.

Florida.--Five live and one dead specimen, Miami (Porter, personal communication, 1958).

=Latiblattella= sp.

At sea.--One female alive in hold of S.S. _Tenadores_ (Hebard, 1917).

=Leucophaea maderae=

Brought from West Africa to West Indies and Brazil by slave ships (Rehn, 1945).

=Nauphoeta cinerea=

Widely disseminated by sailing ships (Rehn, 1945).

=Neoblattella fratercula=

At sea.--Two females found dead on S.S. _Tenadores_ (Hebard, 1917).

=Neoblattella fraterna=

At sea.--One male found dead in hold of S.S. _Tenadores_ (Hebard, 1917).

=Neoblattella nahua=

At sea.--One female dead in hold of S.S. _Tenadores_ (Hebard, 1917).

=Neoblattella= sp.

Florida.--Five dead specimens recovered from ships, Miami (Porter, personal communication, 1958).

=Neostylopyga rhombifolia=

Widely distributed by sailing ships (Rehn, 1945).

England.--Captured on a sugar vessel from Java (Lucas, 1920).

=Nyctibora noctivaga=

At sea.--One male and one female nymph found dead on S. S. _Tenadores_ (Hebard, 1917).

=Nyctibora= sp.

Florida.--Two dead specimens recovered from ships at Miami (Porter, personal communication, 1958).

=Oxyhaloa buprestoides=

Spread from Africa to New World by ships (Rehn and Hebard, 1927; Rehn, 1945).

=Panchlora nivea=

At sea.--One female dead in hold of S.S. _Tenadores_ (Hebard, 1917).

Florida.--Fifteen dead specimens taken from ships, Miami (Porter, personal communication, 1958).

From the numerous records of this species as an adventive taken on bananas (p. 150), it may be presumed to be a frequent traveler on banana boats.

=Periplaneta americana=

At sea.--Lewis (1836). Hebard (1933a) stated that this is "often a serious pest on the smaller ships sailing the South Seas."

_U.S.A._--San Francisco (Simanton, 1946). Port of New York (Richardson, 1947). At Miami, 62 live and 123 dead specimens (Porter, personal communication, 1958).

=Periplaneta australasiae=

Migrated from West Africa to America in slave ships (Rehn, 1945).

=Pycnoscelus surinamensis=

Probably in part reached the New World by way of Africa in slave ships (Rehn, 1945).

=Supella supellectilium=

Reached America from West Africa by slave ship (Rehn, 1945).

=Xestoblatta festae=

At sea.--One female found dead in hold of S.S. _Tenadores_ (Hebard, 1917).

AIRCRAFT

Michel (1935) stated that the development of air transportation brought the same insect dispersal problems that exist in land and water transportation; in addition, the problem of cockroach infestation had become a very serious one, quite aside from the hygienic point of view, because it had been discovered that these insects seek out the wings of airplanes, where they subsisted on the glue and dope used in airplane construction. However, Dethier (1945) found no cockroaches in dismantled or wrecked wing and tail structures of metal aircraft in central Africa. In fact, all-metal aircraft would seem to provide little in the way of food or water for stowaway cockroaches.

Laird (1951, 1952, 1956a) found living specimens of _Blattella germanica_, _Periplaneta americana_, and _Periplaneta australasiae_ in baggage compartments and/or kitchens in aircraft. Other species which have been recovered from undisclosed spaces in aircraft are listed below. Some of the cockroaches that were reported as dead may not have died from exposure during flight but may have been killed by insecticide applied by inspecting personnel at the airports.

COCKROACHES ASSOCIATED WITH AIRCRAFT

In the following list we include some previously unpublished data on cockroaches that were recovered from aircraft in Miami, Fla., International Airport from 1 July 1956 through 30 June 1957 (Porter, personal communication, 1958). These data were lumped under the entry Orthoptera without breakdown to species in Porter (1958).

Species reported by Hughes (1949), and cited below as from southern United States, were recovered from aircraft that arrived at Brownsville, Fort Worth, Miami, New Orleans, and San Juan. There was no way of linking a specific record with any particular city.

The comments we made above about species that are infrequently encountered on ships apply with equal validity to similar species found on aircraft.

=Anaplecta= sp.

_U.S.A._--One live and 15 dead specimens recovered from 16 aircraft at Miami (Denning et al., 1947).

=Blatta orientalis=

_U.S.A._--Six live and four dead specimens recovered from six aircraft at Miami (Denning et al., 1947). Southern U.S. (Hughes, 1949).

=Blattella germanica=

Hawaii.--Williams (1946a).

Khartoum.--Whitfield (1940).

New Zealand.--Laird (1951, 1952, 1956a).

_U.S.A._--At Miami 193 live and 184 dead specimens were recovered from 141 aircraft (Denning et al., 1947). Recovered at airports in southern U.S. (Hughes, 1949). Recovered at Miami, 51 live and 24 dead specimens (Porter, personal communication, 1958). Exposed experimentally in jet aircraft (Sullivan et al., 1958).

=Blattella= sp.

Khartoum.--Whitfield (1940).

Southern U.S.--Hughes (1949).

=Cariblatta= ssp.

_U.S.A._--One live and three dead specimens recovered from three aircraft at Miami (Denning et al., 1947). Southern U.S. (Hughes, 1949).

=Epilampra= sp.

_U.S.A._--One dead specimen, Miami (Denning et al., 1947). Southern U.S. (Hughes, 1949).

=Eublaberus posticus=

Southern U.S.--Hughes (1949).

=Ischnoptera rufa rufa=

_U.S.A._--Two dead specimens recovered from two aircraft at Miami (Denning et al., 1947).

=Ischnoptera= sp.

_U.S.A._--Two live and one dead specimen recovered from three aircraft, Miami (Denning et al., 1947). Southern U.S. (Hughes, 1949). One dead specimen, Miami (Porter, personal communication, 1958).

=Leucophaea maderae=

_U.S.A._--Three dead specimens recovered from three aircraft at Miami (Denning et al., 1947). Southern U.S. (Hughes, 1949).

=Nauphoeta cinerea=

_U.S.A._--One live and one dead specimen, Miami (Denning et al., 1947).

=Neoblattella= sp.

_U.S.A._--One dead specimen recovered at Miami (Denning et al., 1947).

=Panchlora nivea=

_U.S.A._--Two dead specimens recovered from two aircraft, Miami (Denning et al., 1947).

=Periplaneta americana=

_U.S.A._--Five live and three dead specimens recovered from seven aircraft, Miami (Denning et al., 1947). Southern U.S. (Hughes, 1949). Three live and five dead, Miami (Porter, personal communication, 1958). Experimentally exposed in jet aircraft, U.S. (Sullivan et al., 1958).

New Zealand.--Laird (1951, 1952).

=Periplaneta australasiae=

_U.S.A._--Two dead specimens recovered from two aircraft, Miami (Welch, 1939). Five live and three dead from five aircraft, Miami (Denning et al., 1947). Southern U.S. (Hughes, 1949). Three live and two dead, Miami (Porter, personal communication, 1958).

New Zealand.--Laird (1952).

=Periplaneta= spp.

_U.S.A._--One live and three dead specimens from four aircraft, Miami (Denning et al., 1947). Southern U.S. (Hughes, 1949). One live and six dead, Miami (Porter, personal communication, 1958).

=Pycnoscelus surinamensis=

_U.S.A._--Two live and three dead specimens from five aircraft, Miami (Denning et al., 1947). Southern U.S. (Hughes, 1949).

New Zealand.--Laird (1956a).

=Supella supellectilium=

_U.S.A._--Two live and one dead specimen from one aircraft, Miami (Denning et al., 1947). Southern U.S. (Hughes, 1949).

Khartoum.--Whitfield (1940).

=Supella= sp.

_U.S.A._--Two live specimens from two aircraft, Miami (Welch, 1939). Southern U.S. (Hughes, 1949).

=Symploce= sp.

_U.S.A._--Two live and one dead specimen from three aircraft, Miami (Denning et al., 1947). Southern U.S. (Hughes, 1949).

=Unidentified cockroaches=

Anglo-Egyptian Sudan.--At Khartoum (Whitfield, 1940).

Brazil.--From flying boats, 62 specimens; from land planes, 45 specimens (Carneiro de Mendonça and Cerqueira, 1947).

Central Africa.--Dethier (1945).

Kenya.--At Kisumu (Symes in Whitfield, 1940).

New Zealand.--Laird (1956a).

_U.S.A._--Four live and 14 dead specimens from 4 aircraft, Miami (Welch, 1939). One live and 15 dead specimens (adults?) from 16 aircraft; 8 oöthecae from 7 planes; 147 live and 83 dead nymphs from 108 planes, Miami (Denning et al., 1947). Southern U.S. (Hughes, 1949). Five live and 7 dead specimens, Miami (Porter, personal communication, 1958).

IV. CLASSIFICATION OF THE ASSOCIATIONS

Asano (1937) classified the natural enemies of cockroaches into two types as follows:

1. Enemies that feed mainly on cockroaches (certain ripiphorid beetles and certain chalcid, evaniid, and ampulicid wasps).

2. Organisms which, in their search for food, devour cockroaches that may be encountered (certain species of scorpions, spiders, ticks, centipedes, Strepsiptera, ants, birds, rats, and "parasitic bacteria").

Cameron (1955) arranged the associates of cockroaches in two groups as follows:

Group A. Parasites and predators.

1. Parasites: Hymenoptera (Evaniidae, Eulophidae, Eupelmidae, Encyrtidae, Pteromalidae, Cleonymidae) and Coleoptera (Ripiphoridae).

2. Predators: Hymenoptera (Ampulicidae), Hemiptera (Reduviidae), Coleoptera (Dermestidae), Arachnida (Araneae, Acarina).

Group B. Parasites and symbionts.

1. Protozoa (including examples of both parasites and commensals).

2. Nematoda (including both primary "parasites" and secondary parasites).

3. Bacteria (including the mutualistic bacteroids).

4. Algae [_Arthromitis_ (= _Hygrocrocis_) _intestinalis_; see p. 124].

Asano's arrangement, although essentially true, is limited; Cameron's system is divided into arthropods (group A) and lower forms (group B), but does not include higher animals. Both attempts at classification need amplification; this we have endeavored to do below.

In classifying the biotic associates of cockroaches, we were immediately confronted with a problem in semantics. The concepts parasitism, predatism, and symbiosis have all been used with various shades of meaning by different authors. The problem is not solved merely by accepting as authoritative specific definitions, however apt they seem to be, because, unfortunately, these concepts are not mutually exclusive. For example, among the entomophagous insects, as Sweetman (1936) has pointed out, there can be no definite line of separation between parasitism and predatism: the two intergrade, only the extremes being quite distinct. In fact, Andrewartha and Birch (1954) generalized these relationships by calling both categories predatism. These authors divided natural populations of associated organisms into nonpredators and predators. Although this simplifies their presentation of the general principles of ecology, for our purpose more narrowly defined terms have proved useful.

In the main we have followed Sweetman (1936) and Allee et al. (1949) in arriving at the following definitions:

=Symbiosis= is the living together in more or less intimate association of organisms of different species; it includes virtually all relationships between cockroaches and other organisms, such as parasitism, predatism, commensalism, and mutualism. Allee et al. (1949) apparently do not include predatism in symbiosis.

=Mutualism= is symbiosis in which both members benefit by the association. The smaller partner has commonly been called a symbiont or symbiote by authors.

=Commensalism= includes associations in which neither party appears to benefit or be harmed. One partner may live on the surplus food or wastes of the other; shelter and transport may be involved.

=Parasitism= is the state of symbiosis in which one of the members feeds upon the other during the whole of either the immature or mature feeding stage; the host is harmed in some way and may be killed.

=Predatism= is an association in which one member attacks and feeds upon, or stores as food for its progeny, one or more other organisms; the predator spends less than the immature or mature feeding period on the prey. This category includes a few invertebrates and all the vertebrates that capture, kill, and feed on cockroaches. This association may be divided into interspecies predatism, in which the predator preys upon a different species, and intraspecies predatism (cannibalism) in which the predator preys upon its own species.

Although we have attempted to adhere to these definitions throughout this discussion, we realize that in doing so we may have tended to oversimplify complex relations. Some questionable interpretations stem from insufficient knowledge of the basic relationships between cockroaches and their associates. Only further study will clarify these relationships. Some of the problems are discussed below.

Probably many of the so-called parasites (e.g., Protozoa like _Nyctotherus_, and intestinal nematodes of the family Thelastomatidae), which do not invade the host's tissues and seem to have no effect on the activity and vitality of the host, are

commensals. Although we consider these forms to be commensals, we realize that they might actually affect the host in some way even though this has not been shown. It is possible that Rothschild and Clay's (1957) statement about bird parasites may well apply to the apparently harmless organisms found in the cockroach. These authors wrote, "It cannot be too strongly emphasized that the effect of all types of parasites on the host is detrimental. If we find that a bird seems little, if at all, inconvenienced by the presence of Protozoa or worms or lice, or a cuckoo in the nest, we can nevertheless assume that it would be better off without them.... Small effects such as lack of vitality, loss of voice, excessive blinking, or perverted habits like dirt eating are extremely difficult to gauge. Nevertheless, it is only a question of degree. Potentially all parasites are harmful." It should also be pointed out that some workers would consider certain of our commensals of cockroaches to be parasites. Thus Faust (1955) stated that "A truly successful parasite is one which has developed a state of equilibrium with its host, so that no detectable damage is produced which endangers the health or life of the host. In a suitable host the parasite may obtain food and shelter without any evidence of trauma or toxicity. The damage produced may be so slight that repair and functional readjustment keep pace with the injury." Faust's successful parasite would be indistinguishable from a commensal, but there is undeniably a difference between an organism causing slight, and undetectable, damage to a host and one causing none. Certain of the organisms we list as commensals may eventually be shown to be parasites.

Certain organisms which live in cockroaches appear to have no effect on the vitality of the host even though the tissues of the host are invaded. Gregarines may penetrate the intestinal wall of the cockroach without seeming to injure the host. Fungi of the genus _Herpomyces_ invade the cuticle of cockroaches producing pathological changes; yet the insects' behavior is apparently unaffected (see p. 129). We consider these organisms to be parasites because the host's tissues are invaded and, as far as we know, no benefit to the host results.

In the literature certain insects have been considered to be either parasites or predators or both. Among these are the ensign wasps (Evaniidae), whose larvae feed on the eggs of cockroaches within the oötheca, and the ampulicid wasps, which capture, paralyze, and store in their nests (as food for their larvae) nymphs and adults of cockroaches. Clausen (1940) claimed that the evaniid _Zeuxevania splendidula_ is a true egg parasite when it destroys the first egg in a cockroach oötheca; but after the wasp larva molts and proceeds to devour the other eggs, he considered it to be a predator. Clausen's definition of an entomophagous parasite is an insect that in its larval stage develops either internally or externally upon a _single_ host which is eventually killed; with few exceptions the adults are free-living and their food is usually different from that of the larvae. A predatory insect, by Clausen's definition, is principally free-living in the larval as well as adult stage, kills the host immediately by direct attack, and requires a number of victims to reach maturity; the predator is of greater size than the prey, and the food sources of the adults and immature stages are frequently the same.

It is apparent, as Clausen and other writers have pointed out, that there are instances of a particular species showing characteristics which fit both the definitions for predator and parasite. Thus, among the evaniids one wasp larva destroys all the eggs in an oötheca, but in spite of this the larva has more of the characteristics of a parasite than of a predator; the adult wasp does not utilize the same food as the larva (adults have been taken on flowers and on honeydew from scale insects). It is questionable whether the evaniid larva kills the cockroach egg outright. The wasp larva, being restricted to the inside of the oötheca, is not free-living. Probably the only criterion by which the evaniid could be judged to be a predator, by Clausen's definition, is that more than a single egg is devoured by the maturing wasp larva.

Among the other wasp parasites (Encyrtidae, Eulophidae, Eupelmidae) of cockroach eggs many individuals develop in a single oötheca. When a hundred or more wasps emerge from an oötheca which contained less than 20 eggs, it is obvious that a single cockroach egg supported more than one wasp, yet it is possible that one particular wasp larva may have fed upon more than one cockroach egg before becoming an adult. We consider all entomophagous wasps that develop in cockroach oöthecae to be parasites rather than predators.

On the other hand, even though _Anastatus floridanus_, _A. tenuipes_, and _Tetrastichus hagenowii_ are egg parasites as larvae, the adult females are, in a sense, predators when they sometimes eat part of the cockroach egg that oozes through the oviposition puncture (Roth and Willis, 1954a, 1954b). Williams (1929) has seen the female of _Ampulex canaliculata_ imbibe blood that oozed from the cut ends of the cockroach's antennae after she had clipped them off before leading the prey to her nest. Yet despite this evident predatism on the part of the adult, the larva feeds as a parasite on the stored cockroaches in accordance with Sweetman's (1936) (though not Clausen's 1940) definition of parasitism, which is "that form of symbiosis in which one symbiont lives in or on the host organism and feeds at its expense during the whole of either the immature or mature feeding stage." The ampulicid larva, as the evaniid, is not free-living and does not kill the host immediately by direct attack, even though it may require more than one victim to reach maturity. Thus, within one individual both parasitic and predatory behavior are operant during different stages of its life history.

With the above discussion in mind we have summarized below the various biotic associations of cockroaches. Only a few examples are given for each section, but all organisms with similar habits presumably would be classified in the same categories.

Class A. Associations in which cockroaches serve as hosts, vectors, or prey for other organisms.

Type I. Obligate associates. Animals and plants that normally develop only on or in the cockroach; in general, these organisms depend entirely upon the cockroach for survival.

Group 1. Mutuals (symbiotes or symbionts of authors).

(a) Bacteria-like organisms (bacteroids which are found in the fat body of all cockroaches that have been examined; p. 96).

(b) Bacteria (wood-digesting forms in _Panesthia_, and possibly certain bacteria in the intestines, of other cockroaches; p. 100).

(c) Protozoa (several genera and species found in _Cryptocercus_; p. 101).

Group 2. Commensals.

(a) Protozoa (_Nyctotherus_, _Herpetomonas_, _Lophomonas_, etc.; p. 172).

(b) Nematodes (Thelastomatidae; p. 193).

Group 3. Parasites.

(a) Fungi (Laboulbeniales; p. 134).

(b) Protozoa (gregarines, _Plistophora_, etc.; p. 181).

(c) Helminths.

(1) Primary parasites (mermithids and gordian worms; p. 201).

(2) Secondary parasites (_Gongylonema neoplasticum_, _Oxyuris mansoni_, _Moniliformis_ spp.; p. 206).

(d) Arthropods.

(1) Mites (_Pimeliaphilus podapolipophagus_; p. 219).

(2) Insects (larvae of ripiphorids, evaniids, and ampulicids; p. 231).

Type II. Facultative associates. Animals and plants that prey on cockroaches or are incidentally or accidentally picked up by the cockroach, but which can survive or propagate readily on some other host or prey. Steinhaus (1946) emphasized the importance of the environment in determining the type of microbial flora associated with the cockroach which may carry one type of flora in an area which is exposed to filth and a different type in other areas. Because many of these organisms survive passage through or on the cockroach, the blattid may act as a vector of these animals and plants.

Group 1. Commensals.

(a) Viruses (strains of poliomyelitis virus; p. 103).

(b) Bacteria (Enterobacteriaceae, Pseudomonadaceae, Micrococcaceae, etc.; p. 111).

(c) Fungi (_Aspergillus_; p. 130).

(d) Protozoa (_Iodamoeba_, _Dobellina_, and cysts of _Entamoeba coli_ and _Entamoeba histolytica_; p. 179).

(e) Helminths (cysts of various helminths parasitic in vertebrates; p. 208).

(f) Arthropods.

(1) Mites (_Tyrophagus lintneri_; p. 218).

Group 2. Parasites.

(a) Bacteria (_Serratia marcescens_; p. 117).

(b) Helminths (_Protospirura_ spp.; p. 206).

(c) Arthropods.

(1) Mites (_Locustacarus_ sp.; p. 219).

(2) Insects (_Melittobia chalybii_; p. 248).

Group 3. Predators, active.

(a) Arthropods.

(1) Spiders (p. 214).

(2) Scorpions (p. 212).

(3) Centipedes (p. 222).

(4) Mites (_Rhizoglyphus tarsalus_; p. 218).

(5) Insects (dermestids, reduviids, and on occasion adult females of _Tetrastichus_, _Anastatus_, _Ampulex_; p. 234).

(b) Vertebrates.

(1) Amphibia (p. 269).

(2) Reptilia (p. 272).

(3) Aves (p. 276).

(4) Mammalia (p. 283).

Group 4. Predators, passive: Pitcher plants (p. 154).

Class B. Associations in which cockroaches serve as commensals or predators.

Group 1. Commensal cockroaches.

(a) Associates of social insects (_Attaphila_ spp., etc.; p. 315).

(b) Obscure associates (p. 316).

Group 2. Predatory cockroaches.

(a) Interspecies predators (p. 319).

(b) Intraspecies predators (p. 322).

Class C. Associations of cockroaches with other cockroaches.

Group 1. Intraspecies associations.

(a) Familial associations (p. 325).

(b) Other conspecific associations (aggregations and fighting) (p. 336).

Group 2. Interspecies associations.

(a) Compatible associations (p. 337).

(b) Antagonistic associations (p. 329).

Class D. Ecological associations of cockroaches with higher plants.

Group 1. Benign associations (p. 139).

Group 2. Associations detrimental to plants (p. 162).

V. MUTUALISM

BACTEROIDS

Blochmann (1887, 1888) discovered intracellular particles (the bacteroids or symbiotes of authors) that resembled bacteria in the fat body of males and females of _Blatta orientalis_ and _Blattella germanica_ (pl. 26), in the ova of these insects, and in their embryos. Bacteroids have since been found in at least 25 species and 19 genera of cockroaches. Presumably such microorganisms are universally distributed throughout the Blattaria. General reviews of the bacteroids of cockroaches and other insects have been published by Glaser (1930b), Schwartz (1935), Steinhaus (1946, 1949), Buchner (1952, 1953), Brooks (1954), and Richards and Brooks (1958). The reader is referred to these papers, and those of authors cited in the list at the end of this section, for discussions of the morphology of the bacteroids, their distribution within the host, and attempts to culture them in vitro.

It has long been assumed, without proof, that cockroaches and their bacteroids form a mutually beneficial association. As it has not been possible to cultivate bacteroids apart from their cockroach hosts, it may be assumed that the host is essential to the continued existence of the microorganism, which also derives from the association other obvious benefits as well. Experiments to show that the host also benefits from the association have centered around rendering cockroaches bacteroid free. Starvation, parasites, electromagnetic radiation, heat or cold, or chemicals have all been used in attempts to eliminate the bacteroids. Of these, only chemical treatment has provided a satisfactory technique.

Few chemicals other than antibiotics have proved to be useful in the elimination or reduction of bacteroids. Yetwin (1932) injected various dilutions of 22 compounds into _Blattella germanica_. He observed decreases in the bacteroids of the fat body only following injection of methylene blue, but did not pursue this lead further. Gier (_in_ Steinhaus, 1946) observed reduction in the numbers of bacteroids after cockroaches were injected with crystal violet, hexylresorcinol, or metaphen. Bode (1936) reported that injection of irritants such as lithium salts or quinine hydrochloride had no apparent effect on the symbiotes.

Brooks (1957) reared _Blattella germanica_ on diets containing different concentrations of inorganic ions. On a manganese-deficient diet the cockroaches grew poorly and some of their progeny lacked normal bacteroids; about 10 percent of the aposymbiotic generation grew and reproduced on a diet fortified with yeast. Varying the concentrations of other salts in the diets gave results in which the progeny were either aposymbiotic or the fat body was abnormal but the mycetocytes

were abundant; all these cockroaches soon died even on fortified diets.

The administration of certain antibiotic drugs has produced cockroaches very nearly free of bacteroids. Brues and Dunn (1945) found that although sulfa drugs had no effect on the bacteroids, penicillin in large doses reduced the number of bacteroids in _Blaberus craniifer_, but the cockroaches died within a few days. Death was attributed to lack of bacteroids rather than effect of the drug; this is perhaps an unwarranted conclusion in view of the survival of aposymbiotic cockroaches that have been produced by other drugs (Brooks, 1954; Brooks and Richards, 1955). Glaser (1946) found that the bacteroids of _Periplaneta americana_ could be adversely affected or destroyed by sulfathiazole, or sodium or calcium penicillin. Fraenkel (1952) questioned the conclusions of Brues and Dunn (1945) and of Glaser (1946) because of the high mortality in their experiments, and he suggested that the described phenomena "were due rather to direct toxic effects on the host than to loss of the symbionts." Noland (_in_ Brooks, 1954; Brooks and Richards, 1955) confirmed Glaser's results with penicillin and extended sulfa treatments to include _Blattella germanica_. Every female whose bacteroids were reduced to the vanishing point resorbed her ovaries and was incapable of reproduction. Brooks (1954; Brooks and Richards, 1955) found that administration of several antibiotics did not eliminate the bacteroids from the fat body of _B. germanica_ unless the dose was so high that it caused excessive mortality. Frank (1955, 1956) was able to eliminate bacteroids from _Blatta orientalis_ by injecting or feeding chlortetracycline, oxytetracycline, or penicillin; survival of treated insects was not good and reproduction was poor; the aposymbiotic individuals were smaller than normal. As Richards and Brooks (1958) have pointed out, it is uncertain how much of this difference was the result of loss of bacteroids and how much the effect of the drug. It is obvious that in all these experiments the action of the drugs on the bacteroids was accompanied by equivocal side effects which confused interpretation of the results. The effect on the cockroach of a loss of bacteroids cannot be separated from a possible toxic effect of the drug.

Fortunately Brooks (1954; Brooks and Richards, 1955) obtained completely aposymbiotic offspring from _Blattella germanica_ that had been reared on aureomycin. These bacteroid-free nymphs were practically incapable of growth on a natural diet that was adequate for nymphs with symbiotes. However, the addition of large amounts of dried yeast to the diet enabled aposymbiotic nymphs to mature in two to three times the period required by normal nymphs. Final proof of the function of the bacteroids was obtained by reestablishing them in aposymbiotic cockroaches. The insects that received implants of normal fat body of _B. germanica_ showed a slow, steady gain in weight over the controls (Brooks, 1954; Brooks and Richards, 1956). Obviously, the intracellular symbiotes subserve the normal nutrition of the cockroach. Whether the bacteroids produce only vitaminlike substances, as suggested by Keller (1950), or function in some other way is still to be determined. Brooks (1954) concluded that the amount of vitamin-containing food required for increased growth by aposymbiotic cockroaches is much greater than the known vitamin requirements; hence the factor(s) needed is unknown and present in low concentration, or it serves as a precursor of a second factor(s) whose synthesis is aided by the bacteroids. Brooks (_in_ Richards and Brooks, 1958) has since found that

the bacteroids of _Blattella germanica_ "can supply the insect with B vitamins, amino acids and some larger protein fragment."

Gier (1947) stated that the symbiotes of cockroaches are generically all the same. However, as the symbiotes are presumed to have been associated with cockroaches for over 300,000,000 years (Buchner, 1952) they may be assumed to have developed specific differences that link them inseparably to their respective hosts. Ries (1932) transplanted symbiote-containing fat body from _Blatta orientalis_ into the mealworm and larva of _Ephestia kühniella_, and from _Blattella germanica_ and _Stegobium paniceum_ (=_Sitodrepa panicea_) into _B. orientalis_. The implants did not become established in the new host, although most of the transplantations were successful in that the hosts survived and the implants remained intact for some time before they were encapsulated by host tissue. Brooks (1954; Brooks and Richards, 1956) transplanted fat body of _Periplaneta americana_ and _B. orientalis_ into aposymbiotic _B. germanica_. The growth of the cockroaches injected with foreign tissue was not different from that of aposymbiotic controls. Sections of host insects did not contain mycetocytes and no bacteroids were found. Haller (1955a) injected bacteroids or implanted mycetocytes of _B. germanica_ into gryllids, acridids, and locustids. These implants and innoculations were rapidly destroyed by the hosts. But as Richards and Brooks (1958) have pointed out, none of these experiments provide information about the specificity of the bacteroids themselves.

COCKROACHES IN WHICH BACTEROIDS HAVE BEEN FOUND

Bantua stigmosa. Fraenkel (1921)

Blaberus craniifer. Brooks (1954); Brues and Dunn (1945); Hoover (1945).

Blaberus giganteus. Blochmann (1892).

Blatta orientalis. Blochmann (1887, 1888, 1892); Bode (1936); Brooks (1954); Buchner (1912); Cuénot (1896); Fraenkel (1921); Frank (1955, 1956); Gier (1936, 1947); Glaser (1920); Gropengiesser (1925); Gubler (1948); Heymons (1895); Hollande and Favre (1931); Hoover (1945); Hovasse (1930); Javelly (1914); Keller (1950); Koch (1949); Menel (1907)?; Mercier (1906a, 1907, 1907b, 1907c); Ries (1932); Ronzoni (1949); Tacchini (1946); Wolf (1924, 1924a).

Blattella germanica. Blochmann (1887, 1888, 1892); Bode (1936); Borghese (1946, 1948, 1948a); Brooks (1954); Brooks and Richards (1954, 1955, 1955a, 1956); Fraenkel (1921); Gier (1936, 1947); Glaser (1920, 1930a); Gropengiesser (1925); Haller (1955, 1955a); Heymons (1892, 1895); Hoover (1945); Koch (1949); Lwoff (1923); Milovidov (1928); Neukomm (1927, 1927a, 1932); Pérez Silva (1954, 1954a); Ries (1932); Rizki (1954); Ronzoni (1949); Tacchini (1946); Wollman (1926); Yetwin (1932, 1953).

Cryptocercus punctulatus. Gier (1936, 1947); Hoover (1945).

Derocalymma cruralis. Fraenkel (1921).

Ectobius lapponicus. Heymons (1892); Cuénot (1896); Koch (1949).

Ectobius pallidus. Heymons (1892, 1895); Cuénot (1896).

Epilampra grisea. Fraenkel (1921).

Eurycotis floridana. Gier (1936, 1947).

Loboptera decipiens. Buchner (1930).

Nauphoeta cinerea. Fraenkel (1921).

Nyctibora noctivaga. Gier (1947).

Parcoblatta lata. Gier (1936, 1947).

Parcoblatta pensylvanica. Brooks (1954); Gier (1936, 1947).

Parcoblatta uhleriana. Gier (1936, 1947); Hoover (1945).

Parcoblatta virginica. Glaser (1920); Gier (1947).

Periplaneta americana. Baudisch (1956); Bode (1936); Cuénot (1896); Gier (1936, 1937, 1947); Glaser (1920, 1930, 1946); Gubler (1948); Hertig (1921); Hoover (1945); Ketchel and Williams (1953).

Periplaneta australasiae. Gier (1936, 1947); Koch (1949).

Platyzosteria armata. Fraenkel (1921).

Polyphaga aegyptiaca. Fraenkel (1921).

Pseudoderopeltis aethiopica. Fraenkel (1921).

Pycnoscelus surinamensis. Bode (1936); Koch (1949).

Supella supellectilium. Brooks (1954).

BACTERIA

Evidence showing that intestinal bacteria contribute to the nutrition of cockroaches is meager. Cleveland et al. (1934) isolated a bacterial organism from the foregut of the wood-feeding cockroach _Panesthia angustipennis_. The bacterium digested cellulose rapidly in vitro and these workers believe that this cockroach and other related wood-feeding species are dependent on symbiotic bacteria for the digestion of their food.

Mencl (1907) described cell nuclei in "symbiotic," not closely defined types of bacilli that he found in abundance in the digestive tract of the Küchenschabe, _Periplaneta_ (presumably _Blatta orientalis_). Unfortunately, he was more concerned about the morphology of the bacteria than the stated mutualistic relationship, so nothing is known of their physiology.

The growth rates of _Periplaneta americana_ and _Blattella germanica_ were retarded when the insects were reared aseptically, which suggests that microorganisms normally found in the digestive tract supply certain necessary

dietary constituents (Gier, 1947a; House, 1949). Noland et al. (1949) suggested that microorganisms in the digestive tract of _B. germanica_ synthesized riboflavin since the nymphs reared on a low riboflavin diet accumulated more of the vitamin than could have been ingested in the diet. However, Metcalf and Patton (1942) found little or no bacterial synthesis of riboflavin in _P. americana_. Noland and Baumann (1951) suggested that methionine, one of the amino acids essential for rapid growth of _B. germanica_, was synthesized by intestinal microorganisms in the insects.

PROTOZOA

It is probable that with few exceptions protozoa found in the digestive tract are not necessary for survival of the cockroach. However, very few experiments have been performed to determine the importance, if any, of these microorganisms to the host. Cleveland (1925) removed the protozoa from the cockroach (possibly _Periplaneta americana_) by oxygenation at 3.5 atmospheres. The ciliates _Nyctotherus_ and _Balantidium_, flagellates _Lophomonas_ and _Polymastix_, the amoeba _Endamoeba blattae_, and three unidentified protozoa were killed by this treatment, yet the insects lived normally after defaunation.

Armer (1944) studied the effects of high-carbohydrate, high-fat, and high-protein diets, as well as starvation, on the intestinal protozoa (_Nyctotherus ovalis_, _Endamoeba blattae_, _Endolimax blattae_, _Lophomonas striata_, and _Lophomonas blattarum_) in _Periplaneta americana_. Starvation of the host lowered the incidence or eliminated most of the protozoa, but a high-carbohydrate diet maintained them at a relatively high level. _Lophomonas blattarum_ was eliminated by a high-protein diet, and practically eliminated by a high-fat diet. _Lophomonas striata_ was eliminated from some hosts that were kept on high-fat and high-protein diets. _Endamoeba blattae_ showed a decrease in infection rate when the cockroaches were maintained on high-fat and high-protein diets. The effects of diets on _Endolimax blattae_ were not uniform.

It has been shown by Cleveland (1930, 1948) and Cleveland et al. (1931, 1934) that the wood-feeding cockroach _Cryptocercus punctulatus_ depends upon certain intestinal protozoa for survival; these protozoa utilize as food the wood ingested by this cockroach. The wood is broken down into compounds the cockroach can utilize by the protozoa which elaborate a cellulase and possibly a cellobiase (Trager, 1932). Only molting nymphs of _Cryptocercus_ can pass the protozoa on to the newly hatched young, so that molting and hatching must happen concurrently each year or the young die.

The sexual cycles in species of protozoa in the genera _Trichonympha_, _Saccinobaculus_, _Oxymonas_, _Monocercomonoides_, _Hexamita_, _Eucomonympha_, _Leptospironympha_, _Urinympha_, _Rhynchonympha_, _Macrospironympha_, and _Barbulanympha_ (fig. 3, B) are induced by hormones produced by _Cryptocercus_ only during its molting period (Cleveland, 1931, 1947, 1947a, 1949-1956a). Perhaps the prothoracic gland hormone of the host may be responsible for initiation of the flagellate sexual cycles (Cleveland and Nutting,

1955). The protozoan sexual cycles may be used as indicators of the onset of molting in _Cryptocercus_; thus different species of protozoa begin their sexual cycles from 35 days before to 2 days after molting of the cockroach (Cleveland and Nutting, 1954). Hollande (1952) and Grassé (1952) have reviewed the roles and the evolution of the flagellates in _Cryptocercus_ and in termites.

The protozoa of cockroaches and termites are clues to the relationship between these two groups of insects. Kirby (1927) pointed out similarities between _Endamoeba blattae_ of _Periplaneta_ and the amoebae of the termite _Mirotermes_, suggesting that these protozoans were probably derived from an amoeba in an ancestor common to both blattid and termite. Kirby (1932, _in_ Kidder, 1937) found a species of _Nyctotherus_ in _Amitermes_ that resembles _Nyctotherus ovalis_ from domestic cockroaches. The belief that the termites and cockroaches had a common origin is also strengthened by the similarities between the hypermastigotes of both _Cryptocercus_ and termites (Cleveland et al., 1934).

The cockroaches _Cryptocercus_ and _Panesthia_ both feed on wood, but the protozoa found in _Panesthia_ resemble more closely the species in domestic cockroaches than those in _Cryptocercus_. The Clevelandellidae (from _Panesthia_) are closely related to _Nyctotherus_ and have probably evolved from common ancestors. However, the separation of the Clevelandellidae from _Nyctotherus_ must have taken place at a later date than the divergence of their hosts, otherwise representatives of that family would probably also be found in _Periplaneta_ and _Blatta_ (Kidder, 1937).

The protozoa of _Cryptocercus_ can be transferred from one individual to another (Nutting and Cleveland, 1954). They can also be transferred to the termite _Zootermopsis_ where they survive only until the host molts; the reverse is also true, _Zootermopsis_ Protozoa can survive in _Cryptocercus_ until the cockroach molts (Nutting and Cleveland, 1954a).

VI. VIRUSES ASSOCIATED WITH COCKROACHES

Annotations on some of the following observations may be found in Roth and Willis (1957a). Use of asterisk is explained in footnote 3, page 4.

POLIOMYELITIS VIRUSES

* =Lansing strain=

Experimental vectors.--_Blattella germanica_, U.S.A. (Hurlbutt, 1949, 1950).

Periplaneta americana, U.S.A. (Hsiang et al., 1952).

* =Brunhilde type, Minnesota and Mahoney strains=

Experimental vectors.--_Periplaneta americana_, U.S.A. (Fischer and Syverton, 1951; Syverton et al., 1952).

* =Columbia SK virus=

Experimental vectors.--_Periplaneta americana_, Great Britain? (Findlay and Howard, 1951): Results with _Blattella germanica_ were negative.

* =Four unspecified strains=

Natural vectors.--_Blattella germanica_ and/or _Blattella vaga_, _Periplaneta americana_ and/or _Periplaneta brunnea_, and _Supella supellectilium_, U.S.A. (Syverton et al., 1952; Dow, 1955; Dow _in_ Roth and Willis, 1957a).

OTHER VIRUSES

* =Coxsackie viruses=

Experimental vectors.--_Periplaneta americana_, U.S.A. (Fischer and Syverton, 1951a, 1957): Recently Fischer and Syverton (1957) found that after feeding a single meal of Coxsackie virus to _Periplaneta americana_, the gastrointestinal tracts of the insects, which were removed at 5-day intervals up to 20 days, contained sufficient virus to paralyze and kill test mice. Cockroach salivary glands, removed 5 days after the insects had fed, contained the virus which caused paralysis and death in test mice; mice were also infected by virus obtained from salivary glands removed from the insects 10 and 20 days after the cockroaches had fed once on the virus. The virus was also isolated from the cockroaches' feces and rarely from the fat bodies and reproductive organs. Fischer and Syverton concluded that it is possible that cockroaches could acquire the virus, by feeding on mammalian excreta, maintain it for a period of time, and transmit it by contamination of food. The virus could also be transmitted through the feces of wild mice if the mice happened to feed on virus-infected cockroaches.

* =Mouse encephalomyelitis virus=

Experimental vectors.--_Periplaneta americana_, U.S.A. (Syverton and Fischer, 1950).

* =Yellow-fever virus=

Experimental vectors.--_Blattella germanica_, Great Britain? (Findlay and MacCallum, 1939).

VII. BACTERIA ASSOCIATED WITH COCKROACHES

Classification of the bacteria follows Breed et al. (1948). Synonymy in most cases was taken from the same source. Names of bacteria preceded by the symbol [cross] are either not listed by Breed et al. or are stated by them to be insufficiently characterized for definite classification. Use of asterisk is explained in footnote 3, page 4.

Phylum SCHIZOPHYTA

Class SCHIZOMYCETES

Order EUBACTERIALES

Family PSEUDOMONADACEAE

* =Pseudomonas aeruginosa= (Schroeter) Migula

Natural vectors.--_Blaberus craniifer_, U.S.A. (Wedberg et al., 1949).

Blatta orientalis, U.S.A. (Olson and Rueger, 1950).

Blattella germanica, U.S.A. (Olson and Rueger, 1950; Janssen and Wedberg, 1952).

Periplaneta americana, U.S.A. (Bitter and Williams, 1949, 1949a; Olson and Rueger, 1950).

Experimental vectors.--_Blattella germanica_, U.S.A. (Herms and Nelson, 1913).

Cockroaches, U.S.A. (Longfellow, 1913).

* =Pseudomonas eisenbergii= Migula

Natural and experimental vectors.--_Blatta orientalis_, Italy (Cao, 1906).

* =Pseudomonas fluorescens= Migula

Natural and experimental vectors.--_Blatta orientalis_, Italy (Cao, 1898, 1906; Spinelli and Reitano, 1932).

Periplaneta americana, U.S.A. (Gier, 1947): The organism was pathogenic to the cockroach when injected.

[cross] =Spirillum periplaneticum= Kunstler and Gineste

Habitat.--_Periplaneta americana_, France? (Kunstler and Gineste, 1906): From intestinal tract.

[cross] =Spirillum α, β, and γ= Dobell

Habitat.--_Blatta orientalis_, England (Dobell, 1911, 1912): From hind gut.

=Spirillum= sp.

Habitat.--_Blatta orientalis_, U.S.S.R. (Zasukhin, 1930): From intestinal tract.

Cockroaches, Venezuela (Tejera, 1926): From digestive tract.

* =Vibrio comma= (Schroeter) Winslow et al.

Experimental vectors.--_Blatta orientalis_, Italy (Cao, 1898; Spinelli and Reitano, 1932).

Blattella germanica, Orient (Toda, 1923); Germany (Jettmar, 1927).

Periplaneta americana, Philippine Islands (Barber, 1914); Netherlands (Akkerman, 1933); Formosa (Morischita and Tsuchimochi, 1926).

Periplaneta australasiae, Formosa (Morischita and Tsuchimochi, 1926).

* =Vibrio metschnikovii= Gamaléia

Experimental vectors.--_Blatta orientalis_, Italy (Cao, 1898, 1906).

=Vibrio tyrogenus= (Flügge) Holland

Synonymy.--_Vibrio_ of Deneke.

Experimental vectors.--_Blatta orientalis_, Italy (Cao, 1898): The organism passed through the intestinal tract unchanged.

[cross] =Vibrio Types I and II= Heiberg

Habitat.--Water.

Natural vectors.--Cockroaches, India (Pasricha et al., 1938): The vibrios were found in 16 or 17 percent of 94 cockroaches and resembled _Vibrio comma_ in their morphology and their main biochemical reactions; however, serum-agglutination reactions differed.

=Vibrio= sp.

Habitat.--_Blatta orientalis_, U.S.A. (Leidy, 1853): From intestine.

Family RHIZOBIACEAE

=Chromobacterium violaceum= (Schroeter) Bergonzini

Synonymy.--_B. violaceus._

Habitat.--Water.

Experimental vectors.--Cockroach, U.S.A. (Longfellow, 1913): Recovered from outer part of body and intestinal tract.

Family MICROCOCCACEAE

* =Micrococcus aurantiacus= (Schroeter) Cohn

Natural vectors.--_Blattella germanica_, U.S.A. (Janssen and Wedberg, 1952).

* =Micrococcus citreus= Migula

Natural vectors.--Cockroaches, U.S.A. (Longfellow, 1913).

Experimental vectors.--_Blatta orientalis_, Italy (Cao, 1898).

* =Micrococcus epidermidis= (Winslow and Winslow) Hucker

Natural vectors.--_Blattella germanica_, U.S.A. (Janssen and Wedberg, 1952).

[cross] =Micrococcus nigrofaciens= Northrup

Source.--Diseased June beetle larvae.

Experimental infection.--_Periplaneta americana_, U.S.A. (Northrup, 1914): Three of four adults were infected by feeding them bread saturated with a broth culture of the _Micrococcus_. After 11 days the tarsi of the cockroaches became infected, and the hind legs split and broke off. Antennae and setae also were affected and micrococci were recovered from the feces.

* =Micrococcus pyogenes= var. =albus= (Rosenbach) Schroeter

Natural vectors.--_Blaberus craniifer_, U.S.A. (Wedberg et al., 1949).

Blatta orientalis, U.S.A. (Tauber, 1940; Tauber and Griffiths, 1942).

Blattella germanica, U.S.A. (Herms and Nelson, 1913; Herms, 1939; Janssen and Wedberg, 1952).

Experimental vectors.--_Blatta orientalis_, Italy (Cao, 1898); U.S.A. (Tauber and Griffiths, 1942).

Micrococcus pyogenes var. _albus_ (=_Staphylococcus albus_) and an unidentified short rod form were found by Tauber (1940) in the hemolymph of _B. orientalis_. These microorganisms were never found together in the same insect and caused loss of appetite, sluggishness, irregular respiratory movements, and paralysis in the cockroach; in the final stages of the disease the legs were folded under the body, the head was tucked beneath the forelegs, the whole insect became arched and maintained this position until death. In some cockroaches infected with the rod pathogen, conjunctival folds, particularly those between the dorsal abdominal sclerites, and the joints of the metathoracic legs ruptured liberating thick white hemolymph filled with bacteria. Tauber suggested that the infection might be spread by contact, especially to newly molted individuals or by actual ingestion of the bacteria by the cockroaches feeding on dead or dying individuals. All the roaches died after successful inoculation with the _Micrococcus_. The bacterial infection was associated with high total hemocyte counts and high percentages of mitotically dividing hemolymph cells (Tauber, 1940); these responses of the insect were interpreted as a mechanism whereby the number of hemocytes increases resulting in an increase in the number of phagocytes for combating the bacteria (Tauber and Griffiths, 1942).

* =Micrococcus pyogenes= var. =aureus= (Rosenbach) Zopf

Natural vectors.--_Blaberus craniifer_, U.S.A. (Wedberg et al., 1949).

Blatta orientalis, Italy (Cao, 1906).

Blattella germanica, U.S.A. (Herms, 1939).

Cockroaches, U.S.A. (Longfellow, 1913).

=Micrococcus ureae= Cohn

Habitat.--Stale urine and soil containing urine.

Natural vectors.--_Blattella germanica_, U.S.A. (Janssen and Wedberg, 1952): From intestinal tract and feces.

* =Micrococcus= spp.

These organisms were obtained from pus or were designated as staphylococci [i.e., pathogenic micrococci (Blair _in_ Dubos, 1948)].

Natural vectors.--_Blatta orientalis_, Italy (Spinelli and Reitano, 1932); Germany (Jettmar, 1935).

Blattella germanica, Germany (Jettmar, 1935).

Experimental vectors.--_Blattella germanica_, on shipboard (Morrell, 1911); Germany (Vollbrechtshausen, 1953).

=Micrococcus= sp.

Natural vectors.--_Periplaneta americana?_ ("_Blatella americana_"), England (Shrewsbury and Barson, 1948): From intestinal tract.

[cross] =Sarcina alba= Zimmermann

Habitat.--Water.

Natural and experimental vectors.--_Blatta orientalis_, Italy (Cao 1898, 1906): From intestinal contents.

=Sarcina aurantiaca= Flügge

Habitat.--Air and water.

Experimental vectors.--_Blatta orientalis_, Italy (Cao, 1906): Intestinal contents.

=Sarcina lutea= Schroeter

Habitat.--Air, soil, water, skin surfaces.

Experimental vectors.--_Blatta orientalis_, Italy (Cao, 1906): From intestinal contents.

[cross] =Sarcina symbiotica= Pribram

Habitat.--Oöthecae of _Blatta orientalis_ and/or _Blattella germanica_, Germany (Gropengiesser, 1925): It was described as "eine gelbe _Sarcina_"; Pribram (1933) named the organism.

=Sarcina ventriculi= Goodsir

Habitat.--Garden soil, dust, mud; isolated from a diseased stomach.

Natural vectors.--_Blatta orientalis_, Poland (Nicewicz et al., 1946): From intestinal tract.

=Sarcina= sp.

Natural and experimental vectors.--_Periplaneta americana_, U.S.A. (Gier, 1947): Organism not pathogenic to the cockroach when injected.

[cross] =Sarcina= sp.

Synonymy.--_Sarcina_ "blanche" of Sartory and Clerc.

Natural vectors.--_Blatta orientalis_, France (Sartory and Clerc, 1908): Isolated from intestinal tract.

*[cross] =Sarcina= sp.

Synonymy.--_Sarcina alba_ "patogena" of Cao.

Natural vectors.--_Blatta orientalis_, Italy (Cao, 1898, 1906).

*[cross] =Sarcina= sp.

Synonymy.--_Sarcina_ "bianca" and "gialla" of Cao.

Experimental vectors.--_Blatta orientalis_, Italy (Cao, 1898).

Family NEISSERIACEAE

* =Neisseria meningitidis= (Albrecht and Ghon) Holland

Experimental vectors.--Cockroaches, U.S.A. (Longfellow, 1913).

* =Veillonella parvula= (Veillon and Zuber) Prévot

Natural vectors.--_Periplaneta americana_, U.S.A. (Hatcher, 1939).

Family LACTOBACTERIACEAE

* =Diplococcus pneumoniae= Weichselbaum

Experimental vectors.--_Blatta orientalis_, Italy (Cao, 1898).

Cockroaches, U.S.A. (Longfellow, 1913).

[cross] =Enterococcus= sp.

Natural vectors.--_Blatta orientalis_, Italy (Spinelli and Reitano, 1932): From intestinal tract.

=Lactobacillus fermenti= Beijerinck

Habitat.--Fermenting plant or animal products.

Natural vectors.--_Blatta orientalis_, Poland (Nicewicz et al., 1946): From intestinal canal.

[cross] =Pneumococcus Type I, No. 1231=

Experimental vectors.--_Blattella germanica_, Germany (Vollbrechtshausen, 1953).

* =Streptococcus faecalis= Andrewes and Horder

Natural vectors.--_Blatta orientalis_, Poland (Nicewicz et al., 1946).

Blattella germanica, U.S.A. (Steinhaus, 1941).

Periplaneta americana? ("_Blatella americana_"), England (Shrewsbury and Barson, 1948).

Cockroaches [presumably any or all of the above three species], Egypt (El-Kholy and Gohar, 1945).

* =Streptococcus liquefaciens= Sternberg emend. Orla-Jensen

Natural vectors.--_Blatta orientalis_, Poland (Nicewicz et al., 1946).

*[cross] =Streptococcus microapoika= Cooper, Keller, and Johnson

Natural vectors.--_Blatta orientalis_, Poland (Nicewicz et al., 1946).

*[cross] =Streptococcus non-hemolyticus II= Holman

Natural vectors.--_Periplaneta americana?_ ("_Blatella americana_"), England (Shrewsbury and Barson, 1948).

* =Streptococcus pyogenes= Rosenbach

Natural vectors.--_Blatta orientalis_, Italy (Cao, 1906).

Experimental vectors.--Cockroaches, U.S.A. (Longfellow, 1913).

* =Streptococcus= sp. (pyogenic group)

Experimental vectors.--_Blatta orientalis_, Germany (Jettmar, 1935).

* =Streptococcus= sp. (_viridans_ group)

Experimental vectors.--_Blatta orientalis_, Germany (Jettmar, 1935).

* =Streptococcus= spp.

Natural vectors.--_Blatta orientalis_, Germany (Jettmar, 1935).

Blattella germanica, Germany (Jettmar, 1935); U.S.A. (Janssen and Wedberg, 1952).

Cockroaches, U.S.A. (Longfellow, 1913).

Experimental vectors.--_Blatta orientalis_, Italy (Cao, 1898).

Family CORYNEBACTERIACEAE

* =Corynebacterium diphtheriae= (Flügge) Lehmann and Neumann

Experimental vectors.--_Blatta orientalis_, Italy (Cao, 1898).

Cockroaches, U.S.A. (Longfellow, 1913).

Family ACHROMOBACTERIACEAE

[cross] =Achromobacter hyalinum= (Jordan) Bergey et al.

Natural vectors.--_Periplaneta americana_, U.S.A. (Hatcher, 1939): Isolated from feces.

=Achromobacter= sp.

Natural vectors.--_Blattella germanica_, U.S.A. (Janssen and Wedberg, 1952).

=Alcaligenes faecalis= Castellani and Chalmers

Synonymy.--_Bacillus faecalis alkaligenes_; _Bacillus alcaligenes faecalis_; _B. alcaligenes faecalis_.

Habitat.--Intestinal canal of man. Has been isolated from feces, abscesses related to intestinal canal, and occasionally in the bloodstream. However, it is generally considered nonpathogenic.

Natural vectors.--_Blaberus craniifer_, U.S.A. (Wedberg et al., 1949): Isolated from feces.

Blatta orientalis, Poland (Nicewicz et al., 1946): Isolated from intestinal tract.

Blattella germanica, U.S.A. (Janssen and Wedberg, 1952): From intestinal tract and feces.

Periplaneta americana, U.S.A. (Bitter and Williams, 1949): Isolated from intestinal tract.

Cockroaches [presumably _Blatta orientalis_, _Blattella germanica_ and/or _Periplaneta americana_], Egypt (El-Kholy and Gohar, 1945): From suspensions of macerated whole insects.

=Alcaligenes recti= (Ford) Bergey et al.

Synonymy.--_B. alcaligenes recti._

Habitat.--Intestinal canal.

Natural vectors.--_Blatta orientalis_, Poland (Nicewicz et al., 1946): Isolated from intestinal tract.

=Alcaligenes viscosus= (Weldin and Levine) Weldin

Habitat.--Water, dairy utensils; produces ropiness in milk.

Natural vectors.--_Blattella germanica_, U.S.A. (Janssen and Wedberg, 1952): Isolated from intestine and feces.

Family ENTEROBACTERIACEAE

=Aerobacter aerogenes= (Kruse) Beijerinck

Synonymy.--_Bacillus lactis aerogenes._

Habitat.--Grains, plants, intestinal tract of man and other animals.

Natural vectors.--_Blaberus craniifer_, U.S.A. (Wedberg et al., 1949): Isolated from feces.

Blatta orientalis, Poland (Nicewicz et al., 1946): Isolated from intestinal tract.

Blattella germanica, on shipboard (Morrell, 1911): Isolated from feces.

Periplaneta americana, U.S.A. (Bitter and Williams, 1949): Isolated from intestinal tract.

Cockroaches [presumably _Blatta orientalis_, _Blattella germanica_ and/or _Periplaneta americana_], Egypt (El-Kholy and Gohar, 1945): Isolated from outer surface of body, intestinal tract, and suspensions of macerated whole insects.

=Aerobacter cloacae= (Jordan) Bergey et al.

Synonymy.--_Bacillus cloacae._

Habitat.--Sewage, soil, water, human and other animal feces.

Natural vectors.--_Blattella germanica_, on shipboard (Morrell, 1911): From feces. U.S.A. (Janssen and Wedberg, 1952; Steinhaus, 1941): From intestinal tract,

feces, and oötheca.

Periplaneta americana, U.S.A. (Bitter and Williams, 1949): From intestinal tract.

=Aerobacter= sp.

Natural vectors.--_Periplaneta americana_, U.S.A. (Bitter and Williams, 1949): Isolated from intestines.

[cross] =Eberthella oedematiens= Assis

Habitat.--Intestinal canal.

Natural vectors.--_Periplaneta americana_, U.S.A. (Bitter and Williams, 1949): Intestinal tract.

* =Escherichia coli= (Migula) Castellani and Chalmers

Natural vectors.--_Blatta orientalis_, Italy (Cao 1898, 1906; Spinelli and Reitano, 1932); France (Sartory and Clerc, 1908); Europe (Jettmar, 1935); Poland (Nicewicz et al., 1946).

Blattella germanica, U.S.A. (Steinhaus, 1941).

Periplaneta americana, U.S.A. (Bitter and Williams, 1949, 1949a).

Cockroaches [presumably one or all of the above three species], Egypt (El-Kholy and Gohar, 1945).

Cockroaches, U.S.A. (Longfellow, 1913).

Experimental vectors.--_Blattella germanica_, Germany (Vollbrechtshausen, 1953): When injected anally or orally the bacteria invaded the intestinal cells and in heavy infections killed the cockroaches.

=Escherichia coli= var. =acidilactici= (Topley and Wilson) Yale

Synonymy.--_Bacillus acidi lactici._

Source.--Diseased nun moth larvae.

Experimental vectors.--_Blatta orientalis_, Europe (Filatoff, 1904): Organism pathogenic to cockroach when injected but not when fed.

=Escherichia coli= var. =communior= (Topley and Wilson) Yale

Natural vectors.--_Blaberus craniifer_, U.S.A. (Wedberg et al., 1949): From feces.

=Escherichia freundii= (Braak) Yale

Habitat.--Soil, water, intestinal canal of man and other animals.

Natural vectors.--_Blaberus craniifer_, U.S.A. (Wedberg et al., 1949): From feces.

Blattella germanica, U.S.A. (Janssen and Wedberg, 1952): From intestinal canal and feces.

Periplaneta americana, U.S.A. (Bitter and Williams, 1949): From intestinal tract.

=Escherichia intermedium= (Werkman and Gillen) Vaughn and Levine

Habitat.--Soil, water, intestinal canal of man and other animals.

Natural vectors.--_Periplaneta americana_, U.S.A. (Bitter and Williams, 1949): From intestinal canal.

* =Klebsiella pneumoniae= (Schroeter) Trevisan

Experimental vectors.--_Blatta orientalis_, Italy (Cao, 1898).

* =Paracolobactrum aerogenoides= Borman, Stuart and Wheeler

Natural vectors.--_Blattella germanica_, U.S.A. (Janssen and Wedberg, 1952).

Periplaneta americana, U.S.A. (Bitter and Williams, 1949).

* =Paracolobactrum coliforme= Borman, Stuart and Wheeler

Natural vectors.--_Blattella germanica_, U.S.A. (Janssen and Wedberg, 1952).

Periplaneta americana, U.S.A. (Bitter and Williams, 1949).

* =Paracolobactrum= spp.

Natural vectors.--_Blatta orientalis_, U.S.A. (Olson and Rueger, 1950).

Blattella germanica, U.S.A. (Olson and Rueger, 1950).

Periplaneta americana, U.S.A. (Bitter and Williams, 1949; Olson and Rueger, 1950).

* =Paracolon bacilli=

Natural vectors.--Cockroaches [presumably _Blatta orientalis_, _Blattella germanica_ and/or _Periplaneta americana_], Egypt (El-Kholy and Gohar, 1945).

* =Proteus mirabilis= Hauser

Natural vectors.--_Periplaneta americana_, U.S.A. (Bitter and Williams, 1949, 1949a).

* =Proteus morganii= (Winslow et al.) Rauss

Natural vectors.--_Periplaneta americana_, U.S.A. (Bitter and Williams, 1949, 1949a).

Cockroaches [presumably _Blatta orientalis_, _Blattella germanica_, and/or _Periplaneta americana_], Egypt (El-Kholy and Gohar, 1945).

* =Proteus rettgeri= (Hadley et al.) Rustigian and Stuart

Natural vectors.--_Periplaneta americana_, U.S.A. (Bitter and Williams, 1949).

* =Proteus vulgaris= Hauser

Natural vectors.--_Blaberus craniifer_, U.S.A. (Wedberg et al., 1949).

Blatta orientalis, Italy (Spinelli and Reitano, 1932).

Periplaneta americana, U.S.A. (Bitter and Williams, 1949, 1949a).

Cockroaches, U.S.A. (Longfellow, 1913).

* =Proteus= spp.

Natural vectors.--_Blatta orientalis_, U.S.A. (Olson and Rueger, 1950).

Periplaneta americana, U.S.A. (Bitter and Williams, 1949; Olson and Rueger, 1950).

* =Salmonella anatis= (Rettger and Scoville) Bergey et al.

Natural vectors.--_Periplaneta americana_, U.S.A. (Eads et al., 1954).

* =Salmonella choleraesuis= (Smith) Weldin

Experimental vectors.--_Polyphaga saussurei_, U.S.S.R. (Zmeev _in_ Pavlovskii, 1948).

* =Salmonella enteritidis= (Gaertner) Castellani and Chalmers

Experimental vectors.--_Blatta orientalis_, U.S.S.R. (Rozengolts and Îûdina _in_ Pavlovskii, 1948).

Blattella germanica, U.S.A. (Olson and Rueger, 1950); U.S.S.R. (Rozengolts and Îûdina _in_ Pavlovskii, 1948).

Polyphaga saussurei, U.S.S.R. (Zmeev _in_ Pavlovskii, 1948).

* =Salmonella morbificans= (Migula) Haupt

Natural vectors.--_Periplaneta americana_, Australia (Mackerras and Mackerras, 1948, 1949).

Experimental vectors.--_Nauphoeta cinerea_, _Periplaneta australasiae_, _Periplaneta ignota_, and _Supella supellectilium_, Australia (Mackerras and Pope, 1948).

* =Salmonella paratyphi= (Kayser) Castellani and Chalmers

Experimental vectors.--_Polyphaga saussurei_, U.S.S.R. (Zmeev _in_ Pavlovskii, 1948).

* =Salmonella schottmuelleri= (Winslow et al.) Bergey et al.

Natural vectors.--_Periplaneta americana_, U.S.A. (Bitter and Williams, 1949, 1949a).

Experimental vectors.--_Periplaneta americana_, Gold Coast Colony (Macfie, 1922).

Polyphaga saussurei, U.S.S.R. (Zmeev _in_ Pavlovskii, 1948).

* =Salmonella= sp. (=Type Adelaidae=)

Experimental vectors.--_Nauphoeta cinerea_, _Periplaneta australasiae_, _Periplaneta ignota_, and _Supella supellectilium_, Australia (Mackerras and Pope, 1948).

* =Salmonella= sp. (=Type Bareilly=)

Natural vectors.--_Periplaneta americana_, U.S.A. (Eads et al., 1954).

* =Salmonella= sp. (=Type Bredeny=)

Natural vectors.--_Periplaneta americana_, U.S.A. (Bitter and Williams, 1949, 1949a).

* =Salmonella= sp. (=Type Derby=)

Experimental vectors.--_Nauphoeta cinerea_, _Periplaneta australasiae_, and _Supella supellectilium_, Australia (Mackerras and Pope, 1948).

* =Salmonella= sp. (=Type Kentucky=)

Natural vectors.--_Periplaneta americana_, U.S.A. (Eads et al., 1954).

* =Salmonella= sp. (=Type Kottbus=)

Experimental vectors.--_Periplaneta australasiae_, Australia (Mackerras and Pope, 1948).

* =Salmonella= sp. (=Type Meleagris=)

Natural vectors.--_Periplaneta americana_, U.S.A. (Eads et al., 1954).

* =Salmonella= sp. (=Type Montevideo=)

Experimental vectors.--_Periplaneta americana_, U.S.A. (Jung and Shaffer, 1952).

* =Salmonella= sp. (=Type Newport=)

Natural vectors.--_Periplaneta americana_, U.S.A. (Eads et al., 1954).

* =Salmonella= sp. (=Type Oranienburg=)

Natural vectors.--_Periplaneta americana_, U.S.A. (Bitter and Williams, 1949, 1949a; Eads et al., 1954).

Experimental vectors.--_Blatta orientalis_, _Blattella germanica_, and _Periplaneta americana_, U.S.A. (Olson and Rueger, 1950).

* =Salmonella= sp. (=Type Panama=)

Natural vectors.--_Periplaneta americana_, U.S.A. (Eads et al., 1954).

* =Salmonella= sp. (=Type Rubislaw=)

Natural vectors.--_Periplaneta americana_, U.S.A. (Eads et al., 1954).

* =Salmonella= sp. (=Type Tennessee=)

Natural vectors.--_Periplaneta americana_, U.S.A. (Eads et al., 1954).

* =Salmonella typhimurium= (Loeffler) Castellani and Chalmers

Natural vectors.--_Blattella germanica_, Belgium (Graffar and Mertens, 1950).

Nauphoeta cinerea, Australia (Mackerras and Mackerras, 1948).

Experimental vectors.--_Blaberus craniifer_, U.S.A. (Wedberg et al., 1949).

Blatta orientalis, U.S.S.R. (Rozengolts and Îûdina _in_ Pavlovskii, 1948).

Blattella germanica, Belgium (Graffar and Mertens, 1950); U.S.A. (Olson and Rueger, 1950; Janssen and Wedberg, 1952; Beck and Coffee, 1943); U.S.S.R. (Rozengolts and Îûdina _in_ Pavlovskii, 1948).

Nauphoeta cinerea, Australia (Mackerras and Pope, 1948).

Periplaneta americana, U.S.A. (Beck and Coffee, 1943; Jung and Shaffer, 1952).

Periplaneta australasiae and _Supella supellectilium_, Australia (Mackerras and Pope, 1948).

Polyphaga saussurei, U.S.S.R. (Zmeev _in_ Pavlovskii, 1948).

* =Salmonella typhosa= (Zopf) White

Natural vectors.--_Blatta orientalis_, Italy (Antonelli, 1930, 1943).

Experimental vectors.--_Blaberus craniifer_, U.S.A. (Wedberg et al., 1949).

Blatta orientalis, Italy (Spinelli and Reitano, 1932); U.S.A. (McBurney and Davis, 1930); U.S.S.R. (Rozengolts and Îûdina _in_ Pavlovskii, 1948).

Blattella germanica, U.S.A. (Janssen and Wedberg, 1952); Germany (Jettmar, 1927); U.S.S.R. (Rozengolts and Îûdina _in_ Pavlovskii, 1948).

Periplaneta americana, Gold Coast Colony (Macfie, 1922); Netherlands (Akkerman, 1933); Formosa (Morischita and Tsuchimochi, 1926); U.S.A. (Olson _in_ Roth and Willis, 1957a).

Periplaneta australasiae, Formosa (Morischita and Tsuchimochi, 1926).

Polyphaga saussurei, U.S.S.R. (Zmeev _in_ Pavlovskii, 1948).

Cockroaches [presumably _Blatta orientalis_, _Blattella germanica_, and/or _Periplaneta americana_], Egypt (El-Kholy and Gohar, 1945).

=Serratia marcescens= Bizio

Synonymy.--_Bacillus prodigiosus_, _Bacterium prodigiosum_.

Habitat.--Water, soil, milk, foods, and various insects.

Natural hosts.--_Blatta orientalis_, Poland (Nicewicz et al., 1946): From intestinal tract. Italy (Spinelli and Reitano, 1932).

Blattella germanica, Canada (Heimpel and West, 1959).

Diploptera punctata, _Nauphoeta cinerea_, _Neostylopyga rhombifolia_, _Panchlora nivea_, _Pycnoscelus surinamensis_, and _Supella supellectilium_, U.S.A. (Roth and Willis, unpublished data, 1958): The organism was isolated and identified by Dr. Hillel Levinson, Quartermaster bacteriologist, from dead specimens found in our laboratory colonies which showed the red coloration characteristic of insects that have died with infections of _S. marcescens_ (pl. 16, A, B).

Leucophaea maderae, U.S.A. (Levinson, personal communication, 1958): The organism was isolated from the hemolymph of living insects while attempting to determine the cause of unexplained mortality in our laboratory colony of this insect.

Leucophaea maderae or _Periplaneta americana_, Philippine Islands (Barber, 1912): From hemolymph.

Periplaneta americana, U.S.A. (Gier, 1947; Steinhaus, 1959).

Periplaneta australasiae and _Periplaneta brunnea_ (Roth and Willis, unpublished data [1954]): In laboratory colonies. Isolated from suspensions of ground insects. In 1954 we received a culture of _Periplaneta brunnea_ from the Department of Public Health, University of Minnesota. These insects began to die off rapidly and the normally lightly pigmented parts of the body became red. Dr. Hillel Levinson, Quartermaster bacteriologist, cultured _Serratia marcescens_ from several moribund individuals. The Department of Public Health of Minnesota had at times in the past cultured _S. marcescens_ but had discarded the cultures and was unaware that it might be surviving in the cockroach colonies (Richards, personal communication, 1954).

Periplaneta sp., U.S.A. (Olson _in_ Roth and Willis, 1957a): Isolated from an undetermined species of _Periplaneta_, received in a shipment from the South, a strain of _S. marcescens_ which was toxic to mice when administered intraperitoneally.

Experimental hosts.--_Blaberus craniifer_, U.S.A. (Wedberg et al., 1949): When fed in small numbers, _S. marcescens_ increased to such an extent that the insect's extremities and upper halves of their bodies turned deep red. The insects died after this color appeared and practically pure cultures of _Serratia_ were recovered from the reddened areas.

Blatta orientalis, Italy (Cao, 1898): Isolated from intestinal contents. Passed unchanged through the gut.

Blattella germanica, Canada (Heimpel and West, 1959): Not normally pathogenic per os; LD_{50} by injection, is approximately 38,000 bacteria per insect.

Periplaneta americana, U.S.A. (Gier, 1947): Organism toxic to the cockroach when injected.

Cockroaches, U.S.A. (Longfellow, 1913): Isolated from legs and viscera after feeding experiments.

* =Shigella alkalescens= (Andrewes) Weldin

Natural vectors.--_Periplaneta americana_, U.S.A. (Bitter and Williams, 1949, 1949a).

* =Shigella dysenteriae= (Shiga) Castellani and Chalmers

Experimental vectors.--_Blatta orientalis_, Italy (Spinelli and Reitano, 1932).

Periplaneta americana, Formosa (Morischita and Tsuchimochi, 1926).

Polyphaga saussurei, U.S.S.R. (Zmeev _in_ Pavlovskii, 1948).

* =Shigella paradysenteriae= (Collins) Weldin

Natural vectors.--_Blatta lateralis_, Tadzhikistan (Zmeev, 1940).

Experimental vectors.--_Periplaneta americana_, Gold Coast Colony (Macfie, 1922).

Polyphaga saussurei, U.S.S.R. (Zmeev _in_ Pavlovskii, 1948).

Cockroaches, Venezuela (Tejera, 1926).

Family PARVOBACTERIACEAE

=Bacteroides uncatus= Eggerth and Gagnon

Habitat.--Probably intestinal canal of mammals; from human feces.

Natural vectors.--_Blatta orientalis_, Poland (Nicewicz et al., 1946): From intestinal canal.

* =Brucella abortus= (Schmidt and Weis) Meyer and Shaw

Experimental vectors.--_Periplaneta americana_, U.S.A. (Ruhland and Huddleson, 1941).

=Fusiformis lophomonadis= Grassé

Habitat.--Surface of a flagellate (_Lophomonas striata_) which lives in the intestine of cockroaches (Breed et al., 1948; Grassé 1926, 1926a).

* =Malleomyces mallei= (Zopf) Pribram

Experimental vectors.--_Blatta orientalis_, Italy (Cao, 1898).

* =Pasteurella multocida= (Lehmann and Neumann) Rosenbusch and Merchant

Experimental vectors.--_Blatta orientalis_, Germany (Küster, 1902, 1903); Italy (Cao, 1906).

* =Pasteurella pestis= (Lehmann and Neumann) Holland

Natural vectors.--_Blatta orientalis_, Hongkong (Hunter, 1906).

Experimental vectors.--_Blatta orientalis_, Italy (Cao, 1898); Germany (Küster, 1903).

Blattella germanica, Germany (Jettmar, 1927).

Leucophaea maderae and _Periplaneta americana_, Philippine Islands (Barber, 1912).

Family BACTERIACEAE

[cross] =Bacterium alkaligenes= Nyberg

Natural vectors.--_Periplaneta americana?_ ("_Blatella americana_"), England (Shrewsbury and Barson, 1948): From intestinal tract.

[cross] =Bacterium delendae-muscae= Roubaud and Descazeaux

Source.--Diseased fly larvae.

Experimental infection.--Cockroach, France (Roubaud and Descazeaux, 1923): Organism pathogenic to cockroach when injected.

=Bacterium haemophosphoreum= Pfeiffer and Stammer

Habitat.--Diseased larvae of _Mamestra oleracea_.

Experimental infection.--_Blatta orientalis_ and _Blattella germanica_, Germany (Pfeiffer and Stammer, 1931): Organism pathogenic, when injected, to eight _B. orientalis_ and two _B. germanica_.

=Coccobacillus cajae= Picard and Blanc

Experimental host.--_Blatta orientalis_, France (Picard and Blanc, 1913): The organism was pathogenic to _B. orientalis_ when injected.

Family BACILLACEAE

* =Bacillus anthracis= Cohen emend. Koch

Experimental vectors.--_Blatta orientalis_, Italy (Cao, 1898, 1906); Germany (Küster, 1903).

[cross] =Bacillus bütschlii= Schaudinn

Habitat.--_Blatta orientalis_, Germany (Schaudinn, 1902): Isolated from intestinal tract. Three percent of the cockroaches from Berlin bakeries were infected.

=Bacillus cereus= Frankland and Frankland

Synonymy.--_Bacillus albolactis._

Habitat.--Soil, dust, milk, plants.

Natural vectors.--_Blaberus craniifer_, U.S.A. (Wedberg et al., 1949): From feces.

Periplaneta americana, U.S.A. (Hatcher, 1939): In feces.

Experimental host.--_Periplaneta americana_, U.S.A. (Babers, 1938): The cockroaches died within 96 hours after being injected with 10^{-3} ml. of a 24-hour broth culture.

=Bacillus circulans= Jordan

Habitat.--Soil, water, dust.

Natural vectors.--_Blattella germanica_, U.S.A. (Janssen and Wedberg, 1952): From intestine and feces.

[cross] =Bacillus flacheriae= (Hoffman)

Source.--Diseased nun moth larvae.

Experimental infection.--_Blatta orientalis_, Europe (Filatoff, 1904): The organism was not pathogenic when fed to the cockroach, but killed the insects when

injected into the body cavity; after the insects died Filatoff reisolated this pathogen together with another bacillus from the cadavers. He succeeded in culturing the new microorganism and found it to be pathogenic when injected into, but not when fed to, the cockroaches. The diseased insects became sluggish, failed to eat or drink, turned over on their backs, their extremities became totally paralyzed, and they finally died.

=Bacillus megaterium= De Bary

Habitat.--Soil, water, decomposing materials.

Natural vectors.--_Periplaneta americana?_ ("_Blatella americana_"), England (Shrewsbury and Barson, 1948): From intestinal tract.

Experimental vectors.--_Blatta orientalis_, Italy (Cao, 1898): Organism recovered, apparently unchanged, from intestinal contents.

[cross] =Bacillus monachae= (von Tubeuf) Eckstein

Synonymy.--_Bacterium monache._

Source.--Diseased larvae of nun moth, _Lymantria monacha_.

Experimental infection.--_Blatta orientalis_, Europe (Filatoff, 1904): Organism pathogenic to the cockroach when injected but not when fed.

[cross] =Bacillus periplanetae= Tichomiroff

Habitat.--_Blatta orientalis_, U.S.S.R.? (Tichomiroff, 1870[?], _in_ Filatoff, 1904): The infected insects suffered from a diarrhea and the liquid feces were yellow-brown.

[cross] =Bacillus stellatus= Hollande

Natural infection.--_Blatta orientalis_, France (Hollande, 1934): Organism observed regularly in the intestine (especially rectum). Extensive description given.

[cross] =Bacillus radiciformis=

Experimental vectors.--_Blatta orientalis_, Italy (Cao 1898): Organism recovered, apparently unchanged, from intestinal contents.

* =Bacillus subtilis= Cohn emend. Prazmowski

Natural vectors.--_Blaberus craniifer_, U.S.A. (Wedberg et al., 1949).

Blatta orientalis, Italy (Cao, 1898, 1906; Spinelli and Reitano, 1932); France (Sartory and Clerc, 1908); Poland (Nicewicz et al., 1946).

Cryptocercus punctulatus, U.S.A. (Hatcher, 1939).

Periplaneta americana? ("_Blatella americana_"), England (Shrewsbury and Barson, 1948).

Cockroaches, U.S.A. (Longfellow, 1913).

Experimental vectors.--_Blatta orientalis_, Italy (Cao, 1898, 1906).

"=Bacillus subtilis= group"

Natural infection.--_Blatta orientalis_, Italy (Ronzoni, 1949): Isolated from oöthecae.

[cross] =Bacillus tritus= Batchelor

Habitat.--Isolated from feces (man?).

Natural vectors.--_Blatta orientalis_, Poland (Nicewicz et al., 1946): Isolated from intestinal tract.

* =Clostridium feseri= Trevisan

Experimental vectors.--_Blatta orientalis_, Italy (Cao, 1898).

=Clostridium lentoputrescens= Hartsell and Rettger

Habitat.--Soil, intestinal tract of man.

Natural vectors.--_Blatta orientalis_, Poland (Nicewicz et al., 1946): Isolated from intestinal tract.

* =Clostridium novyi= (Migula) Bergey et al. or

* =Clostridium sporogenes= (Metchnikoff) Bergey et al.

Natural and experimental vectors.--_Blatta orientalis_, Italy (Cao, 1898).

* =Clostridium perfringens= (Veillon and Zuber) Holland

Natural vectors.--Cockroaches [presumably _Blatta orientalis_, _Blattella germanica_, and/or _Periplaneta americana_], Egypt (El-Kholy and Gohar, 1945).

* =Clostridium tetani= (Flügge) Holland

Experimental vectors.--_Blatta orientalis_, Italy (Cao, 1898).

* =Clostridium= spp.

Natural vectors.--_Periplaneta americana?_ ("_Blatella americana_"), England (Shrewsbury and Barson, 1948).

Order ACTINOMYCETALES

Family MYCOBACTERIACEAE

* =Mycobacterium avium= Chester

Experimental vectors.--_Blatta orientalis_, U.S.S.R. (Ekzempliarskaia _in_ Pavlovskii, 1948).

=Mycobacterium friedmannii= Holland

Habitat.--Parasitic in turtles and possibly sparingly distributed in soils.

Natural vectors.--_Periplaneta americana_, U.S.A., Texas (Micks, in Roth and Willis, 1957a): Organism isolated from batches of intestinal tracts of cockroaches collected at random.

* =Mycobacterium lacticola= Lehmann and Neumann?

Natural vectors.--_Periplaneta americana_, U.S.A. (Leibovitz, 1951).

* =Mycobacterium leprae= (Armauer-Hansen) Lehmann and Neumann

Natural vectors.--_Blattella germanica_, Southern Rhodesia and Kenya (Moiser, 1945, 1946, 1946a; Anonymous, 1946).

Periplaneta americana and _Periplaneta australasiae_, Formosa (Arizumi, 1934, 1934a).

Cockroaches, Venezuela (Tejera, 1926); Belgian Congo (Radna, 1939).

Experimental vectors.--_Blatta orientalis_, Europe (Paldrock _in_ Klingmüller, 1930); Nyasaland (Lamborn, 1940).

Blattella germanica, Europe (Paldrock _in_ Klingmüller, 1930); Southern Rhodesia and Kenya (Moiser, 1945, 1946, 1946a, 1947; Anonymous, 1946).

Nauphoeta cinerea, Nyasaland (Lamborn, 1940).

Periplaneta americana, Gold Coast Colony (Macfie, 1922); Formosa (Arizumi, 1934, 1934a).

Periplaneta australasiae, Formosa (Arizumi, 1934, 1934a).

Cockroaches, Belgian Congo (Radna, 1939); Venezuela (Tejera, 1926).

* =Mycobacterium lepraemurium= Marchoux and Sorel

Experimental vectors.--Cockroaches, Belgian Congo (Radna, 1939).

* =Mycobacterium phlei= Lehmann and Neumann

Natural vectors.--_Periplaneta americana_, U.S.A. (Leibovitz, 1951; Micks _in_ Roth and Willis, 1957a).

* =Mycobacterium piscium= Bergey et al.

Natural vectors.--_Periplaneta americana_, U.S.A. (Leibovitz, 1951).

Experimental vectors.--_Blatta orientalis_, U.S.S.R. (Ekzempliarskaia _in_ Pavlovskii, 1948).

* =Mycobacterium tuberculosis= (Schroeter) Lehmann and Neumann

Experimental vectors.--_Blatta orientalis_, Italy (Cao, 1898); Germany (Küster, 1903); U.S.S.R. (Ekzempliarskaia _in_ Pavlovskii, 1948).

Blattella germanica, on shipboard (Morrell, 1911).

Periplaneta americana, Gold Coast Colony (Macfie, 1922).

Cockroaches, Venezuela (Tejera, 1926); U.S.A. (Read, 1933).

* =Mycobacterium= spp.

Natural vectors.--_Periplaneta americana_, U.S.A. (Leibovitz, 1951; Micks _in_ Roth and Willis, 1957a).

Family ACTINOMYCETACEAE

* =Nocardia= sp.?

Natural vectors.--_Periplaneta americana_, U.S.A. (Leibovitz, 1951).

Family STREPTOMYCETACEAE

=Streptomyces leidynematis= Hoffman

Habitat.--Surface of the nematodes _Hammerschmidtiella diesingi_ and _Leidynema appendiculata_ in _Periplaneta americana_, U.S.A. (Hoffman, 1952, 1953): Eighteen percent of 192 nematodes found in 52 adult cockroaches were infected with the bacterium.

Order CARYOPHANALES

Family ARTHROMITACEAE

=Arthromitus intestinalis= (Valentin) Peshkoff

Synonymy.--_Hygrocrocis intestinalis._

Habitat.--_Blatta orientalis_, Europe (Valentin, 1836; Robin, 1847, 1853; Peshkoff, 1940): Isolated from intestinal tract. The organism appears as fragments in fecal masses or as fibers adhering to the mucous membrane of the large intestine (Robin, 1853).

Cockroach, France? (Chatton and Pérard, 1913).

Order SPIROCHAETALES

Family SPIROCHAETACEAE

[cross] =Spirochaeta blattae= Tejera

Habitat.--_Blaberus atropos_, Venezuela (Tejera, 1926): Isolated from intestinal tract.

* [cross] =Spirochaeta periplanetae= Laveran and Franchini

Habitat.--_Blatta orientalis_, France (Laveran and Franchini, 1920a).

Cockroaches, Venezuela (Tejera, 1926): Tejera reported finding "_Spirochaeta blatarum_ Laveran et Franchini" which may have been a lapsus.

Family TREPONEMATACEAE

[cross] =Treponema parvum= Dobell

Habitat.--_Blatta orientalis_, England (Dobell, 1912); U.S.S.R.? Zasukhin (1930): From intestinal tract.

[cross] =Treponema stylopygae= Dobell

Synonymy.--_Spirochaeta stylopygae_ Zuelzer.

Habitat.--_Blatta orientalis_, England (Dobell, 1912); U.S.S.R.? Zasukhin (1930): From intestinal tract.

=Unidentified spirochaetes=

Habitat.--_Blatta orientalis_, U.S.S.R. (Yakimov and Miller, 1922): Spirochaetes and spirilla were found in the intestines of 70 percent of 124 specimens collected in Petrograd.

Periplaneta americana, Gold Coast Colony (Macfie, 1922).

ADDITIONAL BACTERIA WHOSE TAXONOMIC POSITION IS UNKNOWN

* "=B. aerobio del pseudoedema maligno=" of Cao

Natural vectors.--_Blatta orientalis_, Italy (Cao, 1906).

=B. alcaligenes beckeri=

Natural vectors.--_Blatta orientalis_, Poland (Nicewicz et al., 1946): Isolated from intestinal tract.

* "=B. del pseudoedema maligno=" of Cao

Natural vectors.--_Blatta orientalis_, Italy (Cao, 1906).

* "=Bacillo proteisimile=" of Cao

Natural and experimental vectors.--_Blatta orientalis_, Italy (Cao, 1898, 1906).

* "=Bacillo del barbone dei bufali=" of Cao

Experimental vectors.--_Blatta orientalis_, Italy (Cao, 1898).

* "=Bacillo similcarbonchio=" of Cao

Natural and experimental vectors.--_Blatta orientalis_, Italy (Cao, 1898, 1906).

* "=Bacillo similtifo=" or "=Bacillo tifosimile=" of Cao

Natural and experimental vectors.--_Blatta orientalis_, Italy (Cao, 1898, 1906).

"=Bacillus="

Natural infection.--_Blatta orientalis_, Germany (Heinecke, 1956): Disease organism found in the hemolymph of infected cockroaches. It can be spread by mouth and through wound infection. The animals died with symptoms of paralysis in 85-90 days. The organism has been isolated and is in the culture collection of the Institute for Microbiology and Experimental Therapy, Jena, under the numbers SG 896, Strain A; SG 897, Strain B; SG 898, Strain C.

Experimental infection.--_Blattella germanica_, Germany (Heinecke, 1956): Infected animals died in 26-30 days.

Periplaneta americana was unaffected even by heavy inoculations of the pathogen.

"=Bacterium="

Source.--(I) Diseased silkworm larvae. (II) Diseased _Ocneria dispar_ larvae and blood of _Blatta orientalis_.

Experimental infection.--(I)(II) _Blatta orientalis_, Europe (Filatoff, 1904): Organism pathogenic when injected, nonpathogenic when ingested.

(I) Cockroach, U.S.A. (Glaser, 1925): Organism pathogenic to cockroach when injected.

"=Coccobacillus="

Natural infection.--_Blatta orientalis_, France (Hollande, 1934): Organism described morphologically.

"=Colon bacilli="

Natural vectors.--Cockroaches [presumably _Blatta orientalis_, _Blattella germanica_, and/or _Periplaneta americana_], Egypt (El-Kholy and Gohar, 1945): From the outer surface, intestinal tract, and suspensions of macerated insects.

"=Diplococci="

Natural vectors.--_Blatta orientalis_, Germany (Jettmar, 1935): From intestinal tract.

Blattella germanica, Germany (Jettmar, 1935): From outer surface of body.

"=Diphtheroid I and II="

Source.--_Periplaneta americana._

Natural and experimental infections.--_Periplaneta americana_, U.S.A. (Gier, 1947): Pathogenicity to the cockroach variable when organism injected.

"=Gram positive rods="

Source.--Feces of _Blattella germanica_.

Experimental vector.--_Blattella germanica_, Germany (Vollbrechtshausen, 1953): Nonpathogenic to the insect when injected into the mouth or anus.

"=Silkworm disease bacillus="

Cockroaches that were inoculated with living cultures succumbed in a few days (Glaser, 1925).

[cross] =Spirillochaeta blattae= Hollande

Habitat.--_Blatta orientalis_, France (Hollande, 1934; Hollande and Hollande, 1946): Organism found in hind intestine. This spirillum was stated to be related in external morphology to _Spirillum periplaneticum_ Kunstler and Gineste, but it was believed that _S. blattae_ should be in the Spirochaetaceae rather than the Spirillaceae.

"=Spirochaetoid bacteria="

Habitat.--_Blatta orientalis_, France (Hollande, 1934): Two kinds described but not named.

[cross] =Tetragenous= sp.

Natural and experimental infections.--_Periplaneta americana_, U.S.A. (Gier, 1947): Pathogenicity to the cockroach variable when organism injected.

VIII. FUNGI AND YEASTS

By far the greatest number of fungi known to be associated with cockroaches belong to the Laboulbeniaceae, genus _Herpomyces_, the species of which are restricted to parasitizing cockroaches (Thaxter, 1908). Most species are hyaline, small and inconspicuous (Thaxter, 1931) and are usually, but not exclusively, found on the insects' antennae. Species of _Herpomyces_ are highly, but not completely, host specific (Richards and Smith, 1954). While attached to the host, these fungi appear like minute dark-colored, yellow, or white (e.g., _H. arietinus_) bristles or bushy hairs (pl. 27, A).

[Illustration: FIG. 1.--Diagram illustrating the relationship between a mature plant of _Herpomyces stylopygae_ and the integument of _Blatta orientalis_. (Reproduced from Richards and Smith [1956], through the courtesy of Dr. A. G. Richards.)]

Richards and Smith (1955, 1955a) have studied the life history of _Herpomyces stylopygae_ on the oriental cockroach. The plants grow only on living cockroaches, and the infection is disseminated by contact. The mature plants are found mostly on the antennae (pl. 27, B), either on setae or on hard or soft cuticle. Spores are ejected from perithecia singly or in groups of 2 to 4 spores, although groups as large as 12 spores have been found. The presence of single, paired, or multiple spore groups on the surface of the host was correlated with the presence of single, paired, or multiple plants on infected cockroaches. Development from spore to mature perithecia takes about two weeks. The plant obtains nutriment from the host by means of a tubular haustorium that extends through the cockroach's cuticle and expands into a large bulb in the underlying epidermal cells (fig. 1). Infections on nymphs are lost when the nymph moults, but infections on adults persist throughout life. However, nymphs which have lost the fungus upon moulting are readily reinfected. Collart's (1947) statement that nymphs are never infected with _Herpomyces_ is not true.

Richards and Smith (1956) concluded that there is no evidence of pathogenicity in _Herpomyces_ infections because heavily infected cockroaches appear fully active in laboratory colonies; they can run at the same speed as uninfected cockroaches; they reproduce normally and do not appear to die prematurely. These workers stated that the infections cause a dermatitis which is neither pathogenic nor debilitant. So far as we know there are no comparative data on longevity and reproductive performance of fungus-infected versus normal cockroaches. However, Gunn and Cosway (1938) have shown that the presence of these fungi (identified as _Stigmatomyces_ sp.; see p. 138) on the antennae seemed to interfere with the humidity reactions of _Blatta orientalis_. Although Richards and Smith (1956) admit that humidity receptors and

other sense organs on the antennae may be destroyed by the fungus, they state that "insects possess such a large number of sensilla that the result may well be more distressing to the sensory physiologist than to the insect." Yet it seems to us that the loss of sense organs from fungal infection and concomitant shortening of the antennae (pl. 27, A) might be considerably more of a handicap to free-living cockroaches than those in laboratory colonies.

Bode (1936) studied the flora of _Periplaneta americana_ and cultured Aspergillaceae and Mucorinae from the insect's body surface and intestinal contents; he also found nonsporulating yeasts in _P. americana_. To prevent fungal growth on oöthecae of _P. americana_, Griffiths and Tauber (1942a) autoclaved their rearing containers and dipped the oöthecae in 70-percent alcohol for 10 seconds.

Mercier (1906) isolated and cultured a pathogenic yeastlike parasite which had invaded the fat body and blood of _Blatta orientalis_. The abdomens of the infected insects became swollen, distended, and soft. McShan (unpublished MS., 1953) consistently isolated Saccharomycetes from the feces of _Periplaneta americana_.

FUNGI ASSOCIATED WITH COCKROACHES

The use of the asterisk (*) is explained in footnote 3, page 4.

Phylum THALLOPHYTA

Class FUNGI IMPERFECTI

Order MONILIALES

Family PSEUDOSACCHAROMYCETACEAE

=Candida zeylanoides= (Castellani) Langeron and Guerra

Natural host.--Oötheca of _Blatta orientalis_, Italy (Ronzoni, 1949).

=Torulopsis= sp.

Natural host.--Oötheca of _Blatta orientalis_, Italy (Ronzoni, 1949).

Family MONILIACEAE

=Spicaria prasina= (Maublanc) Sawada

Natural host.--_Ischnoptera rufa rufa_, Puerto Rico (Wolcott, 1950): A dead specimen of this cockroach was found stuck to a leaf and covered with this fungus.

=Aspergillus flavus= Link

Natural hosts.--Oöthecae of _Blattella germanica_ and _Eurycotis floridana_, U.S.A., Pennsylvania (Roth and Willis, unpublished data, 1952): On outer surface. Determination by Miss Mary Downing.

Oöthecae of _Periplaneta americana_, U.S.A., Pennsylvania (Roth and Willis, unpublished data, 1952): Inside oöthecae. Determination by Miss Mary Downing.

* =Aspergillus fumigatus= Fresenius

Natural vector.--_Blatta orientalis_, France (Sartory and Clerc, 1908): From intestine.

* =Aspergillus niger= van Tieghem

Natural vector.--_Periplaneta americana_, U.S.A., Texas (McShan in Roth and Willis, 1957a): From feces.

Experimental vector.--_Blatta orientalis_, Italy (Cao, 1898): Organism passed unchanged through the gut of the insects.

=Aspergillus sydowi= (Bainier and Sartory) Thom and Church

Natural host.--Oötheca of _Eurycotis floridana_, U.S.A., Pennsylvania (Roth and Willis, unpublished data, 1952): On outer surface. Determination by Miss Mary Downing.

=Aspergillus tamarii= Kita

Natural host.--Oöthecae of _Blattella germanica_, U.S.A., Pennsylvania (Roth and Willis, unpublished data, 1952): On exterior surface. Determination by Miss Mary Downing.

=Aspergillus= sp.?

Natural and experimental vector.--_Blattella germanica_, on shipboard (Morrell, 1911): Isolated from feces. Experimentally Morrell also showed that the spores of the fungus could be recovered from feces of cockroaches that had fed on them.

=Aspergillus= sp.

Natural vector.--_Periplaneta americana_, England (Bunting, 1956): The fungus was isolated mostly from imperfectly excreted feces.

=Beauveria bassiana= (Balsamo) Vuillemin

Experimental host.--_Blattella germanica_ and _Periplaneta americana_, U.S.A. (Dresner, 1949, 1950): The nymphs of American cockroaches became infected when they (1) were injected with a 1-percent suspension of spores, (2) ate rat pellets sprayed with the spore suspension, or (3) were dusted with the fungus spores. The symptoms of the fungus infection were paralysis followed by death; some of the infected insects liquefied, others dried up after the appearance of a subcuticular blackening.

=Cephalosporium= sp.

Natural vector.--_Periplaneta americana_, U.S.A., Texas (McShan, unpublished MS., 1953): From feces of cockroaches collected in the basement of a grain elevator at the docks in Galveston.

* =Geotrichum candidum= Link

Experimental vector.--_Blatta orientalis_, Italy (Cao, 1898): Organism retained its pathogenicity after passing through the insect's gut.

=Penicillium= sp.

Natural vector.--_Blaberus craniifer_, U.S.A. (Wedberg et al., 1949): From feces.

Periplaneta americana, England (Bunting, 1956): Mostly from imperfectly excreted feces.

=Metarrhizium anisopliae= (Metschnikoff) Sorokin

Natural hosts.--_Blattidae_, Seymour (1929); Charles (1941).

Panesthia australis, U.S.A., Massachusetts (Roth and Willis, unpublished data, 1957): Growing on adult specimens that were found dead in a laboratory colony. Determination by Miss Dorothy Fennell.

Periplaneta americana, England (Bunting, 1956): Growing on genitalia of females where it prevented oöthecal formation.

Cockroach, Puerto Rico (Johnston, 1915): From a "small roach" in the pathological collection at Rio Piedras (no data).

Family DEMATIACEAE

=Memnoniella echinata= (Rivolta) Galloway

Natural host.--Oötheca of _Blattella germanica_, U.S.A., Pennsylvania (Roth and Willis, unpublished data, 1952): On material that had oozed from a damaged oötheca. Determination by Miss Mary Downing.

=Torula acidophila= Owen and Mobley

Natural host.--_Periplaneta americana_, U.S.A. (Owen and Mobley, 1948): The digestive tract of this cockroach is the normal habitat of this yeast which was transmitted to sirup by the insects. The yeast superimposed a foreign taste, suggestive of malic acid, upon the original flavor of the sirup.

=Torula gropengiesseri= Lodder

Natural host.--_Blatta orientalis_, Germany (Gropengiesser, 1925; Lodder, 1934): Isolated from fat body and oöthecae. Gier (1947) is of the opinion that the so-called yeasts that supposedly may displace the bacteroids in the fat body (Mercier, 1907b; Gropengiesser, 1925) may actually represent poorly fixed and insufficiently stained bacteroids.

=Torula rosea= Preuss

Experimental host.--_Blaberus craniifer_, U.S.A. (Wedberg et al., 1949): Upon repeated feeding of massive doses of this yeast to the cockroach, these workers were able to isolate the organism from the feces up to six days thereafter. There was no evidence that _T. rosea_ was pathogenic for _B. craniifer_.

Class PHYCOMYCETES

Order MUCORALES

Family MUCORACEAE

=Mucor guilliermondii= Nadson and Filippov

Natural host.--_Periplaneta americana_, U.S.S.R. (Nadson and Filippov, 1925; Filippov, 1926): Isolated and cultured from intestine.

=Mucor= sp.

Natural host.--Oötheca of _Periplaneta americana_, U.S.A., Pennsylvania (Roth and Willis, unpublished data, 1952): Inside oötheca. Determination by Miss Mary Downing.

Pycnoscelus surinamensis, Germany (Bode, 1936): Isolated from fat body which it had stained red.

=Rhizopus nigricans= Ehrenberg

Natural vector.--_Blaberus craniifer_, U.S.A. (Wedberg et al., 1949): From feces.

=Rhizopus= sp.

Natural vector.--_Periplaneta americana_, U.S.A., Texas (McShan, unpublished MS., 1953): From feces.

=Syncephalastrum= sp.

Natural vector.--_Periplaneta americana_, U.S.A., Texas (McShan, unpublished MS., 1953): From feces.

Order ENTOMOPHTHORALES

Family BLASTOCYSTIDACEAE

=Blastocystis hominis= Brumpt

Natural vector.--_Blatta orientalis_, U.S.S.R. (Zasukhin, 1930): In hind gut in 40 percent of over 3,000 cockroaches.

=Blastocystis= sp.

Natural vectors.--_Blatta orientalis_, U.S.S.R. (Yakimov and Miller, 1922): Found in the intestinal contents of 29 percent of 124 _B. orientalis_.

Cockroaches, Venezuela (Tejera, 1926).

* * * * *

The placement of the following fungus is problematic.

=Coccidioides periplanetae= Avrech

Natural host.--_Blatta orientalis_, Germany (Avrech, 1931): Found in cells lining the lumen of midgut and caeca. The whole upper part of the epithelium was filled with sporangia and spores.

Class ASCOMYCETES

Order ENDOMYCETALES

Family SACCHAROMYCETACEAE

=Saccharomyces cerevisiae= Hansen

Natural vector.--_Blaberus craniifer_, U.S.A. (Wedberg et al., 1949): In feces.

=Saccharomyces= sp.

Natural vector.--_Blattella germanica_, U.S.A. (Janssen and Wedberg, 1952): Found consistently in alimentary tract of _B. germanica_ fed sucrose solutions.

Order HYPOCREALES

Family HYPOCREACEAE

=Cordyceps amazonica= Hennings

Natural host.--Cockroaches, British Honduras (Mains, 1940).

=Cordyceps blattae= Petch

Natural host.--_Blattella germanica_, Ceylon (Petch, 1924): Collected at Hakgala twice. A slight covering of brown mycelium overran the insect and fastened it to the underside of a living leaf.

Order LABOULBENIALES

Family LABOULBENIACEAE

=Herpomyces amazonicus= Thaxter

Natural host.--_Nyctibora obscura_, Brazil, Natal (Thaxter, 1931): On antennae.

=Herpomyces anaplectae= Thaxter

Natural hosts.--_Anaplecta_ sp., Venezuela, Caracas (Thaxter, 1905, 1908); Trinidad (Thaxter, 1931): On antennae.

Cockroach, Sumatra (Thaxter, 1931).

=Herpomyces appendiculatus= Thaxter

Natural host.--_Platyzosteria scabra_, Australia, N.S.W. (Thaxter, 1931): On antennae.

=Herpomyces arietinus= Thaxter

Natural hosts.--_Ischnoptera_ sp., U.S.A., Georgia (Thaxter, 1908).

Parcoblatta uhleriana, U.S.A., Massachusetts (Roth, unpublished data, 1957): The nymphs were in a culture of _Parcoblatta virginica_ which was infected with this fungus; it is possible that these _P. uhleriana_ became infected by contact with _P. virginica_. Fungus identified by Dr. R. K. Benjamin.

Parcoblatta virginica, U.S.A., Massachusetts (Roth, unpublished data, 1957): Fungus determined by Dr. R. K. Benjamin. Fungus found on antennae, palpi, legs, body surface (pl. 27, A).

Parcoblatta sp., U.S.A., Kentucky, Massachusetts (Thaxter, 1902, 1908): On antennae.

It is likely that Thaxter's host records (certainly those assigned to _Temnopteryx_ and possibly those assigned to _Ischnoptera_) were species of _Parcoblatta_. Hebard (1917) has shown that all the species referred to _Ischnoptera_ in the United States, except _I. deropeltiformis_, now belong in the genus _Parcoblatta_. All species originally referred to the genus _Temnopteryx_ in the United States are now synonymized with species of _Parcoblatta_.

=Herpomyces chaetophilus= Thaxter

Natural hosts.--_Periplaneta americana_, Brazil (Thaxter, 1931).

Periplaneta sp., Zanzibar and Mauritius (Thaxter, 1902, 1908): On spines of legs, antennae, and cerci.

=Herpomyces chilensis= Thaxter

Natural host.--Cockroach, Chile (Thaxter, 1918): On antennae.

=Herpomyces diplopterae= Thaxter

Natural hosts.--_Diploptera punctata_, Ascension Island (Thaxter, 1902, 1908): On antennae. This species also was infected experimentally (Richards and Smith, 1954).

Cockroach, Fiji (Thaxter, 1931).

=Herpomyces ectobiae= Thaxter

Natural hosts.--_Blattella germanica_, U.S.A., Massachusetts (Thaxter, 1902, 1908); Burma, Tenasserim (Spegazzini, 1915); Argentina, Buenos Aires (Spegazzini, 1917): On antennae. U.S.A., Minnesota (Richards and Smith, 1955): Scattered over entire body, wings. France? (Picard, 1913): On tibial spines. Chile and Philippine Islands (Thaxter, 1931).

"_Ectobia_" spp., Zanzibar and Saint Kitts, B.W.I. (Thaxter, 1902, 1908): Possibly on species that are now in the genus _Blattella_ rather than in the genus _Ectobius_ as it is known today, because Thaxter also used the synonym _Ectobia germanica_ for the German cockroach, _Blattella germanica_.

Experimental hosts.--_Blattella germanica_ and _Blattella vaga_, U.S.A. (Richards and Smith, 1954).

=Herpomyces forficularis= Thaxter

Natural hosts.--Cockroaches, Mauritius? and Fiji (Thaxter, 1902, 1908, 1931): On antennae.

=Herpomyces gracilis= Thaxter

Natural host.--_Blattella humbertiana_, Philippine Islands, Luzon (Thaxter, 1931): On antennae.

=Herpomyces grenadinus= Thaxter

Natural host.--Cockroach, Grenada, B.W.I. (Thaxter, 1931): On antennae of a "brown wingless blattid."

=Herpomyces leurolestis= Thaxter

Natural host.--_Leurolestes pallidus_, British Guiana and Trinidad (Thaxter, 1931): On antennae.

=Herpomyces lobopterae= Thaxter

Natural host.--_Loboptera_ sp., Argentina (Thaxter, 1931): On antennae.

=Herpomyces macropus= Spegazzini

Natural host.--_Blaberus_ sp.?, Argentina (Spegazzini, 1917).

Cockroaches, Peru, Puerto Rico, Ecuador, and Haiti (Spegazzini, 1915, 1917): Material previously assigned by Spegazzini (1915) to _H. paranensis_ was also placed by him in this new species. However, Thaxter (1931) believed that _H. macropus_ may be synonymous with _H. paranensis_, but he provisionally retained _H. macropus_ because he had not seen Spegazzini's material.

=Herpomyces nyctoborae= Thaxter

Natural hosts.--_Nyctibora tomentosa_, U.S.A., Texas (Thaxter, 1905, 1908): On antennae. This cockroach is not established in Texas, and the specimen may have been misidentified (Gurney, personal communication, 1958).

Nyctibora sp., Argentina (Spegazzini, 1917): On antennae.

=Herpomyces panchlorae= Thaxter

Natural hosts.--_Panchlora nivea_, Trinidad (Thaxter, 1931): On antennae.

=Herpomyces panesthiae= Thaxter

Natural host.--_Panesthia lobipennis_, Ceylon (Thaxter, 1915): On antennae.

=Herpomyces paranensis= Thaxter

Natural hosts.--_Blaberus_ sp.? Brazil (Thaxter, 1902, 1908): On antennae.

Blaberus sp., Brazil and Argentina (Spegazzini, 1917): On antennae.

Cockroaches, Trinidad and Argentina (Thaxter, 1931).

=Herpomyces periplanetae= Thaxter

Natural hosts.--_Blaberus_ sp.?, Argentina (Spegazzini, 1917).

Blatta orientalis, U.S.A., Massachusetts (Thaxter, 1902, 1908); Locality? (Spegazzini, 1915); France? (Picard, 1913).

Periplaneta americana, Bermuda and U.S.A., Massachusetts (Thaxter, 1902, 1908); Plains of Biajar, Italian Somaliland, and Argentina (Spegazzini, 1915, 1917).

Periplaneta australasiae, Bermuda (Thaxter, 1902, 1908).

Periplaneta brunnea, Brazil (Thaxter, 1931).

Periplaneta sp., Mexico, West Indies, Panama, Brazil, Africa, South Seas, and China (Thaxter, 1902, 1908).

Cockroaches, Belgium (Collart, 1947).

Additional locality records: Grenada, Trinidad, B.W.I., and Tangier (Thaxter, 1931).

The fungus was found growing on spines, tegmina, integument, and antennae, at times abundantly.

Experimental hosts.--All the following data are from Richards and Smith (1954):

Blatta orientalis: A few plants matured.

Neostylopyga rhombifolia: Some development but no mature plants.

Periplaneta americana: Fungus developed prolifically with a density equal to that on original host.

Periplaneta australasiae: Some development but no mature plants.

Periplaneta brunnea: Fungus developed prolifically with a density equal to that on original host.

=Herpomyces phyllodromiae= Thaxter

Natural host.--"_Phyllodromia_" sp., Abyssinia (Thaxter, 1905, 1908): On antennae.

=Herpomyces platyzosteriae= Thaxter

Natural host.--"_Eurycotis floridana_," Mexico (Thaxter, 1905, 1908): On antennal setae.

Since this cockroach is not found in Mexico (J. A. G. Rehn, personal communication, 1957), _E. floridana_ is undoubtedly not the host for this fungus. W. B. Brown (personal communication, 1957) searched the cockroach collection at the Museum of Comparative Zoology but was unable to find Thaxter's insect for reidentification.

=Herpomyces stylopygae= Spegazzini

Natural hosts.--_Blatta orientalis_, Argentina (Spegazzini, 1917); U.S.A. (Richards and Smith, 1955a).

Experimental hosts.--_Neostylopyga rhombifolia_ (Richards and Smith, 1954): A few plants matured.

Pycnoscelus surinamensis (Richards and Smith, 1954): Some development but no mature plants.

The fungus (fig. 1) is found on antennae (pl. 27, B, C), palpi, cerci, and femurs. Thaxter (1931) believed _H. stylopygae_ to be synonymous with _H. periplanetae_. However, Richards and Smith (1954) concluded that _H. stylopygae_ would not grow on _P. americana_ under their laboratory conditions although _H. periplanetae_ would grow on _B. orientalis_. This indicated a strain or species difference between the two fungi. Gunn and Cosway (1938) reported a species of _Stigmatomyces_ on the antennae of _B. orientalis_; this fungus was probably _H. stylopygae_ (Richards and Smith, 1956).

=Herpomyces supellae= (Thaxter)

Natural host.--_Supella supellectilium_, Trinidad (Thaxter, 1931): On antennal spines.

=Herpomyces tricuspidatus= Thaxter

Natural hosts.--_Blaberus craniifer_, U.S.A., Key West (Richards and Smith, 1955).

Blaberus sp. and _Epilampra?_ sp., Panama (Thaxter, 1902, 1908).

Epilampra sp., Saint Kitts, B.W.I., and Haiti (Thaxter, 1902, 1908).

Leucophaea maderae, Fernando Po (Spegazzini, 1915).

Nauphoeta cinerea, Brazil (Thaxter, 1931).

Cockroaches, China? (Thaxter, 1902); Philippine Islands, Mindanao (Thaxter, 1931).

Experimental hosts.--_Blaberus craniifer_, U.S.A. (Richards and Smith, 1955).

Infections on the antennae. Richards and Smith (1954) were unable to secure experimental infections in _L. maderae_ with _H. tricuspidatus_. Experiments with _N. cinerea_ showed some development but no mature plants although identification of the growing fungus was uncertain because of simultaneous exposure to _H. ectobiae_, _H. stylopygae_, and _H. tricuspidatus_.

=Herpomyces zanzibarinus= Thaxter

Natural hosts.--_Eurycotis manni_, Brazil (Thaxter, 1931): On antennae.

Gyna sp.?, Isle of Nias (Spegazzini, 1915): On antennae.

Cockroach, Zanzibar (Thaxter, 1902): On antennae.

=INCERTAE SEDIS=

According to Dr. R. K. Benjamin (personal communication, 1957) and Dr. E. G. Simmons (personal communication, 1957), the phylogenetic position of the following genus is uncertain.

=Amphoromorpha blattina= Thaxter

Natural hosts.--Cockroaches, Grenada, B.W.I. (Thaxter, 1920): On the axis of the antennae of a dark wingless and a pale winged blattid.

=Amphoromorpha= sp.

Natural host.--Cockroach, Grenada, B.W.I. (Thaxter, 1920): On antennal setae.

IX. HIGHER PLANTS

The significance of many observed associations between cockroaches and the higher plants is still obscure. Undoubtedly many associations are ecological, but lack of

adequate supporting evidence makes this conclusion somewhat tentative. The ecological aspects are covered in Section III (p. 14). Other associations may be accidental (e.g., certain unique observations that have never again been confirmed). In the absence of contrary evidence, most associations are presumed to be benign; exceptions to this conclusion are found among the cockroaches that feed on living plants (p. 162) and those allegedly captured as prey by the carnivorous pitcher plants (_Sarracenia_ and _Nepenthes_). In all the records cited below the cockroaches were stated to have been on, in, or feeding on the plant.

The plants are listed below by family according to the taxonomic arrangement of Lawrence (1951). Botanical nomenclature follows Bailey (1925), Fernald (1950), or Dr. R. A. Howard (personal communications, 1958, 1959). We take full responsibility for referring to appropriate taxa certain plants that were reported by common name only in the cited literature.

Division PTERIDOPHYTA

Family CYATHEACEAE

=Alsophila= sp.

Associate.--_Pycnoscelus surinamensis_, Louisiana (Anonymous, 1893): Feeding on heart of tree fern.

Family POLYPODIACEAE

=Asplenium nidus= Linnaeus

Associate.--_Comptolampra liturata_, Malaya (Karny, 1924): Often found between dry foliage of the beakers of this fern.

Division EMBRYOPHYTA SIPHONOGAMA

Family PINACEAE

=Pinus australis= Michaux

Associates.--_Aglaopteryx gemma_ and _Parcoblatta lata_, Alabama (Hebard, 1917): The former species was common under signs on longleaf pines, and _P. lata_ was occasional.

Parcoblatta divisa, Georgia (Rehn and Hebard, 1916): Under signs.

=Pinus caribaea= Morelet

Associates.--_Eurycotis floridana_, _Latiblattella rehni_, and _Parcoblatta fulvescens_, Florida (Hebard, 1917): Many records under signs on the tree trunks.

=Pinus clausa= Vasey

Associate.--_Latiblattella rehni_, Florida (Hebard, 1917): Under sign on tree.

=Pinus echinata= Mill.

Associates.--_Parcoblatta divisa_, Virginia (Rehn and Hebard, 1916): Under signs on shortleaf pine.

Parcoblatta zebra, Mississippi (Hebard, 1917): Under sign.

=Pinus sylvestris= Linnaeus

Associate.--_Ectobius pallidus_, England (Milton, 1899; Burr, 1899b): On Scotch fir.

=Pinus= spp.

Associates.--_Plectoptera lacerna_ and _Plectoptera vermiculata_, Cuba (Rehn and Hebard, 1927).

Latiblattella rehni, Florida (Rehn and Hebard, 1905): Under signs. Cuba (Rehn and Hebard, 1927).

Family TAXODIACEAE

=Cryptomeria= sp.

Associate.--_Diploptera punctata_, Hawaii (Pemberton and Williams, 1938; Zimmerman, 1948).

Family CUPRESSACEAE

=Cupressus macrocarpa= Hartweg

Associate.--_Diploptera punctata_, Hawaii (Hebard, 1922): "The species is common and injurious in the territory infesting particularly the Monterey cypress trees ... and doing particular damage by gnawing away the bark." Similar injury has been cited by Pemberton (1934), Fullaway and Krauss (1945), and Zimmerman (1948).

=Juniperus= sp.

Associate.--_Phyllodromica tartara nigrescens_, Southern Uzbekistan (Bei-Bienko, 1950): Under bark.

Family PANDANACEAE

=Freycinetia= sp.

Associate.--_Graptoblatta notulata_ and _Kuchinga remota_, Tahiti (Hebard, 1933).

=Pandanus= sp.

Associate.--_Hololeptoblatta_ sp., Seychelles (Scott, 1910, 1912).

Family GRAMINEAE

=Aristida pennata= Trin.

Associate.--_Phyllodromica pygmaea_, U.S.S.R. (Bei-Bienko, 1950): Found in the dense turf.

=Bamboo=

Associate.--_Comptolampra liturata_, Malaya (Karny, 1925).

=Chloris gayana= Kunth

Associate.--_Blattella vaga_, Texas (Riherd, 1953): This field cockroach was rather abundant in clumps of Rhodes grass.

=Panicum purpurascens= Raddi

Synonymy.--_Panicum barbinode_ [Hitchcock, 1936].

Associate.--_Epilampra abdomen-nigrum_, Puerto Rico (Seín, 1923; Wolcott, 1936): Abundant in "malojillo" meadow.

=Saccharum officinarum= Linnaeus

Associates.--_Balta quadricaudata_, _Balta scripta_, _Balta torresiana_, _Balta verticalis_, _Ellipsidion simulans_, and _Megamareta verticalis_, Australia, Queensland (Hebard, 1943): All collected by J. F. Illingworth on sugarcane.

Blattella humbertiana, _Ischnoptera schenklingi_, and _Pycnoscelus surinamensis_, Formosa (Box, 1953).

Cariblatta stenophrys, Puerto Rico (Seín, 1923; Wolcott, 1936): Between the leaves and under the leaf sheaths.

Panchlora nivea, Cuba (Rehn and Hebard, 1927): On the leaves.

Pelmatosilpha coriacea, Puerto Rico (Wolcott, 1936).

Phoraspis spp., Brazil and Guiana (Doumerc in Blanchard, 1837).

Plectoptera dorsalis, _Plectoptora infulata_, and _Plectoptera rhabdota_, Puerto Rico (Wolcott, 1950): Under the leaf sheaths.

Symploce ruficollis, Puerto Rico (Wolcott, 1950): Often found living under the leaf sheaths.

Cockroaches, Philippine Islands (Uichanco _in_ Williams et al., 1931): Between cane leaf sheaths.

=Setaria verticillata= (Linnaeus) Beauv.

Synonymy.--_Chaetochloa verticillata_ (Linnaeus) [Howard, personal communication, 1958].

Associate.--_Diploptera punctata_, Hawaii (Severin, 1911): The cockroach was caught on the barbed awns of this grass.

=Wild oats=

Associate.--_Ischnoptera deropeltiformis_, Missouri (Rau, 1937).

=Zea mays= Linnaeus

Associates.--_Cariblatta stenophrys_, Puerto Rico (Seín, 1923; Wolcott, 1936).

Ellipsidion bicolor, Australia, Queensland (Hebard, 1943).

Lophoblatta arawaka, Trinidad (Princis and Kevan, 1955).

Phoraspis sp., Brazil and Guiana (Doumerc _in_ Blanchard, 1837).

Supella supellectilium, New Caledonia (Cohic, 1956).

Family CYPERACEAE

=Cyperus= sp.

Associate.--_Maretina uahuka_, Marquesas Islands, Uahuka (Hebard, 1933a).

Family PALMAE

=Acrocomia aculeata= (Jacq.) Lodd.

Associate.--_Pycnoscelus surinamensis_, Trinidad (Princis and Kevan, 1955): On "gru-gru" fruits.

=Cocos nucifera= Linnaeus

Associates.--_Aglaopteryx gemma_, Florida (Rehn and Hebard, 1912).

Cariblatta lutea minima, Florida, and _Cariblatta delicatula_, San Domingo (Hebard, 1916a).

Eurycotis floridana, Florida (Rehn and Hebard, 1912; Hebard, 1917).

Periplaneta australasiae, Jamaica (Rehn and Hebard, 1927).

Pycnoscelus surinamensis, Florida (Rehn and Hebard, 1912; Hebard, 1917). Jamaica (Rehn and Hebard, 1927).

=Phoenix dactylifera= Linnaeus

Associate.--_Blattella germanica_, California (Herms, 1926): On date palms.

=Pritchardia= sp.

Associate.--_Periplaneta australasiae_, Nihoa Island (Bryan, 1926). Hawaii (Zimmerman, 1948).

=Roystonea regia= O. F. Cook

Associate.--_Cariblatta punctulata_, San Domingo (Hebard, 1916a).

=Sabal palmetto= Lodd.

Associate.--_Eurycotis floridana_, Florida (Scudder, 1879).

Periplaneta australasiae, Florida (Hebard, 1917).

=Undetermined palms=

Associates.--_Euthlastoblatta abortiva_, Texas (Hebard, 1917).

Hormetica laevigata, Brazil (Hancock, 1926).

Panchlora antillarum, Dominican Republic (Rehn and Hebard, 1927).

Periplaneta americana, Texas (Zimmern _in_ Gould and Deay, 1940).

Family ARACEAE

=Arum= sp.

Associate.--_Latiblattella vitrea_, Mexico (Hebard, 1921b): In flower shaft.

=Caladium= sp.

Associate.--_Plectoptera dorsalis_, Puerto Rico (Rehn and Hebard, 1927).

Family BROMELIACEAE

=Aechmaea porteoides= Britton

Associate.--_Dryadoblatta scotti_, Trinidad (Princis and Kevan, 1955).

=Ananas comosus= Merr.

Associates.--_Pycnoscelus surinamensis_, Hawaii (Illingworth, 1927, 1929): Feeding on roots of pineapple.

Blattella humbertiana, Formosa (Takahashi, 1940): Imago and grown nymphs occasionally lie concealed in the leaves.

=Catopsis fulgens= Griseb.

Associates.--Cockroaches, Costa Rica (Calvert and Calvert, 1917).

=Glomeropitcairnia erectiflora= Mez

Associate.--_Dryadoblatta scotti_, Trinidad (Princis and Kevan, 1955).

=Grevisia= sp.

Associate.--_Notolampra antillarum_, Trinidad (Princis and Kevan, 1955): One male only.

=Tillandsia fasciculata= Swartz

Associate.--_Eurycotis floridana_, Florida (Rehn and Hebard, 1914; Hebard, 1917).

=Tillandsia usneoides= Linnaeus

Associates.--_Parcoblatta_ sp., Louisiana (Rainwater, 1941).

Latiblattella rehni, Florida (Blatchley, 1920): By beating.

Cockroaches, Louisiana (Rosenfeld, 1911, 1912): One mature and 39 immature blattids were collected from 8 of 12 samples of Spanish moss.

=Tillandsia uttriculata= Linnaeus

Associate.--_Epilampra mona_, Mona Island, West Indies (Rehn and Hebard, 1927): The type and one paratypic female of _E. mona_ were collected in this bromeliad.

Eurycotis floridana, Florida (Blatchley, 1920).

=Tillandsia= sp.

Associates.--_Aglaopteryx gemma_, Texas (Hebard, 1917).

Dryadoblatta scotti, Trinidad (Scott, 1912): Found in the leaf bases.

=Undetermined bromeliads=

Associates.--_Aglaopteryx diaphana_, Jamaica (Hebard, 1917; Rehn and Hebard, 1927).

Anaplecta azteca and _Anaplecta_ sp., Costa Rica (Picado, 1913).

Anaplecta mexicana, Costa Rica (Calvert and Calvert, 1917).

Audreia bromeliadarum, Panama (Caudell, 1914).

Audreia jamaicana, Jamaica (Rehn and Hebard, 1927).

Blattella sp., Costa Rica (Picado, 1913).

Buboblatta armata, Panama (Caudell, 1914): "Probably not a typical bromeliadicolous species."

Cariblatta insularis, Jamaica (Hebard, 1916a, 1917; Rehn and Hebard, 1927).

Cariblatta nebulicola, Jamaica (Rehn and Hebard, 1927); One immature male.

Dryadoblatta scotti, Trinidad (Princis and Kevan, 1955).

Epilampra conspersa, Dominica (Scott, 1912).

Epilampra maya, Panama (Hebard, 1920).

Epilampra sodalis, Panama (Caudell, 1914).

Epilampra sp. and _Hormetica laevigata_, Brazil (Hancock, 1926).

Eurycotis biolleyi, Costa Rica (Picado, 1913).

Ischnoptera rufa occidentalis, Mexico (Caudell, 1914).

Latiblattella chichimeca, Costa Rica (Picado, 1913).

Litopeltis biolleyi, Costa Rica (Rehn, 1928).

Litopeltis bispinosa, Panama (Caudell, 1914).

Neoblattella brunneriana, Costa Rica (Calvert and Calvert, 1917).

Neoblattella dryas, _Neoblattella eurydice_, _Neoblattella grossbecki_, and _Neoblattella proserpina_, Jamaica (Rehn and Hebard, 1927).

Neoblattella fratercula, Mexico (Hebard, 1921b).

Neoblattella nahua, Mexico (Caudell, 1914).

Nesomylacris relica, Jamaica (Rehn and Hebard, 1927).

Nyctibora brunnea(?), Panama (Caudell, 1914): According to Hebard (1920) Caudell's specimen was almost certainly incorrectly identified. It may have been _Nyctibora noctivaga_ or a smaller species of the genus. Brazil (Hancock, 1926).

Nyctibora laevigata, Jamaica (Hebard, 1917; Rehn and Hebard, 1927).

Nyctibora lutzi, Puerto Rico (Rehn and Hebard, 1927): "in epiphytes with pencil-like leaves."

Pelmatosilpha rotundata, Panama (Caudell, 1914).

Pseudomops laticornis, Costa Rica (Picado, 1913).

Pycnoscelus surinamensis, Costa Rica (Picado, 1913). Mexico (Caudell, 1914). Jamaica (Rehn and Hebard, 1927).

"_Rhicnoda_" sp., Costa Rica (Picado, 1913). This genus is now recognized as not being in the New World fauna. Probably the specimen was a species of _Epilampra_ or _Hyporhicnoda_ as suggested by Gurney (personal communication, 1959) and confirmed by Rehn (p.c., 1959).

Cockroaches, Costa Rica (Calvert, 1910): Cockroaches were said to be common in bromeliads on the moist Atlantic slope.

Family LILIACEAE

=Yucca elata= Engelman

Associate.--_Latiblattella lucifrons_, Arizona (Ball et al., 1942).

=Easter lilies=

Associate.--_Pycnoscelus surinamensis_, Connecticut (Zappe, 1918).

Family MUSACEAE

=Bananas=

Cockroaches have been captured in bunches of bananas, in bracts of banana flowers, under banana leaves, and burrowing in rotten banana stalks. Although many of the species associated with bananas are indigenous to the banana-growing areas of the American Tropics, most of the specimens cited below were captured elsewhere as adventitious insects that had been imported with the fruit. It is obvious that many of these insects must have been closely associated with bananas on the plantations, where, undoubtedly, the growing plants provided attractive ecological niches. Bunting (1956) deduced, from the presence of healthy cockroaches on bananas allegedly sprayed with copper arsenate, that the insects did not feed on stems or fruit but hid among the bananas and foraged elsewhere; however, certain reports are of cockroaches actually feeding on bananas. Some of the records cited by Hebard (1917) were compiled from earlier reports not all of which we have seen. Numbers in parentheses following certain citations indicate the number of times the association had been observed. Known or suspected adventive material is so indicated.

Aglaopteryx diaphana, Jamaica (Rehn and Hebard, 1927): Found in bracts of banana blossoms. England (Bunting, 1955): Adventive, on bananas from Dominica.

Aglaopteryx vegeta, Finland (Princis, 1947): Adventive, in banana box.

Amazonina emarginata, Trinidad (Princis and Kevan, 1955): In banana bunch.

Archimandrita marmorata, Denmark (Henriksen, 1939): Adventive (2), in bananas from Jamaica(?). As Princis (1947) and Gurney (personal communication, 1959) point out, this is a Central American species, so Jamaica may be an error.

Archimandrita tessellate, Sweden (Princis, 1947): Adventive, from Honduras.

Blaberus atropos(?), Denmark (Henriksen, 1939): Adventive, from Jamaica. Princis (1947) pointed out that this species was more likely to have been _Blaberus craniifer_ or _Blaberus discoidalis_, which are West Indian species, than _B. atropos_ which is a South American species.

Blaberus boliviensis, Ecuador (Princis, 1952): In a shipment of bananas from near Puna.

Blaberus discoidalis, Puerto Rico (Rehn and Hebard, 1927): From banana ripening room. Great Britain (Pearce, 1929): Adventive. England (Bunting, 1955, 1956): Adventive, from Dominica.

Capucinella delicatula, California (Caudell, 1931): Adventive.

Cariblatta delicatula, Cuba (Rehn and Hebard, 1927).

Cariblatta hylaea, Honduras (Rehn, 1945a): Shaken from hanging dead banana leaves.

Cariblatta insularis, Finland (Frey, 1948): Adventive.

Cariblatta landalei, Jamaica (Rehn and Hebard, 1927): All specimens taken from under drying bracts of banana blossoms.

Cariblatta punctipennis and _Chorisoneura barbadensis_, England (Bunting, 1956): Adventive, from Dominica.

Epilampra abdomen-nigrum and _Epilampra_ sp., England (Bunting, 1955): Adventive, from Dominica.

Epilampra maya, Massachusetts (Hebard, 1917): Adventive.

Epilampra mexicana(?), Denmark (Henriksen, 1939): Adventive (2), from Danish West Indies. Princis (1947) suggested that this should be _Epilampra_ sp., because _E. mexicana_ is not a West Indian species.

Eudromiella calcarata and _Eurycotis bananae_, U.S.S.R., Leningrad (Bei-Bienko, 1947): Adventive, from Colombia.

Euphyllodromia angustata, Sweden (Princis, 1947): Adventive.

Eurycotis caraibea, New York (Hebard, 1917): Adventive.

Eurycotis dimidiata, Washington, D. C. (Caudell, 1931): Adventive.

Eurycotis lixa, New York (Rehn, 1930): Adventive, on banana ship from Jamaica.

Graptoblatta notulata, Marquesas Islands, Uahuka (Hebard, 1933a): In banana leaves.

Holocompsa nitidula, Trinidad (Princis and Kevan, 1955): Eating banana pulp.

Hormetica laevigata, Wales (Sandemann, 1934): Adventive, in pile of banana sacks.

Hormetica ventralis, Sweden (Princis, 1947): Adventive, in local warehouse of banana company.

Hormetica spp., Europe and North America (Bei-Bienko, 1950): Adventive, introduced with bananas and other tropical fruits.

Ischnoptera rufa rufa, Puerto Rico (Wolcott, 1950): Brought into houses on bunches of bananas.

Kuchinga remota, Society Islands, Moorea (Hebard, 1933a): In dead banana leaves.

Lamproblatta albipalpus, Panama Canal Zone (Hebard, 1920): Several under decayed banana stem.

Latiblattella sp., Finland (Frey, 1948): Adventive.

Leucophaea maderae, New York (Hebard, 1917): Adventive. Dominica (Rehn and Hebard, 1927): Under banana sheaths. England (Palmer, 1928): Adventive, captured at railroad station after bananas had been unloaded. England (Bunting, 1955): Adventive, from Dominica. Trinidad (Princis and Kevan, 1955): Nymph, eating bananas in cupboard. Puerto Rico (Seín, 1923): Seín stated that bananas are the favorite food of _L. maderae_.

Litopeltis bispinosa, Panama Canal Zone (Hebard, 1920): From rotting banana stalks at bases of leaves.

Litopeltis musarum, Costa Rica (Rehn, 1928): Shaken from dead banana leaves.

Nauclidas nigra, England (Bunting, 1955, 1956): Adventive, from Dominica.

Nauphoeta flexivitta, Denmark (Vestergaard, 1958): Adventive.

Neoblattella carcinus, _Neoblattella celeripes_, and _Neoblattella laodamia_, England (Bunting, 1956): Adventive, from Dominica. Bunting (1955) first reported these as _Neoblattella_ spp. and stated that they were common.

Neoblattella detersa, Jamaica (Rehn and Hebard, 1927): From under the bracts of banana blossoms. Sweden (Princis, 1947): Adventive.

Neoblattella detersa and _Neoblattella tridens_, Finland (Frey, 1948): Adventive.

Neoblattella fratercula, Nebraska (Hebard, 1916b): Adventive.

Neoblattella semota, Jamaica (Rehn and Hebard, 1927): From under drying bracts of banana blossoms.

Neoblattella vatia, Cuba (Rehn and Hebard, 1927).

Neoblattella sp., Finland (Princis, 1947): Adventive, from Jamaica.

Nyctibora azteca, England (Bunting, 1955): Adventive, from Dominica. Bunting reported this species as _Nocticola azteca_. Dr. A. B. Gurney called our attention to

the fact that _Nocticola_ is an Old World genus, presumably combined in error with the New World species _azteca_. The true identity of the specimen was confirmed by Dr. D. Ragge (personal communication, 1958), who examined it at the British Museum (Natural History).

Nyctibora holoserica, Canada (Walker, 1912): Adventive.

Nyctibora laevigata, Canada, Maine, Massachusetts, Pennsylvania (2) (Hebard, 1917): Adventive. Taken from banana boat _Annetta_ at Philadelphia (Rehn and Hebard, 1927). England (Bunting, 1956): Adventive, from Dominica. Sweden, Denmark (Princis, 1947): Adventive.

Nyctibora mexicana(?), Denmark (Henriksen, 1939): Adventive (5), from Jamaica and West Indies. Princis (1947) suggested that these specimens were probably the West Indian _Nyctibora noctivaga_, because _N. mexicana_ is not a West Indian insect.

Nyctibora noctivaga, Canada, Idaho, Illinois, Massachusetts, Nebraska (4), Virginia (Hebard, 1917): Adventive. Nebraska (Hauke, 1949): Adventive (2). Panama Canal Zone (Hebard, 1920): From banana stalks. England (Blair _in_ Turner, 1930): Adventive, from Costa Rica. Washington (Hatch, 1938): Adventive. Sweden (Princis, 1947): Adventive (2). Finland (Princis, 1947): Adventive, from Jamaica.

Nyctibora obscura, Trinidad (Princis and Kevan, 1955): In banana bunch.

Nyctibora sericea, Canada (Stevenson, 1905; Walker, 1912): Adventive; Hebard (1917) synonymized Walker's specimen under _N. laevigata_. Isle of Wight (Meade-Waldo, 1910): Adventive, from Jamaica. England (Tulloch, 1939): Adventive, in banana crates from Brazil.

Nyctibora sp., England (Welch, 1935): Adventive, in railway truck that had carried bananas. England (Tulloch, 1939): Adventive, from Brazil.

Oxyhaloa deusta, U.S.S.R., Leningrad (Bei-Bienko, 1947): Adventive, from Colombia.

Panchlora antillarum, England (Bunting, 1955): Adventive, from Dominica.

Panchlora exoleta, Scotland (Distant, 1902): Adventive. Great Britain (Shaw, 1902): Adventive. England (Coney, 1918): Adventive. Sweden, Norway (Princis, 1947): Adventive, Norwegian specimen from Brazil. Germany (Zacher, 1917): Adventive, from Jamaica.

Panchlora nivea, Colorado, Nebraska, New Jersey, New York (2), Utah (Hebard, 1917): Adventive. Nebraska (Hauke, 1949): Adventive. Washington (Hatch, 1938): Adventive. Massachusetts (Roth and Willis, 1958): Adventive. England (Bunting, 1955): Adventive, from Dominica. U.S.S.R. (Bei-Bienko, 1947): Adventive, from Colombia. Sweden (13), Norway (3), Finland (3) (Princis, 1947): Adventive, mostly females; origin (where known) Jamaica.

Panchlora fraterna(?) and _Panchlora peruana_(?), Denmark (Henriksen, 1939): Adventive; origin (where known) Danish West Indies and Jamaica; Princis (1947)

suggested that both species were probably _Panchlora nivea_.

Panchlora sagax, Puerto Rico (Wolcott, 1936).

Panchlora virescens, Canada (Walker, 1912): Adventive; this was probably _P. nivea_ as we now know it (Gurney, personal communication, 1959).

Panchlora sp., Canada (Walker, 1912): Adventive. England (Tulloch, 1939): Adventive, from Brazil.

Pelmatosilpha coriacea, Puerto Rico (Wolcott, 1936).

Pelmatosilpha marginalis and _Pelmatosilpha purpurascens_, England (Bunting, 1955, 1956): Adventive, from Dominica; both species common.

Pelmatosilpha vagabunda, New Zealand (Princis, 1954): Adventive, probably from South America.

Periplaneta americana, Belgium (Schepdael, 1931): Adventive, on bananas from the American Tropics.

Periplaneta americana and _Periplaneta brunnea_, England (Bunting, 1955, 1956): Adventive, from Dominica.

Periplaneta americana and _Periplaneta australasiae_, England (Watson, 1907): Adventive; they ate ripening bananas in the tropical plant house of the Royal Botanic Gardens, Kew, where they hid in "the sheathing bases of palm, banana and pandanus leaves." Sweden (Princis, 1947): Adventive.

Periplaneta australasiae, Canada (Walker, 1912): Adventive. Denmark (Henriksen, 1939): Adventive (9); origin mostly Jamaica. England (Tulloch, 1939): Adventive, from Brazil. England (Bunting, 1955, 1956): Adventive, from Dominica; common.

Platyzosteria bifida, Nebraska (Hebard, 1917): Adventive.

Plectoptera dorsalis, Puerto Rico (Rehn and Hebard, 1927): Captured by beating banana plants.

Pycnoscelus surinamensis, Canada (Walker, 1912; Hebard, 1917): Adventive: Marquesas Islands, Nukuhiva (Hebard, 1933a): In banana leaves. England (Goodliffe, 1958): Adventive, doing considerable damage to banana plants growing in a conservatory.

Sibylloblatta panesthoides, Massachusetts (Rehn, 1937a): Adventive, from Jamaica.

Family ZINGIBERACEAE

=Renealmia= sp.

Associate.--_Cariblatta orestera_, Jamaica (Rehn and Hebard, 1927): The male was taken in a head of wild ginger.

Family CANNACEAE

=Canna= sp.

Associate.--_Periplaneta americana_, Hawaii (Zimmerman, 1948).

Family ORCHIDACEAE

=Cattleya= sp.

Associates.--_Periplaneta americana_, U.S.A. (Rau, 1940a).

Periplaneta australasiae, England (Lucas, 1918).

=Vanda= sp.

Associates.--_Periplaneta americana_, U.S.A. (Rau, 1940a).

Periplaneta australasiae, England (Lucas, 1918).

=Undetermined orchids=

Associates.--_Blaberus discoidalis_, _Blatta orientalis_, _Periplaneta americana_, Hawaii (Swezey, 1945).

Blatta orientalis, _americana_, _cinerea_, _maderae_, unidentified cockroaches, England, in bulb from Ecuador (Westwood, 1876).

Graptoblatta notulata, Hawaii (Swezey, 1945): On orchid from India.

Homalopteryx laminata and _Hormetica apolinari_, New York (Hebard, 1912c): In orchids shipped from Colombia.

Pelmatosilpha coriacea, Puerto Rico (Wolcott, 1936).

Periplaneta americana, Germany (Tashenberg, 1884).

Periplaneta australasiae, England (Wainwright, 1898). Pennsylvania (Skinner, 1905). Massachusetts (Morse, 1920).

Pycnoscelus surinamensis, England (Westwood, 1869). Germany (Zacher, 1920). Massachusetts (Morse, 1920). Hawaii (Swezey, 1945).

Family CASUARINACEAE

=Casuarina= sp.

Associate.--_Diploptera punctata_, Hawaii (Zimmerman, 1948).

Family SALICACEAE

=Populus euphratica= Oliv.

Synonymy.--_Populus diversifolia_ Schrenk. [Howard, personal communication, 1959].

Associate.--_Ectobius semenovi_, Kazakhstan (Bei-Bienko, 1950).

=Populus= sp.

Associate.--_Ectobius lapponicus_, U.S.S.R. (Stark _in_ Bei-Bienko, 1950): On aspen.

=Salix= sp.

Associate.--_Ectobius semenovi_, Kazakhstan (Bei-Bienko, 1950): On willow.

Family MYRICACEAE

=Myrica cerifera= Linnaeus

Associate.--_Chorisoneura texensis_, Florida (Rehn and Hebard, 1916): On bayberry. Florida (Blatchley, 1920): Beaten from foliage.

Family FAGACEAE

=Quercus alba= Linnaeus

Associate.--_Parcoblatta pensylvanica_, Virginia (Rehn and Hebard, 1916): Under signs on white oaks.

=Quercus rubra= Linnaeus

Associates.--_Parcoblatta divisa_, Virginia, and _Parcoblatta pensylvanica_, North Carolina (Rehn and Hebard, 1916): Under signs on red oak.

Parcoblatta lata, North Carolina (Hebard, 1917): Under sign.

=Quercus virginiana= Mill.

Associate.--_Eurycotis floridana_, Georgia (Rehn and Hebard, 1916): Under dead bark on live-oak tree. Georgia (Hebard, 1917): In cavity in tree.

=Quercus= spp.

Associates.--_Aglaopteryx gemma_, Alabama, Georgia, Florida, Louisiana, Texas (Hebard, 1917): Under signs on oaks.

Blatta orientalis, England (Donisthorpe, 1918): Under bark.

Cariblatta lutea lutea, Mississippi (Hebard, 1916a): By beating low oaks on hills.

Chorisoneura texensis, Mississippi (Hebard, 1917). Florida (Blatchley, 1920): By beating.

Ectobius pallidus, England (Milton, 1899; Burr, 1899b). Massachusetts (Flint, 1951): Under loose lichens and bark.

Parcoblatta divisa, Georgia, Louisiana, and _Parcoblatta pensylvanica_, Georgia (Hebard, 1917): Under signs.

Periplaneta australasiae, Florida (Rehn and Hebard, 1905): Ten specimens taken from under a tin sign.

Periplaneta brunnea, Georgia (Rehn and Hebard, 1916): Under signs.

Phyllodromica megerlei, U.S.S.R. (Bei-Bienko, 1950): By shaking oak branches.

Plectoptera lacerna, Cuba (Rehn and Hebard, 1927).

Family MORACEAE

=Cecropia= sp.

Associate.--_Cariblatta hylaea_, Honduras (Rehn, 1945a).

Family CHENOPODIACEAE

=Beta maritima= Linnaeus

Associate.--_Ectobius panzeri_, England (Lucas, 1920a).

=Beta vulgaris= var. =cicla= Linnaeus

Associate.--_Ectobius pallidus_, Massachusetts (Flint, 1951): Many specimens collected in the bases of Swiss chard plants.

Family LAURACEAE

=Nectandra coriacea= (Sw.) Griseb.

Synonymy.--_Ocotea catesbyana_ Sarg. [Howard, personal communication, 1959].

Associate.--_Chorisoneura texensis_, Florida (Rehn and Hebard, 1912).

Family SARRACENIACEAE

Only a few records have been found of cockroaches being trapped in the pitchers of carnivorous plants of this and the following family. The insects drown in the fluid within the pitcher where they are apparently digested by proteinases secreted by the plant (Meyer and Anderson, 1939; Lloyd, 1942).

=Sarracenia flava= Linnaeus

Natural prey--_Cariblatta lutea lutea_, _Ischnoptera deropeltiformis_, _Parcoblatta lata_, and nymphs of _Parcoblatta_ sp., North Carolina (Wray and Brimley, 1943): Most of the cockroaches seemed to have been trapped accidentally with the possible exception of _C. lutea lutea_, 11 of which were found in _Sarracenia_ pitchers.

=Sarracenia purpurea= Linnaeus

Natural prey.--_Cariblatta lutea lutea_, North Carolina (Wray and Brimley, 1943).

=Sarracenia minor= Walter

Synonymy.--_Sarracenia variolaris_ Michx. [Howard, personal communication, 1958].

Natural and experimental prey.--_Periplaneta australasiae_, Florida (Treat, 1876): After the insect imbibed some of the fluid in the pitcher it became docile; others became highly active and rushed wildly about before becoming quiescent. See also Treat _in_ Scudder (1877).

Cockroaches, U.S.A. (Riley, 1875).

Family NEPENTHACEAE

=Nepenthes ampularia= Jack

Natural prey.--Cockroaches, Singapore (Dover, 1928).

=Nepenthes gracilis= Korth.

Natural prey.--Cockroaches, Singapore (Dover, 1928).

=Nepenthes= sp.

Natural prey.--Cockroach, Old World Tropics? (Hooker, 1874): The insect was apparently attracted into the pitcher, where it drowned, by a piece of cartilage placed there by Hooker.

Family CUNONIACEAE

=Weinmannia= sp.

Associates.--_Aneurina viridis_, Marquesas Islands, Nukuhiva and Fatuhiva (Hebard, 1933a).

Maretina uahuka, Marquesas Islands, Uahuka (Hebard, 1933a).

Family HAMAMELIDACEAE

=Liquidambar styraciflua= Linnaeus

Associate.--_Parcoblatta divisa_, Georgia (Rehn and Hebard, 1916): Under sign on sweet gum.

Parcoblatta zebra, Louisiana (Hebard, 1917): In decay cavity.

Family ROSACEAE

=Crataegus= sp.?

=Associates.=--_Plectoptera dorsalis_, _Plectoptera infulata_, _Plectoptera rhabdota_, Puerto Rico (Wolcott, 1950): In the dry flower clusters of "espino rubial."

=Rosa= sp.

Associate.--_Pycnoscelus surinamensis_, Connecticut (Zappe, 1918); Rhode Island and Pennsylvania (Caudell, 1925); Pennsylvania (Doucette and Smith, 1926): Feeding on canes in greenhouses.

=Rubus= spp.?

Associate.--_Hololampra chavesi_, Azores (Chopard, 1932): This species is exclusively dendricolous and was found only by beating the bushes on which it abounds. It was very common in hedges, particularly on brambles (ronces).

Family LEGUMINOSAE

=Acacia farnesiana= Willd.

Associate.--_Diploptera punctata_, Hawaii (Bridwell and Swezey, 1915; Zimmerman, 1948): Feeding on pods.

=Acacia= sp.

Associates.--_Ellipsidion australe_, Australia, New South Wales (Hebard, 1943).

Methana curvigera, Australia, Queensland (Pope, 1953a).

=Ceratonia siliqua= Linnaeus

Associate.--_Diploptera punctata_, Hawaii (Pemberton and Williams, 1938): Damaging algarroba.

=Erythrina glauca= Willd.

Associates.--_Aglaopteryx absimilis_, Puerto Rico (Wolcott, 1950): In abandoned cocoon.

Aglaopteryx facies, Puerto Rico (Wolcott, 1936): In empty cocoons.

=Inga laurina= Willd.

Associate.--_Aglaopteryx facies_, Puerto Rico (Wolcott, 1936): On trunk.

=Inga vera= Willd.

Associates.--_Aglaopteryx facies_, Puerto Rico (Wolcott, 1936): In larval tents.

Cariblatta stenophrys, Puerto Rico (Wolcott, 1936): On leaves.

Plectoptera dorsalis, _Plectoptera infulata_, and _Plectoptera rhabdota_, Puerto Rico (Wolcott, 1950): In "butterfly nests" in leaves.

=Mesquite=

Associate.--_Nyctibora stygia_, Haiti (Rehn and Hebard, 1927).

=Samanea saman= Merr.

Associate.--_Aglaopteryx absimilis_, Puerto Rico (Wolcott, 1950).

=Tamarindus indica= Linnaeus

Associate.--_Hemiblabera brunneri_, Puerto Rico (Rehn and Hebard, 1927).

Family GERANIACEAE

=Geraniums=

Associate.--_Diploptera punctata_, Hawaii (Zimmerman, 1948).

Family ZYGOPHYLLACEAE

=Tribulus= sp.

Associates.--_Periplaneta americana_ and _Pycnoscelus surinamensis_, Johnston Island (Bryan, 1926). Zimmerman (1948) lists _Tribulus_ as a host plant for these cockroaches.

Family RUTACEAE

=Citrus aurantifolia= Swingle

Associates.--_Plectoptera dominicae_ and _Plectoptera perscita_, Dominica (Rehn and Hebard, 1927): Beaten from moss-covered lime trees.

=Citrus maxima= Merr.

Associates.--_Plectoptera dorsalis_, _Plectoptera infulata_, _Plectoptera rhabdota_, Puerto Rico (Wolcott, 1950).

Plectoptera rhabdota, Puerto Rico (Rehn and Hebard, 1927).

=Citrus sinensis= Osbeck

Associate.--_Diploptera punctata_, Hawaii (Bridwell and Swezey, 1915; Zimmerman, 1948): Feeding on oranges on tree.

=Citrus= sp.

Associates.--_Diploptera punctata_, Hawaii (Zimmerman, 1948).

Riatia [=_Lissoblatta_] _fulgida_, Panama, Rio Trinidad (Hebard, 1920).

Plectoptera porcellana, Puerto Rico (Sein, 1923).

=Zanthoxylum caribaeum= Lam.

Associates.--_Plectoptera dorsalis_, _Plectoptera infulata_, _Plectoptera rhabdota_, Puerto Rico (Wolcott, 1950): In the dry flower clusters.

Family BURSERACEAE

=Bursera simaruba= (L.) Sarg.

Associate.--_Chorisoneura texensis_, Florida (Rehn and Hebard, 1912; Hebard, 1917): Beaten from the lower branches of gumbo limbo.

Family EUPHORBIACEAE

=Poinsettia= sp.

Associate.--_Pycnoscelus surinamensis_, Connecticut (Zappe, 1918): Ate bark of greenhouse plants.

Family ANACARDIACEAE

=Mangifera indica= Linnaeus

Associate.--_Diploptera punctata_, Hawaii (Bridwell and Swezey, 1915; Zimmerman, 1948): Feeding on mangoes on the tree.

=Spondias mombin= Linnaeus

Associates.--_Plectoptera dorsalis_, _Plectoptera infulata_, _Plectoptera rhabdota_, Puerto Rico (Wolcott, 1950): Living on leaves of "jobo."

=Spondias purpurea= Linnaeus

Associate.--_Eurycotis biolleyi_, Costa Rica (Rehn, 1918): In the crown of dry jocoto.

Family AQUIFOLIACEAE

=Ilex cassine= Linnaeus

Associate.--_Plectoptera poeyi_, Florida (Rehn and Hebard, 1912, 1914; Hebard, 1917).

=Ilex coriacea= (Pursh) Chapm.

Synonymy.--_Ilex lucida_ [Fernald, 1950].

Associate.--_Cariblatta lutea lutea_, Florida (Hebard, 1916a).

Family SAPINDACEAE

=Exothea paniculata= (Juss.) Radlk.

Associate.--_Aglaopteryx gemma_, Florida (Hebard, 1917).

Family MALVACEAE

=Gossypium= spp.

Associates.--_Graptoblatta notulata_, Marquesas Islands, Tahuata (Hebard, 1933a).

Periplaneta australasiae, St. Kitts, B.W.I. (Ballou, 1916).

Periplaneta fuliginosa and _Plectoptera poeyi_, Florida (Rainwater, 1941).

Plectoptera dorsalis, _Plectoptera infulata_, _Plectoptera rhabdota_, Puerto Rico (Wolcott, 1950).

=Hibiscus rosa-sinensis= Linnaeus

Associate.--_Riatia orientis_, Trinidad (Princis and Kevan, 1955).

=Hibiscus= sp.

Associates.--_Amazonina emarginata_, _Cariblatta antiguensis_, _Eurycotis kevani_, and _Rhytidometopum dissimile_, Trinidad (Princis and Kevan, 1955).

=Sida= sp.

Associate.--_Periplaneta australasiae_, Nihoa Island (Bryan, 1926). Hawaii (Zimmerman, 1948).

Family STERCULIACEAE

=Theobroma cacao= Linnaeus

Associate.--_Ceratinoptera picta_, Trinidad (Princis and Kevan, 1955).

Family BIXACEAE

=Bixa= sp.

Associate.--_Notolampra antillarum_, Trinidad (Princis and Kevan, 1955): Nymphs in dry fruits on "annato" tree.

Family FLACOURTIACEAE

=Xylosma suaveolens= Forst.

Associate.--_Graptoblatta notulata_, Marquesas Islands, Uahuka (Hebard, 1933a).

Family PASSIFLORACEAE

=Passiflora= sp.

Associate.--_Aristiger_ [=_Plumiger_] _histrio_, Malaya (Karny, 1924).

Family CARICACEAE

=Carica papaya= Linnaeus

Associate.--_Diploptera punctata_, Hawaii (Bridwell and Swezey, 1915; Zimmerman, 1948): Feeding on papaya fruit on tree.

Family RHIZOPHORACEAE

=Rhizophora mangle= Linnaeus

Associate.--_Aglaopteryx gemma_, Florida (Hebard, 1917).

Family COMBRETACEAE

=Conocarpus erectus= Linnaeus

Associate.--_Plectoptera poeyi_, Florida (Rehn and Hebard, 1914): Running on leaves.

Family MYRTACEAE

=Eucalyptus= sp.

Associate.--_Ellipsidion australe_, Australia, New South Wales (Hebard, 1943).

=Eugenia aromatica= Baill.

Synonymy.--_Syzygium aromaticum_ [Bailey, 1925].

Associate.--_Plectoptera dorsalis_, Puerto Rico (Wolcott, 1936): On flowers of "pomarrosa."

=Metrosideros collina= Gray

Associates.--_Aneurina viridis_, Marquesas Islands: Nukuhiva, Fatuhiva, and Tahuata (Hebard, 1933a)

Aneurina tahuata, Marquesas Islands, Tahuata (Hebard, 1933a).

Graptoblatta notulata, Marquesas Islands, Nukuhiva (Hebard, 1933a).

=Psidium guajava= Linnaeus

Associate.--_Plectoptera rhabdota_, Puerto Rico (Rehn and Hebard, 1927).

Family ONAGRACEAE

=Jussiaea natans= Humb. and Bonpl.

Associate.--_Epilampra abdomen-nigrum_, Panama (Crowell, 1946): In an aquarium the cockroach fed on leaves of this aquatic plant which had been collected

in the lagoon where the insect was captured.

Family ERICACEAE

=Calluna vulgaris= Salisb.

Associates.--_Ectobius lapponicus_, England (Lucas, 1925): "Nymphs of varying size were beaten out of heather ... on 9 February and later."

Ectobius panzeri, England (Lucas, 1927): "numerous imagines of both sexes were swept from heather."

=Vaccinium meridionale= Sw.

Associates.--_Chorisoneura formosella_, _Neoblattella dryas_, _Neoblattella proserpina_, Jamaica (Rehn and Hebard, 1927).

Family SAPOTACEAE

=Sideroxylon foetidissimum= Jacq.

Associate.--_Pelmatosilpha coriacea_, Puerto Rico (Wolcott, 1941): Under bark.

Family APOCYNACEAE

=Vinca minor= Linnaeus

Associate.--_Ectobius pallidus_, Massachusetts (Willis, unpublished observation, 1958).

Family CONVOLVULACEAE

=Ipomoea tiliasea= Choisy

Associate.--_Plectoptera dorsalis_, Puerto Rico (Rehn and Hebard, 1927).

Family BORAGINACEAE

=Cordia dentata= Poiret

Synonymy.--_Calyptracordia alba_ [Howard, personal communication, 1958].

Associates.--_Cariblatta antiguensis_, _Ischnoptera rufa rufa_, _Supella supellectilium_, _Symploce ruficollis_ and _Symploce hospes_, St. Croix, Virgin Islands (Beatty, 1944): On fruits of _C. dentata_ except _S. supellectilium_ which was found at night on the flowers.

Family VERBENACEAE

=Citharexylum villosum= Jacq.

Associate.--_Chorisoneura texensis_, Florida (Rehn and Hebard, 1912).

Family SOLANACEAE

=Nicotiana= sp.

Associate.--_Pycnoscelus surinamensis_, Sumatra (Roeser, 1940).

=Solanum tuberosum= Linnaeus

Associate.--_Pycnoscelus surinamensis_, Haiti (Hoffman, 1927): Feeding on tubers in field.

Family GESNERIACEAE

=Cyrtandra= sp.

Associate.--_Aneurina viridis_, Marquesas Islands, Nukuhiva (Hebard, 1933a).

Family RUBIACEAE

=Canthium barbatum= (Forst.) Seem.

Associate.--_Graptoblatta notulata_, Marquesas Islands, Uahuka (Hebard, 1933a).

=Cinchona pubescens= Vahl.

Associate.--_Periplaneta americana_, Puerto Rico (Plank and Winters, 1949): In greenhouse.

=Coffea= sp.

Associate.--_Plectoptera porcellana_, Puerto Rico (Seín, 1923).

Family COMPOSITAE

=Goldenrod=

Associate.--_Eurycotis floridana_, Florida (Hebard, 1917): "Climbing about on top of goldenrod at night."

=Helianthus= sp.

Associate.--_Pseudomops septentrionalis_, Texas (Hebard, 1917).

=Scorzonera acanthoclada= Franch.

Associate.--_Phyllodromica tartara nigrescens_, Southern Uzbekistan (Bei-Bienko, 1950): On the flowers.

DAMAGE TO PLANTS BY COCKROACHES

Cockroaches characteristically feed on dead plant and animal material. Damage to living plants occurs principally in the Tropics or under subtropical conditions in greenhouses in temperate regions. Among the depredations attributed to cockroaches in text books, damage to plants is seldom emphasized. This is surprising in view of the many records cited below.

Capt. William Bligh (1792), while collecting breadfruit trees in Tahiti to take to the West Indies, wrote in his log during January 1789: "This morning, I ordered all the chests to be taken on shore, and the inside of the ship to be washed with boiling water, to kill the cockroaches. We were constantly obliged to be at great pains to keep the ship clear of vermin, on account of the plants."

Westwood (1869) stated that _Pycnoscelus surinamensis_ was very destructive in orchid houses feeding on buds and young shoots. Later Westwood (1876) exhibited the bulb of an orchid from Ecuador which contained six species of cockroaches: _Blatta orientalis_, [_Periplaneta_?] _americana_, [_Nauphoeta_?] _cinerea_, [_Leucophaea_?] _maderae_, and two others unknown to him. Fullaway (1938) stated that cockroaches damage root tips, buds, and flowers of orchids. _Periplaneta americana_ has been said to eat the root tips and blossoms of orchids (Taschenberg, 1884) and to devour the open flower petals of _Cattleya_ orchids as well as the aerial roots and flower spikes of _Vanda_ orchids (Rau, 1940a). Wainwright (1898) stated that _Periplaneta australasiae_ had been observed in an orchid house in Perthshire where over a period of three years it had caused a good deal of damage. Skinner (1905) reported that _P. australasiae_ in greenhouses in Pennsylvania showed no preference for any one plant but ate both plants and flowers of orchids, roses, and carnations. Lucas (1918) received specimens of _P. australasiae_ which had played havoc with orchids especially _Cattleya_ and _Vanda_. Morse (1920) reported that both _P. australasiae_ and _Pycnoscelus surinamensis_ were obnoxious in a conservatory in Massachusetts where they gnawed the tips of the aerial roots of orchids. Swezey (1945) in Hawaii stated that the following cockroaches have been reported as occasional minor pests on orchids: _Blatta orientalis_, _Blaberus discoidalis_, _P. americana_, and _P. surinamensis_; he further stated that _Graptoblatta notulata_ had been intercepted at Honolulu on orchids from India.

Watson (1907) stated that _Blatta orientalis_, _Periplaneta americana_, and _Periplaneta australasiae_ were injurious in the tropical plant houses at Kew: "at night they come out and run or fly about among the plants, devouring flowers and leaves like rabbits. Such plants as _Eucharis_, _Crinum_ and _Alpinia_, when in flower, have little chance in the palm house, where the cockroaches are most abundant; they also find out the ripening bananas and soon devour them." Raffill (1910) stated that in plant houses in England _B. orientalis_, _P. americana_, and _P. australasiae_ commonly, and _Nauphoeta cinerea_, _Nauphoeta flexivitta_, and _Pycnoscelus surinamensis_ more rarely, are extremely destructive to plants. Flowers having a strong perfume, such as orchids, _Eucharis_, _Crinum_, and _Hedychium_, were often attacked while other flowers nearby were left uninjured.

Plank and Winters (1949) reported that in Puerto Rico the species of Orthoptera most injurious under greenhouse conditions was _Periplaneta americana_. Large nymphs destroyed 25 to 30 percent of freshly planted seed of _Cinchona pubescens_. In Hawaii the host plants of _P. americana_ are blossoms of _Canna_ and _Tribulus_, and the host plants of _Periplaneta australasiae_ are _Pritchardia_ and _Sida_ (Zimmerman, 1948). On St. Kitts, B.W.I., young cotton plants were severely attacked by _P. australasiae_; this caused loss of the stand on a considerable area and necessitated replanting (Ballou, 1916). _P. australasiae_ was reported damaging the _Polystichum aristatum_ Presl [=_Lastrea aristata variegata_] in a greenhouse (Thilow and Riley, 1891). Laing (1946; British Museum [Natural History], 1951) stated that _P. australasiae_ abounds in greenhouses and forcing pits where it may do considerable damage to the plants. _Periplaneta fuliginosa_ is also troublesome in greenhouses because of its tendency to feed on seedlings and succulent plants

(Dodge and Rickett, 1943).

Ectobius lapponicus has been observed feeding in galleries in the thick skin of young aspen in 25 percent of the trees examined (Stark _in_ Bei-Bienko, 1950). The aquatic cockroach _Epilampra abdomen nigrum_ fed on the leaves of _Jussiaea natans_ in an aquarium (Crowell, 1946). _Ischnoptera deropeltiformis_ has been taken while it was feeding on a fleshy fungus (_Agaricus_ sp.) in dense woods in Indiana (Blatchley, 1920).

Diploptera punctata, the cypress roach or beetle roach, has been found in Hawaii feeding on ripening mangoes and papayas, oranges on the tree, and the outer covering of the pods of _Acacia farnesiana_ (Bridwell and Swezey, 1915). Pemberton (1934) stated that _D. punctata_ "disfigures our cypress trees by eating the bark from the young branches, often giving them a dead appearance over much of their leaf area." Fullaway and Krauss (1945) added, "This injury [to cypress] is so severe that sometimes areas of leaves die and turn brown. The Japanese cedar, ironwood, citrus and algaroba (kiawe) trees are attacked in a similar manner." Similar injury to cypress was described by Hebard (1922). In addition to girdling _Cupressus_, _D. punctata_ injures _Cryptomeria_ in the same fashion and also attacks algaroba, lime, and other plants (Pemberton and Williams, 1938). Zimmerman (1948) cited the following host plants for _D. punctata_ in Hawaii: "_Cupressus macrocarpa_, _Casuarina_, _Cryptomeria_, _Citrus_, geraniums, _Acacia farnesiana_ pods, mango fruits, orange fruits, papaya fruits."

In the reports of damage to plants by cockroaches, _Pycnoscelus surinamensis_ has been implicated most often. This species is undoubtedly one of the economically most important cockroaches, being the vector of the chicken eyeworm as well as feeding on plants. In addition to the few reports of damage caused by this species that have already been mentioned, _P. surinamensis_ has been reported to be very destructive in New Orleans to palms and ferns, attacking large alsophilas avidly, eating out the hearts (Anonymous, 1893). Zappe (1918) in Connecticut reported damage in a greenhouse to roses valued, at that time, at several hundred dollars; _P. surinamensis_ had girdled the rose bushes, done much damage to Easter lilies, and in another greenhouse had eaten the bark from the stems of poinsettias. In Germany this species bit off the tips of the aerial roots and ate the petals of orchids (Zacher, 1920). Lucas (1923) reported damage to cucumber plants in a greenhouse in Surrey. Damage by _P. surinamensis_ to the stems of rose bushes has been reported in Rhode Island and Pennsylvania; the canes were attacked both under and above ground (Caudell, 1925). Doucette and Smith (1926) reported a heavy infestation of _P. surinamensis_ in a range of greenhouses in Philadelphia: "The roaches were present literally by the millions.... Although the roaches had been observed in cabinets and trash barrels for several months, it was not until the manager had occasion to go through the house one evening that he discovered that roaches were the cause of the troubles previously attributed to soil condition, watering, fungus, and other agencies.... About 30,000 to 35,000 rose plants from a total of 200,000 in the three more heavily infested houses were so badly injured by the gnawing off of the bark, young buds, and shoots of the main stems, that they were not in condition to be kept

in the beds for another season."

In Haiti _Pycnoscelus surinamensis_ damaged the tubers of growing potatoes (Hoffman, 1927). Illingworth (1927, 1929) reported that in Hawaii _P. surinamensis_ was a minor pest of pineapples, feeding on the roots. This species was very plentiful in a propagating pit in England where it did much damage to various seeds and seedlings (Lucas, 1930). Roeser (1940) summarized some of the above-mentioned damage caused by _P. surinamensis_ and added damage to chrysanthemums in Hawaii and tobacco in Sumatra where this cockroach destroyed 300,000 plants in a few days. Roeser was of the opinion that living plants were eaten only as a substitute when the earth became poor in food material. Zimmerman (1948) listed as host plants of _P. surinamensis_ in Hawaii: "blossoms of _Tribulus_; reported feeding at roots of pineapples, and unconfirmed reports of damage to underground parts of some other plants." Goodliffe (1958) reported damage by this species to banana plants in a conservatory in northern England. Cohic (1956) implied that in New Caledonia "Racines de légumes" were attacked by _P. surinamensis_ and that _Zea mays_ Linnaeus was attacked by _Supella supellectilium_. Wolcott (1924a) reported that _P. surinamensis_ damaged transplanted tobacco plants in Puerto Rico by eating the interior of the stalks. Tobacco planters in Cuba consider _P. surinamensis_ injurious to the roots of tobacco plants (Bruner and Scaramuzza, 1936); this belief was confirmed in the laboratory, where adults and nymphs destroyed the roots and stems of tobacco plants two inches high and ate into the edges of the leaves. Dammerman (1929) reported that in Malaya this species often appeared in large numbers in gardens where it gnawed at the underground parts of vegetables and ornamental plants. Lever (1947) listed it as a pest on the leaves of pineapple.

Blattella vaga may occasionally damage seedlings in the laboratory (Flock, 1941a), but no damage has been reported in the field (Ball et al., 1942). Heer (1864) reported receiving a shipment of cycads from Cuba with all stages of _Periplaneta americana_ living in holes in the branches, apparently subsisting on the starchy tissues. Goldenberg (1877) stated that sago trees provide cockroaches with their favorite nourishment. Scudder (1879) found _Eurycotis floridana_ living in the tops of the cabbage palmetto, on which he presumed it fed. _Parcoblatta americana_ has been observed feeding on an apple 6 feet above ground (Fulton, 1930).

X. PROTOZOA ASSOCIATED WITH COCKROACHES

The classification of the Protozoa follows that of Kudo (1954). The use of the asterisk (*) is explained in footnote 3, page 4.

=Phylum PROTOZOA=

Class MASTIGOPHORA

Order EUGLENOIDINA

Family EUGLENIDAE

=Euglena= sp.

Experimental host.--_Periplaneta americana_, U.S.A. (Hegner, 1929): When fed to the insects in concentrated culture, _Euglena_ could withstand conditions in the crop up to 5 hours and were passed into the stomach in a viable state up to 6 hours. However, the majority were killed in the crop within 2 hours and very few reached the stomach alive.

Order PROTOMONADINA

Family OIKOMONADIDAE

=Oikomonas blattarum Tejera=

Natural host.--Cockroach, Venezuela (Tejera, 1926).

=Oikomonas= sp.

Natural host.--_Blatta orientalis_, U.S.S.R. (Yakimov and Miller, 1922): _Oikomonas_ sp. and _Monas_ sp. were found in the intestines of 83 percent of 124 cockroaches.

Cockroach, Venezuela (Tejera, 1926).

Family TRYPANOSOMATIDAE

=Leptomonas blaberae= Tejera

Natural host.--_Blaberus_ sp., Venezuela (Tejera, 1926).

=Leptomonas= sp.

Natural hosts.--_Parcoblatta lata_, _Parcoblatta pensylvanica_, _Parcoblatta virginica_, U.S.A., Ohio (Semans, 1939, 1941): Hind intestine. Of 70 specimens examined, 86 percent harbored _Leptomonas_ sp.

* =Herpetomonas periplanetae= Laveran and Franchini

Natural host.--_Blatta orientalis_, Italy, France (Laveran and Franchini, 1920, 1920a).

Family MONADIDAE

=Monas= sp.

Natural host.--_Blatta orientalis_, U.S.S.R. (Yakimov and Miller, 1922): _Monas_ sp. and _Oikomonas_ sp. were found in the intestines of 83 percent of 124 cockroaches examined.

Cockroach, Venezuela (Tejera, 1926).

Family BODONIDAE

=Bodo blattae=

Natural host.--_Blatta orientalis_, England (Lankester, 1865).

=Bodo= sp.

Natural host.--_Blattella germanica_ and/or _Periplaneta americana_, South Africa (Porter, 1930).

=Retortamonas blattae= (Bishop)

Synonymy.--_Embadomonas blattae_ Bishop [Wenrich, 1932].

Natural host.--_Blatta orientalis_, England (Bishop, 1931): Hind intestine. The organism occurred in about 40 percent of the cockroaches examined. L. G. Feo (_in_ Wenrich, 1932) successfully cultured this protozoan (fig. 2, F).

=Retortamonas= sp.?

Natural host.--_Leucophaea maderae_, Philippine Islands (Hegner and Chu, 1930).

[Illustration: FIG. 2.--Representative Protozoa associated with cockroaches. A, _Monocercomonoides melolonthae_, × 3094 (after Grassé). B, _Coelosporidium periplanetae_, × 1310 (after Sprague); trophozoite with spores and chromatoid bodies. C, _Endamoeba blattae_, × 273 (after Kudo); trophozoite. D, _Lophomonas striata_, × 330 (after Kudo). E, _Lophomonas blattarum_, × 660 (after Kudo). F, _Retortamonas blattae_, × 3094 (after Wenrich). G, _Nyctotherus ovalis_, × 175 (after Kudo). H, _Gregarina rhyparobiae_, c. × 52: mature trophozoite attached to intestinal wall of _Leucophaea maderae_. (Redrawn from J. M. Watson [1945].) I, _Diplocystis schneideri_, c. × 14.4 (after Kunstler). J, _Gregarina blattarum_, c. × 57 (after Kudo). K, _Protomagalhaesia serpentula_, × 36 (after Pinto). L, _Gamocystis tenax_, magnification not known (after Schneider). (All figures except H redrawn from Kudo [1954] after sources indicated.)]

Order POLYMASTIGINA

Family CHILOMASTIGIDAE

=Chilomastix mesnili= (Wenyon)

Experimental vectors.--_Blatta orientalis_ and _Periplaneta americana_, South Africa (Porter, 1918): The cockroaches were fed human excrement that contained cysts of _C. mesnili_. The cysts passed unharmed through the insects' digestive tract. Rats became infected with this protozoan on eating food that had been contaminated with feces from these cockroaches.

Family POLYMASTIGIDAE

=Eutrichomastix= sp.

Synonymy.--_Trichomastix_ [Kudo, 1954].

Natural host.--_Blattella germanica_ and/or _Periplaneta americana_, South Africa (Porter, 1930).

=Monocercomonoides globus= Cleveland et al.

Natural host.--_Cryptocercus punctulatus_, U.S.A., Appalachian and Pacific coast areas (Cleveland et al., 1934): Organism occurs in practically all hosts.

=Monocercomonoides melolonthae= (Grassi)

(Fig. 2, A)

Natural host.--_Platyzosteria novae seelandiae_, New Zealand (Laird, 1956): Found in the intestinal tracts of the adult cockroaches, and of other species of insects.

=Monocercomonoides orthopterorum= (Parisi)

Synonymy.--_Trichomonas_ (_Trichomastix_) _orthopterorum_ Parisi; _Monocercomonas orthopterorum_ [Bělăr, 1916]; _Trichomastic orthopterum?_ [Zasukhin, 1930]; _Monocercomonoides orthopterorum_ [Travis, 1932; Cleveland et al., 1934]; _Retortamonas orthopterorum_ [Semans, 1943].

Natural hosts.--_Blatta orientalis_, Italy (Parisi, 1910); U.S.S.R. (Zasukhin, 1930).

Ectobius lapponicus, Italy (Parisi, 1910).

Periplaneta americana, Philippine Islands (Hegner and Chu, 1930).

"Küchenschaben," Austria (Bělăr, 1916).

The protozoan is found in the hind gut. Zasukhin (1930) found the organism in 85 percent of over 3,000 _B. orientalis_. Parisi (1910) found the flagellate present in very large numbers.

=Monocercomonoides panesthiae= Kidder

Natural host.--_Panesthia angustipennis_, Philippine Islands (Kidder, 1937): In hind gut.

=Tetratrichomastix blattidarum= Young

Natural hosts.--_Blatta orientalis_, _Blattella germanica_, _Periplaneta americana_, U.S.A. (Young, 1935): The organism when present occurs in large numbers in the posterior part of the intestine near the anus. The protozoan was successfully cultivated in a hemoglobin-saline medium.

Family OXYMONADIDAE

=Oxymonas doroaxostylus= (Cleveland et al.)

Synonymy.--_Saccinobaculus doroaxostylus_ Cleveland et al. [Cleveland, 1950].

Natural host.--_Cryptocercus punctulatus_, U.S.A., Appalachian and Pacific coast areas (Cleveland et al., 1934).

=Oxymonas nana= Cleveland

Synonymy.--_Saccinobaculus minor_ Cleveland et al. [Cleveland, 1950].

Natural host.--_Cryptocercus punctulatus_, U.S.A., Appalachian and Pacific coast areas (Cleveland et al., 1934).

Family DINENYMPHIDAE

=Saccinobaculus ambloaxostylus= Cleveland et al.

Natural host.--_Cryptocercus punctulatus_, U.S.A., Appalachian and Pacific coast areas (Cleveland et al., 1934).

=Saccinobaculus lata= Cleveland

Natural host.--_Cryptocercus punctulatus_, U.S.A. (Cleveland, 1950b): There are at least two other species of _Saccinobaculus_ in _C. punctulatus_ that have not been described.

=Notila proteus= Cleveland

Natural host.--_Cryptocercus punctulatus_, U.S.A., Appalachian and Pacific coast areas (Cleveland, 1950b).

Family TRICHOMONADIDAE

* =Trichomonas hominis= (Davaine)

Experimental vectors.--_Blatta orientalis_, South Africa (Porter, 1918); Italy (Mariani and Besta, 1936).

Periplaneta americana, South Africa (Porter, 1918); U.S.A. (Hegner, 1928).

=Trichomonas= sp.

Natural vector.--Cockroach, Venezuela (Tejera, 1926): Organism found in digestive tract of the cockroach.

Family HEXAMITIDAE

=Hexamita cryptocerci= Cleveland et al.

Natural hosts.--_Cryptocercus punctulatus_, U.S.A. (Cleveland et al., 1934).

Panesthia angustipennis, Philippine Islands (Kidder, 1937).

=Hexamita periplanetae= (Bělăr)

Synonymy.--_Octomitus periplanetae_ Bělăr [Kudo, 1954].

Natural hosts.--_Blatta orientalis_, U.S.S.R. (Zasukhin, 1930): Organism is found in the hind gut. Eighty-five percent of over 3,000 _B. orientalis_ contained this organism.

Periplaneta americana, Philippine Islands (Hegner and Chu, 1930).

"Küchenschaben," Austria (Bělăr, 1916).

=Hexamita= sp.?

Natural host.--_Leucophaea maderae_, Philippine Islands (Hegner and Chu, 1930): The flagellates were present in large numbers.

* =Giardia intestinalis= (Lambl)

Experimental vectors.--_Blatta orientalis_, South Africa (Porter, 1918).

Blattella germanica, Brazil (Pessôa and Corrêa, 1927).

Eurycotis floridana, U.S.A. (Young, 1937).

Leucophaea maderae, Brazil (Pessôa and Corrêa, 1927).

Periplaneta americana, South Africa (Porter, 1918); Gold Coast Colony (Macfie, 1922); Brazil (Pessôa and Corrêa, 1927); U.S.A. (Young, 1937).

Periplaneta brunnea, U.S.A. (Young, 1937).

Cockroaches, Venezuela (Tejera, 1926); Argentina (Bacigalupo, in Tejera, 1926).

* =Giardia= sp.

Natural vectors.--Cockroaches, Venezuela (Tejera, 1926).

Order HYPERMASTIGINA

Family HOLOMASTIGOTIDAE

=Leptospironympha eupora= Cleveland et al.

Natural host.--_Cryptocercus punctulatus_, U.S.A., Appalachian area (Cleveland et al., 1934).

=Leptospironympha rudis= Cleveland et al.

Natural host.--_Cryptocercus punctulatus_, U.S.A., Appalachian area (Cleveland et al., 1934).

=Leptospironympha wachula= Cleveland et al.

Natural host.--_Cryptocercus punctulatus_, U.S.A., Appalachian area (Cleveland et al., 1934).

=Macrospironympha xylopletha= Cleveland et al.

Natural host.--_Cryptocercus punctulatus_, U.S.A., Appalachian area (Cleveland et al., 1934).

Family LOPHOMONADIDAE

=Lophomonas blattarum= Stein

Natural hosts.--_Blatta orientalis_, Czechoslovakia (Stein, 1860); Germany (Bütschli, 1878; Schubotz, 1905; Chen, 1933); U.S.A. (Leidy, 1879a; Kudo, 1922, 1925, 1926, 1926b; McAdow, 1931); Europe (Janicki, 1908); U.S.S.R. (Yakimov and Miller, 1922; Zasukhin, 1930); Poland (Lorenc, 1939).

Blattella germanica, U.S.A., Ohio (McAdow, 1931).

Blattella germanica and/or _Periplaneta americana_, Egypt (DeCoursey and Otto, 1956, 1957).

Periplaneta americana, England (Schuster, 1898); Europe (Janicki, 1910); U.S.A. (Kudo, 1926b; McAdow, 1931; Hatcher, 1939; Armer, 1944); Philippine Islands (Hegner and Chu, 1930).

Periplaneta sp., Goa (Mello and Lima Ribeiro, 1924, 1925).

"Küchenschaben," Austria (Bělař, 1916).

The protozoan (fig. 2, E) is found in the host's colon, particularly anterior portion; encysted stages of organism are found throughout hind gut. Of 1,400 _B. orientalis_ studied, 32 percent harbored this organism (Kudo, 1925, 1926). Yakimov and Miller (1922) found 7 percent of 124 _B. orientalis_ infested. Zasukhin (1930) found 10 percent of over 3,000 _B. orientalis_ infested. The flagellate does not harm the host and is never present in the host tissue; it should be considered a commensal (Kudo, 1926).

=Lophomonas striata= Bütschli

Synonymy.--_Lophomonas sulcata_ Schuster is most probably identical with _L. striata_ (Kudo, 1926b).

Natural hosts.--_Blatta orientalis_, Germany (Bütschli, 1878; Schubotz, 1905); Europe (Janicki, 1908, 1910); U.S.A. (Kudo, 1922, 1926, 1926b; McAdow, 1931); U.S.S.R. (Yakimov and Miller, 1922; Zasukhin, 1930); Poland (Lorenc, 1939).

Blattella germanica, U.S.A., Ohio (McAdow, 1931).

Blattella germanica and/or _Periplaneta americana_, South Africa (Porter, 1930).

Periplaneta americana, Indochina (Weill, 1929); Philippine Islands (Hegner and Chu, 1930); U.S.A. (Kudo, 1926b; McAdow, 1931; Armer, 1944).

Cockroach, Venezuela (Tejera, 1926); England or U.S.A.? (Lucas, 1928).

"Küchenschaben," Austria (Bělař, 1916).

Found in the host's colon, particularly the anterior portion. _L. striata_ (fig. 2, D) was found in 29 percent of 1,400 _B. orientalis_ and in 2 of 30 _P. americana_ (Kudo, 1926, 1926b). Yakimov and Miller (1922) found the organism in 9.6 percent of 124 specimens of _B. orientalis_. Zasukhin (1930) found 8.6 percent of over 3,000 _B. orientalis_ infested.

Grassé (1926, 1926a) identified corrugations on the surface of _L. striata_ as a bacterial parasite which he named _Fusiformis lophomonadis_.

=Prolophomonas tocopola= Cleveland et al.

Natural host.--_Cryptocercus punctulatus_, California, Oregon, Virginia, West Virginia (Cleveland et al., 1934): Not abundant.

Family HOPLONYMPHIDAE

=Barbulanympha estaboga= Cleveland et al.

Natural host.--_Cryptocercus punctulatus_, U.S.A., Appalachian and Pacific coast areas (Cleveland et al., 1934).

Barbulanympha coahoma (Cleveland et al., 1934) represents the diploid form of _B. estaboga_ (Cleveland, 1953).

=Barbulanympha laurabuda= Cleveland et al.

Natural host.--_Cryptocercus punctulatus_, U.S.A., Appalachian and Pacific coast areas (Cleveland et al., 1934): This species, _B. ufalula_, and _Rhynchonympha tarda_ occur in all parts of the colon, especially in the enlarged, flexed part near the ileum.

=Barbulanympha ufalula= Cleveland et al.

Natural host.--_Cryptocercus punctulatus_, U.S.A., Appalachian and Pacific coast areas (Cleveland et al., 1934).

=Barbulanympha wenyoni= Cleveland

Natural host.--_Cryptocercus punctulatus_, U.S.A., Pacific coast area (Cleveland, 1953).

=Rhynchonympha tarda= Cleveland et al.

(Fig. 3, D)

Natural host.--_Cryptocercus punctulatus_, U.S.A., Pacific coast area (Cleveland et al., 1934): Fairly abundant in every specimen examined from Pacific coast.

=Urinympha talea= Cleveland et al.

(Fig. 3, C)

Natural host.--_Cryptocercus punctulatus_, U.S.A., Appalachian and Pacific coast areas (Cleveland et al., 1934): Present in fairly great numbers in every cockroach examined.

Family STAUROJOENINIDAE

=Idionympha perissa= Cleveland et al.

Natural host.--_Cryptocercus punctulatus_, U.S.A., Appalachian area (Cleveland et al., 1934): Present in only a few specimens.

Family TRICHONYMPHIDAE

=Trichonympha acuta= Cleveland et al.

Natural host.--_Cryptocercus punctulatus_, U.S.A., Appalachian and Pacific coast areas (Cleveland et al., 1934).

=Trichonympha algoa= Cleveland et al.

(Fig. 3, E)

Natural host.--_Cryptocercus punctulatus_, U.S.A., Appalachian and Pacific coast areas (Cleveland et al., 1934): Fairly abundant and present in most specimens.

[Illustration: FIG. 3.--Protozoa from the gut of the wood-feeding cockroach _Cryptocercus punctulatus_. A, _Eucomonympha imla_, female above, male below, c. × 375. (From Cleveland [1950c].) B, _Barbulanympha_ sp. (From Cleveland [1953].) C, _Urinympha talea_, c. × 712. (From Cleveland [1951a].) D, _Rhynchonympha tarda_, c. × 450. (From Cleveland [1952].) E, _Trichonympha okolona_ or _T. algoa_, c. × 390. (From Cleveland [1949].) (All drawings reproduced through the courtesy of Dr. L. R. Cleveland.)]

=Trichonympha chula= Cleveland et al.

Natural host.--_Cryptocercus punctulatus_, U.S.A., Appalachian and Pacific coast areas (Cleveland et al., 1934).

=Trichonympha grandis= Cleveland et al.

Natural host.--_Cryptocercus punctulatus_, U.S.A., Pacific coast areas (Cleveland et al., 1934): Fairly abundant in all specimens from Pacific area.

=Trichonympha lata= Cleveland et al.

Natural host.--_Cryptocercus punctulatus_, U.S.A., Appalachian and Pacific coast areas (Cleveland et al., 1934).

=Trichonympha okolona= Cleveland et al.

(Fig. 3, E)

Natural host.--_Cryptocercus punctulatus_, U.S.A., Appalachian and Pacific coast areas (Cleveland et al., 1934): Found in only a few specimens, never abundant.

=Trichonympha parva= Cleveland et al.

Natural host.--_Cryptocercus punctulatus_, U.S.A., Appalachian and Pacific coast areas (Cleveland et al., 1934): This organism is smaller than any known species of _Trichonympha_; it is more resistant to warm weather than the other hypermastigotes.

Family EUCOMONYMPHIDAE

=Eucomonympha imla= Cleveland et al.

Natural host.--_Cryptocercus punctulatus_, U.S.A., Appalachian and Pacific coast areas (Cleveland et al., 1934): Organism (fig. 3, A) sometimes becomes attached to the intestinal wall; attached individuals were seen in 2 to 3 percent of the cockroaches examined.

=Unidentified flagellate=

Natural host.--_Pycnoscelus surinamensis_, Hawaii (Schwabe, 1950): A small flagellate was found in the digestive tract and malpighian tubules.

Class SARCODINA

Order MYCETOZOA

INCERTAE SEDIS

=Peltomyces periplanetae= (Léger)

Synonymy.--_Peltomyces blattellae._ Sprague (1940a) synonymizes _Peltomyces periplanetae_, with _Coelosporidium periplanetae_.

Natural hosts.--_Blatta orientalis_, France (Debaisieux, 1927).

Blattella germanica, France (Léger, 1909; Debaisieux, 1927).

The organism inhabits the malpighian tubules of cockroaches. Léger and Debaisieux concluded that their organism was a mycetozoan, but they may have erred in synonymizing _Plistophora periplanetae_ with the organism they studied. Debaisieux found intracellular stages of _Peltomyces periplanetae_ that have not been found in _Plistophora periplanetae_ or _Coelosporidium periplanetae_.

Order AMOEBINA

Family AMOEBIDAE

=Hartmannella blattae= Ivanić

Natural host.--_Blatta orientalis_, Yugoslavia (Ivanić, 1937): Found in the hind gut.

Family ENDAMOEBIDAE

In the following classification we have accepted the conclusions of Kirby (1945), Kudo (1954), and others that species of _Endamoeba_ are generically different from species of _Entamoeba_ and that the latter genus is not a homonym of _Endamoeba._

=Dobellina= sp.

Natural vectors.--_Blattella germanica_ and/or _Periplaneta americana_, Egypt (DeCoursey and Otto, 1956, 1957): Thirty out of 261 cockroaches examined contained this protozoan.

=Endamoeba blattae= (Bütschli)

Synonymy.--_Amoeba blattae_, _Entamoeba blattae_, _Entamoeba blattarum_.

Natural hosts.--_Blatta orientalis_, Germany (Bütschli, 1878; Schubotz, 1905; Chen, 1933); U.S.A. (Leidy, 1879a, 1880; Kudo, 1922, 1925a, 1926a; Kirby, 1927; McAdow, 1931; Meglitsch, 1938, 1940); France (Mercier, 1907a, 1908, 1909, 1910); Europe? (Janicki, 1908, 1909); U.S.S.R. (Yakimov and Miller, 1922; Zasukhin, 1929, 1930); England (Thomson and Lucas, 1926; Lucas, 1927, 1927a, 1928); Yugoslavia (Ivanić, 1926a).

Blattella germanica and/or _Periplaneta americana_, South Africa (Porter, 1930); Egypt (DeCoursey and Otto, 1956, 1957): Seven out of 217 cockroaches examined harbored the protozoan.

Periplaneta americana, Philippine Islands (Hegner and Chu, 1930); U.S.A. (Morris, 1936; Armer, 1944); Gold Coast Colony (Macfie, 1922).

Periplaneta australasiae, U.S.A. (Morris, 1936).

Cockroaches, Paraguay? (Elmassian, 1909); Austria (Bělăr, 1916); U.S.A. (Morris, 1935, 1936; Balch, 1932); Venezuela (Tejera, 1926).

The habitat of _E. blattae_ (fig. 2, C) is the hind intestine and rectum of the cockroach. The incidence of infection varies: Kudo (1925a) found in 1,255 oriental cockroaches infections in 5 percent in March and 50 percent in the summer; Schubotz (1905) found 5 to 20 percent of the examined cockroaches to be infested; Yakimov and Miller (1922) found 4 percent of 124 oriental cockroaches infested;

Zasukhin (1930) found up to 50 percent of over 3,000 _B. orientalis_ infested; Meglitsch (1938, 1940) found almost 100 percent infection in _B. orientalis_ kept in a crowded culture for several weeks. Chen (1933) developed two synthetic media in which _E. blattae_ could be grown for 45 to 50 days.

Mercier (1907a) observed a fungus, _Nucleophaga_ sp., hyperparasitic in the nucleus of _Endamoeba blattae_.

=Endamoeba javanica= Kidder

Natural hosts.--_Panesthia angustipennis_, Philippine Islands, and _Panesthia spadica_, Japan (Kidder, 1937): Occurred in 50 percent of _P. angustipennis_ examined and in one of four _P. spadica_. The endoplasm of this amoeba contains large amounts of wood and cellulose fibers.

=Endamoeba philippinensis= Kidder

Natural host.--_Panesthia angustipennis_, Philippine Islands (Kidder, 1937): Occurred in about 10 percent of the _Panesthia_ examined. The food vacuoles contained bacteria, no wood.

=Entamoeba coli= (Grassi)

Synonymy.--_Endamoeba coli_, _Amoeba coli_ [Kirby, 1945].

Natural vectors.--_Blaberus atropos_, Venezuela (Tejera, 1926): In a lot of 60 cockroaches captured in latrines, two were found that carried apparently live cysts similar to cysts of _E. coli_.

Blattella germanica or _Periplaneta americana_, Egypt (DeCoursey and Otto, 1956, 1957): One out of 44 cockroaches collected in a village harbored _E. coli_.

Experimental vector.--_Periplaneta americana_, Gold Coast Colony (Macfie, 1922): In nine experiments cysts of _E. coli_ were fed to the cockroaches. In seven of the experiments cysts of _E. coli_ were found in the feces. Cysts were observed in the feces for only one to three days, and eventually disappeared completely. The cysts appeared to be unharmed. No amoebae were found.

* =Entamoeba histolytica= Schaudinn

Natural vectors.--_Blatta orientalis_, _Blattella germanica_, _Periplaneta americana_, _Periplaneta australasiae_, and/or _Supella supellectilium_, Peru (Schneider and Shields, 1947).

Blattella germanica and/or _Periplaneta americana_, Egypt (DeCoursey and Otto, 1956, 1957).

Cockroaches, Venezuela (Tejera, 1926).

Experimental vectors.--_Blatta orientalis_, Italian Somaliland (Mariani and Besta, 1936).

Periplaneta americana, Gold Coast Colony (Macfie, 1922); U.S.A. (Frye and Meleney, 1936).

Cockroaches, Venezuela (Tejera, 1926).

=Entamoeba pitheci= Prowazek?

Experimental vector.--_Periplaneta americana_, Formosa (Morischita and Tsuchimochi, 1926): Eleven of 15 cockroaches fed feces of a monkey [_Macaca cyclopis_ (Swinhoe)] containing cysts of the amoeba voided live cysts in their own feces.

=Entamoeba thomsoni= Lucas

Synonymy.--_Endamoeba thomsoni_ [Kudo, personal communication, 1957].

Natural hosts.--_Blatta orientalis_, England (Lucas, 1927a, 1928); U.S.A. (Taliaferro, 1928; McAdow, 1931); U.S.S.R. (Zasukhin, 1930); Germany (Chen, 1933).

Blattella germanica, U.S.A. (McAdow, 1931).

Periplaneta americana, England (Lucas, 1927a); U.S.A. (Smith and Barret, 1928; McAdow, 1931); Philippine Islands (Hegner and Chu, 1930).

The organism is found in the hind intestine and rectum of the cockroach. Smith and Barret (1928) developed a synthetic medium in which cultures of _E. thomsoni_ were carried through successive transfers for 24 months.

=Entamoeba= sp.

Natural vector.--_Periplaneta americana_, Gold Coast Colony (Macfie, 1922): Under the heading "_Entamoeba histolytica_ and _E. coli_" Macfie (p. 445) stated, "The cockroaches used in these experiments had previously been carefully examined for amoebic infections a precaution which was doubly necessary, because some of these insects at Accra had been found naturally infected."

Experimental vectors.--_Periplaneta americana_, Gold Coast Colony (Macfie, 1922): _Entamoeba_, resembling _E. coli_, from feces of the monkey [_Erythrocebus patas patas_ (Schreber)] were fed to cockroaches, and on the second to fourth days thereafter apparently healthy cysts were recovered in the cockroach feces.

=Endolimax blattae= Lucas

Natural hosts.--_Blatta orientalis_, England (Lucas, 1927, 1927a); U.S.S.R. (Zasukhin, 1930); Germany (Chen, 1933).

Periplaneta americana, England (Lucas, 1927, 1927a); Indochina (Weill, 1929); U.S.A. (Armer, 1944).

Periplaneta australasiae, U.S.A. (Steinhaus, 1946).

Organism is found in the hind gut of the cockroach. Zasukhin (1930) found 3-percent infestation in over 3,000 _B. orientalis_ examined.

=Endolimax nana= (Wenyon and O'Connor)?

Synonymy.--_Entamoeba nana._

Natural host.--_Blaberus atropos_, Venezuela (Tejera, 1926): A small amoeba greatly resembling _E. nana_ was found in the intestinal contents of the cockroach. In inoculations this amoeba was not pathogenic.

=Endolimax= sp.

Natural hosts.--_Blatta orientalis_, U.S.S.R. (Zasukhin, 1930): This organism was found in the hind gut of 0.3 percent of over 3,000 cockroaches examined.

Blattella germanica and/or _Periplaneta americana_, Egypt (DeCoursey and Otto, 1956, 1957): Seventy-four out of 261 cockroaches examined harbored this protozoan.

=Iodamoeba= sp.

Natural vectors.--_Blattella garmanica_ and/or _Periplaneta americana_, Egypt (DeCoursey and Otto, 1956, 1957): Fifty-nine of 261 cockroaches examined contained this protozoan. _Iodamoeba_ sp. was common in human feces in villages in which the cockroaches were collected.

=Undetermined species of Amoeba=

Natural host.--_Panesthia angustipennis_, Philippine Islands. (Kidder, 1937): Found in only one specimen.

Class SPOROZOA

Order GREGARINIDA

Family DIPLOCYSTIDAE

=Diplocystis schneideri= Kunstler

(Fig. 2, I)

Natural hosts.--_Blatta orientalis_, England (Woodcock, 1904; Jameson, 1920).

=Periplaneta americana=, France (Kunstler, 1884, 1887); England (Jameson, 1920); Germany (Foerster, 1939).

In body cavity of host. Cysts containing spores are ingested during cannibalistic feeding on infected cockroaches. Sporozoites penetrate the gut wall which later ruptures, freeing the gregarines into the coelom. There is no apparent pathogenic effect. Jameson (1920) found 81 percent of _P. americana_ infested with _D. schneideri_.

=Diplocystis= sp.

Natural host.--_Periplaneta americana_, U.S.A. (Hertig, 1921): Heavy infections in body cavity.

Cockroach, India (Ray and Dasgupta, 1955): A large number of cockroaches, both adults and nymphs, collected in Calcutta were all infected.

=Diplocystis= sp.?

Natural host.--_Blaberus craniifer_, U.S.A. (Nutting, 1953): From 1 to 12 or more paired trophozoites or cysts may be found in the hemocoele and occasionally in the

thorax.

Family STENOPHORIDAE

=Stenophora= sp.

Natural host.--_Blatta orientalis_, India (Bal and Rai, 1955): Organism found in the midgut of the cockroach.

Family GREGARINIDAE

=Gregarina blattarum= von Siebold

Synonymy.--_Gregarina blattae orientalis_; _Clepsidrina blattarum_.

Natural hosts.--_Blatta orientalis_, Germany (Siebold, 1837, 1839; Stein, 1848; Bütschli, 1881; Wolters, 1891; Marshall, 1892; Wellmer, 1910, 1911; Foerster, 1938; Schubotz, 1905); U.S.A. (Leidy, 1853a; Crawley, 1903; Watson, 1917; Kudo, 1922; McAdow, 1931; Sprague, 1940, 1941); England (Lankester, 1863); France (Schneider, 1875; Cuénot, 1901; Laveran and Franchini, 1920a); Brazil (Magalhães, 1900; Pinto, 1919); U.S.S.R. (Zasukhin, 1929, 1930).

Blattella germanica, U.S.A. (Crawley, 1903); South Africa (Fantham, 1929; Porter, 1930: these appear to be the same record).

Blattella germanica and/or _Periplaneta americana_, Egypt (DeCoursey and Otto, 1956, 1957).

Periplaneta americana, Brazil (Magalhães, 1900); U.S.A. (Crawley, 1903; 1907; McAdow, 1931); South Africa (Fantham, 1929; Porter, 1930: these appear to be the same record); Gold Coast Colony (Macfie, 1922).

Parcoblatta pensylvanica, U.S.A., Michigan (Ellis, 1913a).

Cockroaches, Germany (Schiffmann, 1919: probably used the oriental cockroach); Venezuela (Tejera, 1926).

Organism usually found in the intestinal tract of cockroaches where it is attached to the gut cells. Cysts are passed in the feces. Occasionally, _G. blattarum_ (fig. 2, J) is found in the body cavity (Leidy, 1853a; Hall, 1907). Though this is considered to be one of the commonest of the Sporozoa encountered in cockroaches, DeCoursey and Otto (1956) found only 10 of 217 _P. americana_ and _B. germanica_, collected in restaurants in Egypt, infested with this species. Watson (1917) found a dozen or more in one specimen of _Blatta orientalis_. Zasukhin (1929, 1930) found 2.6 percent of 3,000 oriental cockroaches infected with this parasite.

=Gregarina fastidiosa= Harrison

Natural host.--_Aptera fusca_, South Africa (Harrison, 1955): All mature females were heavily infected; in all specimens there were over 100 parasites in the gut. All nymphs were infected, the earlier instars more lightly than the later instars. Gregarines were found in all parts of the gut except the crop and gizzard.

=Gregarina gibbsi= Harrison

Natural host.--_Temnopteryx phalerata_, South Africa (Harrison, 1955): Although the cockroaches were found in groups, only 32 percent were infected and only 10 percent heavily. The gregarines were found in the anterior mesenteron but none in the hepatic caeca. All cysts were found in the hind gut or rectum.

=Gregarina illinensis= M. E. Watson

Natural host.--_Parcoblatta pensylvanica_, U.S.A., Illinois (Watson, 1915, 1916): The intestine of one cockroach was found to contain 25 of these gregarines.

=Gregarina impetuosa= Harrison

Natural host.--_Melanosilpha capensis_, South Africa (Harrison, 1955): All specimens of this gregarine were found in the anterior mesenteron of the host.

=Gregarina légeri= Pinto

Natural host.--_Periplaneta americana_, Brazil (Pinto, 1918, 1918a, 1919): Intestinal canal.

=Gregarina neo-brasiliensis= Al. Cunha

Natural host.--_Periplaneta americana_, Brazil (R. de Almeida Cunha _in_ Pinto, 1919; Cunha, 1919).

=Gregarina ohioensis= Semans

Natural host.--_Parcoblatta virginica_, U.S.A., Ohio (Semans, 1939): The protozoan was present in large numbers in the insect's midgut.

=Gregarina panchlorae= Frenzel

Natural host.--_Panchlora exoleta_, Argentina (Frenzel, 1892): Midgut.

=Gregarina parcoblattae= Semans

Natural hosts.--_Parcoblatta pensylvanica_ and _Parcoblatta uhleriana_, U.S.A., Ohio (Semans, 1939): Midgut.

=Gregarina rhyparobiae= J. M. Watson

Natural host.--_Leucophaea maderae_, Uganda (Watson, 1945): Midgut. Trophozoites could be seen in sections attached to cells of the intestinal wall (fig. 2, H).

=Gregarina sandoni= Harrison

Natural host.--_Melanosilpha capensis_, South Africa (Harrison, 1955): This gregarine was found in the anterior and middle parts of the mesenteron and in the hepatic caeca.

=Gregarina thomasi= Semans

Natural host.--_Parcoblatta pensylvanica_, U.S.A., Ohio (Semans, 1939): Enteric caeca and midgut.

=Protomagalhaesia serpentula= (de Magalhães)

Synonymy.--_Gregarina serpentula_ [Pinto, 1918a, 1919; Semans, 1943].

Natural host.--_Periplaneta americana_, Brazil (Magalhães, 1900): In the coelom and alimentary canal. The host of this parasite (fig. 2, K) was incorrectly cited as _Blatta orientalis_ by Watson (1916).

=Gamocystis tenax= Schneider

(Fig. 2, L)

Natural hosts.--_Ectobius lapponicus_, France (Schneider, 1875); Germany (Wellmer, 1910, 1911; Foerster, 1938).

Ectobius pallidus, Germany (Foerster, 1938): In intestine.

Family ACTINOCEPHALIDAE

=Pileocephalus blaberae= (Frenzel)

Synonymy.--_Gregarina blaberae_ [Watson, 1916].

Natural hosts.--_Blaptica dubia_ and related forms, Argentina (Frenzel, 1892): In midgut.

=Unidentified Gregarinida=

Natural hosts.--_Blaberus craniifer_, U.S.A. (Roth and Willis, unpublished data, 1953): Possibly _Diplocystis_ sp. (pl. 28, A, B).

Cryptocercus punctulatus, U.S.A. (Cleveland et al., 1934).

Leucophaea maderae, Philippine Islands (Hegner and Chu, 1930): In intestines of host. U.S.A. (Roth and Willis, unpublished data, 1958): Cysts in feces (pl. 28, C).

Gromphadorhina portentosa, U.S.A., in laboratory colony (Roth and Willis, unpublished data, 1958): In intestine of adult female.

Pycnoscelus surinamensis, Hawaii (Schwabe, 1950): A cephaline gregarine was found in the cockroach's digestive tract; it was also claimed to be present in new-born nymphs.

Order COCCIDIA

Family ADELEIDAE

=Adelina cryptocerci= Yarwood

Natural host.--_Cryptocercus punctulatus_, U.S.A. (Yarwood, 1937): This intracellular parasite was found in the fat body in light infestations. In heavy infections the coccidia were found in the head, antennae, mouthparts, muscles, legs, salivary glands, nerve cord, as well as fat body. Infection in freshly collected specimens was about 3 percent; when large numbers of cockroaches were kept together in culture, the rate of infection increased because the insects ate their dead companions.

Cleveland et al. (1934) mentioned a coccidium which was sometimes generally distributed through the body (head, legs, antennae, etc.) of _C. punctulatus_; this

parasite was probably the species described by Yarwood.

Order HAPLOSPORIDIA

=Haplosporidium periplanetae= Georgévitch

Natural host.--_Blatta orientalis_, Yugoslavia (Georgévitch, 1953): This organism was described from the malpighian tubules of the cockroach where it apparently occurred in a mixed infection with the microsporidian _Plistophora periplanetae_. See synonymy under _Plistophora periplanetae_.

=Coelosporidium periplanetae= (Lutz and Splendore)

Synonymy.--_Nosema periplanetae_, _Coelosporidium blattellae_, _Bertramia blatellae_ [after Semans, 1943]. Some of the observations cited under _Plistophora periplanetae_ may pertain to _C. periplanetae_ (see Sprague, 1940). See also _Haplosporidium periplanetae_.

Natural hosts.--_Blatta orientalis_, U.S.S.R. (Epshtein, 1911); U.S.A. (Kudo, 1922; Sprague, 1940); Yugoslavia (Ivanić, 1926).

Blattella germanica, U.S.A. (Crawley, 1905); Germany (Wellmer, 1910, 1911).

Periplaneta americana, Brazil (Lutz and Splendore, 1903).

This organism (fig. 2, B) passes its life cycle living free in the lumina of the malpighian tubules of cockroaches. The elongate trophozoite is firmly attached to the wall of the tubule as are clusters of immature spores. Mature spores are freed into the lumina of the tubules from whence they pass to the exterior. Sprague (1940) examined about 200 wild-caught _B. orientalis_ and found them to be practically 100 percent infected.

Order MICROSPORIDIA

Family NOSEMATIDAE

=Plistophora kudoi= Sprague and Ramsey

Natural host.--_Blatta orientalis_, U.S.A., Illinois, West Virginia, Kentucky (Sprague and Ramsey, 1941, 1942): Found in the epithelial cells of caeca and midgut. Considerable damage is done to these cells. Seventy-five percent of 52 _B. orientalis_ harbored the parasite.

=Plistophora periplanetae= (Lutz and Splendore)

Synonymy.--_Nosema periplanetae_, _Pleistophora periplanetae_ [after Semans, 1943]. Georgévitch (1953) has pointed out that one may find in the malpighian tubules of cockroaches a mixed infection of Microsporidia, Haplosporidia, and Mycetozoa, and that some of the discrepancies in the earlier literature may be attributed to attempts to combine in one organism disparate stages belonging to different orders. See also comments under _Coelosporidium periplanetae_, _Haplosporidium periplanetae_, and _Peltomyces periplanetae_.

Natural hosts.--_Blatta orientalis_, France (Mercier, 1906a; Debaisieux, 1927); England (Perrin, 1906, 1906b); U.S.S.R. (Zhivago, 1909); Yugoslavia (Georgévitch,

1925, 1926, 1926a, 1927); Germany (Wellmer, 1910, 1911).

Blattella germanica, France (Léger, 1909; Debaisieux, 1927); U.S.S.R. (Zhivago, 1909).

Periplaneta americana, Brazil (Lutz and Splendore, 1903).

This organism lives in the lumen of the malpighian tubules of cockroaches. The cited authors appear to have been convinced that this organism was a microsporidian. Georgévitch (1927, 1953) described the polar capsule and filament characteristic of this order.

=Plistophora= sp.

Natural host.--_Blatta orientalis_, France (Mercier, 1908a): The organism parasitized the fat body of the cockroach. Mitoses, often abnormal, were induced in the fat cells. Infected cockroaches were easily recognizable by their distended abdomens. The fat body became chalky white and showed through the intersegmental membranes.

Porter (1930) reported finding an unidentified microsporidian in the fat bodies of _Blattella germanica_ and _Periplaneta americana_ collected in South Africa. It may or may not have been a species of _Plistophora_.

Class CILIATA

Order HOLOTRICHA

Family PARAMECIIDAE

=Paramecium= sp.

Natural associate.--Cockroaches, U.S.A., Maryland (Cleveland, 1927): Three of 30 cockroaches collected in the basement of a department store had paramecia in their stomachs but none in the rectum.

Experimental associate.--_Periplaneta americana_, U.S.A. (Hegner, 1929): Paramecia fed to the cockroaches were recovered from the crop at intervals from one-half to six and one-half hours. In no case were the protozoa recovered from the stomach alive.

Cockroaches, U.S.A., Maryland (Cleveland, 1927). About 200 starved cockroaches were fed a culture of _Paramecium_. Few, if any, of the protozoa were killed in the stomach during the first two hours, but all were killed within 5 to 6 hours after ingestion.

Family ISOTRICHIDAE

=Isotricha caulleryi= Weill

Natural host.--_Periplaneta americana_, Indochina (Weill, 1929): Alimentary canal.

Order SPIROTRICHA

Family BURSARIIDAE

=Balantidium blattarum= Ghosh

Natural host.--_Periplaneta americana_, India (Ghosh, 1922; Bhatia and Gulati, 1927); Gold Coast Colony (Macfie, 1922): Intestinal tract.

* =Balantidium coli= (Malmsten)

Experimental vector.--Cockroach, Venezuela (Tejera, 1926).

=Balantidium ovatum= Ghosh

Natural host.--_Periplaneta americana_, India (Ghosh, 1922a; Bhatia and Gulati, 1927); Indochina, Saigon (Weill, 1929): Intestinal tract.

=Balantidium praenucleatum= Kudo and Meglitsch

Natural host.--_Blatta orientalis_, U.S.A., Illinois (Kudo and Meglitsch, 1938; Meglitsch, 1940): This protozoan is found in the lumen of the anterior region of the colon in association with several other species of protozoa. Only 7.6 percent of 500 cockroaches examined contained _B. praenucleatum_. The largest number encountered in a single host was 59, but as a rule each host harbored a smaller number.

=Balantidium= sp.?

Natural host.--_Periplaneta americana_, Brazil (Magalhães, 1900): These organisms were numerous in the intestine.

Family SPIROSTOMIDAE

=Nyctotherus buissoni= Pinto

Natural host.--"Barata sylvestre," Brazil (Pinto, 1926): Organism found in the cockroach's intestine.

=Nyctotherus ovalis= Leidy

Synonymy.--_Bursaria blattarum_; _Plagiotoma blattarum_.

Natural hosts.--_Blatta orientalis_, U.S.A. (Leidy, 1850, 1853, 1853b, 1879a; Kudo, 1922, 1926, 1936; McAdow, 1931; Kudo and Meglitsch, 1938; Meglitsch, 1940); Germany (Stein, 1860; Schubotz, 1905; Chen, 1933); England (Lankester, 1865; Schuster, 1898; Lucas, 1927a, 1928); Spain (Zulueta, 1916); U.S.S.R. (Yakimov and Miller, 1922; Zasukhin, 1928, 1930; Ostroumov, 1929); Portugal (Lima Ribiero, 1924); Brazil (Pinto, 1926); Venezuela (Tejera, 1926).

Blattella germanica, South Africa (Porter, 1930); U.S.A. (Balch, 1932; McAdow, 1931).

Parcoblatta pensylvanica, U.S.A. (Semans, 1939, 1941).

Periplaneta americana, India (Bhatia and Gulati, 1927); Indochina (Weill, 1929); Philippine Islands (Hegner and Chu, 1930); South Africa (Porter, 1930); U.S.A. (McAdow, 1931; Hatcher, 1939; Meglitsch, 1940; Armer, 1944); Goa (Mello et al., 1934); China (Pai and Wang, 1947); Czechoslovakia (Low, 1956).

"Barata sylvestre," Brazil (Pinto, 1926).

"Küchenschaben," Austria (Bělař, 1916).

Nyctotherus ovalis (fig. 2, G) inhabits the hind gut of cockroaches, where it occurs almost always in the anterior half of the colon in association with other species of Protozoa (Kudo, 1936). Ninety percent of 500 _B. orientalis_ contained _N. ovalis_ (Kudo and Meglitsch, 1938). Yakimov and Miller (1922) found _N. ovalis_ in 68 percent of 124 _B. orientalis_. Zasukhin (1930) found this organism in 63 percent of over 3,000 _B. orientalis_. Zasukhin (1928, 1934) found a fungus and possibly a bacterium hyperparasitic in the cytoplasm of _N. ovalis_. _N. ovalis_ has been cultured outside the cockroach in several media (Lucas, 1928; Balch, 1932; Chen, 1933; Low, 1956).

=Nyctotherus uichancoi= Kidder

Natural hosts.--_Panesthia angustipennis_, Philippine Islands, and _Panesthia spadica_, Japan (Kidder, 1937): About 90 percent of all _P. angustipennis_ harbored this ciliate in their hindguts.

=Nyctotherus viannai= Pinto

Natural host.--"Barata sylvestre," Brazil (Pinto, 1926): In the intestine of the cockroach.

Family CLEVELANDELLIDAE

Most of the Clevelandellidae are parasitized by rod-shaped or spherical bacteria-like organisms usually in clusters (Kidder, 1937).

Synonymy.--Clevelandiidae (Kidder, 1938).

Genus CLEVELANDELLA

Synonymy.--The generic name _Clevelandia_ Kidder (1937) is preoccupied; it was therefore changed to _Clevelandella_ by Kidder in 1938. All of the following species of _Clevelandella_ were originally described as _Clevelandia_.

=Clevelandella constricta= (Kidder)

Natural hosts.--_Panesthia angustipennis_, Philippine Islands, and _Panesthia spadica_, Japan (Kidder, 1937): In the posterior end of hindgut.

=Clevelandella contorta= (Kidder)

Natural hosts.--_Panesthia angustipennis_, Philippine Islands, and _Panesthia spadica_, Japan (Kidder, 1937).

=Clevelandella elongata= (Kidder)

Natural host.--_Panesthia angustipennis_, Philippine Islands (Kidder, 1937).

=Clevelandella hastula= (Kidder)

Natural host.--_Panesthia angustipennis_, Philippine Islands (Kidder, 1937): Common in hindgut.

=Clevelandella nipponensis= (Kidder)

Natural host.--_Panesthia spadica_, Japan (Kidder, 1937).

=Clevelandella panesthiae= (Kidder)

Natural hosts.--_Panesthia angustipennis_, Philippine Islands, and _Panesthia spadica_, Japan (Kidder, 1937): In the hindgut. This protozoan is commonly parasitized by the microorganism _Sphaerita_.

=Clevelandella parapanesthiae= (Kidder)

Natural host.--_Panesthia angustipennis_, Philippine Islands (Kidder, 1937).

=Paraclevelandia brevis= Kidder

Natural hosts.--_Panesthia angustipennis_, Philippine Islands, and _Panesthia spadica_, Japan (Kidder, 1937): Present in 100 percent of _P. angustipennis_ and in nearly all _P. spadica_.

=Paraclevelandia simplex= Kidder

Natural hosts.--_Panesthia angustipennis_, Philippine Islands (Kidder, 1937, 1938): Incidence of infection about 50 percent.

Panesthia spadica, Japan (Kidder, 1937).

=Unidentified ciliate=

Natural host.--_Pycnoscelus surinamensis_, Hawaii (Schwabe, 1950): A large ciliate was found in the digestive tract and malpighian tubules.

NEGATIVE FINDINGS

In a recent experimental study Schmidtke (1955) failed to demonstrate a host-parasite relationship between _Periplaneta americana_ and the haemosporidian _Toxoplasma gondii_ Nicolle and Manceaux. This protozoan is a blood parasite in a rodent in North Africa (Kudo, 1954).

XI. HELMINTHS

Intestinal nematodes of the family Thelastomatidae have no apparent pathological effect on their cockroach hosts. Galeb (1878) has shown experimentally that oxyurids eat the same food as the host insect and that if one starves them, by withholding food from the host, the oxyurids die and disappear. In other words, these worms are not parasites, in the sense that we use the term in this paper, but commensals. Dobrovolny and Ackert (1934) stated that "all observations seemed to indicate that the health, fertility and activity of the heavily infested cockroaches were comparable with those of the non-parasitised specimens."

Very few papers have dealt with the ecology of the oxyurid parasites of cockroaches. According to Galeb (1878), usually one species of nematode is found in a single cockroach, but sometimes two species live together in the same host (e.g., in

Blatta orientalis and _Polyphaga aegyptiaca_) where they compete for food. Galeb claimed that _Hammerschmidtiella diesingi_ would replace _Leidynema appendiculata_; he observed that _H. diesingi_ surpassed _L. appendiculata_ in numbers and the latter became uncommon in the intestines of the cockroaches. On the other hand, Sobolev (1937) found that 48 percent of his oriental cockroaches were infected with both of the above species of nematodes. The average number of both species was 7.5, and the maximum number was 97; the mean number of _H. diesingi_ was 5.1 and the maximum 64; the mean number of _L. appendiculata_ was 2.4 and the maximum 33. More than 40 nematodes were found in each cockroach of 1.3 percent of those examined. These results apparently contradict Galeb's conclusions inasmuch as the number of each species in mixed infections was essentially the same as the number found in cockroaches infected by only one species (see pp. 195 and 197). Dobrovolny and Ackert (1934) found that 29 percent of 222 _Periplaneta americana_ contained both of the above species of nematodes; whereas 40 percent contained _L. appendiculata_ only, and 21 percent contained _H. diesingi_ only. The infestation ranged from 1 to 36 worms per cockroach with averages of 3.8 per male, 5.1 per female, and 2.7 per nymph.

The eggs of some helminths pass unharmed through the guts of cockroaches that serve as vectors of these ova and have no effect on the insect. However, helminths that are secondary parasites in cockroaches damage the insect to varying degrees depending upon the extent of the infection. Thus the larvae of _Moniliformis moniliformis_ pass through the gut wall and some may become embedded in the fat tissue (Moore, 1946). First stage larvae of _Oxyspirura mansoni_ also burrow through the midgut wall into the fat body; Sanders (1929) believed that _Pycnoscelus surinamensis_ could be killed if at one time a sufficient number of migrating larvae of _O. mansoni_ penetrated the cockroach's intestinal wall. _Gongylonema neoplasticum_ migrates through the digestive tract and encysts in the muscles of the thorax and legs of the host (Fibiger and Ditlevsen, 1914). Infective larvae of _Protospirura muricola_, after hatching from ingested eggs, pass through the cockroach's gut wall and encyst mainly in the thorax, around the crop, and at the bases of the large muscles of the prothoracic legs (Foster and Johnson, 1939). It is probably generally true that nematodes which are secondary parasites in cockroaches do some damage to the host's intestinal tract at least, and they probably also damage other organs in which they may encyst.

Cockroach tissues may react defensively to infections by parasitic nematodes. For example, encysted third-stage larvae of _Physaloptera turgida_ have been found enclosed in a thin, brown, chitinous substance that was undoubtedly deposited by the tissue of the cockroach (Alicata, 1937). Cysts of similar appearance have been found in cockroaches infected with _Physaloptera rara_, _P. maxillaris_, _P. hispida_ (Petri, 1950; Hobmaier, 1941; Schell, 1952), and _Gongylonema pulchrum_ (Schell, 1952a); in the latter species the deposit eventually completely surrounded the nematode larva which was killed and "chitinized." Apparently these pigmented cysts surround unhealthy or dead larvae and are secreted as a defensive mechanism by the host (Schell, 1952a). Oswald (1958) has reported finding similar pigmented cysts in _Blatta orientalis_ and _Periplaneta americana_ that were experimentally infected

with _Rictularia coloradensis_.

Our classification of the helminths follows Hyman (1951, 1951a).

HELMINTHS FOR WHICH COCKROACHES SERVE AS PRIMARY HOSTS

Phylum ASCHELMINTHES

Class NEMATODA

Order MERMITHOIDEA

Family MERMITHIDAE

=Undetermined mermithids=

Natural Hosts.--_Ectobius pallidus_, U.S.A., Plymouth, Massachusetts (Roth and Willis, 1957): This mermithid lies coiled in the body cavity of the host and one end may extend into the thorax. Apparently, the host is eventually killed and the worms may leave the cockroach ventrally between the thorax and abdomen (pl. 29, A) or thorax and head.

Periplaneta americana, Germany (Bode, 1936): Attacked by "_Mermis_" or "_Gordius_." It has been suggested that the name _Mermis_ is often applied without critical identification to immature Nematoda found in insects (Buxton, 1955).

Order RHABDITOIDEA

Family DIPLOGASTERIDAE

=Diplogaster= sp.

Synonymy.--_Lycolaimus_ [Goodey, 1951].

Experimental Host.--_Blattella germanica_, U.S.A. (Christie and Crossman, 1933).

Family STEINERNEMATIDAE

=Neoaplectana= sp.

Experimental hosts.--_Blattella germanica_, _Nauphoeta cinerea_, and _Periplaneta americana_. U.S.A. (Dutky and Hough, 1955): This nematode, found in codling moth larvae, is close to _Neoaplectana chresima_ Steiner but apparently is a new species. _Nauphoeta cinerea_ was very susceptible to infection; _B. germanica_ and _P. americana_ were less susceptible.

Order OXYUROIDEA

Family THELASTOMATIDAE

These nematodes are found in the intestinal tract of cockroaches.

=Aorurus philippinensis= Chitwood and Chitwood, 1934

Natural host.--_Panesthia angustipennis_, Philippine Islands (Chitwood and Chitwood, 1934).

=Binema mirzaia= (Basir, 1940) Basir, 1956

Synonymy.--_Periplaneticola mirzaia_ Basir, 1940.

Natural host.--_Periplaneta americana_, India, Aligarh (Basir, 1940).

=Blattelicola blattelicola= Basir, 1940

Natural host.--_Blattella germanica_, India, Aligarh (Basir, 1940).

=Blatticola blattae= (Graeffe, 1860) Chitwood, 1932

Synonymy.--_Oxyuris blattae_ Graeffe, 1860; _Oxyuris blatticola_ Galeb, 1878; _Blatticola blatticola_ (Galeb, 1877) Schwenck, 1926 [Chitwood, 1930, 1932].

Natural hosts.--_Blattella germanica_, Brazil (Pessôa and Corrêa, 1926; Schwenck, 1926); U.S.A. (Chitwood, 1930; Bozeman, 1942); Egypt? (Galeb, 1877, 1878); U.S.S.R. (Sobolev, 1937; Sondak, 1935); Czechoslovakia (Groschaft, 1956).

Ectobius lapponicus, _Ectobius pallidus_, Egypt? (Galeb, 1877, 1878).

Polyphaga aegyptiaca, France (Graeffe, 1860).

The life cycle has been studied by Bozeman (1942): He found never more than four worms in the large intestine of each cockroach. Embryos developed to "resting" stage in vitro. The resting stage was infective while the active stage was not. The worms seemed to have no effect on the vital activities of the host. Alicata (1934b) found that the embryo undergoes a molt before hatching.

Chitwood (1930) found 75 percent of the German cockroaches examined from houses in Washington infected. As a rule, one adult female, one or two males, and possibly two larval females are found in a single individual, apparently only in the rectum.

Sobolev (1937) found 72 percent of _Blattella germanica_ collected in Gorkov (U.S.S.R.) infected with _Blatticola blattae_. The mean number of worms per host was 1.8, the maximum 5. Sondak (1935) found about 30 percent of 788 _B. germanica_ collected in Leningrad to be infected with _B. blattae_. Groschaft (1956) regularly found only single worms in _B. germanica_, collected in a laboratory in Prague, except for two females that contained 2 and 3 worms each.

=Blattophila sphaerolaima= Cobb, 1920

Synonymy.--_Aorurus sphaerolaima_ (Cobb, 1920) Travassos, 1929. Although Chitwood (1932) indicated that the taxonomic position of this nematode is questionable, Chitwood and Chitwood, 1934, apparently accepted it as a valid species in describing the variety _javanica_.

Natural host.--_Panesthia laevicollis_ [Cobb recorded the host as _Panesthia brevicollis_, but no such cockroach exists. Van Zwaluwenburg (1928) and Caudell (in Chitwood, 1932) believed that Cobb meant _Panesthia laevicollis_. According to Gurney (personal communication, 1957) Caudell's notes show that in 1933 he wrote

to Dr. Chitwood and explained that he had compared Cobb's figure of the cockroach with _laevicollis_ Saussure (figures and description) and had found them the same.] Australia, New South Wales (Cobb, 1920).

=Blattophila sphaerolaima var. javanica= Chitwood and Chitwood, 1934

Natural host.--_Panesthia angustipennis_, Philippine Islands (Chitwood and Chitwood, 1934).

=Blattophila supellaima= Basir, 1941

Natural host.--_Supella supellectilium_, India, Aligarh (Basir, 1941).

=Cephalobellus brevicaudatum= (Leidy, 1851) Christie, 1933

Synonymy.--_Thelastoma brevicaudatum_ Leidy, 1851 [Christie, 1933]. _Thelastoma indiana_ Basir, 1940 [Basir, 1949].

Natural host.--_Leucophaea_ sp., India, Aligarh (Basir, 1940, 1949).

=Cephalobellus magalhāesi= (Schwenck, 1926) Basir, 1956

Synonymy.--_Bulhõesi magalhāesi_ Schwenck, 1926; _Thelastoma magalhāesi_ (Schwenck, 1926) Travassos, 1929 [Basir, 1956].

Natural host.--"Barata selvagem," Brazil, São Paulo (Schwenck, 1926).

=Euryconema paradisa= Chitwood, 1932

Natural host.--_Eurycotis floridana_, U.S.A., Florida (Chitwood, 1932).

=Galebia aegyptiaca= (Galeb, 1878) Chitwood, 1932

Synonymy.--_Oxyuris aegyptiaca_ Galeb, 1878; _Blatticola aegyptiaca_ (Galeb, 1878) Schwenck, 1926 [Chitwood, 1932].

Natural hosts.--_Blattella germanica_, Brazil (Schwenck, 1926).

Polyphaga aegyptiaca, Egypt? (Galeb, 1878).

=Hammerschmidtiella diesingi= (Hammerschmidt, 1838) Chitwood, 1932

Synonymy.--_Anguillula macrura_ Diesing, 1851; _Aorurus diesingi_ (Hammerschmidt, 1838) Travassos, 1929; _Streptostomum gracile_ Leidy, 1850; _Oxyuris diesingi_ Hammerschmidt, 1838; _Oxyuris blattae orientalis_ Hammerschmidt, 1838 [Chitwood, 1932]. _Oxyuris macrura_ of Lankester (1865).

Natural hosts.--_Blatta orientalis_, Europe (Hammerschmidt 1838, 1847; Bütschli, 1871); Egypt? (Galeb, 1878); England (Lankester, 1865; Lee, 1958); U.S.A. (Leidy, 1850a); U.S.S.R. (Yakimov and Miller, 1922; Sobolev, 1937; Sondak, 1935); Brazil (Travassos, 1929); China (Chitwood, 1932); Czechoslovakia (Groschaft, 1956).

Leucophaea maderae, Brazil (Pessôa and Corrêa, 1926).

Periplaneta americana, Brazil (Magalhães, 1900; Pessôa and Corrêa, 1926). U.S.A.: Texas (Todd, 1943); Kansas (Dobrovolny, 1933; Dobrovolny and Ackert, 1934); North Carolina (Hatcher, 1939); Iowa, North Dakota, Michigan (Hoffman,

1953). China (Chitwood, 1932). India (Basir, 1940). Czechoslovakia (Groschaft, 1956). England (Lee, 1958).

Periplaneta australasiae, Brazil (Pessôa and Corrêa, 1926).

Polyphaga aegyptiaca (Linstow, 1878).

Cockroaches (_Blatta orientalis_, _Blattella germanica_, and/or _Periplaneta americana_), U.S.A. (McAdow, 1931).

Cockroach, Venezuela (Tejera, 1926).

According to Hammerschmidt (1847) this worm may be found throughout the intestinal canal but especially in the small intestine. It is frequently found in adults and seldom in the nymphs. There were seldom more than 5 to 10 worms in one cockroach and female worms were found more frequently than males; the male worms were found only in winter and spring while the females were present at all times of the year. Bütschli (1871) stated that all stages from those just hatching to mature males and females are found.

Yakimov and Miller (1922) found _H. diesingi_ in 50.8 percent of 124 _B. orientalis_ collected in Petrograd. Sobolev (1937) found 96 percent of _B. orientalis_ infected with _H. diesingi_ with a mean number of 5.6 and maximum number of 22 in one cockroach. Groschaft (1956) found 18 in one specimen of _B. orientalis_. Dobrovolny and Ackert (1934) found about 50 percent of 222 _P. americana_ infected with _H. diesingi_. Sondak (1935) found about 36 percent of 412 _B. orientalis_ infected with either or both _H. diesingi_ and _Leidynema appendiculata_.

Two molts occur during development of the eggs; the first takes place outside the host resulting in a resting or infective stage. After the egg in the infective stage is eaten by the host, the second molt occurs before the egg hatches. Completion of the second molt and hatching perhaps are connected with ammonia present in the digestive tract; the ammonia seems to arise from the bacteria present in the gut. There appears to be a relationship between the intestinal bacteria of the cockroach and development and hatching of nematode eggs (Todd, 1944).

At the time of oviposition the nematode eggs are in the very earliest stages of cleavage. In 36 hours a motile, tadpole-like stage is reached and in a few days the embryo becomes quiescent and nonmotile. This nonmotile stage is infective whereas the motile embryonic stage is not. Feeding experiments proved that transmission of the nematode is direct. The worm reaches sexual maturity in 20 or 30 days after being ingested by the cockroach (Dobrovolny, 1933).

The bacterium _Streptomyces leidynematis_ Hoffman grows on the cuticle of _H. diesingi_ (Hoffman, 1953). The bacterium apparently is only anchored to the nematode and probably obtains its food from the intestinal contents of the cockroach. See notes under _Leidynema appendiculata_.

=Hammerschmidtiella neyrai= Serrano Sánchez, 1945

Synonymy.--_Hammerschmidtiella neyrae_ Serrano Sánchez, 1947. [According to M. B. Chitwood, personal communication, 1957, Serrano Sánchez's emendation is

apparently an error.]

Natural host.--_Blatta orientalis_, Spain, Grenada (Serrano Sánchez, 1947): Of 2,943 specimens examined, 1,143 were parasitized by oxyurids and of these 45 percent contained _H. neyrai_.

=Leidynema appendiculata= (Leidy, 1850) Chitwood, 1932

Synonymy.--_Oxyuris blattae orientalis_ Hammerschmidt, 1847, of Bütschli, 1871, and _Oxyuris blattae-orientalis_ of Magalhães, 1900; _Oxyuris blattae_ Hammerschmidt, 1847, of Galeb, 1878; _Aorurus_ (_Thelastoma_) _appendiculatus_ Leidy, 1850. [Chitwood, 1932.] Serrano Sánchez (1947) has divided this species into three geographical varieties as follows: _L. appendiculata_ (Leidy, 1852) (Dobrovolny and Ackert, 1934) var. _indiana_; _L. appendiculata_ (Leidy, 1852) (Chitwood, 1932) var. _americana_; _L. appendiculata_ (Serrano Sánchez, 1947) var. _hispana_. However, Basir (1956) does not recognize these varieties. The Russians recognize _hispana_ (M. B. Chitwood, personal communication, 1957).

Natural hosts.--_Blaberus atropos_, South America (Chitwood, 1932).

Blatta orientalis, Egypt? (Galeb, 1878); Europe (Bütschli, 1871); U.S.S.R. (Sobolev, 1937; Sondak, 1935); U.S.A., Nebraska (Todd, 1944); Spain (Serrano Sánchez, 1947): Recorded as var. _hispana_. Czechoslovakia (Groschaft, 1956). England (Lee, 1958a).

Blatta orientalis or _Periplaneta americana_, Brazil (Magalhães, 1900).

Periplaneta americana, U.S.A.: Texas (Todd, 1943); Nebraska (Todd, 1944); Kansas (Dobrovolny, 1933; Dobrovolny and Ackert, 1934); North Carolina (Hatcher, 1939); Iowa, North Dakota, Michigan (Hoffman, 1953). Czechoslovakia (Groschaft, 1956). England (Lee, 1958a).

Cockroach, Venezuela (Tejera, 1926).

Cockroaches (_Blatta orientalis_, _Blattella germanica_, and/or _Periplaneta americana_), U.S.A. (McAdow, 1931).

Chitwood (1932) also listed China for distribution of the worm, but we could not tell which host was involved.

The worms are found in the colon and rectum of the host. Galeb (1878) found as many as 20 individuals in a single _B. orientalis_. Sobolev (1937) found 52 percent of _B. orientalis_ infected with _L. appendiculata_; the mean number of worms per roach was 1.5 and the maximum 2. Dobrovolny and Ackert (1934) found 69 percent of 222 _P. americana_ infected with this species.

Two molts occur within the egg during development of the larva. The first molt occurs outside the host resulting in the formation of an infective resting stage. The second molt occurs inside the cockroach (Todd, 1941, 1944).

Transmission of the nematode is direct, eggs in the resting embryonated stage being infective (Dobrovolny and Ackert, 1934).

Hoffman (1953) described a filamentous bacterium, _Streptomyces leidynematis_ Hoffman, which grows on the cuticle of _L. appendiculata_ in _P. americana_. Leidy (1853) noted the presence of simple, inarticulate, amorphous filaments, growing from nematodes infecting _B. orientalis_. Bütschli (1871) and Magalhães (1900) described similar filaments adhering to the surface of oxyurids from cockroaches.

=Leidynema appendiculata= (Leidy, 1850) Chitwood, 1932?

Natural host.--_Eurycotis floridana_, U.S.A., Massachusetts (Roth and Willis, unpublished data, 1955): Determined by Dr. G. Steiner who wrote us, "In _Eurycotis floridana_ there were ten specimens of the nematode _Leidynema appendiculata_ (Leidy, 1850). This cockroach is obviously a new host for this nematode. I am not sure that the nematode exactly agrees with the description as given in the literature."

=Leidynema cranifera= Chitwood, 1932

Natural hosts.--_Blaberus craniifer_, U.S.A., Florida (Chitwood, 1932); Massachusetts (Roth and Willis, unpublished data, 1955). Determined by Dr. G. Steiner.

Blaberus atropos?, U.S.A., Florida (Chitwood, 1932): _B. craniifer_ has generally been recorded as _B. atropos_ of Stoll which is a closely related but distinct South American species (Rehn and Hebard, 1927).

=Leidynema delatorrei= Chitwood, 1932

Natural host.--_Leucophaea maderae_, Cuba, Havana (Chitwood, 1932).

=Leidynema nocalum= Chitwood and Chitwood, 1934

Natural host.--_Panesthia angustipennis_, Philippine Islands (Chitwood and Chitwood, 1934).

=Leidynemella fusiformis= Cobb, 1934

Natural hosts.--_Panesthia laevicollis_?, Philippine Islands (Cobb _in_ Chitwood and Chitwood, 1934).

Panesthia angustipennis, Philippine Islands (Chitwood and Chitwood, 1934).

=Leidynemella panesthiae= (Galeb, 1878) Chitwood and Chitwood, 1934

Synonymy.--_Oxyuris panesthiae_ Galeb, 1878, in part; _Thelastoma panesthiae_ (Galeb, 1878) Travassos, 1929. [Chitwood, 1932; Chitwood and Chitwood, 1934.]

Natural host.--_Panesthia_ sp., New Guinea (Galeb, 1878): About 40 nematodes may be found in one insect.

=Leidynemella paracranifera= Chitwood and Chitwood, 1934

Natural host.--_Panesthia angustipennis_, Philippine Islands (Chitwood and Chitwood, 1934).

=Oxyuris= (?) =heterogamiae= Galeb, 1878

Synonymy.--_Thelastoma heterogamiae_ (Galeb, 1878) Travassos, 1929. The taxonomic position of this species is questionable; it might possibly belong in _Blatticola_ or _Protrellina_ (Chitwood, 1932). Basir (1956) placed it in an appendix to the family Thelastomatidae.

Natural host.--_Polyphaga aegyptiaca_, Egypt? (Galeb, 1878).

=Protrelleta floridana= Chitwood, 1932

Natural host.--_Blaberus craniifer_, U.S.A., Florida (Chitwood, 1932).

=Protrellus aureus= Cobb, 1920

Synonymy.--The taxonomic position of this nematode is questionable (Chitwood, 1932).

Natural host.--_Polyzosteria melanaria_?, Australia, New South Wales (Cobb, 1920). [Caudell (_in_ Chitwood, 1932) stated that this host was probably _Platyzosteria analis_.]

=Protrellus aurifluus= (Chitwood, 1932) Chitwood, 1933

Synonymy.--_Protrellina aurifluus_ Chitwood, 1932.

Natural hosts.--_Parcoblatta lata_, U.S.A., North Carolina, Maryland (Chitwood, 1932).

Parcoblatta uhleriana, North Carolina (Hatcher, 1939).

=Protrellus australasiae= (Pessôa and Corrêa, 1926) Travassos, 1929

Synonymy.--_Oxyuris australasiae_ Pessôa and Corrêa, 1926; _Protrellina australasiae_ (Pessôa and Corrêa, 1926) Chitwood, 1932 [Chitwood, 1933].

Natural host.--_Periplaneta australasiae_, Brazil (Pessôa and Corrêa, 1926, 1927).

=Protrellus galebi= Schwenck, 1926

Synonymy.--_Protrellina galebi_ (Schwenck, 1926) Chitwood, 1932 [Chitwood, 1933].

Natural host.--"Barata selvagem," Brazil (Schwenck, 1926).

=Protrellus künckeli= (Galeb, 1878) Travassos, 1929

Synonymy.--_Oxyuris künckeli_ Galeb, 1878; _Protrellina künckeli_ (Galeb, 1878) Chitwood, 1932 [Chitwood, 1933].

Natural hosts.--_Periplaneta americana_, Egypt? (Galeb, 1877, 1878). [Chitwood (1932) questioned the determination of this host because he failed to find this nematode in a large number of specimens from U.S.A. and China.] Brazil (Pessôa and Corrêa, 1926).

Periplaneta australasiae, Brazil (Pessôa and Corrêa, 1926).

=Protrellus manni= (Chitwood, 1932) Chitwood, 1933

Synonymy.--_Protrellina manni_ Chitwood, 1932.

Natural host.--_Aglaopteryx diaphana_, Cuba (Chitwood, 1932).

=Protrellus phyllodromi= (Basir, 1942) Basir, 1956

Synonymy.--_Protrellina phyllodromi_ Basir, 1942.

Natural host.--_Blattella humbertiana_, India, Aligarh (Basir, 1942): Found in the rectum.

=Protrelloides paradoxa= Chitwood, 1932

Natural host.--_Eurycotis floridana_, U.S.A., Florida (Chitwood, 1932).

=Schwenkiella icemi= (Schwenck, 1926) Basir, 1956

Synonymy.--_Bulhõesia icemi_ Schwenck, 1926; _Thelastoma icemi_ (Schwenck, 1926) Travassos, 1929; _Thelastoma aligarhica_ Basir, 1940. [Basir, 1956.]

Natural hosts.--"Barata selvagem," Brazil, São Paulo (Schwenck, 1926).

Periplaneta americana, India, Aligarh (Basir, 1940); U.S.A., Nebraska (Todd, 1943).

Periplaneta brunnea, U.S.A., Louisiana (Todd, 1943).

=Severianoia magna= Pereira, 1935

Natural host.--"Blattidae sylvestres," Brazil (Pereira, 1935).

=Severianoia severianoi= (Schwenck, 1926) Travassos, 1929

Synonymy.--_Bulhõesia severianoi_ Schwenck, 1926 [Travassos, 1929].

Natural hosts.--"Baratas de pau podre," Brazil (Schwenck, 1926).

Pycnoscelus surinamensis, U.S.A., Florida (Chitwood, 1932).

=Suifunema caudelli= Chitwood 1932

Natural host.--_Steleopyga? sinensis_, Asia: Suifu, Szchuen, China (Chitwood, 1932).

=Thelastoma pachyjuli= (Parona, 1896) Travassos, 1929

Synonymy.--_Oxyuris bulhõesi_ de Magalhães, 1900; _Bulhõesia bulhõesi_ (Magalhães, 1900) Schwenck, 1926 [Travassos, 1929; Chitwood, 1932]; _Thelastoma bulhõesi_ (Magalhães, 1900) Travassos, 1929; although this last combination (from Chitwood, 1932) is not given by Basir (1956), it is implied by the synonymy that he does cite under _T. pachyjuli_.

Natural hosts.--_Blatta orientalis_, Czechoslovakia (Groschaft, 1956).

Periplaneta americana, Brazil (Magalhães, 1900); North America (Chitwood, 1932); U.S.A., North Carolina (Hatcher, 1939).

=Thelastoma palmettum= Chitwood and Chitwood, 1934

Natural host.--_Panesthia angustipennis_, Philippine Islands (Chitwood and Chitwood, 1934).

=Thelastoma riveroi= Chitwood, 1932

Natural host.--_Periplaneta_ sp., Cuba (Chitwood, 1932).

=Undetermined nematodes=

Natural host.--_Cutilia_ sp. near _sedilloti_, U.S.A. (hosts imported from New Zealand) (Roth, unpublished data, 1957).

Class NEMATOMORPHA

Order GORDIOIDEA

The immature stages of the following gordian worms have been found in the body cavity of cockroaches.

Family CHORDODIDAE

=Chordodes morgani= Montgomery, 1898

Synonymy.--_Chordotes_ [sic] _puerilis_ Montgomery, 1898 [Ward, 1918].

Natural host.--Cockroach, U.S.A. (Montgomery, 1898); Pennsylvania, Maryland, Michigan, Ohio, Florida, Iowa, Nebraska (Ward, 1918).

Family GORDIIDAE

=Gordius aquaticus= Linnaeus, 1758

Natural host.--_Blatta_ sp., U.S.A. (Stiles and Hassall, 1894).

Leidy (1879) identified a 9-inch-long nematode which came from a cockroach (_Blatta orientalis?_) as probably being _Gordius aquaticus_. Ransom (_in_ Pierce, 1921) states that _G. aquaticus_ may be an accidental parasite of man. Faust (1955) summarizes the few reported cases of human parasitism. Dorier (1930) reported that the regurgitated liquid of _Blatta orientalis_ had no effect on cysts of _G. aquaticus_ after one hour.

=Gordius blattae orientalis= Diesing, 1851

Synonymy.--_Gordius orientalis_ of Lankester (1865).

Natural host.--_Blatta orientalis_, Germany (Siebold, 1842; Linstow, 1878): Found in abdomen. Von Siebold called this "Filarien" but did not otherwise name or describe the worm.

=Gordius pilosus= (Möbius, 1855) Diesing, 1861

Synonymy.--_Chordodes pilosus_ Möbius, 1855 [Diesing, 1861.]

Natural host.--_Blaberus giganteus_, Venezuela (Möbius, 1855): From the insect's abdomen.

=Gordius= sp.

Natural hosts.--_Periplaneta americana_, South Africa (Porter, 1930); Germany (Bode, 1936): Bode's record may have referred to a _Mermis_ or other nematode.

Cockroaches, Venezuela (Miall and Denny, 1886; Burr, 1899a; Tejera, 1926).

=Parachordodes raphaelis= (Camerano, 1893) Camerano, 1897

Synonymy.--_Gordius raphaelis_ Camerano, 1893 [Camerano, 1897].

Natural hosts.--_Symploce parenthesis_ and _Kuchinga hemerobina_, French Equatorial Africa (Camerano, 1893, 1897).

=Undetermined gordian worms=

Natural hosts.--_Eurycotis floridana_, Florida (T. Eisner, personal communication, 1958): See plate 29, B.

Parahormetica bilobata, Brazil (Pessôa and Corrêa, 1929): Worm referred to as "gordiaceo."

Cockroaches, Australia (E. F. Riek, personal communication, 1953): Three undescribed gordian worms were found in undetermined cockroaches of the subfamily Blattinae.

HELMINTHS[4] FOR WHICH COCKROACHES SERVE AS INTERMEDIATE HOSTS

The use of the asterisk (*) is explained in footnote 3, page 4.

Phylum ACANTHOCEPHALA

Order ARCHIACANTHOCEPHALA

Family OLIGACANTHORHYNCHIDAE

* =Prosthenorchis elegans= (Diesing, 1851) Travassos, 1915

Natural host.--_Blattella germanica_, France (Brumpt and Urbain, 1938, 1938a; Brumpt et al., 1939).

Experimental hosts.--_Blaberus atropos_ and _Leucophaea maderae_, France (Brumpt and Desportes, 1938).

* =Prosthenorchis spirula= (Olfers in Rudolphi, 1819) Travassos, 1917

Natural host.--_Blattella germanica_, France (Brumpt and Urbain, 1938, 1938a; Brumpt et al., 1939); Netherlands (Thiel and Wiegand Bruss, 1946).

Experimental hosts.--_Blattella germanica_, Netherlands (Thiel and Wiegand Bruss, 1946).

Blaberus atropos and _Leucophaea maderae_, France (Brumpt and Desportes, 1938).

Family MONILIFORMIDAE

* =Moniliformis dubius= Meyer, 1932

Natural hosts.--_Periplaneta americana_, Brazil (Magalhães, 1898; Travassos, 1917); Gold Coast (Southwell, 1922); India (Pujatti, 1950); U.S.A. (Burlingame and Chandler, 1941; Moore, 1946).

Periplaneta australasiae, India (Pujatti, 1950).

Experimental hosts.--_Blattella germanica_, Japan (Yamaguti and Miyata, 1942).

Periplaneta americana, U.S.A. (Chandler, 1941; Moore, 1946); Japan (Yamaguti and Miyata, 1942).

* =Moniliformis kalahariensis= Meyer, 1931

Natural host.--_Blattella germanica_, India (Meyer, 1931, 1932).

* =Moniliformis moniliformis= (Bremser _in_ Rudolphi, 1819) Travassos, 1915

Natural hosts.--_Periplaneta americana_, Argentina (Bacigalupo, 1927, 1927a, 1928); Brazil (Pessôa and Corrêa, 1929); Algeria (Seurat, 1912); Burma (Subramanian, 1927); South Africa (Porter, 1930); Madras (Sita, 1949).

Periplaneta spp., New Caledonia (Rageau, 1956).

Cockroaches, Venezuela (Tejera, 1926).

Experimental hosts.--_Blaberus atropos_, _Blatta orientalis_, _Blattella germanica_, _Leucophaea maderae_, France (Brumpt and Urbain, 1938a).

Periplaneta americana, Japan (Yamaguti and Miyata, 1942); France (Brumpt, 1949); Madras (Sita, 1949).

Phylum ASCHELMINTHES

Class NEMATODA

Order OXYUROIDEA

Family SUBULURIDAE

* =Subulura jacchi= (Diesing, 1861) Railliet and Henry, 1914

Synonymy.--_Subulura jacchi_ (Marcel, 1857) [Dr. J. T. Lucker, personal communication, 1957].

Experimental host.--_Blaberus atropos_, France (Chabaud and Larivière, 1955).

Order SPIRUROIDEA

Family THELAZIIDAE

* =Oxyspirura mansoni= (Cobbold, 1879) Ransom, 1904

Natural hosts.--_Pycnoscelus surinamensis_, Australia (Fielding, 1926, 1927, 1928, 1928a); U.S.A. (Sanders, 1927, 1928, 1929; Shealy, 1927); Formosa (Kobayashi, 1927); Antigua (Hutson, 1938, 1943); Hawaii (Illingworth, 1931; Schwabe, 1950, 1950a, 1950b, 1951); New Caledonia (Rageau, 1956).

We have recently found (Roth and Willis, 1960) that two strains of _Pycnoscelus surinamensis_ exist; a parthenogenetic strain (from Florida), and a bisexual strain (from Hawaii) which does not reproduce parthenogenetically. The parthenogenetic strain is undoubtedly the form that has been shown to be the host of _O. mansoni_ in the United States and Antigua, because only this form is found in the New World. Probably the parthenogenetic strain was the form involved in most Pacific areas. However, from internal evidence in his papers, we concluded that Schwabe, in Hawaii, may well have been working with the bisexual strain and possibly also with the parthenogenetic strain; if this is true, then both parthenogenetic and bisexual strains of _Pycnoscelus surinamensis_ may serve as intermediate hosts of the eyeworm.

Experimental hosts.--_Periplaneta americana_, Antigua (Hutson, 1943).

Pycnoscelus surinamensis, U.S.A. (Sanders, 1929); Australia (Fielding, 1927, 1928a); Hawaii (Schwabe, 1951).

=Rictularia coloradensis= Hall, 1916

Natural hosts.--_Parcoblatta pensylvanica_ and _Parcoblatta virginica_, U.S.A., Ohio (Oswald, 1958): Of 49 wood roaches collected, one of each species contained a single larva each.

Experimental hosts.--_Blatta orientalis_, _Blattella germanica_, _Parcoblatta pensylvanica_, _Parcoblatta virginica_, _Periplaneta americana_, and _Supella supellectilium_, U.S.A. (Oswald, 1958): The larvae underwent normal development in all species of cockroaches except _B. orientalis_ and _P. americana_ in which cysts developed that contained a reddish-brown pigment; larvae in such cysts were dead or dying. Eggs of _R. coloradensis_ hatched in the midgut of _B. germanica_ and first-stage larvae entered the hindgut epithelium within 24 hours. The larvae underwent two molts within a cyst formed by tissues of the host's gut, the second molt occurring during the twelfth or thirteenth day. In _Parcoblatta_, cysts were found free in the body cavity as well as attached to the hindgut. In _B. germanica_ and _S. supellectilium_ the cysts remained attached to the hindgut. Usually over 20 cysts developed in each infected _Parcoblatta_; fewer than 10 per insect developed in the other species. Larvae became infective to the definitive host, the white-footed mouse [_Peromyscus leucopus noveboracensis_ (Fischer)], as early as the tenth day.

Family SPIRURIDAE

* =Agamospirura parahormeticae= Pessôa and Corrêa, 1929

Natural host.--_Parahormetica bilobata_, Brazil (Pessôa and Corrêa, 1929).

* =Gongylonema ingluvicola= Ransom, 1904

Experimental host.--_Blattella germanica_, U.S.A. (Cram, 1935).

* =Gongylonema neoplasticum= (Fibiger and Ditlevsen, 1914) Ransom and Hall, 1916

Natural hosts.--_Blatta orientalis_, Netherlands (Baylis, 1925).

Blattella germanica, U.S.A. (Hitchcock and Bell, 1952).

Periplaneta americana, Denmark and St. Croix (Fibiger, 1913, 1913a; Fibiger and Ditlevsen, 1914); Netherlands (Baylis, 1925); Argentina (Bacigalupo, 1930); England (Leiper, 1926); South Africa (Porter, 1930); U.S.A. (Hitchcock and Bell, 1952); Formosa (Yokagawa, 1924, 1925, 1925a).

Periplaneta australasiae, Formosa (Yokagawa, 1924, 1925, 1925a).

Experimental hosts.--_Blattella germanica_, Denmark (Fibiger and Ditlevsen, 1914); U.S.A. (Hitchcock and Bell, 1952); France (Brumpt, 1949).

Blatta orientalis, Denmark (Fibiger and Ditlevsen, 1914).

Periplaneta americana, Denmark and St. Croix (Fibiger, 1913; Fibiger and Ditlevsen, 1914); U.S.A. (Hitchcock and Bell, 1952).

* =Gongylonema pulchrum= Molin, 1857

Experimental hosts.--_Blattella germanica_, U.S.A. (Ransom and Hall, 1915, 1916, 1917; Stiles and Baker, 1927; Schwartz and Lucker, 1931; Lucker, 1932; Alicata, 1934a, 1935); Europe (Baylis et al., 1925, 1926, 1926a; Sambon, 1926).

Parcoblatta sp., Alicata (1934, 1935).

* =Gongylonema= sp.

Natural host.--_Periplaneta americana_, Brazil (Magalhães, 1900); Algeria (Seurat, 1916); England? (Leiper, 1926).

* =Microtetrameres helix= Cram, 1927

Experimental host.--_Blattella germanica_, U.S.A. (Cram, 1934).

* =Protospirura bonnei= Ortlepp, 1924

Natural host.--_Leucophaea maderae_, Venezuela (Brumpt, 1931).

Experimental hosts.--_Blatta orientalis_, _Blattella germanica_, _Leucophaea maderae_, France (Brumpt, 1931).

* =Protospirura columbiana= Cram, 1926

Experimental host.--_Blattella germanica_, U.S.A. (Cram, 1926).

* =Protospirura muricola= Geodoelst, 1916

Natural host.--_Leucophaea maderae_, Panama (Foster and Johnson, 1938, 1939).

* =Seurocyrnea colini= (Cram, 1927) Cram, 1931

Experimental host.--_Blattella germanica_, U.S.A. (Cram, 1931, 1931a, 1933a).

* =Spirura gastrophila= (Müller, 1894) Seurat, 1913

Natural hosts.--_Blatta orientalis_, Europe? (Deslongchamps, 1824, _in_ Seurat, 1911); Italy (Grassi, 1888); Algeria (Seurat, 1916).

Periplaneta americana, Brazil (Pessôa and Corrêa, 1929).

Cockroach, Venezuela (Tejera, 1926).

Experimental hosts.--_Blatta orientalis_, France (Galeb, 1878a); "Cafards," Algeria (Roger, 1906, 1907).

* =Tetrameres americana= Cram, 1927

Natural host.--_Blattella germanica_, U.S.A. (Cram, 1931b, personal communication, 1956); Hawaii (Alicata, 1938, 1947).

Experimental host.--_Blattella germanica_, U.S.A. (Cram, 1931b).

* =Tetrameres pattersoni= Cram, 1933

Experimental host.--_Blattella germanica_, U.S.A. (Cram, 1933).

Family PHYSALOPTERIDAE

* =Physaloptera hispida= Schell, 1950

Experimental host.--_Blattella germanica_, U.S.A. (Schell, 1952, 1952a).

* =Physaloptera maxillaris= Molin, 1860

Experimental host.--_Blattella germanica_, U.S.A. (Hobmaier, 1941).

* =Physaloptera praeputialis= von Linstow, 1889

Experimental host.--_Blattella germanica_, U.S.A. (Petri and Ameel, 1950).

* =Physaloptera rara= Hall and Wigdor, 1918

Experimental host.--_Blattella germanica_, U.S.A. (Petri and Ameel, 1950; Petri, 1950).

* =Physaloptera turgida= Rudolphi, 1819

Experimental host.--_Blattella germanica_, U.S.A. (Alicata, 1937; Schell, 1952).

HELMINTHS WHOSE EGGS HAVE BEEN CARRIED BY COCKROACHES

The use of the asterisk (*) is explained in footnote 3, page 4.

Phylum PLATYHELMINTHES

Class TREMATODA

Order DIGENEA

Family SCHISTOSOMATIDAE

* =Schistosoma haematobium= (Bilharz, 1852) Weinland, 1858

Experimental vector.--_Periplaneta americana_, Gold Coast Colony (Macfie, 1922).

Class CESTODA

Order TAENIOIDEA

Family HYMENOLEPIDIDAE

* =Hymenolepis= sp.

Natural vectors.--_Periplaneta americana_, Formosa (Morischita and Tsuchimochi, 1926).

Polyphaga saussurei, Tadzhikistan (Zmeev, 1936).

Family TAENIIDAE

* =Taenia saginata= Goeze, 1782

Experimental vector.--_Periplaneta americana_, Gold Coast Colony (Macfie, 1922).

=Echinococcus granulosis= (Batsch, 1786) Rudolphi, 1805

Synonymy.--_Taenia echinococcus_ (Zeder, 1803) [Faust, 1939].

Experimental vector.--_Periplaneta americana_, Uruguay (Pérez Fontana, 1955): Eggs were recovered from the feces of artificially infested cockroaches under "natural" conditions.

Family Unknown

* =Undetermined tapeworm ova=

Natural vector.--_Polyphaga saussurei_, Tadzhikistan (Zmeev, 1936).

Phylum ASCHELMINTHES

Class NEMATODA

Order OXYUROIDEA

Family OXYURIDAE

* =Enterobius vermicularis= (Linnaeus, 1758) Leach in Baird, 1853

Natural vectors.--_Blatta orientalis_ and _Blattella germanica_, U.S.S.R. (Sondak, 1935).

Order ASCAROIDEA

Family ASCARIDAE

* =Ascaris lumbricoides= Linnaeus, 1758

Natural vector.--_Periplaneta americana_, South Africa (Porter, 1930): The eggs may have been those of _A. suum_ Goeze, 1782.

Experimental vectors.--_Periplaneta americana_, Gold Coast Colony (Macfie, 1922); India (Chandler, 1926).

Periplaneta americana, _Periplaneta australasiae_, _Neostylopyga rhombifolia_, Formosa (Morischita and Tsuchimochi, 1926).

* =Ascaris= sp.

Natural vector.--_Blatta orientalis_, Italian Somaliland (Mariani and Besta, 1936).

Experimental vector.--_Periplaneta americana_, Uruguay (Pérez Fontana, 1955): Eggs recovered from the insects' feces.

Order STRONGYLOIDEA

Family ANCYLOSTOMIDAE

* =Ancylostoma caninum= (Ercolani, 1859) Hall, 1913

Experimental vector.--_Periplaneta americana_, Netherlands (Akkerman, 1933).

* =Ancylostoma ceylanicum= (Looss, 1911) Leiper, 1915

Experimental vector.--_Periplaneta americana_, Gold Coast Colony (Macfie, 1922); Netherlands (Akkerman, 1933).

* =Ancylostoma duodenale= (Dubini, 1843) Creplin, 1845

Natural vector.--_Periplaneta americana_, South Africa (Porter, 1929, 1930).

Experimental vector.--_Periplaneta americana_, Gold Coast Colony (Macfie, 1922).

* =Necator americanus= (Stiles, 1902) Stiles, 1906

Natural vector.--_Periplaneta americana_, India (Chandler, 1926).

Experimental vector.--_Periplaneta americana_, Gold Coast Colony (Macfie, 1922).

* =Hookworm ova=

Experimental vectors.--_Periplaneta americana_, _Periplaneta australasiae_, _Neostylopyga rhombifolia_, Formosa (Morischita and Tsuchimochi, 1926).

Family TRICHOSTRONGYLIDAE

* =Trichostrongylus= sp.

Natural vector.--_Blatta orientalis_, Italian Somaliland (Mariani and Besta, 1936).

Order TRICHUROIDEA

Family TRICHURIDAE

* =Capillaria hepatica= (Bancroft, 1893) Travassos, 1915

Experimental vector.--_Blatta orientalis_, Italy? (Giordano, 1950).

* =Trichuris trichiura= (Linnaeus, 1771) Stiles, 1901

Natural vectors.--_Blatta orientalis_, Italian Somaliland (Mariani and Besta, 1936); U.S.S.R. (Sondak, 1935).

Blattella germanica, U.S.S.R. (Sondak, 1935).

Periplaneta americana, Gold Coast Colony (Macfie, 1922); Formosa (Morischita and Tsuchimochi, 1926).

Experimental vectors.--_Periplaneta americana_, Gold Coast Colony (Macfie, 1922); India (Chandler, 1926); Uruguay (Pérez Fontana, 1955).

Periplaneta americana, _Periplaneta australasiae_, and _Neostylopyga rhombifolia_, Formosa (Morischita and Tsuchimochi, 1926).

XII. ARTHROPODA

The classification follows Brues et al. (1954) with the following exceptions. The Acarina are arranged according to Dr. J. H. Camin (personal communication, 1955). Family Eupelmidae of the Hymenoptera follows the classification of Peck (1951).

Class ARACHNIDA

In this class, representatives of at least four orders have utilized cockroaches as food: the whip scorpions, scorpions, spiders, and mites. Apparently none of these feed exclusively on cockroaches, but the Philippine forest scorpion _Heterometrus_ (=_Palamnaeus_) _longemanus_ seems to prefer blattids to other insects (Schultze, 1927).

Order PEDIPALPIDA

Family THELYPHONIDAE

=Mastigoproctus giganteus= (Lucas)

Synonymy.--_Thelyphonus giganteus_ Lucas [Dr. R. E. Crabill, personal communication, 1958].

Experimental prey.--Cockroaches, U.S.A. (Marx, 1892, 1894): Immature whip scorpion captured and fed on one or two cockroaches a week. It lived on this diet for about two years.

=Mastigoproctus= sp.

Common name.--Whip scorpion.

Experimental prey.--_Blattella germanica_, U.S.A., Florida [Dr. B. J. Kaston, personal communication, 1953].

Order SCORPIONIDA

Pocock (1893) noticed that a scorpion whose pectines had come in contact with a cockroach immediately turned back and ate the insect. He concluded that the scorpion detected the cockroach by means of the pectines. However, Cloudsley-Thompson (1955) has demonstrated that the main function of the pectines is probably the detection of ground vibrations. He accounted for Pocock's observation by the presence of sensory spines (presumably tactile) which project from beneath the pectines. In a house in Arizona, Lyon (1951) observed over 60 scorpions living in

a kitchen cabinet that enclosed a sink. They were apparently thriving on a heavy infestation of cockroaches. Stahnke (1953) stated that he used _Periplaneta americana_ as the principal food for scorpions at the Poisonous Animals Research Laboratory of Arizona State College. Cloudsley-Thompson (1955a) cited cockroaches as one of the arthropods that scorpions feed upon.

Family BUTHIDAE

=Buthus australis= (Linnaeus)

Synonymy.--_Androctonus australis_ [Crabill, personal communication, 1957].

Experimental prey.--Cockroaches, England (Cloudsley-Thompson, 1955a): This African species ate at least one cockroach per week during the summer months. It can, however, survive four months' starvation and is particularly adapted to a dry climate (Cloudsley-Thompson, personal communication, 1956).

=Centruroides gracilis= (Latreille)

Experimental prey.--_Periplaneta americana_, U.S.A. (Roth, unpublished data, 1953): Scorpion collected in Florida by Roth and identified by Dr. M. H. Muma.

=Centruroides hentzi= (Banks)

Experimental prey.--_Periplaneta australasiae_ and _Pycnoscelus surinamensis_, U.S.A. (Muma, personal communication, 1953): This scorpion occurs in large numbers in the Florida citrus groves, together with _P. australasiae_ which is probably an important natural prey.

=Centruroides vitattus= (Say)

Natural prey.--_Parcoblatta pensylvanica_ (?), U.S.A., Florida (Muma, personal communication, 1953). This may have been another species of this genus, possibly _P. divisa_, as _P. pensylvanica_ is not known from Florida (Rehn, personal communication, 1958).

Experimental prey.--_Blatta orientalis_, _Blattella germanica_, _Periplaneta americana_, and _Pycnoscelus surinamensis_, U.S.A., Florida (Muma, personal communication, 1953).

=Parabuthus capensis= (Hemprich and Ehrenberg)

Experimental prey.--"Common house-cockroach" (Pocock, 1893): The scorpions were collected in Cape Town, South Africa.

Family CHACTIDAE

=Euscorpius germanus= (Koch)

Synonymy.--_Euscorpius carpathicus_ [Cloudsley-Thompson, 1955a].

Experimental prey.--_Blattella germanica_, England? (Pocock, 1893).

Periplaneta americana, nymphs, England? (Cloudsley-Thompson, personal communication, 1956): This scorpion is found naturally in southern Europe and North Africa.

=Euscorpius italicus italicus= (Herbst)

Experimental prey.--Cockroaches including nymphs of _Periplaneta_, England? (Cloudsley-Thompson, 1951): The cockroaches had to be disabled before the scorpion would feed on them. Prey is apparently detected by tactile and auditory senses, sight being poorly developed and not used. The scorpion is found in southern Europe and North Africa.

Family VEJOVIDAE

=Hadrurus arizonensis= Ewing

Experimental prey.--_Periplaneta americana_, U.S.A. (Stahnke, 1949): This record is a photograph showing the scorpion eating the cockroach.

Family ISCHNURIDAE

=Hormurus caudicula= (Koch)

Experimental prey.--Cockroach, Australia (McKeown, 1952): This record is a photograph showing the scorpion feeding on a cockroach.

Family SCORPIONIDAE

=Heterometrus longimanus= (Herbst)

Synonymy.--_Palamnaeus longimanus_ [Cloudsley-Thompson, personal communication, 1956].

Natural prey.--"Large wood cockroach," Philippine Islands (Schultze, 1927): On several occasions Schultze found fragments of wings and legs of the large wood cockroach in a scorpion cavity, under a rotten log.

Experimental prey.--_Leucophaea maderae_, _Periplaneta americana_, and other species of Blattidae, Philippine Islands (Schultze, 1927): Blattids seemed to be the favored food. This scorpion is usually found in humid, damp places in forest and jungle. Schultze describes in detail feeding behavior of the scorpion and method of capturing its prey.

=Urodacus novaehollandiae= Peters

Experimental prey.--_Periplaneta americana_, Australia (Glauert, 1946): An injured cockroach was accepted at once by the scorpion, which held the insect in its claws and tore it with the alternately moving chelicerae. The scorpion ate all the soft parts and most of the sclerotized exoskeleton.

Order ARANEIDA

Many observations of spiders feeding on cockroaches are quite general, and many observers have failed to identify either the spider or its prey. Belt (1874) stated that "the cockroaches that infest houses in the tropics ... have numerous enemies--birds, rats, scorpions, and spiders." When Belt tried to drive a cockroach toward a large cockroach-eating spider, the insect rushed away from him until it came within a foot of the spider when it would double back, never advancing nearer.

Beebe (1925) watched a giant "wood roach," which was in the grasp of a 2-inch ctenid spider, fly through the window of his British Guiana laboratory. While the spider ate the cockroach, the insect gave birth to 51 nymphs. Sonan (1924) reported that large gray spiders devour nymphs and adults of _Periplaneta americana_ and _P. australasiae_ in Formosa; this spider also occurs on Hiyakejima Island and Okinawa. Passmore (1936), who has produced some excellent photographs of tarantulas, stated that they destroy cockroaches. Rau (1940) stated that American and oriental cockroaches were the principal item of diet of a friend's pet tarantula for several years. Kaston (personal communication, 1953) successfully fed a tarantula with _Periplaneta americana_.

Bristowe (1941) found that the British species of _Ectobius_ are readily accepted by _Xysticus_, _Clubiona_, _Drassodes_, _Zelotes_, _Tarantula_, and the web-builders _Ciniflo_ and _Aranea_. The British domestic cockroaches were accepted by _Tegenaria_ and _Ciniflo_, spiders large enough to overpower them, and were useful as food for tropical avicularids, ctenids, and sparassids in captivity.

Family THERAPHOSIDAE

=Avicularia avicularia= (Linnaeus) and =Avicularia= sp.?

Common name.--Bird-eating spider.

Natural prey.--_Periplaneta americana_, Trinidad (Main, 1924, 1930): The remains of the host were compressed into globular form by the spider after it had extracted the nutritive parts.

=Phormictopus cancerides= (Latreille)

Experimental prey.--Cockroach, West Indies (Wolcott, 1953).

Family SPARASSIDAE

=Heteropoda venatoria= (Linnaeus)

Synonymy.--_Heteropoda regia_ Fabricius.

Common names.--Banana spider (Comstock, 1912); huntsman spider (Gertsch, 1949); big brown house spider (Bryan, 1915).

Natural and experimental prey.--Cockroaches, Bermuda (Verrill, 1902); Puerto Rico (Sein, 1923; Wolcott, 1924a; Petrunkevitch, 1930a); Hawaii (Bryan, 1915, Williams et al., 1931); British Guiana (Moore _in_ Williams et al., 1931); Panama (Gertsch, 1949); New Zealand (adventive) (Parrott, 1952); England (Cloudsley-Thompson, 1953); Comstock (1912); Hawaii (Pemberton, 1917).

This (pl. 30, A) is a tropical species frequently imported into northern localities with bunches of bananas (Comstock, 1912; Cloudsley-Thompson, 1953). Adults measure 3 to 4 inches across with bodies over an inch long. They seldom leave their resting places during the day, but are active at night and search for food. The female does not spin a web (Bryan, 1915; Gertsch, 1949). The spider turns the cockroach over onto its back at the instant of seizure and holds it firmly against the substrate. The cockroach dies in 10 minutes and is gradually rolled up by the spider as it sucks

out the nutriment (Moore _in_ Williams et al., 1931). The spider does not attempt to bite when captured, but if it does, its bite is said to be painful but not dangerous (Cloudsley-Thompson, 1953). Zimmerman (1948) found scores of _Periplaneta australasiae_ breeding in rock piles in Hawaii; also present were large numbers of these spiders and centipedes which presumably preyed upon the cockroaches.

Family THERIDIIDAE

=Latrodectus indistinctus= Pickard-Cambridge

Common name.--Button spider.

Natural prey.--_Karnyia discoidalis_, South Africa, Western Cape Province (Hesse, 1942): The nest is constructed on the ground among grass stems or other vegetation. Preferred sites are slight hollows, hoof imprints, etc. Nests are roughly tubular. The remains of insects are entangled in the walls of the nest where they form dense accumulations. Predatory activities of the spider are limited to an area close to the tubular entrance to the nest and do not extend beyond the trapping strands near the entrance. Capture is dependent upon accidental contact of the insect with sticky threads surrounding the entrance. This spider apparently attacks any insect or arachnid that becomes entangled in the nest. In an examination of 40 nests, remains of 6 _K. discoidalis_ were found.

=Latrodectus mactans= (Fabricius)

Common names.--Black widow, hourglass, or shoe-button spider.

Natural prey.--Cockroaches, Puerto Rico (Petrunkevitch, 1930). U.S.A., Florida, on shipboard (Anonymous, 1939): This is a presumptive host record, as the spiders were not reported as having been seen eating cockroaches; however, heavy infestations of both were found together.

Family LYCOSIDAE

=Lycosa helluo= Walckenaer

Experimental prey.--Young nymphs of _Diploptera punctata_, U.S.A. (Eisner, 1958): Larger nymphs and adults repelled the spider by ejecting a repellent secretion, which has been identified as a mixture of _p_-benzoquinone and its derivatives by Roth and Stay (1958).

=Lycosa= sp.

Experimental prey.--_Supella supellectilium_, U.S.A. (Roth and Willis, unpublished data, 1953): The lycosid (pl. 30, B-E) was probably _L. avida_ Walckenaer (tentatively identified by Dr. B. J. Kaston from a photograph).

Order ACARINA

Family PHYTOSEIIDAE

=Blattisocius tineivorus= (Oudemans)

Synonymy.--_Blattisocius triodons_ Keegan [Baker and Wharton, 1952].

Natural host.--_Blattella germanica_, U.S.A. (Keegan, 1944): Three mites found on 238 cockroaches examined; others taken in debris from floor of cockroach cage (Keegan, 1944). Members of this family are predaceous (Baker and Wharton, 1952).

Family LAELAPTIDAE

=Blattilaelaps nauphoetae= Womersley

Natural host.--_Nauphoeta cinerea_, Australia, Brisbane (Womersley, 1956).

=Coleolaelaps= (?) sp.

Natural host.--_Gromphadorhina portentosa_, the hosts were imported into U.S.A. from Madagascar via Europe (Roth and Willis, unpublished data, 1958): The mites (pl. 12, C) were tentatively determined by Dr. E. W. Baker.

=Hypoaspis= sp.

Natural host.--_Panesthia australis_, imported into U.S.A. from Australia (Roth and Willis, unpublished data, 1955): Cockroach determined by J. A. G. Rehn. Generic determination of mite made by Dr. R. W. Strandtmann (Camin, personal communication, 1955).

Family UROPODIDAE

=Uropoda= sp.

Natural host.--_Blattella humbertiana_, Formosa (Takahashi, 1940). Nymphs of the cockroach may be destroyed (Takahashi, 1940). Uropodids frequently attach themselves to insects, especially in nymphal stages but probably are harmless (Baker and Wharton, 1952).

Family DIPLOGYNIIDAE

=Undetermined diplogyniid=

Natural host.--_Panesthia australis_, imported into U.S.A. from Australia (Roth and Willis, unpublished data, 1955): Cockroach determined by J. A. G. Rehn. According to Dr. J. H. Camin (personal communication, 1955) this is a new genus and new species in the subfamily Diplogyniinae, and is most closely related to the genus _Lobogynioides_. Mites of this family live as ectoparasites and commensals on beetles and possibly other insects (Baker and Wharton, 1952).

Family ANOETIDAE

=Histiostoma feroniarum= (Dufour)

Natural host.--_Pycnoscelus surinamensis_, Germany (Roeser, 1940): Though not parasitic, the mites at times became so numerous that the insects were hindered in their movement, were unable to feed, and died. The mites were introduced with soil and leaves and had originally been attached to millipedes, waterfleas, and sowbugs.

The deutonymphs, hypopial forms, or travelers are found on insects; the other stages are found in decaying organic matter (Baker and Wharton, 1952).

Family ACARIDAE

=Caloglyphus spinitarsus= (Hermann)

Natural host.--_Pycnoscelus surinamensis_, Germany (Roeser, 1940): See notes following _Histiostoma feroniarum_ above.

=Caloglyphus= sp.

Natural hosts.--_Blattella germanica_ and _Periplaneta americana_, U.S.A. (Piquett and Fales, 1952): Mite feeds on organic matter but can reduce the vigor of a cockroach colony.

=Tyrophagus lintneri= (Osborne)

Common name.--Mushroom mite.

Associate.--_Pycnoscelus surinamensis_, U.S.A. (Roth and Willis, unpublished data, 1953): Mite determined by Dr. E. W. Baker (personal communication, 1953). Although this mite was found on the cockroach, it is a known pest in stored foods (Baker and Wharton, 1952) and probably was brought into the culture with food. Rau (1924) reported that the food of _Blatta orientalis_ often became infested with this species, but it did not affect the health or mortality of the cockroaches in his culture.

=Tyrophagus noxius= A. Z.

Natural host.--_Periplaneta americana_, U.S.A. (Roth and Willis, unpublished data, 1953): Mite determined by Dr. E. W. Baker (personal communication, 1953). Mites were found in the oöthecal cavity of a female cockroach that had been isolated for her entire adult life. The mites were in a closely packed mass behind a plug of what appeared to be feces, disintegrated eggs, and dried blood; none of the mites were visible until this plug was removed. Baker (personal communication, 1953) stated that the mite is probably not parasitic and that species of the genus feed on organic matter.

=Rhizoglyphus tarsalus= Banks

Natural host.--_Periplaneta americana_, U.S.A. (Rau, 1940a): Not normally parasitic on cockroaches, but the mites became so numerous at times they would attack living as well as dead and dying cockroaches.

Family GLYCIPHAGIDAE

=Chaetodactylus= sp.

Synonymy.--_Trichotarsus_ sp. [Baker and Wharton, 1952].

Natural host.--_Leucophaea maderae_, Puerto Rico (Seín, 1923): Mites found on cockroach's thorax and particularly among the folds of the wings (Seín, 1923). Mites of this genus are found infesting organic matter (Baker and Wharton, 1952).

Family PODAPOLIPODIDAE

=Locustacarus= sp.

Natural hosts.--_Diploptera punctata_ and _Nauphoeta cinerea_, U.S.A. (Roth and Willis, unpublished data, 1954): Mite genus determined by Dr. E. W. Baker (personal

communication, 1954). The mites cluster thickly on intersegmental membranes, particularly around the coxae and neck. Despite a heavy infestation, the colony of _Nauphoeta_ thrived for several years. This mite was found first on _N. cinerea_ and possibly transferred to _D. punctata_ when the latter was brought into the laboratory from Hawaii.

Family IOLINIDAE

=Iolina nana= Pritchard

Natural hosts.--_Blaberus craniifer_ (originally from a culture at Harvard University) and _Diploptera punctata_ (originally from Hawaii), U.S.A., Pennsylvania (Roth and Willis, unpublished data, 1953; Pritchard, 1956): The mites usually attached near the wing bases of the insects. Morphologically, the species is intermediate between certain predaceous and phytophagous mites (Pritchard, 1956).

Family PTERYGOSOMIDAE

=Pimeliaphilus podapolipophagus= Trägårdh

Common name.--Cockroach mite.

Natural hosts.--_Parcoblatta_ sp., U.S.A. (Edmunds, 1953a).

Periplaneta americana, U.S.A. (Piquett and Fales, 1952).

Cockroaches. U.S.A. (Baker and Wharton, 1952).

Experimental hosts.--_Blatta orientalis_, _Blattella germanica_, and _Periplaneta americana_, U.S.A. (Cunliffe, 1952).

Eggs of this mite (fig. 4) are usually laid indiscriminately in the rearing cages, rarely on the host. Eggs are coated with a sticky secretion which enables those laid on the host to adhere. Hatching occurs in 6-11 days at 90-95° F., and in 9-11 days at 80° F. The newly hatched larva starts to feed immediately on the cockroach. Larval stage lasts 4-6 days, rests 2-3 days, and molts. During the single nymphal instar, the mite feeds on the host and moves about for 6-7 days. The mite then rests 3-4 days before molting. Entire life cycle covers a period of 28-32 days. Adult mite lives 2-3 weeks, during which time it can produce 2-3 batches of from 1 to 20 eggs; the usual batch is about 12 eggs. The mites are unable to live on cockroach feces, cast skins, or dead cockroaches. Mites died within 4-5 days unless live cockroaches were supplied. Parasitism was proved by detecting radioactivity in mites that had fed on cockroaches which had been previously fed radioactive NaCl (Cunliffe, 1952).

The mites can destroy laboratory cultures of cockroaches (Piquett and Fales, 1952; Edmunds, 1953a). A cockroach attacked by 25 mites succumbed after about an hour, falling on its back; it died after 5 hours (Cunliffe, 1952).

When found in homes and offices, these mites are an indication of the presence of cockroaches; the mite has been twice accused of biting people (Baker et al., 1956).

RECORDS OF UNIDENTIFIED MITES

Natural hosts.--_Aglaopteryx facies_, Puerto Rico (Seín, 1923): Four red "tick" nymphs found under wings of female.

Blaberus craniifer, U.S.A., Florida (Hebard, 1917): "A number of lice [mites] are present on many of these specimens [28[F][F]]."

Blaberus discoidalis, adventive from West Indies, taken in Scotland (Stewart, 1925): A considerable number of mites were all over the body and hind wings.

Blatta orientalis, Germany (Cornelius, 1853): Ex sexual organs of male.

Blattella germanica, U.S.A., in laboratory (Parker, 1939): Under conditions of high humidity, the cockroaches became heavily infested with mites. In cages where the infestation was heavy, an abnormally large number of females dropped their oöthecae, and the percentage of eggs hatching was low.

Parcoblatta uhleriana, U.S.A., North Carolina (Hatcher, 1939): Hypopi of mites were found deeply embedded in the fat body of two individuals.

Mites in the hypopial stage attach to insects by which they are dispersed. Hypopi have been found in the gill chambers of a mollusk and in the gonads of a millipede (Baker and Wharton, 1952).

Periplaneta americana, U.S.A., in laboratory (Fisk, 1951): The insects were sluggish and molted with difficulty. Gold Coast Colony (Macfie, 1922): Larvae of a tarsonemid mite were found in the feces.

Pycnoscelus surinamensis, Hawaii (Illingworth, 1915): During the summer the soil was literally swarming with young of various stages. Early in September most of the adults were dead and all were covered with mites. U.S.A., Connecticut, in laboratory (Zappe, 1918a). Hawaii, in laboratory (Schwabe, 1950): Some of the cockroaches apparently died from mite infestations.

[Illustration: FIG. 4.--The cockroach mite, _Pimeliaphilus podapolipophagus_. (From Baker et al. [1956]; reproduced through the courtesy of Dr. E. W. Baker and the National Pest Control Association.)]

Cockroach, England? (Ealand, 1915): Cockroaches may carry the hypopial stage of the cheese mite.

CONTROL OF MITES IN COCKROACH COLONIES

Fisk (1951) eliminated the mites [possibly _Pimeliaphilus podapolipophagus_ (Baker et al., 1956)] in his cockroach colony by using a 5-percent spray and a 5-percent dust of p-chlorophenyl, p-chlorobenzene sulfonate. The exterior of the cockroach containers were sprayed with the solution and the interior, including the insects, were dusted. Within a month the mites had disappeared and the vigor of the cockroach colony improved. Piquett and Fales (1952) used flowers of sulfur and general sanitary procedures for eliminating the mites in laboratory colonies of _Blatta orientalis_; they cleaned the dishes every few days and applied grease around the edges of the containers to prevent new mite invasions. Qadri (1938) employed similar

control measures.

Class CHILOPODA

Large centipedes which entered houses in India probably sought out cockroaches (Maxwell-Lefroy, 1909). In Puerto Rico, centipedes entered homes to which they were attracted by cockroaches (Seín, 1923). In Hawaii, centipedes preyed on insects generally but especially on cockroaches (Bryan, 1915). Sonan (1924) reported that in Formosa and Okinawa Islands a species of centipede 5 to 6 inches long comes into the houses and devours both adults and nymphs of _Periplaneta americana_ or _P. australasiae_. Zimmerman (1948) found _P. australasiae_ breeding by scores in rock piles in Hawaii accompanied by large numbers of _Scolopendra_ and large spiders that probably preyed upon the cockroaches.

Order SCUTIGEROMORPHA

Family SCUTIGERIDAE

=Scutigera coleoptrata= (Linnaeus)

Synonymy.--_Scutigera forceps_ Rafinesque [Crabill, 1952].

Common name.--House centipede.

Natural prey.--Cockroaches, U.S.A. (Felt, 1909; Back, 1947; Auerbach, 1951; Crabill, 1952; and others): This predator-prey relationship seems to be based on good circumstantial evidence (Crabill, personal communication, 1953).

Experimental prey.--_Blattella germanica_, newly hatched nymphs and adult female, U.S.A. (Snodgrass, 1930; Roth and Willis, unpublished data, 1953).

Periplaneta americana, U.S.A. (Roth and Willis, unpublished data, 1953).

Supella supellectilium, U.S.A. (Roth and Willis, unpublished data, 1953): See plate 31.

Our specimen caught a small American cockroach nymph that we placed in its jar. Before it had finished its meal, it caught and held two other nymphs with its legs while it continued to feed on the first. The body of this centipede reaches a maximum length of 27 mm. and it is usually found in basements, dark corners, or in spaces in the walls (Auerbach, 1951). Introduced from Europe, this species is now widespread in the United States (Crabill, 1952).

=Allothereua maculata= (Newport)

Synonymy.--_Scutigera maculata_ [Crabill, personal communication, 1957].

Natural prey.--Cockroaches, Malay peninsula, Batu caves (Ridley in Annandale et al., 1913): This is a presumptive host record.

Order SCOLOPENDROMORPHA

Family SCOLOPENDRIDAE

=Scolopendra cingulata= Latreille

Experimental prey.--Cockroaches, England (Cloudsley-Thompson, 1955): After capture in France, this specimen was kept for four weeks without food. She was then fed medium-sized nymphal cockroaches of which she ate an average of about one per week throughout the summer. Adult cockroaches were attacked only after they had been disabled.

=Scolopendra morsitans= Linnaeus

Natural prey.--Cockroaches, Guadeloupe (Lherminier, 1837).

Experimental prey.--Cockroaches, India, Nagpur (Jangi, 1955): As soon as the centipede became aware of its prey, it rapidly embraced the cockroach within its legs and with its fangs gripped the insect's thorax. The predator continued to hold the prey with its fangs while its mouth parts prodded the victim's body. After feeding on an adult cockroach, the centipede is not inclined to kill another for 2-3 days.

=Scolopendra subspinipes= Leach

Natural prey.--Cockroaches, Hawaii (Williams et al., 1931): This is a common species with a body length of 6 or more inches. It is reported to be a great enemy of cockroaches.

=Scolopendra= sp.

Natural prey.--_Ectobius panzeri_, England (Lucas, 1911, 1920): When captured, the centipede was holding a live cockroach which it had apparently just caught. The insect was held beneath its captor's body, ventral surface upward, by several of the anterior legs while the centipede fed.

Class INSECTA

We have found representatives of only 10 orders that have preyed on or parasitized cockroaches: Beetles, flies, bugs, ants, wasps, stylops, and cockroaches occurred in nature; the others resulted from feeding cockroaches to captive insects or were laboratory observations.

Order ODONATA

Family AESHNIDAE

=Anax strenuus= Hagen

Common name.--Giant Hawaiian dragonfly.

Experimental prey.--Cockroaches, Hawaii (Williams, 1936): The dragonfly nymph was fed with medium large cockroaches and other insects.

Order BLATTARIA

In this chapter the relations of other arthropods to cockroaches are either as parasites or as predators. Certain cockroaches have turned the tables on their adversaries and become predators themselves. This aspect of cockroach behavior is discussed in chapter XVI. Other associations of cockroaches, as commensals with other insects and as associates of other cockroaches, are discussed in chapters XV and XVII.

Order ORTHOPTERA

Family MANTIDAE

=Hierodula tenuidentata= (Saussure) (?) (Serville)

(Pl. 32)

Experimental prey.--_Blatta orientalis_, _Diploptera punctata_, _Eurycotis floridana_, _Leucophaea maderae_, _Nauphoeta cinerea_, _Neostylopgya rhombifolia_, and _Periplaneta americana_, U.S.A. (Rilling, personal communication, 1957): Mrs. Rilling wrote us that with the exception of _N. rhombifolia_, all the above cockroaches were readily eaten. All the mantids initially rejected _N. rhombifolia_ after grasping and making a brief attempt to chew the cockroaches. However, if specimens of _N. rhombifolia_ were left in the jars with the mantids, the cockroaches were usually eaten within the next 24 hours. _N. rhombifolia_ ejects an odorous substance when seized and the mantids probably ate these insects after most of this secretion had been depleted. It is highly probable that the secretion of _N. rhombifolia_ may deter the mantid's attack, but it should be pointed out that, with the possible exception of _N. cinerea_, all the other species fed to these mantids give off odorous substances when seized or disturbed. Apparently, certain naturally repellent compounds will deter this mantid, whereas others that are presumed to be repellent will not; however, the nutritional state of the mantid is undoubtedly a factor which may limit the effectiveness of certain repellent secretions against this predator.

Byrsotria fumigata, teneral males, and _Periplaneta australasiae_, nymphs, U.S.A. (Roth and Willis, unpublished data, 1958).

Diploptera punctata, U.S.A. (Eisner, 1958).

=Mantis religiosa= Linnaeus

Common name.--European mantis.

Experimental prey.--_Nauphoeta cinerea_, and _Periplaneta americana_, U.S.A. (Rilling, personal communication, 1957).

=Metallyticus semiaeneus= Westwood

Experimental prey.--Cockroaches, Borneo (Shelford, 1916).

=Sphodromantis viridus= (Forskål)

Synonymy.--_Sphodromantis bioculata_ Burmeister [Gurney, personal communication, 1958].

Experimental prey.--_Blatta orientalis_, Egypt (Adair, 1923): This species of cockroach was apparently used regularly as food for the mantid in the laboratory.

=Stagmomantis carolina= (Johansson)

Common name.--Carolina mantis.

Experimental prey.--_Blattella germanica_ and _Periplaneta americana_, U.S.A. (Breland, 1941): The mantids were fed 1-2 German cockroaches daily. One female

mantid consumed 10 adult German cockroaches plus one oötheca and part of another in 2.5 hours. An adult German cockroach was consumed in an average of 8.5 minutes (range 5.5-15 minutes).

Blatta orientalis, nymphs, and _Diploptera punctata_, U.S.A. (Roth and Willis, unpublished data, 1953).

=Tarachodes maurus= (Stal)

Experimental prey.--Cockroaches, South Africa (Faure, 1940).

=Tenodera aridifolia sinensis= Saussure

Common name.--Chinese mantis.

Experimental prey.--_Nauphoeta cinerea_ and _Periplaneta americana_, U.S.A. (Rilling, personal communication, 1957).

Family GRYLLACRIDIDAE

=Diestrammena apicalis= Br. v. Wattenwyl

and

=Diestrammena japanica= Blatchley

Natural prey.--Cockroach eggs, Japan (Asano, 1937): These are questionable records. Asano found _D. apicalis_ and _D. japanica_ beneath his house near several empty cockroach oöthecae which appeared to have been eaten into. He assumed from the condition of the oöthecae and the proximity of the stone crickets that the insects had devoured the cockroach eggs.

Experimental prey.--Eggs of _Blattella germanica_ and _Periplaneta japanica_, Japan (Asano, 1937): Seven eggs of _B. germanica_ (obtained from an oötheca being carried by a female) and eggs of _P. japanica_ (presumably in oöthecae) were fed to both species of stone crickets in the evening. The eggs were devoured by the next morning.

Order DERMAPTERA

Family FORFICULIDAE

=Undetermined earwigs=

Experimental prey.--Cockroaches, France (Chopard, 1938): According to Chopard, Brisout de Barneville in 1848 indicated that earwigs in captivity can be fed small cockroaches.

Order HEMIPTERA

Family LYGAEIDAE

=Clerada apicicornis= Signoret

Natural prey.--Cockroach, Hawaii (Illingworth, 1917): This predaceous bug is commonly found about buildings. Illingworth says that Kirkaldy suspected that it fed on small blattids and that Dr. Perkins saw it feeding on a dead cockroach.

Family REDUVIIDAE

=Spiniger domesticus= Pinto

Natural prey.--_Periplaneta americana_, Brazil, Matto Grosso (Pinto, 1927, 1927a): This bug preys principally on cockroaches and was observed infesting the walls of dwellings where it preyed on _P. americana_.

=Triatoma arthurneivai= Lent and Martins

Natural prey.--_Monastria_ sp., Brazil, Minas Gerais (Martins, 1941): This bug probably feeds on cockroaches of this genus, as well as on rodents.

=Undetermined reduviids=

Natural prey.--_Arenivaga roseni_ and _Polyphaga saussurei_, Turkmen S.S.R. (Vlasov and Miram, 1937): These desert cockroaches are found in burrows of rodents and desert turtles around Ashkhabad. Reduviids are their main enemies. Vlasov (1933) found nymphs of _Reduvius christophi_ Jak. and _R. fedtschenkianus_ Osch. in similar burrows in this same area, although he did not specifically cite them as enemies of the desert cockroaches.

Family NEPIDAE

=Ranatra= sp.

Experimental prey.--Cockroaches, U.S.A. (Hoffman, 1924).

Order NEUROPTERA

Family ASCALAPHIDAE

=Undetermined larva=

Experimental prey.--_Blattella germanica_, Kenya Colony (Someren, 1924).

Order DIPTERA

From the few observations that have come to our attention, it seems that flies are comparatively rare parasites in cockroaches.

Family PHORIDAE

=Megaselia= sp.

Host.--Eggs of _Parcoblatta_ sp., Ohio (Edmunds, 1952a).

Family CONOPIDAE

=Stylogaster stylata= (Fabricius)

Hosts.--Cockroaches, Brazil (Souza Lopes, 1937): L. Travassos was quoted as having observed this species pursue cockroaches that were escaping columns of the army ant _Eciton_ sp. Souza Lopes (1937) stated that the female deposits eggs on the cuticle of the host near the end of the body; the egg is barely inserted and two recurrent hooks prevent it from falling off. Souza Lopes (1937) also observed other species of _Stylogaster_ pursue Orthoptera, but he was unable to devote proper

attention to the behavior of the flies.

=Stylogaster= spp.

Hosts.--_Chorisoneura_ sp., Brazil (Souza Lopes, 1937): An adult specimen was found in a museum collection with an egg of _Stylogaster_ attached to the posterior end of its abdomen.

Cockroaches, Panama (C.W. Rettenmeyer, personal communication, 1959): "Seven species were collected hovering over army ant swarms and a few flies were seen apparently attacking cockroaches that had been flushed by the ants."

Family LARVAEVORIDAE

=Calodexia= (?) =venteris= Curran

Hosts.--_Periplaneta americana_, Brazil (Souza Lopes, 1937): Obtained complete evolution of the parasite in this host. This may have been an experimental host.

=Calodexia= spp.

Hosts.--Cockroaches, Panama (Rettenmeyer, personal communication, 1959): Swarms of army ants are accompanied by about 20 species of _Calodexia_. These flies larviposit on the cockroaches, crickets, and possibly other arthropods that are flushed from cover by the ants. Larvae were found in one(?) cockroach. Larvae from an adult of _Calodexia_ were introduced experimentally into a cockroach and successfully reared.

=Undetermined tachinids=

Hosts.--_Eurycotis floridana_, from Florida (Roth, unpublished data, 1953): Three larvae (det. by W.W. Wirth) were found in a living adult male.

Panesthia australis, from Australia (Roth, unpublished data, 1957): Reared from a wild-caught cockroach that was maintained in a laboratory colony.

Cockroaches, Australia (E. F. Riek, personal communication, 1955): Reared from some of the larger species.

Family MUSCIDAE

=Coenosia basalis= Stein

Host.--Eggs of _Parcoblatta_ sp., Ohio (Edmunds, 1952a).

Family SARCOPHAGIDAE

=Sarcophaga omani= Hall

Host.--_Arenivaga bolliana_, Texas (Wirth, personal communication, 1953): Specimens in U.S. National Museum.

=Sarcophaga lambens= Wied.

Synonymy.--_Sarcophaga sternodontis_ (Towns.).

Hoffman (1927) claimed that approximately 40 percent of some specimens of _Pycnoscelus surinamensis_ collected in southern Haiti were parasitized by _S. lambens_. However, according to entomologists at the University of Puerto Rico Agricultural Experiment Station, Hoffman was incorrect in his observations: _S. lambens_ was never reared from a living insect and had been recovered only from dead cockroaches and other dead insects and was considered saprophytic rather than parasitic (Schwabe, 1950b).

=Sarcophaga= spp.

Sanjean (1957) reared various species of sarcophagid larvae on _Periplaneta americana_ which were freshly killed or chopped up; first instar larvae were also introduced into the body cavity of cockroaches which had their heads and legs removed. Adult sarcophagids were collected and freshly killed American cockroaches used as bait.

Order COLEOPTERA

Family CARABIDAE

=Harpalus pennsylvanicus= De Geer

Experimental prey.--_Cryptocercus punctulatus_, U.S.A. (Cleveland et al., 1934): This beetle is often found in the galleries of _C. punctulatus_ in nature. In the laboratory it killed and devoured cockroaches as large as itself.

Family DYTISCIDAE

=Rhantus pacificus= Boisduval

Experimental prey.--Cockroaches, disabled, Hawaii (Williams, 1936): This beetle, which is common in mountain streams, located wounded cockroaches in an aquarium by sense of smell or taste rather than sight.

Family LAMPYRIDAE

=Undetermined larva=

Experimental prey.--_Parcoblatta virginica_, adult female (pl. 33, C), U.S.A. (Roth and Willis, unpublished data, 1953).

Family RIPIPHORIDAE[5]

=Neonephrites partiniger= Riek

Natural host.--Cockroach (undescribed genus belonging to the Pseudomopinae), Australia Capital Territory (Riek, 1955).

=Neorhipidius neoxenus= Riek

Natural host.--_Robshelfordia longiuscula_ or _Robshelfordia circumducta_, Australia Capital Territory (Riek, 1955).

=Paranephrites xenus= Riek

Natural host.--_Oniscosoma granicollis_, Australia Capital Territory (Riek, 1955).

=Rhipidioides ableptus= Riek

Natural host.--_Balta patula_, Australia, Victoria (Riek, 1955): Pupal stage lasted only 3 days.

=Rhipidioides adynatus= Riek

Natural host.--_Escala_ sp. or an undescribed genus of Pseudomopinae, Australia, Victoria (Riek, 1955).

=Rhipidioides fuscatus= Riek

Natural host.--_Ellipsidion affine_, Australia, New South Wales (Riek, 1955).

=Rhipidioides helenae= Riek

Natural host.--_Robshelfordia longiuscula_ or _Robshelfordia circumducta_, Australia Capital Territory (Riek, 1955).

=Rhipidioides mollis= Riek

Natural host.--_Robshelfordia longiuscula_ or _Robshelfordia circumducta_, Australia Capital Territory (Riek, 1955).

=Rhipidioides rubricatus= Riek

Natural host.--_Choristima_ sp. and _Choristimodes_ sp., Australia Capital Territory (Riek, 1955).

=Riekella australis= (Riek)

Synonymy.--_Nephrites australis_ Riek [Selander, 1957].

Natural host.--_Cutilia_ sp., Australia Capital Territory (Riek, 1955): Two females emerged from one host.

=Riekella nitidioides= Selander

Synonymy.--_Nephrites nitidus_ of Riek not Shuckard [Selander, 1957].

Natural host.--_Platyzosteria_ sp., Tasmania (Riek, 1955).

=Riekella= sp.

Synonymy.--_Nephrites_ sp. [Selander, 1957].

Natural host.--_Platyzosteria castanea_, Australia Capital Territory (Riek, 1955).

Biology of Australian Ripidiini.--The Australian species of Ripidiini are parasites of apparently endemic, ground-dwelling species of cockroaches. There is some correlation between host subfamily and parasite genus: _Riekella_ spp. [= _Nephrites_] have only been bred from Blattinae. _Rhipidioides_ spp. occur only in the closely related Ectobiinae and Pseudomopinae. _Neonephrites_ and _Neorhipidius_ also occur in the Pseudomopinae. _Paranephrites_ occurs in the Panchlorinae. There is some evidence that the parasitized cockroaches migrate onto trees when the larval parasite is mature, as pupae have only been found on the trunks of eucalyptus trees. In all species the larva leaves the host dorsally through an

intersegmental membrane. The host continues to live for a few days after the parasite emerges. The larva attaches itself to bark on the tree trunk by a few strands of silk before pupating. The larviform, wingless female remains near the pupal skin and is sought out by the winged male. The eggs are laid in a mass around the pupal skin (Riek, 1955).

=Ripidius[5] boissyi= Abeille

Balduf (1935) lists _Ripidius boissyi_ as parasitic on nymphs of _Ectobius pallidus_ giving Abeille de Perrin (1909) as a source for this information. However, Abeille de Perrin simply presumed that _R. boissyi_ parasitized _E. pallidus_ because he collected this cockroach in the same habitat as the beetle. Abeille de Perrin suggested that the species of the genus _Ripidius_ lived in the bodies of cockroaches, but there are no rearing records, as far as we know, of _R. boissyi_ from cockroach hosts.

=Ripidius denisi= Chobaut

Chobaut (1919), in France, collected both _R. denisi_ and _Ectobius pallidus_ when beating an oak tree. Because of the known association of other species of _Ripidius_ with cockroaches, he presumed that this beetle was parasitic on _E. pallidus_, a cockroach common in this beetle's habitat.

=Ripidius pectinicornis= Thunberg

Synonymy.--_Symbius blattarum_ Sundevall [Leng, 1920].

Natural hosts.--_Blattella germanica_, on shipboard (Sundevall, 1831); Germany (Aclogue and Fowler, _in_ Burr, 1899a); on steamship "Samui" (Stamm, 1936); on cruiser "Duguay-Trouin" (Barbier, 1947); Hawaii (Williams, 1946a): This last record was based on a specimen dissected from an adult German cockroach collected on an airplane from the South Pacific. The parasite was reported as _Ripidius_ sp. by Williams, but Weber (1948) made the specific identification.

Ectobius pallidus? Abeille de Perrin (1909) stated that _R. pectinicornis_ was first described by Sunders [sic] as _blattarum_ because it had been captured in the body of _Ectobia livida_. We presume that Abeille de Perrin was referring to Sundevall's work in which the host was given as _Blattella germanica_.

Periplaneta americana, on shipboard (Sundevall, 1831): One nymph only.

With the exception of the single nymph of _P. americana_, _R. pectinicornis_ apparently attacks only adult females and nymphs of _B. germanica_. Barbier (1947) found only _B. germanica_ parasitized, although both _Blatta orientalis_ and _Supella supellectilium_ were prevalent on board the ship. Primary larvae of the parasite failed to parasitize _Supella_.

Adult behavior.--The winged male is relatively active compared to the apterous female; it runs around, flies well, and jumps on the female when in her vicinity. The female remains stationary and lays eggs around her by bending her long ovipositor (Sundevall, 1831). The eggs (50-100) are laid among a network of silk fibers secreted by the female. The female dies after completing oviposition (Barbier, 1947).

Development.--The eggs hatch after 14 days, and the primary (triungulin) larvae ascend the host's legs to its body; the larvae then cut the intersegmental membrane between the metasternum and first abdominal segment of the cockroach, in order to enter the host's abdomen (Barbier, 1947). Chobaut (1892) first suggested this method of attack by the ripiphorid larva. As the parasites develop, the abdomen of the host becomes swollen. Developing larvae apparently eat the host's fat body, leaving the vital organs until the last. Parasitized female hosts were sterile and the eggs, when formed, never hatched. Development of the oötheca was also inhibited. There were usually two larval parasites per host, but three or four were found several times (Barbier, 1947). Sundevall (1831) found only one larva per cockroach except one host which, when crushed, yielded five. Stamm (1936) found three hosts infested with five larvae each. In a little over 100 cockroaches, Stamm found 10 that were parasitized.

The day before the parasite leaves the host, the cockroach shows an abrupt uneasiness and runs about, finally falling over on its back. The parasite larva emerges from the host through an opening it makes in the membrane between penultimate and last tergite. The host dies a few hours after the larva has left. The larva seeks a sheltered area and pupates within 48 hours. Adults emerge in 9 days (females) and 13 days (males) (Barbier, 1947).

Distribution.--Adult males have been collected in light traps in Hawaii (Van Zwaluenburg, 1946), and the first female was reported by Weber (1948); the parasite is now established in the islands around Pearl Harbor (Dr. F. X. Williams, personal communication, 1953). The U. S. National Museum has specimens of _R. pectinicornis_ from England, Guatemala, Hawaii, Panama, and from Florida and Georgia in the U. S. (Dr. E. A. Chapin, personal communication, 1953). Kono (_in_ Asano, 1937) reported two species in Japan. It is noteworthy that all these records are from localities adjacent to oceans and on ships; none are from interiors of continents. The only biological data were obtained from parasites found on board ships. Sundevall (1831) believed that the parasites boarded his ship with their hosts during loading in Calcutta, since before that not any were seen on board. Barbier (1947) suggested that the parasite must be spread very easily in ports between neighboring ships by parasitized cockroaches in baskets or sacks of provisions.

=Ripidius scutellaris= Heller

Natural hosts.--Blattidae, Philippine Islands (Schultze, 1925).

Family DERMESTIDAE

=Dermestes ater= De Geer

Common name.--Black larder beetle.

Natural prey.--_Blatta orientalis_, U.S.A. (Roth and Willis, unpublished data, 1953): _Dermestes ater_ is generally a scavenger, but we have seen adult beetles, which had developed in our cockroach colony, clinging to and feeding on living oriental cockroaches, eventually killing them; the beetles probably attack only the weakened or injured cockroaches in a culture. This was a natural infestation of a laboratory culture by a predator.

Experimental prey.--_Blattella germanica_, oöthecae, U.S.A. (Roth and Willis, 1950): The beetle larvae can penetrate unhatched oöthecae of the German but not those of the American or oriental cockroaches.

=Dermestes= sp.

Natural prey.--_Blatta orientalis_, oöthecae, U.S.A., Missouri: Rau (1924) stated that _Dermestes_ larvae often infest the egg cases of this cockroach; it is probable that Rau was referring to cockroaches in laboratory cultures.

Order STREPSIPTERA

Pierce (1909) predicted that the Blattoidea and the Grylloidea would be the only groups of the Orthoptera which would be parasitized by Strepsiptera. Essig (1926) made the statement that certain cockroaches are among the hosts of Strepsiptera. E. F. Riek (personal communication, 1952) found a strepsipteron in a late nymph of _Cutilia_ sp. from Waroona, Western Australia; he wrote us, "The female parasite is extruded between a pair of sternites towards the base of the abdomen and appears to belong to the family Halictophagidae." This is the only record that we have been able to find of a strepsipteron parasitizing cockroaches.

Order HYMENOPTERA

PREDATORS AND PARASITES OF COCKROACH EGGS

Wasps from at least six families of Hymenoptera have been recorded as developing on cockroach eggs. All the Evaniidae are presumed to be parasitic in the egg capsules of cockroaches (Clausen, 1940; Townes, 1951), although hosts for many of the described species have yet to be discovered. The presence of evaniids in dwellings indicates the presence of cockroaches (Gross, 1950). At times these wasps may become a nuisance; a family in Worthington, Ohio, complained of the evaniid wasps that they found on the windows and in other areas of their home, but they were apparently not annoyed by the oriental cockroaches in the basement (Edmunds, 1953).

The known parasites of cockroach eggs are listed below with summaries of their biology.

Family EVANIIDAE

=Acanthinevania princeps= (Westwood)

Synonymy.--_Evania princeps_ [Dr. H. Townes, personal communication, 1956].

Natural host.--Cockroach eggs, Australia (Froggatt, 1906).

=Brachygaster minutus= (Olivier)

Synonymy.--_Evania minuta_ Olivier [Kieffer, 1920].

Natural hosts.--_Blattella germanica_, Europe? (Schletterer, 1889; Kiefer, 1912; Crosskey, 1951.)

Ectobius lapponicus, Europe? (Schletterer, 1889; Kieffer, 1912; Crosskey, 1951.)

Ectobius panzeri var. _nigripes_? Great Britain (Blair, 1952): This is a presumptive record. The wasp was collected at Niton and Headon Hill, Isle of Wight, an area in which this variety of _E. panzeri_ was the only species of cockroach known to occur.

Ectobius sp., England (Cameron, 1955, 1957): Natural History Museum records.

Adult wasps have been collected on _Asparagus officinalis_ Linnaeus (Schmiedeknecht _in_ Schletterer, 1889; Crosskey, 1951). Thompson's (1951) citation of records of _B. minutus_ and _Evania appendigaster_ from _Blatta orientalis_ and _Blattella germanica_, and Cameron's (1957) citation of these records and one from _Ectobius lapponicus_, all attributed to Kadocsa (1921), are almost certainly in error. Kadocsa (1921, p. 33) listed these wasps as egg parasites of cockroaches but not necessarily in Hungary and did not name specific cockroach hosts.

The present writers have found no information, other than host reports, on the biology of _Brachygaster minutus_. The records of this wasp parasitizing _B. germanica_ may trace back to Schletterer, but his listing may not have been an original observation. Since the female of _B. germanica_ carries its oötheca attached to the abdomen until or just before the eggs hatch, it would seem that the female of _B. minutus_ (if the host records are valid) must oviposit into the oötheca of this species while it is still being carried by the female; this would not necessarily be true for the other hosts which drop the egg case long before the eggs hatch.

Distribution.--Europe: Sweden, Russia, England, France, Germany, Austria, Hungary, Switzerland, Italy (Kieffer, 1920).

=Evania appendigaster= (Linnaeus)

Synonymy.--_Evania desjardinsii_ Bordage, _Evania laevigata_ Latreille [Dalla Torre, 1901-1902].

Natural hosts.--_Blatta_, "exotic species" (Westwood, 1854, 1954a).

Blatta orientalis, Europe? (Schletterer, 1886; Howard, 1888, Kieffer, 1912); Egypt? (Alfieri, 1914; Adair, 1923). [Girault (1907, 1914) erroneously attributed another record to Marlatt (1902);[see footnote 6]. See also notes under _Brachygaster minutus_ with respect to Kadocsa.]

Blattella germanica? (Girault 1907, 1914). [This record is obviously an error. Girault attributed the record to Marlatt (1902); see footnote 6.]

Cutilia soror, Hawaii (Swezey, 1929; Zimmerman, 1948).

Leucophaea maderae (Schletterer, 1889; Bordage, 1896; Kieffer, 1912): These records are probably erroneous inasmuch as this cockroach incubates its eggs internally (Roth and Willis, 1954). Later, after finding that _L. maderae_ is ovoviviparous, Bordage (1913) admitted having misidentified a parasitized oötheca from some other species; he concluded that the developing eggs of this species are protected against egg parasites because they are carried within the female. Clausen

(1940), in classifying the placement of parasitic wasp eggs in relation to the host, erected the category: Egg placed in the embryo while the latter is still within the parent. He stated that although this behavior was not definitely known to occur, it probably could occur. However, the records cited above do not indicate that the alleged parasitization followed this pattern.

Neostylopyga rhombifolia, Hawaii (Swezey, 1929).

Periplaneta americana, Europe (Schletterer, 1889; Bordage, 1896; Kieffer, 1912); Réunion Island (Bordage, 1913); Puerto Rico (Seín, 1923); Jamaica (Gowdey, 1925); Hawaii (Swezey, 1929); Palestine (Bodenheimer, 1930); U.S.A., Florida (Ashmead, 1900); Maryland (Piquett and Fales, 1952); Saudi Arabia, Jedda (Cameron, 1957); Canton Island and Samoa (Dumbleton, 1957).

Periplaneta americana or _P. australasiae_, Formosa (Sonan, 1924).

Periplaneta australasiae, U.S.A., Florida (Ashmead, 1900); Hawaii (Swezey, 1929; Zimmerman, 1948). [Girault (1914) erroneously attributed another record to Marlatt (1902); see footnote 6, above.]

Experimental host.--_Blatta orientalis_, U.S.A. (Haber, 1920).

Relatively little detailed information was known about this wasp (fig. 5), one of the earliest parasites of cockroach eggs to be discovered, until Cameron (1957) studied its biology. Arnold (Kirby and Spence, 1826) discovered that the genus _Evania_ parasitized _Blatta_, but did not know whether the wasp developed on the cockroach eggs or in the nymphs. MacLeay (Westwood, 1843) determined that _Evania_ developed within the oöthecae of cockroaches. Westwood (1854a) found the larvae, pupae, and adults of _E. appendigaster_ in egg cases of an unidentified species of cockroach found on orchids received from Calcutta.

Adult behavior.--Adult wasps visited flowers of parsley, _Petroselium crispum_, and fennel, _Foeniculum vulgare_ (Margretti _in_ Schletterer, 1886; Crosskey, 1951). In Hawaii the adult wasps have been seen resting on leaves coated with honey dew (Williams et al., 1931); _Evania_ sp. were attracted to the honey dew secreted by a diaspine scale insect (Williams, 1931). Adults lived two to three weeks in captivity with ample food and water (Cameron, 1957).

Oviposition.--Shelford (1912, 1916) erroneously supposed that _Evania_, by means of her cleaverlike abdomen, opened the oötheca at the crista and then deposited her egg or eggs on the eggs of the cockroach. Haber (1920) observed and described oviposition. The female wasp crawled over the surface of the oötheca, actively vibrating her antennae, and settled with the axis of her body parallel to the axis of the egg case as it lay upon its right side. Lying on her right side, the wasp extended her ovipositor and punctured the oötheca in the fifth cell on the left side; she remained in this position for about 15 minutes. Cameron (1957) described similar oviposition behavior that lasted about half an hour. Kieffer (1912) and Crosskey (1951) stated that the female deposits her eggs before the walls of the oötheca harden.

[Illustration: FIG. 5.--_Evania appendigaster._ Left, dorsal view, × 8. Right, side view, × 5. (Reproduced with permission. British Museum [Natural History], 1951, figs. 1A and 1B.)]

Development.--Kieffer (1912) stated that the larvae in this family eat the cockroach eggs and pupate in the oötheca without forming a cocoon. Smith (1945) stated that the larva feeds on one cockroach egg after another until all are destroyed; by that time it is full grown and it pupates within the oötheca. Cameron (1957) found that there are five larval instars and that in material from Saudi Arabia there are three or possibly four generations a year.

Distribution.--Tropical and subtropical parts of the world as far north as New York City, and all of Europe except the northern part (Kieffer, 1920; Townes, 1949). The wide distribution of _Evania_ has been attributed to the abundance of host cockroaches on ships between the Tropics (Haldeman, 1847). Kieffer (1903) appears to have shown some correlation between the numbers of species of cockroaches found in various geographical regions and the numbers of species of evaniids found in similar regions. However, the number of blattids he listed is small.

=Evania dimidiata= Fabricius

Synonymy.--_Evania abyssinica_ Westwood [Schletterer, 1889].

Natural host.--_Blatta orientalis_, Egypt? (Alfieri, 1914).

=Evania subspinosa= Kieffer

Natural host.--_Periplaneta_ sp., Fiji (Lever, 1946): Although Lever (1946) listed this species as a cockroach-egg parasite, he did not state that he actually reared it from _Periplaneta_ oöthecae.

=Hyptia dorsalis= of Ashmead

Synonymy.--Dr. H. Townes, (personal communication, 1956) believes that this wasp was probably either _H. reticulata_, _H. harpyoides_, or _H. thoracica_; it is not possible to tell which without reexamining Ashmead's specimens; these apparently have been lost.

Natural host.--_Parcoblatta pensylvanica_, U.S.A., Mississippi (Ashmead, 1900).

=Hyptia harpyoides= Bradley

Natural hosts.--_Parcoblatta virginica_, U.S.A., Ohio (Edmunds, 1952a, 1953a, 1954).

Parcoblatta pensylvanica, U.S.A. (Muesebeck, 1958).

Parcoblatta uhleriana, U.S.A., Natick, Mass.: Oötheca collected by L. Roth, May 17, 1956; wasp emerged June 12, 1956 (pl. 33, B); determined by Dr. H. Townes. The keel region of the oötheca of _P. uhleriana_ (pl. 18, B) is different from that of any other species of _Parcoblatta_ (Hebard, 1917; Lawson, 1954) so there can be no doubt as to the species of cockroach parasitized by this wasp.

Development.--The last instar larva overwinters inside the cockroach oötheca (Edmunds, 1954). Five oöthecae yielded one parasite each (Edmunds, 1953a).

Distribution.--Canada, Ontario. U.S.A.: New Hampshire and Minnesota to South Carolina, Mississippi, Texas, and Kansas. Upper and Lower Austral Zones (Townes, 1951).

=Hyptia reticulata= Say

Natural host.--_Parcoblatta pensylvanica_, U.S.A., Missouri (Rau, 1940).

Adult wasps have been taken on parsnip, _Pastinaca sativa_ (Robertson, 1928).

Distribution.--U.S.A.: Pennsylvania to Florida and Louisiana. Mexico. Upper Austral to Tropical Zones (Townes, 1951).

=Hyptia thoracica= (Blanchard)

Natural host.--_Parcoblatta pensylvanica_, U.S.A., Ohio (Edmunds, 1952a, 1953a, 1954).

Adult behavior.--Copulation was rapid, lasting only a few seconds. Blooms of _Asmorrhiza longistylis_ were placed in a cage with adult wasps. The insects were attracted to and fed on the flowers (Edmunds, 1954).

Development.--Entire contents of oötheca are eaten by the single larva. Last instar larva overwinters inside the oötheca. Emergence in Ohio was around the middle of June. The emergence hole made by this genus was about 2 mm. in diameter. The hole was made at the top side of the oötheca near one end. Adult took about 65 minutes to emerge from the time its mandibles first broke through the oöthecal wall. (Edmunds, 1954.)

Distribution.--Canada, Ontario, U.S.A.: Connecticut to Wisconsin, south to Florida and Texas. Upper Austral to Tropical Zones. (Townes, 1951.)

=Hyptia= sp.

Natural host.--_Cariblatta delicatula_, Cuba (Hebard, 1916a); Parasite identified by Ashmead.

=Hyptia= sp. (undescribed)

Natural host.--_Parcoblatta_ sp., U.S.A., Ohio (Edmunds, 1952a).

=Prosevania punctata= (Brullé)

Synonymy.--_Evania punctata_ Brullé [Townes, 1949].

Natural and experimental hosts.--_Blatta orientalis_, Istrian Peninsula (Fahringer, 1922); Algeria (Cros, 1942); U.S.A., Ohio (Edmunds, 1954).

Blattella germanica? Europe? (Girault 1907, 1914); Europe (Fahringer, 1922). [The records on this host are extremely doubtful. Girault erroneously cited Marlatt (1902) as the source of this record; see footnote 6, page 236. Fahringer, however, claimed that he obtained seven female parasites from oöthecae of _Blattella germanica_. He placed female parasites with adults of _B. germanica_ in a glass cage.

As soon as oöthecae could be seen between folds of a woolen rag, he removed all the larger cockroaches and held the oöthecae until the parasites emerged. Fahringer may have been dealing with a different species of cockroach, because placing oöthecae in crevices (or between folds of rag) is a habit foreign to _B. germanica_, the female of which usually carries her oötheca until hatching or until about a day before. Edmunds (1953b) could not induce this wasp to parasitize eggs of _B. germanica_.]

Periplaneta americana, Istrian Peninsula (Fahringer, 1922); Palestine (Bodenheimer, 1930); U.S.A., Ohio (Edmunds, 1952, 1953b, 1954).

Adult behavior.--The wasps (pl. 33, A) are very active; they walk about a great deal and fly short distances. They are often found in abundance in buildings infested with the larger domiciliary cockroaches where they may reproduce for many generations without leaving the premises. Specimens have also been collected outdoors. (Edmunds, 1953, 1954.) As the adult walks about, the laterally compressed abdomen moves up and down like a waving flag; because of this behavior, these insects are commonly known as ensign-flies. Cros (1942) maintained adults 17 days without food. Edmunds (1954) fed adults on unidentified flowers in the laboratory. He also maintained them for 20 days after capture on a 5-percent honey solution.

Oviposition.--A female _P. punctata_ selected oöthecae of _P. americana_ for oviposition and ignored those of _B. orientalis_ and _Parcoblatta pensylvanica_ in the same cage. Oviposition was accomplished as described for _Evania appendigaster_. One oötheca was turned over onto its right side by the wasp before she oviposited. (Edmunds, 1952.) Although there seemed to be a "preferred" position for oviposition, it was not obligatory. The usual position was for the female to face the keel of the oötheca, but she also oviposited from the opposite side or, rarely, directly down into the side of the oötheca. The average time spent by females in 10 ovipositions was 29 minutes (range 16-62 minutes). The wasp apparently could not determine whether the eggs had been previously parasitized. The wasp laid her egg between the cockroach eggs rather than in them and she oviposited into oöthecae that had just been dropped and those two weeks old or older. On three occasions nymphal cockroaches emerged within a few hours after the wasp had oviposited. (Edmunds, 1954.) Apparently, for successful parasitization the wasp must oviposit before the cockroaches have reached the final stages of preemergence development. Edmunds (1954) placed females of _Periplaneta americana_ that were carrying oöthecae, into cages with _Prosevania_; some of the female wasps showed considerable interest in the attached oöthecae, but he observed oviposition only into egg cases that had been dropped by the cockroaches.

Cros (1942) described an interesting reaction that he called "instinctive hostility" of the oriental cockroach toward _Prosevania_. A wasp was placed in a jar in which a cockroach had just deposited its oötheca. The wasp tried to oviposit into the egg case but was upset and pursued by the cockroach. The cockroach placed herself over the oötheca, standing high on her legs, and remained there motionless. The wasp then approached from the rear, slipped under the cockroach, and, unnoticed by the cockroach, climbed on the oötheca and oviposited successfully.

Development.--In _Blatta orientalis_: The developmental period was completed in 40-57 days in summer and fall (Cros, 1942). Time from oviposition to emergence of adult varied from 45-177 days; three parthenotes from an oviposition by an unfertilized female wasp developed in 45-53 days (Edmunds, 1954). In _Blattella germanica_: Almost 4 weeks spent in development (Fahringer, 1922). In _Periplaneta americana_: Three wasps developed in 127 days (Edmunds, 1952). Only one parasite develops in each oötheca. There were three generations a year in Ohio. (Edmunds, 1954.) In Algeria there were two to three generations per year. The adult emerged from the oötheca through a hole 4 mm. in diameter. (Cros, 1942.) Parthenogenesis exists; the unfertilized eggs produced only males (Edmunds, 1954).

Distribution.--Eastern U.S.A., from New York and Ohio south to Georgia (Townes, 1949). Europe, Syria, Palestine (Kieffer, 1920).

=Szepligetella sericea= (Cameron)

Synonymy.--_Evania sericea_ Cameron [Townes, 1949, personal communication, 1956]. _Evania impressa_ Schletterer [Townes, p. c., 1956].

Natural hosts.--_Cutilia soror_ and _Neostylopyga rhombifolia_, Hawaii (Swezey, 1929).

Periplaneta americana and _Periplaneta australasiae_, Hawaii (Swezey, 1929; Zimmerman, 1948).

Periplaneta sp., Fiji (Lever, 1943, 1946).

Adults are sometimes found resting on leaves covered with honey dew (Williams et al., 1931).

=Zeuxevania splendidula= Costa

Natural hosts.--_Loboptera decipiens_, France (Lavagne, 1914; Genieys, 1924).

Picard (1913) believed that _Z. splendidula_ parasitized _L. decipiens_ and not its eggs; however, Lavagne (1914) explained the true relationship by dissecting two specimens of _Z. splendidula_ from oöthecae of _L. decipiens_.

The following information is taken from Genieys (1924): _Oviposition._--Wasp egg is introduced into the still-soft oötheca before the wall hardens. Some oöthecae had four oviposition scars but never contained more than two parasite eggs. _Development._--Larva commences development in July or August. Only one larva completes development, but it eats all the eggs in the oötheca. The wasp passes the winter as a last instar larva and pupates in the spring; the adult emerges during the spring or in June. _Hyperparasitism._--About 10 percent of the oöthecae of _Loboptera decipiens_ that were parasitized by _Z. splendidula_ were also hyperparasitized by an eulophid (see _Syntomosphyrum ischnopterae_, p. 249).

Family CLEONYMIDAE

=Agamerion metallica= Girault

Natural hosts.--_Ellipsidion australe_, Australia, Queensland (Dodd, 1917): "the parasite when ready to emerge fully occupies the whole space of the destroyed eggs."

Cockroach, Australia, New South Wales (Dr. B. D. Girault, 1915a).

Family ENCYRTIDAE

=Blatticida pulchra= Ashmead

Natural host.--Cockroach eggs on orange leaves, Australia, New South Wales (Gahan and Peck, 1946). According to Dr. A. B. Gurney the oötheca associated with the type specimens of the wasps in the United States National Museum is possibly _Balta_ sp. (Burks, personal communication, 1956).

=Blatticidella ashmeadi= (Girault)

Synonymy.--_Blatticida ashmeadi._ _Blatticida_ Girault, 1915, is preoccupied by _Blatticida_ Ashmead, 1904. In 1923 Gahan and Fagan renamed _Blatticida_ Girault, _Blatticidella_. [Burks, p. c., 1956.]

Natural host.--Cockroach, Australia, Queensland (Girault, 1915).

=Cheiloneurus viridiscutum= (Girault)

Synonymy.--_Cristatithorax_ Girault = _Cheiloneurus_ Westwood [Mercet, 1921].

Natural host.--_Ellipsidion australe_, Australia, Queensland (Dodd, 1917).

=Comperia merceti= (Compere)

Synonymy.--_Comperia merceti_ var. _falsicornis_ Gomes [Peck, 1951].

Natural hosts.--_Blattella germanica_, Brazil, Distrito Federal (Gomes, 1941): In the English summary of his paper, Gomes states that _C. merceti_ var. _falsicornis_ was reared from _B. germanica_. However, in the body of the paper, he states that the _supposed_ origin of the parasite was the oötheca of _B. germanica_. Burks (personal communication, 1956) does not believe that this wasp parasitizes the eggs of _B. germanica_. We (unpublished data, 1957) exposed six oöthecae of _B. germanica_ to _C. merceti_. In order to retard water loss the oöthecae were removed from the females by cutting the insects in two so that each oötheca remained attached to the posterior part of the abdomen. No wasps developed in these oöthecae.

Supella supellectilium, U.S.A., Kansas (Lawson, 1954a); Hawaii (Zimmerman, 1944; Compere, 1946; Keck, 1951).

Adult behavior.--Males and nonovipositing females showed a flea-like jumping tendency. Adults were attracted to light and were found near windows. Both sexes pursued an erratic course in walking and continually touched the surface with their antennae. (Lawson, 1954a.)

Oviposition.--The wasp (pl. 34, B) selected a site on an oötheca with the sheath of her ovipositor; it was uncertain whether there was a definite preference for oviposition sites. Wasp tended to choose a nearly horizontal position for oviposition. She preferred to oviposit into eggs about 2 weeks old, although she would place eggs

in oöthecae less than a week old and in embryos in the green band stage. There were 1-50 oviposition punctures per oötheca. (Lawson, 1954a.)

Development.--If enough wasp larvae were present, they ate all eggs in an oötheca. Occasionally wasps developed in one end of an oötheca while cockroaches developed in the other; when this occurred, the cockroach nymphs always emerged last. The developmental period was 30-41 days at room temperature. There were 5-25 parasites per oötheca. The single exit hole in the oötheca varied from 0.6 to 0.9 mm. in diameter. (Lawson, 1954a.)

Distribution.--U.S.A.: New Jersey south to Florida, west to Illinois, Kansas, and Arizona. West Indies; Central and South America; Hawaii. (Burks, personal communication, 1956.)

=Dicarnosis alfierii= Mercet

Natural hosts.--"_Phyllodromia_" sp., Egypt (Mercet, 1930): According to Mercet, Dr. Alfieri claimed that this wasp parasitized one of the species of "_Phyllodromia_" found in Egypt, namely, _Phyllodromia_ [= _Blattella_] _germanica_, _Phyllodromia_ [= _Supella_] _supellectilium_ and/or _Phyllodromia treitliana_. We do not know to which modern genus the host of this wasp belonged.

Cockroach, Egypt? (Mercet in Compere, 1938.)

=Eutrichosomella blattophaga= Girault

Natural host.--Cockroach, Australia, Queensland (Girault, 1915).

Family EUPELMIDAE[7]

=Anastatus blattidifurax= Girault

Natural host.--Cockroach, Australia, Queensland (Girault, 1915).

=Anastatus floridanus= Roth and Willis

Natural host.--_Eurycotis floridana_, U.S.A., Florida (Roth and Willis, 1954a).

Experimental hosts.--_Blatta orientalis_, _Eurycotis floridana_, and _Periplaneta americana_, U.S.A. (Roth and Willis, 1954a).

Adult behavior.--Female wasps are sexually receptive almost immediately on leaving the oötheca. Mating takes 3-4 seconds. Males mate repeatedly and may fertilize several females; females may also mate more than once. At about 80° F. the female wasps lived 2-4 days, males one day.

Oviposition.--The female wasp first probes the oötheca with her sheathed ovipositor until she finds an acceptable spot; she then drills through the wall of the oötheca with her ovipositor. One female oviposited for 5 hours, but briefer periods were more usual. We have seen six or more females ovipositing simultaneously into an oötheca of _Eurycotis floridana_. One female was seen to feed on material that oozed from the oviposition puncture. The wasp (pl. 34, A) may oviposit into the oötheca of _E. floridana_ while it is still being carried by the female, as well as in oöthecae that have been dropped and which have hard walls. Eggs 36 days old were

successfully parasitized.

Development.--In _Eurycotis floridana_: In the laboratory, development was completed in 34-36 days at about 85° F. This time was regulated to some extent by the number of parasites in the oötheca.

There is evidence that larvae eat unhatched wasp eggs or other larvae. In 34 oöthecae exposed to many female wasps, the maximum number of parasites to emerge was 306; yet an average of 601 wasp larvae were dissected from four oöthecae that had each been exposed to 50 female wasps one week earlier. The larvae usually eat all the host eggs. Cockroach eggs that were not eaten by the wasp larvae sometimes developed but usually failed to hatch. Adult wasps made one to six emergence holes in the oötheca; the average number in 42 oöthecae was two holes.

Number of parasites per oötheca.--In _Blatta orientalis_: One of 111 oöthecae exposed to female wasps yielded 48 parasites. In _Eurycotis floridana_: One oötheca parasitized in the field yielded 68 parasites; 8 oöthecae exposed to single wasps for their entire lifespan yielded an average of 50 ± 6 parasites (range 23-81); 34 oöthecae exposed to many wasps for their entire lifespan yielded an average of 198 ± 8 parasites (range 93-306). In _Periplaneta americana_: Nine oöthecae of 152 exposed to the wasps were found to be parasitized when dissected; 11 adults emerged from one oötheca; no parasites emerged from the other 8 oöthecae.

Sex ratio.--4 [F][F]:1 [M] from ovipositions by isolated females. In the one oötheca collected in the field, the ratio was 21.6 [F][F]:1 [M]. Parthenogenesis exists; the unfertilized eggs produced only males.

=Anastatus tenuipes= Bolívar y Pieltain

Synonymy.--_Anastatus blattidarum_ Ferrière. Dr. C. Ferrière (personal communication, 1957) is of the opinion that his _A. blattidarum_ is a synonym of _A. tenuipes_. He stated "I have never been able to see the unique type of _A. tenuipes_ B. y P., which is in Madrid, but the description agrees with _A. blattidarum_. I had not yet knowledge of Bolívar's description, when describing my species. The parasite of cockroaches eggs [_Supella supellectilium_] should be called _A. tenuipes_ Bol." Mani (1938) synonymized _Solindenia blattiphagus_ Mani with _Anastatus blattidarum_.

Natural hosts.--_Supella supellectilium_, Anglo-Egyptian Sudan (Ferrière, 1930, 1935); U.S.A., Arizona (Flock, 1941); Egypt (Alfieri, _in_ Hafez and Afifi, 1956). Ohio (Hull and Davidson, 1958).

Periplaneta americana, India (Burks _in_ Roth and Willis, 1954a).

Cockroach, Hawaii (Weber, 1951); India (Mani, 1936).

The following is based on parasites that developed on eggs of _Supella supellectilium_ (Flock, 1941): _Adult behavior._--Wasp may be seen running rapidly on walls in buildings infested with the cockroach host. The wasp rarely flies but hops proficiently; when disturbed it can hop from several inches to several feet. The female licks up the drop of fluid that oozes from the oviposition puncture. Females die in a few days, but if fed honey and water may live two weeks. _Oviposition._--The female

selects an oötheca by feeling with her antennae. Flock stated, without citing experimental evidence, that the age of the egg case was apparently the chief factor determining choice. The wasp took 15-45 minutes to oviposit. Three females oviposited simultaneously into a single oötheca; a single female repeatedly oviposited into one oötheca at intervals. _Development._--Completed in an average of 32.6 days at a constant temperature of 82° F.

Number of parasites per oötheca.--Average about 10.7 (range 4-16) (Flock, 1941); 15 (Ferrière, 1935).

Sex ratio.--4 [F][F]:1 [M] (Ferrière, 1935); average of 6 [F][F]:1 [M] (Flock, 1941). Parthenogenesis occurs; the unfertilized eggs produced only males (Flock, 1941).

Distribution.--U.S.A.: Maryland, south to Florida, west to Illinois, Kansas, and Arizona. Guatemala; Hawaii; India; Egypt; Sudan. (Burks, personal communication, 1956).

=Eupelmus atriflagellum= Girault

Natural host.--_Blattella germanica_, Australia, Queensland (Girault, 1924).

=Eupelmus= sp.

Natural host.--"Tree cockroach," U.S.A., Florida (Howard, 1892).

=Solindenia picticornis= Cameron

Natural hosts.--_Allacta similis_, Hawaii (Perkins, 1906, 1913; Timberlake, 1924; Swezey, 1929; Zimmerman, 1948).

Other species of cockroaches, Hawaii (Perkins, 1913).

Family PTEROMALIDAE

=Pteromalus= sp.?

Natural host.--_Leucophaea maderae_?, Jamaica (Westwood, 1839; Sells, 1842). [This host is undoubtedly an error. Sells stated that the oötheca which contained 96 unidentified chalcids had 16 dentations at the edge; the description fits the oötheca of an oviparous cockroach and not that of _L. maderae_ (see Roth and Willis, 1954). Westwood (1839, footnote p. 423) stated that at the meeting of the Entomological Society in 1838 Mr. Sells exhibited 94 specimens of a small _Pteromalus_ (apparently identified by Westwood) obtained from one cockroach oötheca. This same record of Sells was published posthumously in 1842, although in this paper he identified the host oötheca as "_Blaberus_" _maderae_. Cameron (1955) lists a European record of _Pteromalus_ sp. from _Periplaneta americana_ citing Girault (1914) as the source of the record. Girault's record was apparently taken from Westwood's footnote mentioned above.]

=Systellogaster ovivora= Gahan

Natural hosts.--_Blatta orientalis_, U.S.A., Illinois (Gahan, 1917).

Parcoblatta pensylvanica, Canada, Ontario (Judd, 1955).

Parcoblatta sp., U.S.A., Ohio (Edmunds, 1952a, 1953a).

"Blattid," U.S.A., Maryland (Gahan, 1917).

One oötheca of _P. pensylvanica_ yielded 14 parasites with a sex ratio of 2.5 [F][F]: 1[M] (Judd, 1955). The average number of parasites in 11 oöthecae of _Parcoblatta_ sp. collected in 1950-51 was 27 wasps (Edmunds, 1952a, 1953a). The adults made two to three emergence holes in the oötheca (Edmunds, 1953a; Judd, 1955).

Family EULOPHIDAE

=Melittobia chalybii= Ashmead

Natural host.--_Periplaneta americana_, U.S.A., Missouri (Rau, 1940a): _M. chalybii_ is normally a parasite of Coleoptera and Hymenoptera (Peck, 1951). This is the only record from cockroach eggs. Burks (personal communication, 1956) stated that this species will attack any insect to which it is exposed and can be a serious pest in insect cultures of practically any insect order. In nature it seems to prefer the nests of aculeate Hymenoptera; Rau suggested that the parasites were probably brought into his laboratory with mud nests of _Sceliphron caementarium_ (Drury).

=Mestocharomyia oophaga= Dodd

Natural host.--_Ellipsidion australe_, Australia, Queensland (Dodd, 1917).

=Syntomosphyrum blattae= Burks

Natural hosts.--_Parcoblatta_ sp., U.S.A., Ohio (Burks, 1952; Edmunds, 1952a, 1953a): Ten oöthecae yielded an average of 92 wasps (Edmunds, 1952a). Five oöthecae, collected a year later, yielded an average of 74 wasps; adults sometimes made two to three exit holes in the oötheca (Edmunds, 1953a).

Cockroach, U.S.A., West Virginia (Burks, 1952).

=Syntomosphyrum ischnopterae= (Girault)

Synonymy.--_Epomphaloides ischnopterae_ Girault [Peck, 1951].

Parker and Thompson (1928) called their hyperparasite _Tetrastichus_ sp. However, Dr. B. D. Burks (personal communication, 1955) has examined the teneral specimens which Parker and Thompson deposited in the U.S. National Museum; he stated that the species is apparently _Syntomosphyrum ischnopterae_. In view of the experimental work by Parker and Thompson (see below), this wasp may prove to be a hyperparasite on evaniids in cockroach oöthecae rather than a primary parasite on cockroach eggs. (See _Zeuxevania splendidula_, p. 243.)

Natural hosts.--_Ischnoptera_ sp. [probably _Parcoblatta_ sp. (Rehn, personal communication, 1958)]. U.S.A., Maryland (Girault, 1917).

Zeuxevania splendidula Costa (an evaniid in the oöthecae of _Loboptera decipiens_), France (Parker, 1924; Parker and Thompson, 1928).

The following information is from Parker and Thompson (1928): _Adult behavior._--Courtship and mating were accomplished as soon as adults emerged, and in a manner similar to that in other chalcids. The females oviposited only into

oöthecae that were parasitized by _Zeuxevania_, never into normal, nonparasitized oöthecae. _Oviposition._--Oviposition occurred two days after mating. The female wasp stroked the oötheca with her antennae, selected a site, and bored into the oötheca with her ovipositor. She inserted the ovipositor deeply and oviposited for 10-30 minutes. The eggs were deposited randomly on the evaniid larva, some upright and others lying down. _Development._--Eggs of the hyperparasite hatched within 3 days and the larvae commenced feeding on the host larva. There were 30 and 50 hyper-parasites in two oöthecae. _Sex ratio._--5 [F][F]:1 [M] (from 3 oöthecae).

Distribution.--U.S.A., District of Columbia, Maryland (Burks, 1952).

=Tetrastichus australasiae= Gahan

Natural host.--_Periplaneta australasiae_, Sumatra (Gahan, 1923).

=Tetrastichus hagenowii= (Ratzeburg)

Synonymy.--_Entedon hagenowii_ Ratzeburg, _Blattotetrastichus hagenowii_ (Ratzeburg) [Burks, 1943]. _Tetrastichodes asthenogmus_ Waterston. G. J. Kerrich (personal communication, 1957) compared the type of _Tetrastichodes asthenogmus_ Waterston with authentically determined material of _Tetrastichus hagenowii_ and concluded that _T. asthenogmus_ is only a weakly developed specimen of _T. hagenowii_. He stated, "The longitudinal dorsal grooves of the scutellum, which are strongly developed in normal _hagenowii_, are only rather faintly developed in Waterston's type and also the second specimen, which was dissected and mounted on a series of ten microscope slides. No doubt it was this faint development that caused Waterston to describe the species in _Tetrastichodes_, a segregate that has since been recognized by Dr. Burks (Proc. U. S. Nat. Mus., 1943) as being not truly generically distinct from _Tetrastichus_."

Natural hosts.--_Blatta orientalis_, Seychelles (Ratzeburg, 1852); India (Usman, 1949).

Blatta sp., U.S.A., Louisiana (Gahan, 1914).

Blattella germanica (Burks, 1943; Peck, 1951). [In personal communications, Burks and Peck cite Howard (1892) and Marlatt (1902, and the 1908 revision of 1902) as sources for this host record. However, _B. germanica_ is not mentioned specifically as a host of _T. hagenowii_ in the sources cited nor in the 1915 revision of Marlatt's 1902 paper cited by Burks (1943); see footnote 6, p. 236.]

Neostylopyga rhombifolia, Hawaii (Pemberton, 1941): This record is based on one parasitized oötheca. We have exposed, at three different times, groups of 10 to 20 oöthecae of _N. rhombifolia_ to many newly emerged _T. hagenowii_, but none of the eggs was parasitized (Roth and Willis, unpublished data, 1957).

Parcoblatta sp., U.S.A., Ohio (Edmunds, 1953a).

Periplaneta americana, Africa (Crawford, 1910; Nash, 1955): Nash's record was incorrectly attributed to _Syntomosphyrum glossinae_ Wtstn., a parasite of tse-tse fly pupae (Jordan, 1956), Formosa (Takahashi, 1924; Sonan, 1924); Palestine (Bodenheimer, 1930); Puerto Rico (Seín, 1923; Plank, 1947, 1950; Wolcott, 1951); St.

Croix, Virgin Islands (Beatty, 1944); Hawaii (Schmidt, 1937); U.S.A.: Missouri (Rau, 1940a); Ohio (Edmunds, 1955); Florida (parasitized oöthecae were collected near Orlando by members of the Orlando Laboratory, Entomology Research Branch, U.S. Department of Agriculture; the parasites were identified by Burks, personal communication, 1955). Fiji (Lever, 1943); India (Mani, 1936; Usman, 1949); Trinidad and Saudi Arabia (Cameron, 1955). Westwood (1839) stated that 70 parasites belonging to the genus _Eulophus_ emerged from an oötheca of _P. americana_ collected on shipboard. Burks (personal communication, 1955) stated that the wasp was probably _T. hagenowii_.

Periplaneta australasiae, Australia (Shaw, 1925); India (Usman, 1949); Saudi Arabia, Trinidad (Cameron, 1955); Formosa (Sonan, 1924).

Periplaneta brunnea, U.S.A., Florida (parasitized oöthecae were collected near Orlando, by members of the Orlando Laboratory, Entomology Research Branch, U.S. Department of Agriculture. The parasites were identified by Burks, p. c., 1955).

Cockroach eggs, Formosa (Maki, 1937); Ceylon (Waterston, 1914): Taken on an oötheca.

"Domestic cockroaches," U.S.A., Louisiana (Girault, 1917).

"Roach egg cases," Panama Canal Zone (Rau, 1933).

Evania sp., Hawaii (Ashmead, 1901; Perkins, 1913); Guam (Fullaway, 1912); Fiji (Lever, 1946); Europe, Cuba, Florida (Marlatt, 1902, 1915).

Experimental hosts.--_Blatta orientalis_, _Eurycotis floridana_, and _Periplaneta americana_, U.S.A. (Roth and Willis, 1954b): We have maintained _T. hagenowii_ for over two years through more than 30 generations on eggs of both _B. orientalis_ and _P. americana_.

Periplaneta fuliginosa, U.S.A., Pennsylvania (Roth and Willis, 1954b); Massachusetts (Roth and Willis, unpublished data, 1957).

Schmidt (1937) deduced that _T. hagenowii_ was a primary parasite of eggs of _P. americana_ because the parasitized oötheca was obtained from a cage covered with screen too fine to permit entry of a larger parasite, such as an evaniid. As noted above, we have reared _T. hagenowii_ for more than 30 generations on cockroach eggs, none of which was ever exposed to parasitization by an evaniid. If _T. hagenowii_ were ever hyperparasitic on _Evania_, this relationship would be accidental, the eulophid happening to oviposit into an oötheca already containing an evaniid, or vice versa.

Adult behavior.--The male mates soon after becoming adult; he mounts the female from behind, grasps her antennae with his own antennae, and vibrates his wings during copulation. Mating is accomplished in from "several" to 20 seconds (Takahashi, 1924; Edmunds, 1955). The adults are positively phototactic and are capable of hopping for some distance (Edmunds, 1955). The females feed on material that oozes through the oviposition puncture (Roth and Willis, 1954b). Females lived 10 days (Seín, 1923). Without food, females lived 7.8 days and males 3.4 days, but

when fed dilute honey females lived 12.5 days (Usman, 1949). Females lived 5-11 days (Roth and Willis, 1954b). Fed water and sugar, the wasps lived 2-6 weeks at 65°F. (Cameron, 1955). Without food, 9 females lived an average of 3.5 days and 9 males an average of 1.7 days, but when fed on raisins, 9 females lived an average of 25 days and 9 males 15 days (Edmunds, 1955). In Formosa there were six generations from April to December (Maki, 1937).

In Hawaii, Severin and Severin (1915) caught 571 _T. hagenowii_ in 10 kerosene traps that were set up to sample populations of Mediterranean fruitfly. Apparently the parasite is attracted by the odor of kerosene.

Oviposition.--The female wasp explores the surface of the oötheca with vibrating antennae (Edmunds, 1955). She bends her abdomen ventrad and repeatedly touches the surface of the oötheca with her valvae; when she finds an acceptable oviposition site, the wasp unsheathes her ovipositor and bores through the wall of the oötheca (Roth and Willis, 1954b). The wasp deposited her eggs in 2-5 minutes (Edmunds, 1955). Wasps oviposited (pl. 34, C) into young or old eggs of _P. americana_ (Roth and Willis, 1954b). A single wasp parasitized more than one oötheca and more than one wasp oviposited into the same oötheca (Roth and Willis, 1954b; Edmunds, 1955). We found freshly laid wasp eggs in 34 empty but previously parasitized oöthecae from which the wasps had emerged (Roth and Willis, 1954b).

Development.--In _Periplaneta americana_: Development is completed in an average of 36 days (range 29-58 days) (Maki, 1937); 29-40 days (Lever, 1943); average of 23.6 days (range 22-26 days) at 62°-85° F. (Usman, 1949); about 3 months at 60°-65° F. (Cameron, 1955); 31-60 days at 70°-80° F. (Edmunds, 1955). We found that the wasps completed development in 23-56 days at about 85° F., but the period depended on the number of wasps in the oötheca; the larger the number of wasps (up to an average of about 70 wasps per oötheca), the shorter the time required to complete development. Wasps in oöthecae containing 70 or more parasites developed in an average of about 32 days (Roth and Willis, 1954b). Wasp larvae eat the contents of the cockroach egg in which they start development, then rupture the chorion and attack adjoining eggs (Cameron, 1955; Edmunds, 1955). All eggs are consumed when the parasite density is high, but if too few larvae develop per oötheca, some cockroach eggs survive and the embryos complete development (Roth and Willis, 1954b). However, a certain number of cockroach nymphs must complete development to enable the survivors to force open the crista and emerge from the oötheca; fewer than this number of surviving nymphs will be trapped and killed as effectively as if they had been eaten by the parasite. The adult parasites emerge from one to three holes cut through the wall of the oötheca (Usman, 1949; Roth and Willis, 1954b).

Number of offspring per female.--In _Blatta orientalis_: In the laboratory, 5 oöthecae were left with each of 25 female wasps for their entire lifespans; of the 125 oöthecae, 32 were parasitized. The average number of offspring per female was 66 (range 5-164) (Roth and Willis, 1954b). In _Periplaneta americana_: Each of 206 oöthecae was exposed to a single female wasp for 24 hours; the average number of offspring per female was 103 (range 50-139). Five oöthecae were left with each of 38

females for their entire lifespans; of the 190 oöthecae, 81 were parasitized. The average number of offspring per female was 94 (range 45-168 [from original data]) (Roth and Willis, 1954b).

Number of parasites per oötheca.--In _Eurycotis floridana_: In the laboratory, 3 oöthecae that had been exposed to 20 female wasps yielded an average of 648 parasites (range 606 [from original data] to 685) (Roth and Willis, 1954b). In _Neostylopyga rhombifolia_: One oötheca yielded 73 parasites (Pemberton, 1941). In _Parcoblatta_ sp.: Two oöthecae yielded an average of 100 parasites (Edmunds, 1953a). In _Periplaneta americana_: 100 parasites per oötheca (Seín, 1923); 140 (Schmidt, 1937); 25 (Rau, 1940a); 7-38, average 33 (Usman, 1949); 71 (Wolcott, 1951); 4 oöthecae exposed to 20 female wasps yielded an average of 204 wasps (range 164 [from original data] to 261) (Roth and Willis, 1954b); average of 30-40 (Cameron, 1955); 39 oöthecae yielded an average of 93 parasites (range 12-187) (Edmunds, 1955). In _Periplaneta australasiae_: Oöthecae yielded an average of 40-50 adult parasites (Cameron, 1955); about 50 (Shaw, 1925).

Sex ratio.--3 [F][F]:1 [M] (Usman, 1949); 4 [F][F]:1 [M] (Cameron, 1955); 2-8 [F][F]:1 [M] (Roth and Willis, 1954b); 1.2 [F][F]:1 [M] (Edmunds, 1955). Parthenogenesis exists; the unfertilized eggs produced only males (Roth and Willis, 1954b; Edmunds, 1955).

Distribution.--Probably worldwide. Eastern and southern U.S.A.; Central and South America; Europe; Arabia; Africa; India; Formosa; Hawaii.

=Tetrastichus periplanetae= Crawford

Natural hosts.--_Periplaneta americana_, Mozambique (Crawford, 1910); Union of South Africa (parasites reared from oöthecae collected in Durban, Natal, by the City Health Department): The parasites were identified by Burks (personal communication, 1956). Jamaica (Gowdey, 1925); Réunion Island (Bordage, 1913).

"Domestic cockroach," Puerto Rico (Wolcott, 1951).

=Tetrastichus= sp. I

Taxonomy.--Burks (personal communication, 1956) stated that this species (specimens of which are in the U.S. National Museum) is very close to _T. hagenowii_.

Natural hosts.--_Periplaneta americana_, Union of South Africa (parasites reared from oöthecae collected in Durban, Natal, by the City Health Department [Burks, p. c., 1956]).

Periplaneta australasiae, Manila, Philippine Islands (Burks, p. c., 1956).

=Tetrastichus= sp. II

Synonymy.--Because of the war, Cros (1942) could not determine this insect specifically. He designated it provisionally and with reserve under the name _Eulophus_ sp. However, Burks (p. c., 1956) stated that the species is most certainly a _Tetrastichus_ from the description given; but, it is apparently not _T. hagenowii_ because of its brilliant steel-blue color.

Natural host.--_Blatta orientalis_, Algeria (Cros, 1942): _Adult behavior._--Mating began as soon as wasps emerged from an oötheca. Males mated repeatedly. Adults lived up to 5 days in summer and up to 12 days in fall. There were up to four generations per year in the laboratory. _Oviposition._--Wasps oviposited into oöthecae 6, 22, 40, and 43 days old, and the parasites developed successfully. More than one female oviposited into the same oötheca. Oviposition was of long duration. _Development._--From egg to eclosion took an average of 34 days in summer (range: 30-38 days, 5 oöthecae), and an average of 67 days in fall (range: 58-73 days, 3 oöthecae). An average of 55 parasites developed per oötheca (range 21-105, 5 oöthecae); over 130 wasps emerged from a sixth oötheca. _Sex ratio._--10-20 [F][F]:1 [M].

HOST SELECTION BY EGG PARASITES

The nature of the oviposition stimulus(i) for the wasp parasites of cockroach eggs is unknown. Edmunds (1954) noted that _Prosevania punctata_ showed more interest in oöthecae that had been cemented to the substrate than in clean oöthecae that had simply been dropped. Cros (1942) experimented with two females of _P. punctata_ to see if the wasps could find oöthecae that had been buried in sand by the oriental cockroach. After prospecting the sand with their antennae, the wasps dug deep excavations with their front legs but always mistook the location of the oöthecae. Cros suggested that the wasps were misled by the odor left in the jar by the cockroaches. It is quite possible that odor helps the wasp find the host oötheca.

The extent of host selection varies among these parasites; some species will oviposit into the eggs of more than one species of cockroach, but others show some degree of host specificity. Positive selection of specific hosts by certain parasites appears in correlative data from different investigators on pages 235 to 254. There is a small body of data that shows nonacceptance of certain hosts by some of these wasps. For example, _Comperia merceti_ would not parasitize eggs of _Blatta orientalis_ or _Periplaneta americana_ in the laboratory (Lawson, 1954a). We (unpublished data, 1957) exposed a soft oötheca, recently removed from _Eurycotis floridana_, to _C. merceti_; no wasps developed; we had similar negative results with _C. merceti_ and oöthecae of _B. germanica_. We (1954b) could not induce _Tetrastichus hagenowii_ to parasitize eggs of _Blattella germanica_, _B. vaga_, or _Parcoblatta virginica_ in the laboratory. In our experiments, _T. hagenowii_ oviposited into eggs of _Supella supellectilium_, but the wasp eggs either failed to hatch, or if they hatched, the larvae died before completing development. Neither would _T. hagenowii_ parasitize eggs of _N. rhombifolia_ (Roth and Willis, unpublished data, 1957). _Anastatus tenuipes_ would not parasitize the eggs of _Latiblattella lucifrons_ Hebard, _Periplaneta americana_, _B. germanica_, or _B. vaga_ (Flock, 1941). _Anastatus floridanus_ would not oviposit into eggs of _S. supellectilium_ and only rarely into eggs of _P. americana_ or _B. orientalis_ (Roth and Willis, 1954a); in the laboratory, this wasp could not be maintained beyond one generation on the eggs of _P. americana_. Edmunds (1953b) could not induce _Prosevania punctata_ to parasitize eggs of _B. germanica_. Cros (1942) induced _P.

punctata_ to oviposit into a mantid oötheca, but neither mantids nor parasite developed.

COCKROACH-HUNTING WASPS

A number of wasps of the families Ampulicidae, Sphecidae, and a very few species of Pompilidae have been found to provision their nests with nymphal or adult cockroaches. This habit of preying on cockroaches is primitive (Leclercq, 1954); Leclercq (personal communication, 1955) stated that this habit is always associated with the conservation of a number of structures considered as archaic from a purely morphological point of view.

The records of wasps of the genus _Astata_ capturing cockroaches (e.g., Sickmann, 1893; St. Fargeau _in_ Sharp, 1899) "all trace back to a questionable record by Lepeletier (1841) which probably was a misidentification of the predator" (K. V. Krombein, personal communication, 1956). Marshall (1866) suggested that the braconid _Paxylomma buccata_ Bréb., which he found frequenting cockroach runs in Pembrokeshire, was parasitic on _Ectobius nigripes_ Stephens; however, this wasp is undoubtedly parasitic on ants, probably on ant larvae (Donisthorpe and Wilkinson, 1930).

The wasps that are known to capture cockroaches, and summaries of their biology, are listed below.

WASPS THAT PROVISION THEIR NESTS WITH COCKROACHES

Family POMPILIDAE

=Pompilus bracatus= Bingham

Natural hosts.--Cockroaches, India (Bingham, 1900).

=Pompilus= sp.

Natural host.--Cockroach, Nyasaland (Lamborn _in_ Poulton, 1926): The wasp was collected leading a nymph of the cockroach by its antenna. The cockroach was in a stupefied state, and its antennae were bitten off to about half their length.

=Salius verticalis= Smith

Natural hosts.--Cockroaches, India (Bingham, 1900).

Family AMPULICIDAE

The species of _Ampulex_ do not appear to make special nests in which to lay their eggs but drag their prey to any convenient hole, or crack in the ground (Arnold, 1928). Although many species of _Ampulex_ have been described, the prey of only a small number of species have been discovered, but the known prey are all cockroaches.

=Ampulex amoena= Stål

Synonymy.--_Ampulex novarae_ Saussure [Krombein, personal communication, 1957].

Natural hosts.--_Periplaneta americana_ and _Periplaneta australasiae_, both as small nymphs, Formosa (Sonan, 1924, 1927): The wasp stings a nymph about one inch long and carries it to a suitable place, such as bamboo pipes, folds of newspaper, or books (in houses), for oviposition.

Periplaneta picea, Japan (Kamo, 1957; Kohriba, 1957).

Experimental hosts.--_Periplaneta picea_, Japan (Kamo, 1957; Kohriba, 1957).

Kamo (1957) observed that in the field both males and females sucked juices from wounds they made in the stems of _Clerodendron_ _trichotomum_ Thunberg or _Ilex rotunda_ Thunberg. Kohriba (1957), on the other hand, found both sexes sucking sap of _Abies_ sp. and other trees from points injured by the rostrum of cicadas. Kamo (1957) observed that the female wasp grasped the cockroach by a tergum and stung it several times in the thorax. The wasp always amputated the antennae of the prey and sucked up the fluid oozing from the cut antennae. The wasp egg was placed on the mesocoxa of the cockroach. In the laboratory as many as three cockroaches, each with a wasp egg, were stored in artificial nests per day. Kohriba (1957) observed similar behavior in the laboratory and made these additional notes. The paralyzed cockroach could move its legs and was led to the nest by the wasp which seized its antennae. The egg hatched in 2 days, and after sucking up body fluid for 2 days the larva began to devour the prey. Three days later the larva spun its cocoon, and about one month after spinning a female wasp emerged.

=Ampulex assimilis= Kohl

Natural hosts.--_Blatta lateralis_, wingless females, Iraq (Hingston, 1925): Nesting sites are holes in palm trees, galleries of beetles, or tunnels in ground. The wasp first seizes a cockroach by the edge of its thorax and stings it in the thoracic region, then seizes the cockroach by an antenna and pulls and leads it to the nest. The wasp deposits her egg on the outer surface of the femur of the cockroach's midleg. The nest is closed with debris; later the cockroach recovers from the sting. The wasp larva first feeds externally, then bores into the cockroach and devours the internal organs. Pupation occurs inside the exoskeleton of the cockroach.

=Ampulex canaliculata= (Say)

Synonymy.--_Rhinopsis caniculatus._

Natural hosts.--_Ischnoptera_ sp., U.S.A. (Krombein, 1951).

Lobopterella dimidiatipes, Hawaii (Williams, 1928a, 1929).

Parcoblatta pensylvanica? MacNay (1954) referred to a rare sphecoid wasp in eastern Canada which provisioned its nest with nymphs and adults of _P. pensylvanica_. Dr. W. R. M. Mason (personal communication, 1957) wrote us that although this wasp was _Ampulex canaliculata_, it was not reared from the cockroach

but was swept from a pine tree. There are no positive records linking _A. canaliculata_ with _P. pensylvanica_.

Experimental host.--_Parcoblatta virginica_, females, U.S.A., Missouri (Williams, 1928a, 1929): figure 6.

Nesting sites are in twigs (Krombein, 1951). The adult behavior is similar to that of _A. compressa_; the female wasp imbibes blood that oozes from the amputated antennae of the cockroach; the egg hatches in 2-3 days, and the development of one male was completed in 33 days (Williams, 1929).

[Illustration: FIG. 6. _Ampulex canaliculata_ attacking _Parcoblatta virginica_. A, Female wasp stinging her prey, c. × 4.8. B, Wasp's egg attached to the coxa of the mesothoracic leg of the cockroach. C, Larva of _A. canaliculata_ (about three-quarters grown) feeding on the internal organs of the host from the exterior, c. × 4. (Reproduced from F. X. Williams [1929], through the courtesy of Dr. F. X. Williams and F. A. Bianchi, Hawaiian Sugar Planters' Association.)]

Distribution.--U.S.A.: Connecticut south to Georgia; Ohio, Wisconsin, Missouri, Kansas; in open woods (Krombein, 1951).

=Ampulex compressa= (Fabricius)

(Pl. 35)

Synonymy.--Guĕpe ichneumon of Réaumur [Williams, 1929]; _Chlorion_ (_Ampulex_) _compressum_.

Natural hosts.--_Periplaneta americana_, New Caledonia (Lucas, 1879); India (Dutt, 1912); Reunion (Bordage, 1912).

Periplaneta australasiae, Hawaii (Swezey, 1944).

Periplaneta sp., India (Maxwell-Lefroy, 1909).

Cockroach. Mauritius (Réaumur, 1742); Burma (Bingham, 1897).

Experimental hosts.--_Neostylopyga rhombifolia_, _Periplaneta americana_, and _Periplaneta australasiae_, Hawaii (Williams, 1942, 1942a). Zimmerman's (1948) listings probably were taken from Williams.

Nesting sites.--Holes in walls; holes in banyan and fig trees; in houses in drawers and cartons. _Behavior._--Similar to that of _A. assimilis_. Bordage (1912) gives a complete description of capture of prey. The female wasp cuts off part of the cockroach's antennae, legs, and wings; she sticks her egg onto the host's mesothoracic coxa. The wasp frequents houses in search of prey. Five [F][F], supplied with a cockroach per day, stored an average of 57±14 cockroaches; 8 [F][F] stored an average of 45±3 cockroaches; these latter wasps were not supplied with a cockroach per day throughout (mean values computed from Williams, 1942). This wasp will not attack _Nauphoeta cinerea_ (Williams, 1942a) or _Pycnoscelus surinamensis_ (Schwabe, 1950b). On one occasion, _A. compressa_ stung _Diploptera punctata_, but did not oviposit (Williams, 1942a). _Development._--Minimum 34 days, maximum 140 days (Williams, 1942). About 6 weeks (Swezey, 1944). _Longevity of adults._--13

[F][F] lived an average of 110±11 days (minimum 31, maximum 159); several [M][M] lived 2 months (Williams, 1942).

=Ampulex fasciata= Jurine

Natural host.--_Ectobius pallidus_, France (Picard, 1911, 1919): Nesting sites are in brier or bramble stems, or in crevices in fig trees; the female possibly uses old nests of leaf-cutter bees. The feeding of the wasp larva is similar to that of other _Ampulex_. Adult wasp emerges by cutting open a passage through its cocoon and through the anus of the cockroach.

=Ampulex ruficornis= (Cameron)

Natural hosts.--Cockroaches, Oriental region (Rothney _in_ Sharp, 1899): Nesting sites are in crevices in bark. The female grasps the cockroach by an antenna to drag it to her nest.

=Ampulex sibirica= Fabricius

Synonymy.--Perkins referred to this species as _Ampulex sibirica_. Williams (1942a), referring to Perkins's observations, mentions the species as "_A. compressiventris_ Guérin (=_A. siberica_ Sauss.)." Krombein (personal communication, 1956) has commented upon this synonymy as follows: _Ampulex siberica_ Sauss. is apparently a misidentification by Saussure of _sibirica_ Fab. Kohl (1893) in his revision of the genus _Ampulex_ considered _A. compressiventris_ Guérin to be the correct name for this common African species and that _sibirica_, described from Siberia, must be another species. However, Turner (1912) stated that he had seen Fabricius's type specimen and that it was identical with what had been called _compressiventris_; he considered the Siberian locality given by Fabricius as an error. Krombein suggested that Williams's use of the combination _siberica_ Sauss. was a lapsus and that the valid name, if Turner is correct, is _sibirica_ Fab.

Natural hosts.--Cockroaches, West Africa (Perkins _in_ Sharp, 1899): Nesting sites are keyholes. Enters apartments in search of cockroaches. Wasp cocoon protrudes from dead body of cockroach.

=Ampulex sonnerati= Kohl

Synonymy.--"La mouche bleue" of Sonnerat (Kohl, 1893).

Natural host.--"Kakkerlac," Philippine Islands (Sonnerat, 1776): Nesting sites are readymade crevices. The wasp seizes the cockroach by an antenna and stings the host many times in the "abdomen." She drags the cockroach by an antenna to the nest, and, after depositing her egg, plugs the opening with moistened earth.

=Dolichurus bicolor= Lepeletier

Synonymy.--Schulz (1912) considered this to be _Dolichurus corniculus_. Berland (1925) stated that this is possibly a color variety of _D. corniculus_. Soyer (1947), from a study of the behavior of the wasps, believed that both _D. bicolor_ and _D. haemorrhous_ are varieties of _D. corniculus_. Krombein (personal communication, 1956) stated that _D. corniculus_ and _D. bicolor_ differ in characters other than

color alone and that _D. bicolor_ is considered a valid species today.

Natural host.--Cockroach, France (Benoist, 1927): The wasp was observed closing the entrance to its burrow. Its egg was attached to the coxa of the midleg of the cockroach.

Maneval (1932) stated that _D. bicolor_ is found at the edge of dry woods along with _D. corniculus_ and that the wasp will also accept the prey of _D. corniculus_ if presented to it.

=Dolichurus corniculus= (Spinola)

Synonymy.--_Dolichurus haemorrhous_ Costa [Schulz, 1912]. Berland (1925) listed _D. haemorrhous_ separately but stated that it is perhaps a color variety of _D. corniculus_.

Natural hosts.--_Blattella germanica_, France (Benoist, 1927).

Ectobius lapponicus, Germany (Sickmann, 1893); Denmark (Nielsen, 1903); Sweden (Adlerz, 1903); Italy (Grandi, 1931, 1954); France (Benoist, 1927; Maneval, 1928).

Ectobius pallidus, France (Maneval, 1932; Soyer, 1947).

Ectobius panzeri, France (Soyer, 1947).

Ectobius sp., Italy (Grandi, 1954).

Hololampra punctata, Pitten (Handlirsch, 1889).

Loboptera decipiens, France (Ferton, 1894).

Cockroach, Netherlands (Bouwman, 1914).

Nesting sites.--The wasp uses already-made cavities such as rotting dead branches on ground, fissures in the earth, abandoned ant holes, chinks in stone, or the empty cocoon of the ichneumon _Ophion luteus_ (Ferton, 1894; Maneval, 1932).

Behavior.--The prey is immobile while being dragged to the nest but recovers sufficiently from the sting so that if dug up it will run around (Ferton, 1894; Bouwman, 1914; Benoist, 1927; Grandi, 1954). The wasp cuts off about two-thirds of the cockroach's antennae prior to putting its prey in its nest (Adlerz, 1903; Bouwman, 1914; Soyer, 1947). One cockroach is placed in the nest and the wasp's egg is attached to the midcoxa (Ferton, 1894). Oviposition takes 5 to 6 minutes (Maneval, 1939). Wasp fills and seals its nest with bits of earth and stones (Ferton, 1894; Grandi, 1954). The wasp larva feeds externally and devours the entire cockroach, including its exoskeleton (Ferton, 1894).

Development.--Hatching occurs in 3 to 4 days (Ferton, 1894) or longer during cooler weather (Maneval, 1939). Larval development takes 6 days (Grandi, 1954), 8 days (Ferton, 1894), or 10 to 25 days depending on season (Maneval, 1939).

=Dolichurus gilberti= Turner

Natural hosts.--"Small Blattidae," India (Turner, 1917).

=Dolichurus greenei= Rohwer

Natural host.--_Parcoblatta_ sp., U.S.A., Virginia (Krombein, 1951, 1955): Nesting sites are under leaf litter. The prey was a paralyzed third-instar nymph. _Distribution._--Ontario. U.S.A. from Canadian border south to Florida in coastal States (Krombein, 1951).

=Dolichurus ignitus= Sm.

Natural hosts.--Cockroaches, Natal and Southern Rhodesia (Arnold, 1928): The wasp is "usually seen running up and down the trunks of trees searching for small cockroaches in the crevices of the bark."

=Dolichurus stantoni= (Ashmead)

Natural hosts.--_Allacta similis_, nymphs, Hawaii (Williams et al., 1931; Zimmerman, 1948).

Blattella lituricollis, usually nymphs, Philippine Islands, Hawaii (Williams, 1919).

Cutilia soror, nymphs, Hawaii (Williams et al., 1931; Zimmerman, 1948).

"_Phyllodromia_" sp., Philippine Islands, Hawaii (Williams, 1918; Bridwell, 1920).

Experimental hosts.--"Field cockroaches," Philippine Islands (Williams, 1944).

Nesting site.--Readymade crevices or holes in ground; porosity in lava. _Behavior._--The wasp seizes the cockroach by a cercus or leg and stings it in the thorax. She (fig. 7, A) then drags the cockroach to the nest by the base of an antenna. Wasp bites off distal part of host's antennae. She deposits her egg on one of the host's midcoxae. Nest is plugged with lumps of soil. The larva eats the entire host. _Development._--Eggs hatched in about a day and a half. Adults emerged about 3 weeks later. About five generations per year. (Williams, 1918, 1919; Williams et al., 1931.)

=Dolichurus= sp.

Natural hosts.--Cockroaches, nymphs, South Africa (Bridwell, 1917). Adult female cockroach carrying an oötheca, France (Deleurance, 1943).

Nesting site.--Plant stem, or in ground possibly an old abandoned nest of _Ammophile_. _Behavior._--Bridwell noted that one wasp larva ate two cockroach nymphs before pupating; the adult emerged about 4 months after cocoon formation. Deleurance observed the wasp close its nest with small pebbles, balls of earth, and small dead branches. The wasp egg was placed on the femur of the midleg. The prey in the nest is alert when disturbed. Deleurance believed the wasp was a variety of _D. corniculus_.

=Trirhogma caerulea= Westwood

Natural hosts.--_Periplaneta americana_ and _Periplaneta australasiae_, Formosa (Sonan, 1924): The wasp stings a nymph about one inch long and carries it to a suitable place (bamboo pipe) for oviposition.

[Illustration: FIG. 7.--Cockroach-hunting wasps. A, _Dolichurus stantoni_ leading a nymph of _Blattella lituricollis_ to her nest, c. × 4. (Reproduced from F. N. Williams [1919].) B, _Podium haematogastrum_ attaching her egg to an _Epilampra_ sp. while on the side of a termite mound that contains the wasp's nest, c. × 1.6. C, _Epilampra_ sp. parasitized by _P. haematogastrum_ showing the wasp's egg attached to the right fore coxa, c. × 3.2. (B and C reproduced from Williams [1928], through the courtesy of Dr. F. X. Williams and F. A. Bianchi.)]

=Trirhogma= sp.

Natural hosts.--Cockroaches, Oriental region (Williams, 1918, 1928): As far as is known species of this genus of wasps hunt cockroaches.

Family SPHECIDAE

=Tachysphex blatticidus= Williams

Natural hosts.--_Chorisoneura_ sp., adults, Trinidad, St. Augustine (Callan, 1942): The wasps nest gregariously in sandy places. The wasp itself is parasitized by the mutillid _Timulla_ (_Timulla_) _eriphyla_ Mickel.

Cockroaches, Trinidad (Williams, 1941a; Callan, 1950).

=Tachysphex coriaceus= Costa

Natural hosts.--Cockroaches, Italy (Beaumont, 1954).

=Tachysphex fanuiensis= Cheesman

Natural hosts.--_Graptoblatta notulata_, Society Islands (Cheesman, 1927, 1928).

Cockroach ("except for its smaller size [it] much resembles _Graptoblatta notulata_."), New Caledonia (Williams, 1945).

Nesting sites.--Patches of dry soil (Cheesman, 1928); coarse sand at base of a bank (Williams, 1945). _Behavior._--The female wasp pounces on the cockroach and stings it into immobility; she carries her prey in flight to the nest. Two to 13 cockroaches may be found in one nest; and one or more wasp eggs may be deposited in one nest. The egg is attached at one end to the host's thorax behind a forecoxa. Nest is sealed with dry pellets of soil. The cockroaches apparently do not recover from the wasp's sting.

=Tachysphex lativalvis= (Thomson)

Natural hosts.--_Ectobius lapponicus_, adults, Sweden (Adlerz, 1906); France (Maneval, 1932).

Ectobius pallidus, nymphs, France (Ferton, 1894, 1901; Maneval, 1932; Deleurance, 1946); Italy (Grandi, 1928).

Ectobius panzeri, Netherlands (Bouwman, 1914).

Ectobius sp., Denmark (Nielsen, 1933).

Ferton (1914) stated that he had reported in 1912 that this species hunted Hemiptera, but that this observation was a lapsus. _Nesting site._--In the ground of

sandy woodlot or border of dry woods; the nest is a hole 5.5 to 8 cm. long ending in a horizontal cell. Grandi (1928) stated that the entrance to the nest descended obliquely for 5 to 6 cm. and ended 4 cm. below the surface of the ground. _Behavior._--Two cockroaches, either sex, adults or nymphs, were stored in the cell (Adlerz, 1903; Grandi, 1928). The wasp laid her egg on the first prey brought, attaching it behind the front coxa. The cockroaches were not excitable and their antennae had not been injured. Grandi (1928) stated that the claws of the hind tarsi of the victims may be amputated. The hatched larva may consume one of its victims in four days leaving only the head, pronotum, tegmina, wings, and the urosternum.

=Podium abdominale= (Perty)

Synonymy.--_Trigonopsis abdominalis_ Perty [Kohl, 1902].

Natural hosts.--Cockroaches, nymphs, Ecuador (Williams, 1928): These wasps are apparently mainly arboreal mud daubers. The female wasp constructs a mud nest on underside of a palm leaf. Wasp egg is attached behind one of the forecoxae of the cockroach. Several cockroaches are stored in each nest. The prey is not immobilized as a result of the sting, and its antennae are left intact.

=Podium carolina= Rohwer

Natural host.--_Parcoblatta pensylvanica_, nymphs (Rau, 1937): Nesting sites are mud nests of _Sceliphron caementarium_ (Drury). One to three cockroach nymphs are stored per nest; mud partitions are placed in tube; the nest is plugged with mud which is coated with resin. _Distribution._--U.S.A., New York to North Carolina (Murray, 1951); Florida (Krombein and Evans, 1955).

=Podium dubium= Taschenberg

Natural hosts.--Epilamprine cockroaches, Brazil (Williams, 1928): Burrows, lenticular in cross section, are found on shaded trails. The wasp's habits are similar to those of _P. flavipenne_ and _P. haematogastrum_.

=Podium flavipenne= Lepeletier

Natural host.--_Epilampra abdomen-nigrum_, British Guiana (Williams, 1928): _Nesting site._--Burrows, about 2 inches deep and lenticular in cross section, are dug in the ground in well-drained, partly sheltered areas; also old _Podium_ nests are used. _Behavior._--The wasp stings the cockroach to helplessness and flies with it back to her nest where the host may recover from the sting; one or more cockroaches are stored per nest; the egg is deposited behind the forecoxa while the cockroach is still outside the burrow. The nest is sealed with mud. The larva feeds on most of the cockroach and leaves only some heavily sclerotized portions in the cell. In 153 nests examined, there was an average of 2.2 ± 0.08 [standard error computed from cited data] cockroaches per cell; four nests contained five cockroaches apiece. Of the 331 cockroaches in the nests, only 6 percent were adults. _Development._--Egg hatches in about 2 days; larva feeds about 4 days and pupates about 2 weeks later; adult emerges about 10-12 days later.

=Podium haematogastrum= Spinola

Natural host.--_Epilampra_ sp., Brazil, Pará (Williams, 1928): The female wasp (fig. 7, B) burrows into the surface of termite mounds, in banks, and in level ground. This wasp's behavior is similar to that of _P. flavipenne_. There was an average of 1.6 cockroaches (fig. 7, C) per cell in 74 nests examined. Of the 121 cockroaches collected, 28 percent were adults. Under artificial conditions, the life cycle varied from about a month to 45 days or more.

=Podium luctuosum= Smith

Natural host.--_Parcoblatta virginica_, female, U.S.A., New York (Pate, 1949).

Distribution.--U.S.A.: New York to Texas (Murray, 1951).

=Podium rufipes= Fabricius

Natural hosts.--"Wood roaches," British Guiana (Howes, 1917, 1919); Brazil (Williams, 1928): Nesting sites were clay column nests on houses, sides of stumps, or forest trees; banks; termite mound. Variable numbers of cockroaches were placed in the nests with one wasp egg attached behind forecoxa of the last host. The egg hatches in 2 days, the larva pupates about 2 weeks later, and the adult emerges 24 days later.

=Podium= sp.

Natural host.--_Epilampra conferta_, Brazil (Poulton, 1917): The burrow contained several cockroaches of the same species.

ANTS PREDACEOUS ON COCKROACHES

A large roach endeavored to escape by crossing the main front of the army. The creature made several powerful jumps, but each time it touched the ground ... its legs were grasped by the fearless ants.... In the end it fell ... and was instantly torn to bits and carried to the rear.... Another ant with the body of a wood roach was assisted by a worker who held the carrier's abdomen high in the air out of the way of her burden, all the way to the nest.

HOWES (1919)

Family FORMICIDAE

From the known entomophagous habits of the lower ants (Wheeler, 1928), we wonder that there are not more records of ants feeding on cockroaches, because this act must occur frequently. Kirby and Spence (1822) stated that R. Kittoe had observed in Antigua that ants which nested in the roofs would seize a cockroach by the legs so it could not move, kill it, and carry it up to their nest. Hotchkiss (1874) observed ants kill cockroaches on shipboard. Cockroaches attracted to sugar in the pantry were killed and carried off by the ants. The destruction of cockroaches by army ants has been recorded by Bates (1863), Wallace (1891), Beebe (1917, 1919), Howes (1919), and others. Dead and mutilated specimens of _Ischnoptera_ sp. [undoubtedly _Parcoblatta americana_ (Gurney, personal communication, 1958)] are

common in the nests of species of _Formica_ in California (Mann, 1911).

=Aphaenogaster picea= Emery

Natural prey.--_Ectobius pallidus_, U.S.A., Massachusetts (Roth and Willis, 1957).

=Camponotus pennsylvanicus= (De Geer)

Common name.--Carpenter ant.

Natural prey.--_Parcoblatta pensylvanica_, U.S.A. (Rau, 1940): The ants entered traps set up to capture the cockroach and carried off about a dozen adults of both sexes.

=Dorylus (Anomma) nigricans= subsp. =sjöstedi Emery=

Natural prey.--Small cockroach, Belgian Congo (Raignier and van Boven, 1955).

=Dorylus (Anomma) wilverthi= Emery

Natural prey.--Small cockroaches, Belgian Congo (Raignier and van Boven, 1955).

=Dorylus= sp.

Common name.--"Safari ant."

Natural prey.--Cockroaches, Africa, Lake Victoria (Carpenter, 1920): When the "Safari ants" were hunting, many species of cockroaches were driven from hiding among dead leaves in the forest. The cockroaches rushed about but easily fell prey to the ants which tore them to bits.

=Eciton burchelli= (Westwood)

Common name.--Army ant.

Natural prey.--Cockroaches, Panama Canal Zone (Johnson, 1954; Schneirla, 1956).

=Formica omnivora=

Synonymy.--The identity of this form is unknown. There are no species of _Formica_ on Ceylon. There was another _Formica omnivora_ described from tropical America, whose identity is also unknown (W. L. Brown, personal communication, 1956).

Natural prey.--Cockroaches, Ceylon (Kirby and Spence, 1822).

=Iridomyrmex humilis= Mayr

Common name.--Argentine ant.

Natural prey.--Cockroaches, injured individuals only (Ealand, 1915).

=Lasius alienus= (Förster)

Natural prey.--_Ectobius pallidus_, U.S.A., Massachusetts (Roth and Willis, 1957).

=Pheidole megacephala= (Fabricius)

Common name.--Big-headed ant.

Natural prey.--_Holocompsa fulva_, Hawaii (Illingworth, 1916).

Nauphoeta cinerea and _Pycnoscelus surinamensis_, Hawaii (Illingworth, 1914, 1942): The ants followed and killed _N. cinerea_ and _P. surinamensis_ as they burrowed in moist soil and attacked and destroyed _N. cinerea_ in breeding cages.

XIII. VERTEBRATA

Class PISCES

In British Guiana, Beebe (1925a) found undetermined cockroach remains in the stomachs of four species of fish belonging to three families, as follows:

Family POTAMOTRYGONTIDAE

=Potamotrygon humboldti= (Duméril)

(= _Potamotrygon hystrix_)

Family PIMELODIDAE

=Rhamdia sebae= Cuvier and Valenciennes

Family CHARACIDAE

=Cyrtocharax magdalenae essequibensis= (Eigenmann)

(= _Cynopotamus essequibensis_)

=Chalceus macrolepidotus= Cuvier and Valenciennes

The only other records of cockroaches being eaten by fish pertain to the use of cockroaches as bait.[8] Captain William Owen (_in_ Webster, 1834) stated that the Chinese used cockroaches as bait in their fishing excursions. At Reelfoot Lake, Tennessee, _Blatta orientalis_ were kept in large numbers by bait dealers and were sold to fishermen who used them for catching _Lepomis pallidus_, a sunfish locally known as bream, blue bream, or bluegill (Rau, 1944). In Indiana, oriental cockroaches were collected at a city dump by fishermen (Gould, 1941). Peterson (1956) states that cockroaches are satisfactory bait for bluegills, crappies, channel cat, blue heads, and large mouth black bass.

Class AMPHIBIA[9]

Order CAUDATA

Family PLETHODONTIDAE

=Plethodon glutinosus= (Green)

Natural prey.--_Cryptocercus punctulatus_, U.S.A. (Honigberg, 1953): Protozoa which are normally only found in _C. punctulatus_ were present in the intestine of the salamander indicating that this cockroach had been eaten by the amphibian.

Order SALIENTIA

Family BUFONIDAE

=Bufo funereus= Bocage

Natural prey.--Cockroaches, Belgian Congo (Noble, 1924): The stomachs of 62 out of 72 specimens contained food; this included 3 cockroaches.

=Bufo ictericus= Spix

Natural prey.--Cockroaches, Brazil (Valente, 1949): Stomach contents revealed the prothorax, legs, and wings of cockroaches, and fragments of wood-cockroaches. This toad frequently feeds at night.

=Bufo marinus= (Linnaeus)

Common name.--Giant toad, marine toad, Surinam toad.

Natural prey.--_Epilampra abdomen-nigrum_, Trinidad (Weber, 1938): Found in the stomachs of two toads.

Diploptera punctata, Hawaii (Pemberton and Williams, 1938).

Periplaneta sp., Fiji (Lever, 1939): Many householders in Suva have seen the toad eat considerable numbers of these cockroaches.

Pycnoscelus surinamensis, Hawaii (Alicata, 1938; Illingworth, 1941).

Cockroaches, Nicaragua (Noble, 1918): Stomach contents of toads captured at street lamps in Rio Grande consisted chiefly of large cockroaches. Puerto Rico (Wolcott, 1937).

=Bufo valliceps= Wiegmann

Experimental prey.--_Periplaneta americana_, U.S.A. (Moore, 1946): Cockroaches containing infective acanthellas of _Moniliformis dubius_ were fed to three toads.

Family HYLIDAE

=Hyla cinerea= (Schneider)

Common name.--Green tree frog.

Natural prey.--_Ischnoptera deropeltiformis_, _Periplaneta americana_, and undetermined cockroaches, U.S.A., Georgia (Haber, 1926): Cockroaches were found in 11 of 100 stomachs.

Family RANIDAE

=Arthroleptis variabilis= Matschie

Natural prey.--Cockroaches, Belgian Congo (Noble, 1924): Of 52 specimens examined, the stomach contents of 17 contained food, including 3 cockroaches.

=Hyperolius picturatus= Peters

Natural prey.--Cockroach, Belgian Congo (Noble, 1924): The stomachs of 12 of 56 specimens examined contained food, including one cockroach.

=Leptodactylus albilabris= (Günther)

Natural prey.--Cockroach, Puerto Rico (Schmidt, 1920): One of 25 stomachs contained a medium-sized cockroach.

=Leptodactylus pentadactylus= (Laurenti)

Common name.--"Smoky jungle frog" or "pepper frog."

Natural prey.--Cockroaches, Nicaragua (Noble, 1918): Cockroach wings were found in the stomach of a frog caught around human habitation. Brazil (Valente, 1949).

=Leptopelis calcaratus= (Boulenger)

Natural prey.--Cockroaches, Belgian Congo (Noble, 1924): The stomachs of 35 specimens were examined of which 13 contained food, including 2 cockroaches.

=Leptopelis rufus= Reichenow

Natural prey.--Cockroaches, Belgian Congo (Noble, 1924): Forty-five of 83 stomachs examined contained food, including 2 cockroaches.

=Megalixalus fornasinii= (Bianconi)

Natural prey.--Cockroaches, Belgian Congo (Noble, 1924): The stomachs of 3 of 40 specimens contained food, including 2 cockroaches.

=Rana catesbeiana= Shaw

Common name.--Bullfrog.

Natural prey.--Cockroaches, Puerto Rico (Derez, 1949).

=Rana mascareniensis= Duméril and Bibron

Natural prey.--Cockroaches, Belgian Congo (Noble, 1924): The stomach contents of 138 specimens were examined, 39 of which contained food, including 2 cockroaches.

=Rana pipiens= Schreber

Common name.--Leopard frog.

Experimental prey.--_Periplaneta americana_, U.S.A. (Moore, 1946): Cockroaches containing infective acanthellas of _Moniliformis dubius_ were fed to two frogs.

Neostylopyga rhombifolia, U.S.A. (Dr. T. Eisner, personal communication, 1958.)

=Frogs=

Natural prey.--_Blatta orientalis_, U.S.A. (Rau, 1924): Frogs which escaped from a tank in the cellar consumed quantities of this cockroach.

Parcoblatta pensylvanica, U.S.A. (Frost, 1924): One adult specimen recovered from alimentary canal of a frog, probably _Rana_ sp.

=Unidentified batrachians=

Experimental prey.--_Blattella germanica_, Germany, Frankfurt am Main, Zoological Garden (Lederer, 1952): These insects were preferred by all the insect eaters in the zoo.

Periplaneta americana, Germany, Frankfurt am Main, Zoological Garden (Lederer, 1952): Newly molted individuals were accepted as food, but others were usually passed by or consumed unwillingly.

Class REPTILIA[10]

Order CHELONIA

Family EMYDIDAE

=Chrysemys picta= (Schneider)

Common name.--Painted turtle.

Natural prey.--_Periplaneta australasiae_, England (Lucas, 1916, 1920): The cockroach, apparently injured, fell into water in the tortoise house, Zoological Gardens, Regent's Park, and the terrapin ate it.

Order SAURIA

Family GEKKONIDAE

=Gekko gecko= (Linnaeus)

Natural prey.--Cockroaches, Philippine Islands, Laguna (Villadolid, 1934): The geckos frequent holes in trees and underside of bark which are favorable haunts of cockroaches. Stomach contents mostly Blattidae and "Locustidae."

=Hemidactylus frenatus= Duméril and Bibron

Common name.--House lizard.

Natural prey.--Cockroaches, Philippine Islands, Laguna (Villadolid, 1934): Bulk of stomach contents of 22 lizards consisted of Orthoptera, mostly cockroaches.

=Sphaerodactylus= sp.

Natural prey.--Cockroaches, British Guiana (Beebe, 1925a): The above lizard is found in houses.

=Thecadactylus= sp.

Natural prey.--Cockroaches, British Guiana (Beebe, 1925a): The above lizard is found in houses.

=Undetermined geckos=

Natural prey.--Cockroaches, Australia, Flinders River (Froggatt, 1906): The lizard lived in the walls of the hut and hunted cockroaches upon the roof at night. Arno Atoll (Usinger and La Rivers, 1953).

Family IGUANIDAE

=Anolis carolinensis= Voigt

Experimental prey.--_Diploptera punctata_, U.S.A. (Eisner, 1958).

=Anolis cristatellus= Duméril and Bibron

Natural prey.--_Blattella_ sp., _Cariblatta delicatula_, _Epilampra wheeleri_, _Periplaneta americana_, _Periplaneta australasiae_, and _Symploce flagellata_, Puerto Rico (Wolcott, 1924): The last-named cockroach may have been _S. ruficollis_ Rehn and Hebard, the females of which are hard to distinguish from _flagellata_. Rehn and Hebard (1927) stated that in all probability _flagellata_ does not occur on the island of Puerto Rico. Wolcott (1950) stated that _Symploce ruficollis_ [= _bilabiata_] serves as food for the crested lizard.

Cockroaches, Puerto Rico (Schmidt, 1920): Of 100 stomachs examined, 16 contained Orthoptera, including cockroaches. Puerto Rico (Wolcott, 1924): One hundred _A. cristatellus_ had eaten 8 cockroaches, 4.14 percent of the total food, or 25 percent of the food for 8 lizards.

=Anolis pulchellus= Duméril and Bibron

Natural prey.--Cockroaches, Puerto Rico (Wolcott, 1924): Two small cockroaches found in 50 lizards examined.

=Anolis equestris= Merrem

Experimental prey.--_Neostylopyga rhombifolia_, U.S.A. (Eisner, personal communication, 1958.)

=Anolis grahami= Garman

Natural prey.--_Periplaneta_ spp. and Blattidae, Bermuda (Simmonds, 1958): Stomachs of 176 lizards yielded 6 cockroaches.

=Anolis leachi= Duméril and Bibron

Natural prey.--_Periplaneta_ spp. and Blattidae, Bermuda (Simmonds, 1958): Stomachs of 46 lizards yielded 31 cockroaches.

=Anolis sagrei= Cocteau

Natural and experimental prey.--_Pycnoscelus surinamensis_, Cuba (Darlington, 1938): This species was eaten both in captivity and in nature. The lizard ate most readily soft, immature cockroaches. _Pycnoscelus surinamensis_ is probably a staple food of the lizard in nature, as Darlington observed wild lizards catch the nymphs.

=Anolis stratulus= Cope

Natural prey.--_Aglaopteryx facies_, Puerto Rico (Wolcott, 1924): One cockroach was found in 50 lizards examined.

Cockroach, Puerto Rico (Schmidt, 1920): One of 25 stomachs contained a cockroach.

=Anolis= sp.

Natural prey.--"Wood roaches," British Guiana (Beebe, 1925a): The above lizard is arboreal on foliage in low jungle.

Family SCINCIDAE

=Leiolopisma laterale= Say

Common name.--Brown skink.

Natural prey.--Woodroaches, U.S.A., Louisiana (Slater, 1949): Analysis of stomach contents of 84 adult skinks showed that nymphal and adult woodroaches comprised the majority of Orthoptera.

=Tropidophorus grayi= Günther

Common name.--Spiny lizard.

Natural prey.--Cockroaches, Philippine Islands, Laguna (Villadolid, 1934): Food of this species was mostly Blattidae.

=Unidentified skinks=

Natural prey.--Cockroaches, Arno Atoll (Usinger and La Rivers, 1953).

Family AGAMIDAE

Experimental prey?--Cockroaches, Australia (Lee and Mackerras, 1955): A general statement was made that in captivity Agamidae were observed feeding avidly on cockroaches and other insects. Three agamids studied by these workers were _Amphibolurus barbatus_ (Gray), _Physignathus lesueurii_ Gray, and _Chlamydosaurus kingii_ Gray.

Family CHAMAELEONTIDAE

=Chamaeleon chamaeleon= (Linnaeus) and =Chamaeleon oustaleti= Mocquard

Experimental prey.--Cockroaches, Amsterdam (Portielje, 1914): Large cockroaches were fed to these lizards in the reptile house of Artis.

Family TEIIDAE

=Ameiva exsul= Cope

Common name.--Iguana, ground lizard.

Natural prey.--Cockroach (nymph), _Epilampra wheeleri_, and _Periplaneta americana_, Puerto Rico (Wolcott, 1924): Stomach contents of 15 lizards were analyzed. _E. wheeleri_ formed 30 percent of the food of one lizard. The cockroach nymph formed 5 percent of the food of one lizard. One _P. americana_ formed 20

percent of the food of one lizard; another formed 50 percent of the food of a second lizard.

Experimental prey.--Cockroach nymphs, Puerto Rico (Wolcott, 1924).

=Ameiva= sp.

Natural prey.--Cockroaches, British Guiana (Beebe, 1925a): The above lizard is terrestrial and found near clearings. The stomach contents of 18 out of 40 reptiles contained cockroach remains.

=Cnemidophorus= sp.

Natural prey.--Cockroaches, British Guiana (Beebe, 1925a): This is a terrestrial lizard found near clearings. The stomach contents of 4 out of 40 lizards contained cockroaches.

=Unidentified lizards=

Natural prey.--Cockroaches, West Indies (H., 1800).

Experimental prey.--_Blatta orientalis_, U.S.A. (Rau, 1924): Rau called the predator a common gray lizard.

Periplaneta americana, Germany, Frankfurt am Main, Zoological Garden (Lederer, 1952): Newly molted cockroaches were accepted as food, but others were usually passed by or consumed unwillingly.

Order SERPENTES

Family COLUBRIDAE

=Heterodon platyrhinos= Latreille

Synonymy.--_Heterodon contortrix_ [Dr. Doris M. Cochran, personal communication, 1957].

Common name.--Hog-nosed snake.

Experimental prey.--_Periplaneta americana_, U.S.A. (Moore, 1946): Cockroaches containing infective acanthellas of _Moniliformis dubius_ were fed to one snake.

=Garter Snake=

Experimental prey.--_Blatta orientalis_, U.S.A. (Rau, 1924).

Class AVES

The cockroach is always wrong when arguing with a chicken.

Spanish proverb (HARTNACK, 1939)

Arboreal cockroaches hidden in and under bark are much more likely to be encountered by birds than by other predators, and insectivorous birds undoubtedly consume many more cockroaches than the few records would indicate. Most of the records we have located identify the birds at least by common name. Where possible

we have given the scientific names for those birds whose common names are recognizably specific. We have followed the systematic classification of Wetmore (1940).

Figuier (1869) stated that poultry and owls are very fond of cockroaches. Perkins (1913) made the general statement that some of the native birds of Hawaii are partial to the endemic _Allacta similis_. Asano (1937) stated that in Japan natural enemies of cockroaches may be found in the Galliformes, Strigiformes, Passeriformes, and Piciformes. Although Lederer (1952) successfully fed newly molted _Periplaneta americana_ to insectivorous birds in the Zoological Garden, Frankfurt am Main, _Blattella germanica_ were preferred by these birds. The following records are of specific birds feeding on cockroaches.

Order ANSERIFORMES

Family ANATIDAE

=Domestic duck=

Natural prey.--_Pycnoscelus surinamensis_, Australia (Fielding 1926): The ducks became infected with Manson's eye worm of which _P. surinamensis_ is the only known intermediate host.

Cockroach, Bermuda (Jones, 1859): "All kinds of poultry feed greedily upon the cockroach; tame ducks spending entire moonlight nights in their capture."

Order GALLIFORMES

Family PHASIANIDAE

=Bambusicola thoracica= Temminck

Common name.--Kojukei.

Natural prey.--Cockroaches, Japan (Asano, 1937).

=Gallus= sp.

Common name.--Jungle fowl.

Natural prey.--_Periplaneta australasiae_, Hawaii (Schwartz and Schwartz, 1949).

=Phasianus calchicus karpowi= Buturlin

Common name.--Korean pheasants.

Experimental prey.--_Blattella germanica_ and _Periplaneta picea_, Japan (Asano, 1937): Adults of these cockroaches were devoured at once when they were fed with the heads cut off.

=Phasianus= sp.

Common name.--Pheasant.

Natural prey.--Blattidae, unidentified (below 1 percent of the diet), _Cutila soror_ (below 1 percent of the diet), _Diploptera punctata_ (above 6 percent of the diet), and

Pycnoscelus surinamensis (6 percent of the diet), Hawaii (Schwartz and Schwartz, 1949).

=Coturnix coturnix japonica= (Temminck and Schlegel)

Common name.--Japanese quail.

Natural prey.--Blattidae (unidentified) and _Lobopterella dimidiatipes_, Hawaii (Schwartz and Schwartz, 1949).

=Domestic chicken=

Natural and experimental prey.--_Blaberus craniifer_, U.S.A., Key West, Florida. J.A.G. Rehn in 1912 (personal communication) observed chickens feeding on nymphs of _B. craniifer_ which had dropped to the ground from among stacked coffins in an undertaker's shack.

Blatta orientalis, U.S.A. (Rau, 1924): The chickens ate cockroaches that were caught in traps.

Hebardina concinna, Japan (Asano, 1937): Experimental feeding to white Leghorn chickens.

Periplaneta americana, Surinam (Stage, 1947): Several cockroaches ran off the floor of a house, which was being sprayed with DDT, and were eaten by chickens. Although some chickens had DDT tremors the next day, all appeared normal two days later.

Pycnoscelus surinamensis, Australia (Fielding, 1926); Formosa, experimental feeding (Kobayashi, 1927); Australia, experimental feeding (Fielding, 1927, 1928); U.S.A., Florida, experimental feeding (Sanders, 1928); Antigua (Hutson, 1943); Hawaii (Illingworth, 1931; Schwabe, 1949, 1950a, 1950b). This cockroach is the intermediate host of _Oxyspirura mansoni_, the chicken eye worm.

Cockroaches, Guadeloupe (Dutertre, 1654); Africa (Moiser, 1947): "Poultry" ate cockroaches which had been killed by DDT and sodium fluoride. Hawaii (Zimmerman, 1948).

=Partridge=

Natural prey.--Cockroaches, British Guiana (Beebe, 1925a): The food of two small species of leaf-colored partridges that lived on the jungle floor, consisted chiefly of cockroaches and beetles.

Family MELEAGRIDIDAE

=Meleagris gallopavo= (Linnaeus)

Common name.--Turkey.

Natural prey.--_Pycnoscelus surinamensis_, Antigua (Hutson, 1943): Turkeys were found heavily infected with Manson's eye worm of which _P. surinamensis_ is the only known intermediate host. These turkeys therefore were presumed to have fed on this cockroach.

Order COLUMBIFORMES

Family COLUMBIDAE

=Streptopelia chinensis= (Scopoli)

Common name.--Chinese dove.

Natural prey.--_Pycnoscelus surinamensis_, Hawaii (Schwabe, 1950b).

=Pigeon=

Experimental prey.--_Pycnoscelus surinamensis_, Australia (Fielding, 1927); U.S.A., Florida (Sanders, 1928).

Order STRIGIFORMES

Family STRIGIDAE

=Gymnasio nudipes= (Daudin)

Common name.--Bare-legged owl.

Natural prey.--_Epilampra_ sp., Puerto Rico (Wetmore, 1916): One specimen identified in stomach of a wild-caught owl.

Cockroaches, Puerto Rico (Wetmore, 1916): These insects were found in stomachs of five owls.

Order CORACIFORMES

Family TODIDAE

=Todus mexicanus= Lesson

Common name.--Porto Rican tody.

Natural prey.--_Plectoptera poeyi?_, Puerto Rico (Wetmore, 1916): The stomachs of 89 birds were examined; a single bird had eaten the above cockroach. According to Wolcott (1950) _P. poeyi_ could be _Plectoptera dorsalis_, _P. rhabdota_, or _P. infulata_.

Family BUCEROTIDAE

=Tockus birostris= (Scopoli)

Synonymy.--_Lophocerus birostris_ [Dr. H. Friedmann, personal communication, 1957].

Common name.--Common gray hornbill.

Natural prey.--Cockroaches, India, Central Provinces (D'Abreu, 1920).

Order PICIFORMES

Family PICIDAE

=Dendrocopus mahrattensis= (Latham)

Synonymy.--_Liopicus mahrattensis_ [Friedmann, p. c. 1957].

Common name.--Yellow-fronted pied woodpecker.

Natural prey.--Cockroaches, India, Central Provinces (D'Abreu, 1920).

=Melanerpes portoricensis= (Daudin)

Common name.--Puerto Rican woodpecker.

Natural prey.--_Pycnoscelus surinamensis_, Puerto Rico (Wetmore, 1916): One specimen found in 59 bird stomachs examined.

Order PASSERIFORMES

Family FORMICARIIDAE

=Gymnopithys leucaspis= (Sclater)

Common name.--Bicolored antbird.

Natural prey.--Cockroaches, Panama Canal Zone (Johnson, 1954): This bird feeds on small cockroaches, and other arthropods, which are flushed from their hiding places by swarms of the army ant, _Eciton burchelli_.

Family ORIOLIDAE

=Icterus portoricensis= (Bryant)

Common name.--Puerto Rican oriole.

Natural prey.--Cockroaches, Puerto Rico (Wetmore, 1916): Cockroaches and oöthecae found in the birds' stomachs.

Family CORVIDAE

=Aphelocoma coerulesens= (Bosc)

Common name.--Florida jay.

Experimental prey.--_Pycnoscelus surinamensis_, U.S.A., Florida (Sanders, 1928).

=Cyanocitta cristata= (Linnaeus)

Common name.--Blue jay.

Experimental prey.--_Diploptera punctata_, U.S.A. (Eisner, 1958).

Eurycotis floridana, _Neostylopyga rhombifolia_, and _Periplaneta americana_, U.S.A. (Eisner, personal communication, 1958): _E. floridana_ was only eaten after the odor of 2-hexenal, which was released by the insect on being attacked by the bird, had dissipated.

Family PARADISEIDAE

=Paradisea papuana= Bechstein

Experimental prey.--Cockroaches, Malaya and on shipboard (Wallace, 1869): Two adult males fed voraciously on rice, bananas, and cockroaches. Wallace collected cockroaches every night on board ship to feed the birds. "At Malta ... I got plenty of

cockroaches from a bakehouse, and when I left, took with me several biscuit-tins full, as provision for the voyage home."

Family TROGLODYTIDAE

=Troglodytes aedon= Vieillot

Common name.--House wren.

Natural prey.--Cockroaches, U.S.A. (Greenewalt and Jones, 1955): The wren carried three small cockroaches to nestlings; the records probably represent incidental captures.

=Troglodytes audax= Tschudi

Common name.--Cucarachero.

Natural prey.--Cockroach (called Chilicabra by Peruvian Indians), Peru (Tschudi, 1847): The bird seized the cockroach and bit off its head then devoured the body discarding the wings.

Family LANIIDAE

=Lanius ludovicianus= Linnaeus

Common name.--Loggerhead shrike.

Experimental prey.--_Pycnoscelus surinamensis_, U.S.A., Florida (Sanders, 1928).

Family STURNIDAE

=Acridotheres tristis= Bonnaterre and Vieillot

Common name.--Myna, mynah.

Natural prey.--Cockroaches, Hawaii, Lanai (Illingworth, 1928): Illingworth reported that he had never seen as many cockroaches anywhere else in Hawaii. The birds followed tractors that were destroying cactus and kept close to the chain that turned over the stumps. The following species were collected: _Allacta similis_, _Blattella germanica_, _Cutilia soror_, _Diploptera punctata_, _Leucophaea maderae_, _Periplaneta americana_, _Periplaneta australasiae_, _Pycnoscelus surinamensis_. Illingworth did not state whether the birds ate all these species indiscriminately.

Pycnoscelus surinamensis, Hawaii (Williams et al., 1931; Schwabe, 1950b): In many places this species forms an important fledgling food for mynah birds.

Family VIREONIDAE

=Vireo latimeri= Baird

Common name.--Latimer's vireo.

Natural prey.--_Periplaneta_ sp., Puerto Rico (Wetmore, 1916): Cockroaches were found in one of 43 stomachs examined.

Family ICTERIDAE

=Agelaius xanthomus= (Sclater)

Common name.--Yellow-shouldered blackbird.

Natural prey.--Cockroaches, Puerto Rico (Wetmore, 1916): Oöthecae and remains of adult cockroaches found in stomachs.

=Dolichonyx oryzivorus= (Linnaeus)

Common name.--Bobolink.

Experimental prey.--_Pycnoscelus surinamensis_, U.S.A., Florida (Sanders, 1928).

=Holoquiscalus brachypterus= (Cassin)

Common name.--Puerto Rican blackbird.

Natural prey.--Cockroaches, Puerto Rico (Wetmore, 1916): A few eggs (oöthecae) of cockroaches in stomachs.

=Black bird=

Experimental prey.--_Pycnoscelus surinamensis_, U.S.A., Florida (Sanders, 1928).

Family FRINGILLIDAE

=Passer domesticus= (Linnaeus)

Common name.--English sparrow.

Natural prey.--_Pycnoscelus surinamensis_, Hawaii (Illingworth, 1931; Schwabe, 1950b): Remains of this cockroach were found in the stomach of the sparrow.

=Sparrow=

Natural prey.--_Periplaneta americana_, England (Lucas, 1908, 1920).

Cockroaches, Japan (Asano, 1937).

=Tiaris bicolor omissa= (Jardine)

Common name.--Carib grassquit.

Natural prey.--Cockroaches, Puerto Rico (Wetmore, 1916): Animal food was found in 5 of 72 stomachs examined; one bird had eaten two cockroaches among other insects.

Class MAMMALIA[11]

Order MARSUPIALIA

Family DIDELPHIDAE

=Monodelphis= sp.

Natural prey.--Cockroaches, British Guiana (Beebe, 1925a): The above opossum is nocturnal and arboreal but nests on the ground in grass.

Order INSECTIVORA

Family ERINACEIDAE

=Erinaceus europaeus= Linnaeus

Experimental prey.--_Blattella germanica_, France (Brumpt and Urbain, 1938): Two hedgehogs were fed cockroaches infested with _Prosthenorchis elegans_ and _P. spirula_.

=Erinaceus= sp.

Common name.--Hedgehog.

Natural prey.--Cockroaches, England (Samouelle, 1841; Cowan, 1865).

Order CHIROPTERA

Family MOLOSSIDAE

=Molossus= sp.

Natural prey.--Cockroaches, British Guiana (Beebe, 1925a): The above bat is a common house bat of the area.

Order PRIMATES

Family LEMURIDAE

=Lemur coronatus= Gray

Natural prey.--_Blattella germanica_, France (Brumpt and Urbain, 1938): The monkey apparently became infested naturally with _Prosthenorchis spirula_ for which _B. germanica_ was the intermediate host in the monkey house.

=Lemur fulvus= E. Geoffroy

Natural prey.--_Blattella germanica_, France (Brumpt and Urbain, 1938): See comment under _Lemur coronatus_.

Family LORISIDAE

=Loris tardigradus= (Linnaeus)

Synonymy.--_Lemur tardigradus_ [Dr. D. H. Johnson, personal communication, 1957].

Natural prey.--Cockroaches, on board ship (Cowan, 1865).

=Perodicticus potto= (P. L. S. Müller)

Common name.--Potto.

Natural prey.--_Blattella germanica_, France (Brumpt and Urbain, 1938): See comment under _Lemur coronatus_.

Experimental prey.--Blattidae, East Africa (Pitman, 1931): Both sexes of the potto ate freely of all types of cockroaches.

Family TARSIIDAE

=Tarsius= sp.

Experimental prey.--Cockroaches, Borneo (Shelford, 1916).

Family CEBIDAE

=Aotes zonalis= Goldman

Synonymy.--_Aotus_ [Simpson, 1945].

Common name.--Canal Zone night monkey.

Natural prey.--_Leucophaea maderae_, Panama (Foster and Johnson, 1939): Captive monkeys became naturally infested with _Protospirura muricola_ by eating cockroaches that contained infective larvae of the worm.

=Ateles dariensis= Goldman

Common name.--Darien black spider monkey.

Natural prey.--_Leucophaea maderae_, Panama (Foster and Johnson, 1939): See comment under _Aotes zonalis_.

=Cebus apella= (Linnaeus)

Natural prey.--_Blattella germanica_, France (Brumpt and Urbain, 1938): See comment under _Lemur coronatus_.

=Cebus capucinus= (Linnaeus)

Common name.--White-faced monkey.

Natural prey.--_Leucophaea maderae_, Panama (Foster and Johnson, 1939): Favorite item of food in the laboratory. See comment under _Aotes zonalis_.

=Saimiri sciurea= Linnaeus

Natural prey.--_Blattella germanica_, France (Brumpt and Urbain, 1938): See comment under _Callithrix chrysoleucos_.

Family CALLITHRICIDAE

=Callithrix chrysoleucos= (Natterer)

Synonymy.--_Callithrix chrysolevea_ [Johnson, personal communication, 1957].

Natural prey.--_Blattella germanica_, France (Brumpt and Urbain, 1938): The monkey apparently became infested naturally with _Prosthenorchis elegans_ for which _B. germanica_ was the intermediate host in the monkey house.

=Callithrix jacchus= (Linnaeus)

Synonymy.--_Simia jacchus._

Natural prey.--_Blattella germanica_, France (Brumpt and Urbain, 1938): See comment under _Lemur coronatus_.

Cockroaches, on board ship (Neill, 1829; also cited by Samouelle, 1841, and Cowan, 1865): "It was quite amusing to see it at its meal. When he had got hold of one of the largest cockroaches, he held it in his fore paws, and then invariably nipped the head off first; he then pulled out the viscera and cast them aside, and devoured the rest of the body, rejecting the dry elytra and wings, and also the legs of the insect, which are covered with short stiff bristles. The smaller cockroaches he eat[s] without such fastidious nicety."

=Leontocebus oedipus= (Linnaeus)

Natural prey.--_Blattella germanica_, France (Brumpt and Urbain, 1938): See comment under _Callithrix chrysoleucos_.

=Leontocebus rosalia= (Linnaeus)

Synonymy.--_Midas rosalia_ [Simpson, 1945].

Natural prey.--_Blattella germanica_, France (Brumpt and Urbain, 1938): See comments under _Lemur coronatus_ and _Callithrix chrysoleucos_.

=Leontocebus ursulus= (E. Geoffroy)

Natural prey.--_Blattella germanica_, France (Brumpt and Urbain, 1938): See comments under _Callithrix chrysoleucos_.

Family CERCOPITHECIDAE

=Cercopithecus= sp.

Experimental prey.--Cockroaches, East Africa (Carpenter, 1921, 1925): The monkey rarely tasted and usually ignored cockroaches offered to it. In one experiment the monkey had to be deprived of food before it would eat the cockroach.

=Macaca mulatta= (Zimmermann)

Synonymy.--_Macaca rhesus_ [Johnson, personal communication, 1957].

Experimental prey.--_Blattella germanica_, France (Brumpt and Urbain, 1938, 1938a): The macaque was fed cockroaches infested with _Prosthenorchis elegans_ and _P. spirula_.

=Macaca sylvanus= (Linnaeus)

Synonymy.--_Inuus sylvanus_ [Simpson, 1945].

Common name.--Macaque.

Natural prey.--_Blattella germanica_, France (Brumpt and Urbain, 1938): See comment under _Lemur coronatus_.

=Papio papio= (Desmarest)

Experimental prey.--_Blattella germanica_, France (Brumpt and Urbain, 1938, 1938a): This baboon was fed cockroaches infested with _Prosthenorchis elegans_ and _P. spirula_.

Family PONGIDAE

=Pan= sp.

Common name.--Chimpanzee.

Natural prey.--_Blattella germanica_, Netherlands (Thiel and Wiegand Bruss, 1946): Indirect evidence for this relationship was shown by these workers who found two animals heavily infected with _Prosthenorchis spirula_ in a zoo in Rotterdam; the intermediate host of the worm was shown to be _B. germanica_.

Family HOMINIDAE

=Homo sapiens= Linnaeus

Natural prey.--Oöthecae of _Blatta orientalis_ and _Neostylopyga rhombifolia_, Thailand (Bristowe, 1932).

Periplaneta americana, Formosa (Takahashi, 1924).

Periplaneta americana and _Periplaneta australasiae_, Australia, China, and Japan (Bodenheimer, 1951).

Cockroaches, Annam and French Guinea (Brygoo, 1946).

In addition to the above records of cockroaches being used as food by man these insects have also been eaten for medicinal purposes (see Roth and Willis, 1957a).

Order EDENTATA

Family DASYPODIDAE

=Dasypus novemcinctus= Linnaeus

Synonymy.--_Tatu novemcinctum_ [Johnson, personal communication, 1958].

Natural prey.--_Ischnoptera deropeltiformis_, Texas (Hebard, 1917): A specimen of this cockroach in the U. S. National Museum was taken from the stomach of the armadillo.

Order RODENTIA

Family MURIDAE

=Mus musculus= Linnaeus

Experimental prey.--_Diploptera punctata_, U.S.A. (Eisner, 1958).

=Rattus norvegicus= (Berkenhout)

Synonymy.--_Mus decumanus; Epimys norvegicus._

Natural prey.--_Leucophaea maderae_, Venezuela (Brumpt, 1931): Rats infested with _Protospirura bonnei_ presumably ate this cockroach which is the intermediate host of the worm.

Periplaneta americana, Brazil (Magalhães, 1898): Remains found in the stomachs of brown rats. Denmark (Fibiger and Ditlevsen, 1914): This cockroach was found to be the intermediate host of _Gongylonema neoplasticum_, a parasite of rats.

=Rattus rattus= (Linnaeus)

Natural prey.--_Periplaneta americana_, Denmark (Fibiger and Ditlevsen, 1914): See comment after these authors under _Rattus norvegicus_.

=Rattus= spp.

Natural prey.--Cockroaches, India (Maxwell-Lefroy, 1909); Burma (Subramanian, 1927).

Family CAVIIDAE

=Cavia= sp.

Experimental prey.--_Blattella germanica_, U.S.A. (Hobmaier, 1941): Guinea pigs were fed cockroaches infested with _Physaloptera maxillaris_.

Order CARNIVORA

Family CANIDAE

=Canis familiaris= Linnaeus

Experimental prey.--_Blattella germanica_, U.S.A. (Hobmaier, 1941): Dogs were fed cockroaches infested with _Physaloptera maxillaris_. U.S.A. (Petri and Ameel, 1950): Cockroaches infested with _Physaloptera rara_ were fed to a dog.

=Canis latrans= Say

Experimental prey.--_Blattella germanica_, U.S.A. (Petri and Ameel, 1950): Cockroaches infested with _Physaloptera rara_ were fed to a coyote.

=Vulpes= sp.

Experimental prey.--_Blattella germanica_, France (Brumpt and Urbain, 1938a): A fox was successfully infected when fed cockroaches infested with _Prosthenorchis elegans_ and _P. spirula_.

Family PROCYONIDAE

=Bassariscus astutus= (Lichtenstein)

Common names.--Cacomistle, ring-tailed cat.

Natural prey.--Cockroaches, U.S.A., Arizona (Dr. H. Stahnke, personal communication, 1953): The ring-tailed cat enters dwellings located on the desert and feeds on cockroaches and other arthropods.

=Nasua narica= (Linnaeus)

Natural prey.--_Blattella germanica_, France (Brumpt and Urbain, 1938): The coati apparently became infested naturally with _Prosthenorchis spirula_ for which _B. germanica_ was the intermediate host in the laboratory.

=Nasua nasua= (Linnaeus)

Natural prey.--_Blattella germanica_, on board ship (Myers, 1931): This insect was eaten when other insects were absent.

Cockroach, a small outdoor species, Trinidad (Myers, 1931).

=Nasua= sp.

Natural prey.--Cockroaches, British Guiana (Beebe, 1925a).

Family MUSTELIDAE

=Meles= sp.

Experimental prey.--_Blattella germanica_, France (Brumpt and Urbain, 1938a): A badger was successfully infected when fed cockroaches infested with _Prosthenorchis elegans_ and _P. spirula_.

Family VIVERRIDAE

=Herpestes javanicus auropunctatus= Hodgson

Natural prey.--_Epilampra wheeleri_, _Eurycotis improcera_, _Panchlora nivea_, _Pycnoscelus surinamensis_, and others unidentified to species, St. Croix and Puerto Rico (Wolcott, 1953): Based on 37 or more cockroaches obtained from stomachs of 42 mongooses collected in St. Croix (by Seaman) and 56 collected in Puerto Rico (by Pimentel).

Pimentel (personal communication, 1958) has given us the following percentage occurrence of cockroach species in the total number of mongoose stomachs that he examined in Puerto Rico: _Epilampra wheeleri_ 1.8, _Ischnoptera rufa rufa_ 3.6, _Panchlora nivea_ 1.8, _Periplaneta americana_ 1.8, and _Pycnoscelus surinamensis_ 19.6.

=Herpestes= sp.

Natural prey.--_Periplaneta americana_ and _Periplaneta australasiae_, Hawaii (Perkins, 1913): Large numbers of these cockroaches are devoured.

Cockroach, East Africa (Loveridge, 1923): Cockroach remains found in stomach of mongoose.

Family FELIDAE

=Felis catus= Linnaeus

Natural prey.--_Periplaneta americana_, Hawaii (Williams et al., 1931).

Cockroaches, U.S.A., Arizona (Stahnke, personal communication, 1953).

Experimental prey.--_Blattella germanica_, U.S.A. (Hobmaier, 1941): Cats were fed cockroaches infested with _Physaloptera maxillaris_. U.S.A. (Petri and Ameel, 1950): Cockroaches infested with _Physaloptera rara_ were fed to a kitten. France (Brumpt and Urbain, 1938a): A young cat was fed cockroaches infested with _Prosthenorchis elegans_ and _P. spirula_.

=Felis pardalis mearnsi= J. A. Allen

Natural prey.--Cockroaches, Panama (Dr. H. L. Sweetman, personal communication, 1958): An ocelot was seen collecting and feeding on cockroaches,

possibly _Blaberus_ sp. "The ocelot was quite efficient and seemed to relish the roaches."

XIV. CHECKLIST OF COCKROACHES AND SYMBIOTIC ASSOCIATES

Only naturally occurring associations are included in this list. Commensal cockroaches are listed on page 315. Bacteroids are not listed because they undoubtedly occur in all species. The higher plants were excluded because most of the associations may be too casual to constitute symbiosis; however, many of the plant associations were included in the chapter on ecology. The cockroaches and the associates within each category are arranged alphabetically by genus and species. Page references are to citations in the classified sections where details of the associations and/or sources of the records are given.

=Aglaopteryx facies= Mite: Undetermined, p. 220. Reptile: _Anolis stratulus_, p. 274.

=Aglaopteryx diaphana= Nematode: _Protrellus manni_, p. 200.

=Allacta similis= Insects: _Dolichurus stantoni_, p. 262. _Solindenia picticornis_, p. 247.

=Anaplecta= sp. Fungus: _Herpomyces anaplectae_, p. 134.

=Aptera fusca= Protozoan: _Gregarina fastidiosa_, p. 182.

=Arenivaga bolliana= Insect: _Sarcophaga omani_, p. 229.

=Arenivaga roseni= Insects: Undetermined reduviids, p. 227.

=Balta patula= Insect: _Rhipidioides ableptus_, p. 230.

=Blaberus atropos= Bacterium: _Spirochaeta blattae_, p. 125. Protozoa: _Endolimax nana_, p. 180. _Entamoeba coli_, p. 178. Nematodes: _Leidynema appendiculata_, p. 197. _Leidynema cranifera_, p. 198.

=Blaberus craniifer= Bacteria: _Aerobacter aerogenes_, p. 111. _Alcaligenes faecalis_, p. 111. _Bacillus cereus_, p. 120. _Bacillus subtilis_, p. 121. _Escherichia coli_ var. _communior_, p. 113. _Escherichia freundii_, p. 113. _Micrococcus pyogenes_ var. _albus_, p. 106. _Micrococcus pyogenes_ var. _aureus_, p. 107. _Proteus vulgaris_, p. 114. _Pseudomonas aeruginosa_, p. 104. Fungi: _Herpomyces tricuspidatus_, p. 138. _Penicillium_ sp., p. 131. _Rhizopus nigricans_, p. 133. _Saccharomyces cerevisiae_, p. 133. Protozoan: _Diplocystis_ (?) sp., pp. 181, 184. Nematodes: _Leidynema cranifera_, p. 198. _Protrelleta floridana_, p. 199. Mite: _Iolina nana_, p. 219. Undetermined, p. 220. Bird: Chicken, p. 278.

=Blaberus discoidalis= Mite: Undetermined, p. 220.

=Blaberus giganteus= Hair worm: _Gordius pilosus_, p. 202.

=Blaberus= sp. Fungi: _Herpomyces macropus_, p. 136. _Herpomyces paranensis_, p. 136. _Herpomyces periplanetae_, p. 137. _Herpomyces tricuspidatus_,

p. 138. Protozoan: _Leptomonas blaberae_, p. 167.

=Blaptica dubia= Protozoan: _Pileocephalus blaberae_, p. 184.

=Blatta (Shelfordella) lateralis= Bacterium: _Shigella paradysenteriae_, p. 119. Insect: _Ampulex assimilis_, p. 257.

=Blatta orientalis= Bacteria: _Aerobacter aerogenes_, p. 111. _Alcaligenes faecalis_, p. 111. _Alcaligenes recti_, p. 111. _Arthromitus intestinalis_, p. 124. B. aerobio del pseudoedema maligno, p. 125. _B. alcaligenes beckeri_, p. 125. B. del pseudoedema maligno, p. 125. Bacillo proteisimile, p. 126. Bacillo similcarbonchio, p. 126. Bacillo similtifo (Bacillo tifosimile), p. 126. _Bacillus bütschlii_, p. 120. _Bacillus periplanetae_, p. 121. _Bacillus stellatus_, p. 121. _Bacillus subtilis_, p. 122. _Bacillus tritus_, p. 122. _Bacteroides uncatus_, p. 119. _Clostridium lentoputrescens_, p. 122. _Clostridium novyi_ or _Clostridium sporogenes_, p. 122. _Enterococcus_ sp., p. 109. _Escherichia coli_, p. 112. _Lactobacillus fermenti_, p. 109. _Micrococcus pyogenes_ var. _albus_, p. 107. _Micrococcus pyogenes_ var. _aureus_, p. 107. _Micrococcus_ sp., p. 108. _Paracolobactrum_ sp., p. 113. _Pasteurella pestis_, p. 119. _Proteus_ sp., p. 114. _Proteus vulgaris_, p. 114. _Pseudomonas aeruginosa_, p. 104. _Pseudomonas eisenbergii_, p. 105. _Pseudomonas fluorescens_, p. 105. _Salmonella typhosa_, p. 117. _Sarcina alba_, p. 108. _Sarcina_ sp., pp. 108, 109. _Sarcina symbiotica_ (host may have been _B. germanica_), p. 108. _Sarcina ventriculi_, p. 108. _Serratia marcescens_, p. 117. _Spirillochaeta blattae_, p. 127. _Spirillum_ α, β and γ, p. 105. _Spirillum_ sp., p. 105. _Spirochaeta periplanetae_, p. 125. _Streptococcus faecalis_, p. 109. _Streptococcus liquefaciens_, p. 110. _Streptococcus microapoika_, p. 110. _Streptococcus pyogenes_, p. 110. _Streptococcus_ sp., p. 110. _Treponema parvum_, p. 125. _Treponema stylopygae_, p. 125. _Vibrio_ sp., p. 106. Fungi: _Aspergillus fumigatus_, p. 130. _Blastocystis hominis_, p. 133. _Blastocystis_ sp., p. 133. _Candida zeylanoides_, p. 129. _Coccidioides periplanetae_, p. 133. _Herpomyces periplanetae_, p. 137. _Herpomyces stylopygae_, p. 137. _Torula gropengiesseri_, p. 132. _Torulopsis_ sp., p. 130. Protozoa: _Balantidium praenucleatum_, p. 187. _Bodo blattae_, p. 167. _Coelosporidium periplanetae_, p. 185. _Diplocystis schneideri_, p. 181. _Endamoeba blattae_, p. 177. _Entamoeba thomsoni_, p. 179. _Endolimax blattae_, p. 180. _Endolimax_ sp., p. 180. _Gregarina blattarum_, p. 181. _Haplosporidium periplanetae_, p. 185. _Hartmannella blattae_, p. 177. _Herpetomonas periplanetae_, p. 167. _Hexamita periplanetae_, p. 171. _Lophomonas blattarum_, p. 172. _Lophomonas striata_, p. 173. _Monas_ sp., p. 167. _Monocercomonoides orthopterorum_, p. 169. _Nyctotherus ovalis_, p. 188. _Oikomonas_ sp., p. 166. _Peltomyces periplanetae_, p. 177. _Plistophora kudoi_, p. 185. _Plistophora periplanetae_, p. 186. _Plistophora_ sp., p. 186. _Retortamonas blattae_, p. 167. _Stenophora_ sp., p. 181. _Tetratrichomastix blattidarum_, p. 170. Helminths: _Ascaris_ sp., p. 209. _Enterobius vermicularis_, p. 209. _Gongylonema neoplasticum_, p. 206. _Gordius blattae orientalis_, p. 202. _Hammerschmidtiella diesingi_, p. 195. _Hammerschmidtiella neyrai_, p. 196. _Leidynema appendiculata_, p. 197. _Spirura gastrophila_, p. 207. _Thelastoma pachyjuli_, p. 201. _Trichostrongylus_ sp., p. 210. _Trichuris trichiura_, p. 210. Mite: Undetermined, pp. 220, 222. Insects: _Dermestes ater_, p. 234. _Dermestes_ sp., p. 234. _Evania

appendigaster_, p. 236. _Evania dimidiata_, p. 239. _Prosevania punctata_, p. 240. _Systellogaster ovivora_, p. 248. _Tetrastichus hagenowii_, p. 250. _Tetrastichus_ sp., p. 254. Bird: Chicken, p. 278. Mammal: _Homo sapiens_, p. 286.

=Blatta= sp. Hair worm: _Gordius aquaticus_, p. 201. Insects: _Evania appendigaster_, p. 236. _Tetrastichus hagenowii_, p. 250.

=Blattella germanica= and/or =Blattella vaga= Viruses: Unspecified strains of poliomyelitis virus, p. 103.

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=Blattella lituricollis= Insect: _Dolichurus stantoni_, p. 262.

=Blattella= sp. Reptile: _Anolis cristatellus_, p. 273.

=Cariblatta delicatula= Insect: _Hyptia_ sp., p. 240. Reptile: _Anolis cristatellus_, p. 273.

=Cariblatta lutea lutea= Plants: _Sarracenia flava_, p. 154. _Sarracenia purpurea_, p. 154.

=Chorisoneura= sp. Insects: _Stylogaster_ sp., p. 228. _Tachysphex blatticidus_, p. 264.

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=Choristimodes= sp. Insect: _Rhipidioides rubricatus_, p. 231.

=Cryptocercus punctulatus= Bacterium: _Bacillus subtilis_, p. 122. Protozoa: _Adelina cryptocerci_, p. 184. _Barbulanympha estaboga_, p. 173. _Barbulanympha laurabuda_, p. 174. _Barbulanympha ufalula_, p. 174. _Barbulanympha wenyoni_, p. 174. _Eucomonympha imla_, p. 176. _Hexamita cryptocerci_, p. 171. _Idionympha perissa_, p. 174. _Leptospironympha eupora_, p. 172. _Leptospironympha rudis_, p. 172. _Leptospironympha wachula_, p. 172. _Macrospironympha xylopletha_, p. 172. _Monocercomonoides globus_, p. 169. _Notila proteus_, p. 170. _Oxymonas doroaxostylus_, p. 170. _Oxymonas nana_, p. 170. _Prolophomonas tocopola_, p. 173. _Rhynchonympha tarda_, p. 174. _Saccinobaculus ambloaxostylus_, p. 170. _Saccinobaculus lata_, p. 170. _Trichonympha acuta_, p. 174. _Trichonympha algoa_, p. 174. _Trichonympha chula_, p. 176. _Trichonympha grandis_, p. 176. _Trichonympha lata_, p. 176. _Trichonympha okolona_, p. 176. _Trichonympha parva_, p. 176. _Urinympha talea_, p. 174. Undetermined gregarine, p. 184. Amphibian: _Plethodon glutinosus_, p. 269.

=Cutilia soror= Insects: _Dolichurus stantoni_, p. 262. _Evania appendigaster_, p. 236. _Szepligetella sericea_, p. 242. Bird: _Phasianus_ sp., p. 277.

=Cutilia= sp. Nematode: Undetermined, p. 201. Insects: _Riekella australis_, p. 231. Undetermined strepsipteron, p. 234.

=Diploptera punctata= Bacterium: _Serratia marcescens_, p. 117. Fungus: _Herpomyces diplopterae_, p. 135. Mites: _Iolina nana_, p. 219. _Locustacarus_ sp., p. 219. Insect: _Nauphoeta cinerea_, p. 324. Amphibian: _Bufo marinus_, p. 270. Bird: _Phasianus_ sp., p. 277.

=Ectobius lapponicus= Protozoa: _Gamocystis tenax_, p. 184. _Monocercomonoides orthopterorum_, p. 169. Nematode: _Blatticola blattae_, p. 193. Insects: _Brachygaster minutus_, p. 235. _Dolichurus corniculus_, p. 261. _Tachysphex lativalvis_, p. 264.

=Ectobius pallidus= Protozoan: _Gamocystis tenax_, p. 184. Helminths: _Blatticola blattae_, p. 193. Undetermined mermithid, p. 192. Insects: _Ampulex fasciata_, p.

259. _Aphaenogaster picea_, p. 267. _Dolichurus corniculus_, p. 261. _Lasius alienus_, p. 268. _Ripidius boissyi_ (presumptive record), p. 231. _Ripidius denisi_ (presumptive record), p. 232. _Tachysphex lativalvis_, p. 264.

=Ectobius panzeri= Centipede: _Scolopendra_ sp., p. 224. Insects: _Dolichurus corniculus_, p. 261. _Tachysphex lativalvis_, p. 264.

=Ectobius= sp. Insects: _Brachygaster minutus_, p. 235. _Dolichurus corniculus_, p. 261. _Tachysphex lativalvis_, p. 264.

=Ellipsidion affine= Insect: _Rhipidioides fuscatus_, p. 230.

=Ellipsidion australe= Insects: _Agamerion metallica_, p. 243. _Cheiloneurus viridiscutum_, p. 244. _Mestocharomyia oophaga_, p. 248.

=Epilampra abdomen-nigrum= Insect: _Podium flavipenne_, p. 265. Amphibian: _Bufo marinus_, p. 270.

=Epilampra conferta= Insect: _Podium_ sp., p. 266.

=Epilampra= sp. Fungus: _Herpomyces tricuspidatus_, p. 138. Insect: _Podium haematogastrum_, p. 266. Bird: _Gymnasio nudipes_, p. 279.

=Epilampra wheeleri= Reptiles: _Ameiva exsul_, p. 275. _Anolis cristatellus_, p. 273. Mammal: _Herpestes javanicus auropunctatus_, p. 289.

=Escala= (?) sp. Insect: _Rhipidioides adynatus_, p. 230.

=Eurycotis floridana= Fungi: _Aspergillus flavus_, p. 130. _Aspergillus sydowi_, P. 130. Helminths: _Euryconema paradisa_, p. 194. _Leidynema appendiculata_(?), p. 198. _Protrelloides paradoxa_, p. 200. Undetermined gordian worm, p. 202. Insects: _Anastatus floridanus_, p. 245. Undetermined tachinid, p. 228.

=Eurycotis improcera= Mammal: _Herpestes javanicus auropunctatus_, p. 289.

=Eurycotis manni= Fungus: _Herpomyces zanzibarinus_, p. 138.

=Graptoblatta notulata= Insect: _Tachysphex fanuiensis_, p. 264.

=Gromphadorhina portentosa= Protozoan: Undetermined gregarine, p. 184. Mite: _Coleolaelaps_ ? sp., p. 216.

=Gyna= sp.(?) Fungus: _Herpomyces zanzibarinus_, p. 138.

=Holocompsa fulva= Insect: _Pheidole megacephala_, p. 268.

=Hololampra punctata= Insect: _Dolichurus corniculus_, p. 261.

=Ischnoptera deropeltiformis= Plant: _Sarracenia flava_, p. 154 Amphibian: _Hyla cinerea_, p. 270. Mammal: _Dasypus novemcinctus_, p. 287.

=Ischnoptera rufa rufa= Fungus: _Spicaria prasina_, p. 130. Mammal: _Herpestes javanicus auropunctatus_, p. 289.

=Ischnoptera= sp. Fungus: _Herpomyces arietinus_, p. 134. Insects: _Ampulex canaliculata_, p. 257. _Syntomosphyrum ischnopterae_, p. 249.

=Karnyia discoidalis= Spider: _Latrodectus indistinctus_, p. 215.

=Kuchinga hemerobina= Hair worm: _Parachordodes raphaelis_, p. 202.

=Leucophaea maderae= Bacterium: _Serratia marcescens_, p. 117. Fungus: _Herpomyces tricuspidatus_, p. 138. Protozoa: _Gregarina rhyparobiae_, p. 183. _Hexamita_ (?) sp., p. 171. _Retortamonas_ (?) sp., p. 167. Undetermined gregarine, p. 184. Nematodes: _Hammerschmidtiella diesingi_, p. 195. _Leidynema delatorrei_, p. 198. _Protospirura bonnei_, p. 206. _Protospirura muricola_, p. 206. Mite: _Chaetodactylus_ sp., p. 218. Mammals: _Aotes zonalis_, p. 284. _Ateles dariensis_, p. 284. _Cebus capucinus_, p. 284. _Rattus norvegicus_, p. 287.

=Leucophaea= sp. Nematode: _Cephalobellus brevicaudatum_, p. 194.

=Leurolestes pallidus= Fungus: _Herpomyces leurolestis_, p. 136.

=Loboptera decipiens= Insects: _Dolichurus corniculus_, p. 261. _Zeuxevania splendidula_, p. 243.

=Loboptera= sp. Fungus: _Herpomyces lobopterae_, p. 136.

=Lobopterella dimidiatipes= Insect: _Ampulex canaliculata_, p. 257. Bird: _Coturnix coturnix japonica_, p. 277.

=Melanosilpha capensis= Protozoa: _Gregarina impetuosa_, p. 183. _Gregarina sandoni_, p. 183.

=Monastria= sp. Insect: _Triatoma arthurneivai_, p. 227.

=Nauphoeta cinerea= Bacteria: _Salmonella typhimurium_, p. 116. _Serratia marcescens_, p. 117. Fungus: _Herpomyces tricuspidatus_, p. 138. Mites: _Blattilaelaps nauphoetae_, p. 216. _Locustacarus_ sp., p. 219. Insect: _Pheidole megacephala_, p. 268.

=Neostylopyga rhombifolia= Bacterium: _Serratia marcescens_, p. 117. Insects: _Evania appendigaster_, p. 237. _Szepligetella sericea_, p. 242. _Tetrastichus hagenowii_, p. 250. Mammal: _Homo sapiens_, p. 286.

=Nyctibora obscura= Fungus: _Herpomyces amazonicus_, p. 134.

=Nyctibora= sp. Fungus: _Herpomyces nyctoborae_, p. 136.

=Nyctibora tomantosa= Fungus: _Herpomyces nyctoborae_, p. 136.

=Oniscosoma granicollis= Insect: _Paranephrites xenus_, p. 230.

=Panchlora exoleta= Protozoan: _Gregarina panchlorae_, p. 183.

=Panchlora nivea= Bacterium: _Serratia marcescens_, p. 117. Fungus: _Herpomyces panchlorae_, p. 136. Mammal: _Herpestes javanicus auropunctatus_, p. 289.

=Panesthia angustipennis= Protozoa: _Clevelandella constricta_, p. 189. _Clevelandella contorta_, p. 189. _Clevelandella elongata_, p. 189. _Clevelandella hastula_, p. 189. _Clevelandella panesthiae_, p. 189. _Clevelandella parapanesthiae_, p. 189. _Endamoeba javanica_, p. 178. _Endamoeba philippinensis_, p. 178.

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=Panesthia australis= Fungus: _Metarrhizium anisopliae_, p. 131. Mite: _Hypoaspis_ sp., p. 217. Undetermined diplogyniid, p. 217. Insect: Undetermined tachinid, p. 228.

=Panesthia laevicollis= Nematode: _Blattophila sphaerolaima_, p. 194.

=Panesthia laevicollis= (?) Nematode: _Leidynemella fusiformis_, p. 198.

=Panesthia lobipennis= Fungus: _Herpomyces panesthiae_, p. 136.

=Panesthia spadica= Protozoa: _Clevelandella constricta_, p. 189. _Clevelandella contorta_, p. 189. _Clevelandella nipponensis_, p. 189. _Clevelandella panesthiae_, p. 189. _Endamoeba javanica_, p. 178. _Nyctotherus uichancoi_, p. 188. _Paraclevelandia brevis_, p. 190. _Paraclevelandia simplex_, p. 190.

=Panesthia= sp. Nematode: _Leidynemella panesthiae_, p. 198.

=Parahormetica bilobata= Helminths: _Agamospirura parahormeticae_, p. 205. Undetermined gordian worm, p. 202.

=Parcoblatta lata= Plant: _Sarracenia flava_, p. 154. Protozoan: _Leptomonas_ sp., p. 167. Nematode: _Protrellus aurifluus_, p. 199.

=Parcoblatta pensylvanica= Protozoa: _Gregarina blattarum_, p. 182. _Gregarina illinensis_, p. 183. _Gregarina parcoblattae_, p. 183. _Gregarina thomasi_, p. 184. _Leptomonas_ sp., p. 167. _Nyctotherus ovalis_, p. 188. Nematode: _Rictularia coloradensis_, p. 205. Scorpion: _Centruroides vittatus_ (?), p. 212. Insects: _Camponotus pennsylvanicus_, p. 267. _Hyptia dorsalis_, p. 239. _Hyptia harpyoides_, p. 239. _Hyptia reticulata_, p. 240. _Hyptia thoracica_, p. 240. _Podium carolina_, p. 265. _Systellogaster ovivora_, p. 248. Amphibian: _Rana_ (?) sp., p. 272.

=Parcoblatta uhleriana= Fungus: _Herpomyces arietinus_, p. 134. Protozoan: _Gregarina parcoblattae_, p. 183. Nematode: _Protrellus aurifluus_, p. 199. Mite: Undetermined, p. 220. Insect: _Hyptia harpyoides_, p. 239.

=Parcoblatta virginica= Fungus: _Herpomyces arietinus_, p. 134. Protozoa: _Gregarina ohioensis_, p. 183. _Leptomonas_ sp., p. 167. Nematode: _Rictularia coloradensis_, p. 205. Insects: _Hyptia harpyoides_, p. 239. _Podium luctuosum_, p. 266.

=Parcoblatta= sp. Fungus: _Herpomyces arietinus_, p. 134. Plant: _Sarracenia flava_, p. 154. Mite: _Pimeliaphilus podapolipophagus_, p. 219. Insects: _Coenosia basalis_, p. 229. _Dolichurus greenei_, p. 261. _Hyptia_ sp., p. 240. _Megaselia_ sp., p. 227. _Syntomosphyrum blattae_, p. 248. _Systellogaster ovivora_, p. 248. _Tetrastichus hagenowii_, p. 250.

=Periplaneta americana= Viruses: Unspecified strain(s) of poliomyelitis virus (host may have been _P. brunnea_), p. 103. Bacteria: _Achromobacter hyalinum_, p. 110. _Aerobacter aerogenes_, p. 111. _Aerobacter cloacae_, p. 112. _Aerobacter_ sp., p. 112. _Alcaligenes faecalis_, p. 111. _Bacillus cereus_, p. 120. _Bacillus megaterium_, p. 121. _Bacillus subtilis_, p. 122. _Bacterium alkaligenes_, p. 119. _Clostridium_ sp., p. 122. _Eberthella oedematiens_, p. 112. _Escherichia coli_, p. 112. _Escherichia freundii_, p. 113. _Escherichia intermedium_, p. 113. _Mycobacterium friedmannii_, p. 123. _Mycobacterium lacticola_, p. 123. _Mycobacterium leprae_, p. 123. _Mycobacterium phlei_, p. 123. _Mycobacterium piscium_, p. 124. _Mycobacterium_ sp., p. 124. _Nocardia_ sp. (?), p. 124. _Paracolobactrum aerogenoides_, p. 113. _Paracolobactrum coliforme_, p. 113. _Paracolobactrum_ sp., p. 113. _Proteus mirabilis_, p. 114. _Proteus morganii_, p. 114. _Proteus rettgeri_, p. 114. _Proteus vulgaris_, p. 114. _Proteus_ sp., p. 114. _Pseudomonas aeruginosa_, p. 104. _Pseudomonas fluorescens_, p. 105. _Salmonella anatis_, p. 114. _Salmonella morbificans_, p. 115. _Salmonella schottmuelleri_, p. 115. _Salmonella_ sp. (Type Bareilly), p. 115. _Salmonella_ sp. (Type Bredeny), p. 115. _Salmonella_ sp. (Type Kentucky), p. 115. _Salmonella_ sp. (Type Meleagris), p. 116. _Salmonella_ sp. (Type Newport), p. 116. _Salmonella_ sp. (Type Oranienburg), p. 116. _Salmonella_ sp. (Type Panama), p. 116. _Salmonella_ sp. (Type Rubislaw), p. 116. _Salmonella_ sp. (Type Tennessee), p. 116. _Sarcina_ sp., p. 108. _Serratia marcescens_, p. 118. _Shigella alkalescens_, p. 118. _Spirillum periplaneticum_, p. 105. _Streptococcus faecalis_, p. 109. _Streptomyces leidynematis_, pp. 196, 197. _Tetragenous_ sp., p. 127. _Veillonella parvula_, p. 109. Fungi: _Aspergillus flavus_, p. 130. _Aspergillus niger_, p. 130. _Aspergillus_ sp., p. 131. _Cephalosporium_ sp., p. 131. _Herpomyces chaetophilus_, p. 135. _Herpomyces periplanetae_, p. 137. _Metarrhizium anisopliae_, p. 131. _Mucor guilliermondii_, p. 132. _Mucor_ sp., p. 132. _Penicillium_ sp., p. 131. _Rhizopus_ sp., p. 133. _Syncephalastrum_ sp., p. 133. _Torula acidophila_, p. 132. Protozoa: _Balantidium blattarum_, p. 187. _Balantidium ovatum_, p. 187. _Balantidium_ sp., p. 187. _Bodo_ sp. (host may have been _B. germanica_), p. 167. _Coelosporidium periplanetae_, p. 185. _Diplocystis schneideri_, p. 181. _Diplocystis_ sp., p. 181. _Dobellina_ sp. (host may have been _B. germanica_), p. 177. _Endamoeba blattae_, p. 177. _Endolimax blattae_, p. 180. _Endolimax_ sp. (host may have been _B. germanica_), p. 180. _Entamoeba coli_ (host may have been _B. germanica_), p. 178. _Entamoeba histolytica_ (host may have been _B. germanica_), p. 179. _Entamoeba_ sp., p. 179. _Entamoeba thomsoni_, p. 179. _Eutrichomastix_ sp. (host may have been _B. germanica_), p. 169. _Gregarina blattarum_, p. 182. _Gregarina légeri_, p. 183. _Gregarina neo-brasiliensis_, p. 183. _Hexamita periplanetae_, p. 171. _Iodamoeba_ sp. (host may have been _B. germanica_) p. 180. _Isotricha caulleryi_, p. 187. _Lophomonas blattarum_, p. 172. _Lophomonas striata_, p. 173. _Monocercomonoides orthopterorum_, p. 169. _Nyctotherus ovalis_, p. 188. _Plistophora periplanetae_, p. 186. _Protomagalhaesia serpentula_, p. 184. _Tetratrichomastix blattidarum_, p. 170. Helminths: _Ancylostoma duodenale_, p. 209. _Ascaris lumbricoides_ or _Ascaris suum_, p. 209. _Binema mirzaia_, p. 193. _Gongylonema neoplasticum_, p. 206. _Gongylonema_ sp., p. 206. _Gordius_ sp., p. 202. _Hammerschmidtiella diesingi_, p. 195. _Hymenolepis_ sp., p. 208. _Leidynema appendiculata_, p. 197. _Moniliformis dubius_, p. 203. _Moniliformis moniliformis_, p.

204. _Necator americanus_, p. 210. _Protrellus künckeli_, p. 199. _Schwenkiella icemi_, p. 200. _Spirura gastrophila_, p. 207. _Thelastoma pachyjuli_, p. 201. _Trichuris trichiura_, p. 210. Mites: _Caloglyphus_ sp., p. 218. _Pimeliaphilus podapolipophagus_, p. 219. _Rhizoglyphus tarsalus_, p. 218. _Tyrophagus noxius_, p. 218. Undetermined, p. 220. Spiders: _Avicularia avicularia_, p. 214. _Avicularia_ sp., p. 214. Centipedes: Undetermined, p. 222. Insects: _Ampulex amoena_, p. 256. _Ampulex compressa_, p. 259. _Anastatus tenuipes_, p. 246. _Calodexia_ (?) _venteris_, p. 228. _Evania appendigaster_, p. 237. _Melittobia chalybii_, p. 248. _Prosevania punctata_, p. 241. _Ripidius pectinicornis_, p. 232. _Spiniger domesticus_, p. 227. _Szepligetella sericea_, p. 242. _Tetrastichus hagenowii_, p. 250. _Tetrastichus periplanetae_, p. 253. _Tetrastichus_ sp., p. 254. _Trirhogma caerulea_, p. 262. Amphibian: _Hyla cinerea_, p. 270. Reptiles: _Ameiva exsul_, p. 275. _Anolis cristatellus_, p. 273. Birds: Chicken, p. 278. Sparrow, p. 282. Mammals: _Felis catus_, p. 289. _Herpestea javanicus auropunctatus_, p. 289. _Herpestes_ sp., p. 289. _Homo sapiens_, p. 286. _Rattus norvegicus_, p. 287. _Rattus rattus_, p. 287.

=Periplaneta australasiae= Bacteria: _Mycobacterium leprae_, p. 123. _Serratia marcescens_, p. 118. Fungus: _Herpomyces periplanetae_, p. 137. Plant: _Sarracenia minor_, p. 154. Protozoa: _Endamoeba blattae_, p. 177. _Endolimax blattae_, p. 180. Nematodes: _Gongylonema neoplasticum_, p. 206. _Hammerschmidtiella diesingi_, p. 195. _Moniliformis dubius_, p. 203. _Protrellus australasiae_, p. 199. _Protrellus künckeli_, p. 200. Centipedes: Undetermined, p. 222. Insects: _Ampulex amoena_, p. 256. _Ampulex compressa_, p. 259. _Evania appendigaster_, p. 237. _Szepligetella sericea_, p. 242. _Tetrastichus australasiae_, p. 249. _Tetrastichus hagenowii_, p. 250. _Tetrastichus_, sp., p. 254. _Trirhogma caerulea_, p. 262. Reptiles: _Anolis cristatellus_, p. 273. _Chrysemys picta_, p. 272. Bird: _Gallus_ sp., p. 277. Mammals: _Herpestes_ sp., p. 289. _Homo sapiens_, p. 287.

=Periplaneta brunnea= Viruses: Unspecified strain(s) of poliomyelitis virus (host may have been _P. americana_), p. 103. Bacterium: _Serratia marcescens_, p. 118. Fungus: _Herpomyces periplanetae_, p. 137. Nematode: _Schwenkiella icemi_, p. 200. Insect: _Tetrastichus hagenowii_, p. 251.

=Periplaneta fuliginosa= Insect: _Ampulex amoena_, p. 256.

=Periplaneta= sp. Bacterium: _Serratia marcescens_, p. 118. Fungi: _Herpomyces chaetophilus_, p. 135. _Herpomyces periplanetae_, p. 137. Protozoan: _Lophomonas blattarum_, p. 172. Helminths: _Thelastoma riveroi_, p. 201. _Moniliformis moniliformis_, p. 204. Insects: _Ampulex compressa_, p. 259. _Evania subspinosa_ (presumptive record?), p. 239. _Szepligetella sericea_, p. 242. Amphibian: _Bufo marinus_, p. 270. Reptiles: _Anolis grahami_, p. 274. _Anolis leachi_, p. 274. Bird: _Vireo latimeri_, p. 281.

"=Phyllodromia=" sp. Fungus: _Herpomyces phyllodromiae_, p. 137. Insects: _Dicarnosis alfierii_, p. 245. _Dolichurus stantoni_, p. 262.

=Platyzosteria castanea= Insect: _Riekella_ sp., p. 231.

=Platyzosteria novae seelandiae= Protozoan: _Monocercomonoides melolonthae_, p. 169.

=Platyzosteria scabra= Fungus: _Herpomyces appendiculatus_, p. 134.

=Platyzosteria= sp. Insect: _Riekella nitidioides_, p. 231.

=Plectoptera= sp. Bird: _Todus mexicanus_, p. 279.

=Polyphaga aegyptiaca= Nematodes: _Blatticola blattae_, p. 193. _Galebia aegyptiaca_, p. 195. _Hammerschmidtiella diesingi_, p. 195. _Oxyuris_ (?) _heterogamiae_, p. 199.

=Polyphaga saussurei= Helminths: _Hymenolepis_ sp., p. 208. Undetermined tapeworm ova, p. 208. Insects: Undetermined reduviids, p. 227.

=Polyzosteria melanaria= (?) or =Platyzosteria analis= Nematode: _Protrellus aureus_, p. 199.

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=Steleopyga= (?) =sinensis= Nematode: _Suifunema caudelli_, p. 200.

=Supella supellectilium= Viruses: Unspecified strain(s) of poliomyelitis virus, p. 103. Bacterium: _Serratia marcescens_, p. 117. Fungus: _Herpomyces supellae_, p. 138. Nematode: _Blattophila supellaima_, p. 194. Insects: _Anastatus tenuipes_, p. 246. _Comperia merceti_, p. 244.

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=Temnopteryx phalerata= Protozoan: _Gregarina gibbsi_, p. 182.

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XV. COCKROACHES AS COMMENSALS

These particular associations may well have been accidental and due to a predilection for the same type of nesting site. But this fact in no way detracts from the interest of such records. Chance must play a very considerable part in first bringing symbiotic or commensal partners together. Once such a partnership between species has been firmly established, it is on the whole, fairly obvious, ... On the other hand, in the early stages before the relationship has become fixed as a specific habit, individual cases are generally dismissed as coincidences. It is, however, unwise to disregard such isolated observations or dismiss them lightly.

ROTHSCHILD AND CLAY (1957)

The following social insects have been found harboring cockroaches in a state of commensalism in which the cockroaches presumably benefit by acquiring food from their hosts. Benefits accruing to the hosts are not apparent. Unfortunately, biological details are not always sufficient to substantiate the suspected association. However, it seems significant that the cockroach commensals of the insects listed below have been found only in association with their hosts and, so far as we know, have never been found apart from them. Chopard (1938) has pointed out that the myrmecophilous cockroaches are all small, being only a very few millimeters long; they are apterous or subapterous; their eyes are reduced; and they are all of American origin.

HOSTS OF COMMENSAL COCKROACHES

Order ISOPTERA

Family RHINOTERMITIDAE

=Coptotermes ceylonicus= Holmgren

Commensal.--_Sphecophila ravana_, Ceylon (Fernando, 1957): Six females, 50 males, and nymphs of both sexes were found among decaying timber in the ground in association with a colony of this termite. The antennae of most specimens were mutilated unsymmetrically.

Family TERMITIDAE

=Macrotermes barneyi= Light

Commensal.--_Nocticola sinensis_, Kowloon (Silvestri, 1947): Among specimens of termites collected from a nest.

=Macrotermes bellicosus= (Smeathman)

or

=Macrotermes natalensis= (Haviland)

Synonymy.--_Termes bellicosus_ [Snyder, 1949].

Commensal.--_Sphecophila termitium_, Kibonoto, East Africa (Shelford, 1910): Two males were collected in a termite mound.

=Macrotermes malaccensis= (Haviland)

Synonymy.--_Termes malaccensis_ Haviland [Snyder, 1949].

Commensal.--_Nocticola termitophila_, Tonkin (Silvestri, 1946): The cockroach was found in the termite nest.

=Odontotermes= sp.

Commensals.--_Nocticola sinensis_, Kowloon (Silvestri, 1946): In termite nest.

Nocticola termitophila, Penang (Silvestri, 1946): In termite nest.

=Termes= sp.

Commensal.--_Nocticola sinensis_, Repulse Bay, Australia(?) (Silvestri, 1946): In a termite gallery.

=Termites=

Commensal.--_Ergaula capensis_ [= _Dyscologamia wollastoni_] French Equatorial Africa, Brazzaville (Rehn, 1926; Chopard, 1949).

Order HYMENOPTERA

Family FORMICIDAE

Subfamily FORMICINAE

=Camponotus femoratus= (Fabricius)

Commensal.--_Phorticolea boliviae_, Bolivia, Cachuela Esperanza (Caudell, 1923): Three males collected in the joint nests of _C. femoratus_ and _Crematogaster limata_.

=Camponotus maculatus= (Fabricius)?

Note.--Dr. W. L. Brown (personal communication, 1957) states that this ant is an Old World species only. So presumably Mann's record pertains to a different species.

Commensal.--_Myrmecoblatta rehni_, Mexico (Mann, 1914): "They were very abundant, several occurring in almost every nest, where they are no doubt very efficient scavengers."

=Camponotus rufipes= (Fabricius)

Commensals.--_Atticola mortoni_, _Nothoblatta wasmanni_, and _Phorticolea testacea_, Brazil, San Leopoldo (Bolívar, 1905): Found in the formicaries of _C. rufipes_.

=Formica rufibarbis= Fabricius

and

=Formica subcyanea= Wheeler

Commensal.--_Myrmecoblatta rehni_, Mexico (Mann, 1914): "They were very abundant, several occurring in almost every nest."

Subfamily MYRMICINAE

=Acromyrmex lobicornis= Emery

Commensal.--_Attaphila bergi_, or possibly a variety of this species, Huasán, Argentina? (Bruch, 1916).

=Acromyrmex lundi= (Guérin)

Synonymy.--_Atta lundi_ [Brown, personal communication, 1957].

Commensal.--_Attaphila bergi_, Argentina and Uruguay (Bolívar, 1901): The cockroach was found in the nests of the ants sitting on the back, neck, or head of sexual individuals. It remains attached to the ant during swarming. The antennae seem always to be mutilated. Bruch (1916) stated that in La Plata _A. bergi_ is encountered by hundreds in every nest of _A. lundi_.

=Acromyrmex niger= (F. Smith)

Synonymy.--_Atta nigra_ Schupp [Brown, p.c., 1957].

Commensal.--_Attaphila schuppi_, Brazil, Porto Alegre (Bolívar, 1905): Found outside the nest of the ant and mixed in the columns of ants on the march.

=Acromyrmex octospinosus= (Reich)

Synonymy.--_Atta octospinosa_ [Brown, p.c., 1957].

Commensals.--_Attaphila fungicola_, Panama (Wheeler, 1928): Taken in the fungus gardens of the ant.

Attaphila aptera, Esperanza, Dibulla, Colombia (Bolívar, 1905).

=Acromyrmex silvestrii= Emery

Commensal.--_Attaphila bergi_, or possibly a variety of this species, San Luis Province, Argentina (Bruch, 1916): According to Bruch, the behavior of this species of _Attaphila_ is identical with the one encountered in Huasán in the nests of _Acromyrmex lobicornis_ Emery; it differed from _A. bergi_ in size and color.

=Atta cephalotes= (Linnaeus)

Commensal.--_Attaphila fungicola_, British Guiana (Wheeler, 1928): Taken in the fungus gardens of the ant.

Attaphila sp., British Guiana (Beebe, 1921): 7 of 12 queens in one nest had cockroaches hanging on them.

=Atta sexdens= (Linnaeus)

Commensal.--_Attaphila sexdentis_, Brazil, San Leopoldo (Bolívar, 1905): Found in nests of the ant.

=Atta texana= (Buckley)

Synonymy.--_Atta fervens_ Say [Wheeler, 1910].

Commensal.--_Attaphila fungicola_, U.S.A., Texas (Wheeler, 1900, 1910): The cockroach does not feed on the fungus in the ants' nest, as Wheeler (1900) first supposed, but mounts the back of the soldiers and licks their surfaces. It is tolerated by the ants with no signs of hostility. The antennae of the cockroach are clipped short. Although Wheeler (1910) stated that this is probably accidental or unintentional, it is peculiar that Bolívar (1905) noticed the same invariable mutilation of the antennae of _Attaphila bergi_. Wheeler (1900) had originally suggested that the antennae were probably clipped off by the ants which are continuously trimming the fungus hyphae. Louisiana (Moser, personal communication, 1959): Numerous specimens were encountered in some nests of _A. texana_. This cockroach is the most closely associated inquiline in the nest and maintains very intimate terms with the ants. It is found living in the fungus cavities and tunnels.

=Crematogaster limata parabiotica= Forel

Commensal.--_Phorticolea boliviae_, Bolivia, Cachuela Esperanza (Caudell, 1923): Collected in joint nests of _C. limata_ and _Componotus femoratus_.

=Solenopsis geminata= (Fabricius)

Commensal.--_Myrmecoblatta wheeleri_, Guatemala (Hebard, 1917a): Collected from a colony of this ant under a stone on the shores of Lake Atitlan, altitude 11,719 feet.

=Unknown host=

Cockroach.--_Attaphila flava_, British Honduras (Gurney, 1937): Because the known hosts of the other five species of _Attaphila_ are ants, we presume that this species also lives in the nest of some myrmecine ant.

Subfamily PONERINAE

=Odontomachus affinis= (Guérin)

Commensal.--_Myrmeblattina longipes_, Brazil, Rio de Janeiro (Chopard, 1924, 1924a; Hancock, 1926): Originally described as _Phileciton longipes_ by Chopard (1924) from the nest of an ant mistakenly identified as _Eciton_ sp.

Family VESPIDAE

=Polybia pygmaea= Fabricius

Commensal.--_Sphecophila polybiarum_, French Guiana (Shelford, 1906a): Shelford stated that it was probable that the cockroaches living on the floor of the paper nest fed on small fragments of insects and spiders that were dropped by the wasp larvae feeding in the cells above.

Family MEGACHILIDAE

=Melipona nigra= Lepeletier

Commensal.--_Oulopteryx meliponarum_, Brazil (Hebard, 1921): According to Hebard, this cockroach is the first one to be known to inhabit the nests of bees. Nothing is known of the relationship between the cockroach and the bees. [See comment by Sonan (1924) on page 318.]

CHECKLIST OF COMMENSAL COCKROACHES WITH THEIR HOSTS

The cockroaches are arranged alphabetically by genus and species. The page references are to citations in the classified section above, where details and/or sources of the records are given.

=Attaphila aptera= Ant: _Acromyrmex octospinosus_, p. 313.

=Attaphila bergi= Ants: _Acromyrmex lobicornis_, p. 312. _Acromyrmex lundi_, p. 312. _Acromyrmex silvestrii_, p. 313.

=Attaphila flava= Host unknown, presumably an ant, p. 314.

=Attaphila fungicola= Ants: _Acromyrmex octospinosus_, p. 313. _Atta cephalotes_, p. 313. _Atta texana_, p. 313.

=Attaphila schuppi= Ant: _Acromyrmex niger_, p. 312.

=Attaphila sexdentis= Ant: _Atta sexdens_, p. 313.

=Attaphila= sp. Ant: _Atta cephalotes_, p. 313.

=Atticola mortoni= Ant: _Camponotus rufipes_, p. 312.

=Ergaula capensis= Termites, p. 311.

=Myrmeblattina longipes= Ant: _Odontomachus affinis_, p. 314.

=Myrmecoblatta rehni= Ants: _Camponotus maculatus_(?), p. 312. _Formica rufibarbis_, p. 312. _Formica subcyanea_, p. 312.

=Myrmecoblatta wheeleri= Ant: _Solenopsis geminata_, p. 314.

=Nocticola sinensis= Termites: _Macrotermes barneyi_, p. 311. _Odontotermes_ sp., p. 311.

=Nocticola termitophila= Termites: _Macrotermes malaccensis_, p. 311. _Odontotermes_ sp., p. 311. _Termes_ sp., p. 311.

=Nothoblatta wasmanni= Ant: _Camponotus rufipes_, p. 312.

=Oulopteryx meliponarum= Bee: _Melipona nigra_, p. 314.

=Phorticolea boliviae= Ants: _Camponotus femoratus_, p. 311. _Crematogaster limata parabiotica_, p. 312.

=Phorticolea testacea= Ant: _Camponotus rufipes_, p. 312.

=Sphecophila polybiarum= Wasp: _Polybia pygmaea_, p. 314.

=Sphecophila ravana= Termite: _Coptotermes ceylonicus_, p. 310.

=Sphecophila termitium= Termite: _Macrotermes bellicosus_ or _Macrotermes natalensis_, p. 311.

OBSCURE ASSOCIATIONS

Cockroaches that are sometimes found in the nests of, or in association with, other animals are not necessarily commensals. This is particularly true of cockroaches that normally are found unassociated with other animals or that merely occupy the same habitat with the other animals because of similar microclimatic requirements (see Chopard, 1924c).

McCook (1877) excavated in February a nest of _Formica rufa_ in Pennsylvania. A hundred or more lively cockroaches occupied a part of the nest that contained few ants. Near the cockroaches McCook also found a colony of _Termes flavipes_. _Ischnoptera deropeltiformis_ has been found in the company of ants, but it is probably not myrmecophilous (Donisthorpe, 1900). Mann (1911) found an "_Ischnoptera_" sp. (probably a species of _Parcoblatta_) abundant in the nests of, and tolerated by, _Camponotus maccooki_ Forel in California. Dead and mutilated specimens of this cockroach were common in the nests of "Formicas." "_Ischnoptera_" sp. was also common in the nests of _Veromessor andrei_ (Mayr) [= _Stenamma andrei_]. Hebard (1917) reported that W. M. Wheeler collected _Eremoblatta subdiaphana_ in Arizona as an ant guest. Rehn (1906a; Rehn and Hebard, 1927) reported that _Pholadoblatta inusitata_ had also been taken by Wheeler from the galleries of a jumping ant, _Odontomachus clarus_ Roger [= _O. haematodes insularis_ Guérin var. _pallens_ Wheeler; Brown (personal

communication, 1958)], on Andros Island, Bahamas; Rehn and Hebard (1927) stated that "This genus and species is the only blattid, which is presumably a myrmecophile, known from the West Indies." Rehn (1932a) reported _Dendroblatta sobrina_ as taken in an ant nest in a tree in the Amazon Basin. _Tivia australica_ was taken in an ant nest in Australia (Princis, 1954). The male of _Compsodes schwarzi_ was taken in an ant nest in the Santa Rita Mountains of Arizona (Ball et al., 1942). A male and female of _Stilpnoblatta minuta_ were taken in a migrating column of the ant _Myrmicaria natalensis_ Sm. subsp. _eumenoides_ Gerst. in Nyasaland (Princis, 1949). Princis cautioned that it is premature to derive any inference from this, possibly accidental, association. Four females of _Parcoblatta desertae_ were taken about a nest of an ant, _Ischnomyrmex_ sp. (Hebard, 1943a). A nymph of _Parcoblatta virginica_ was found in a nest of _Formica_ sp. (Hauke, 1949).

Chorisoneura texensis has been found in nests of webworm in Florida (Rehn and Hebard, 1916). Karny (1924) in Malaya found an oötheca of _Aristiger histrio_ (sp.?) between leaves (_Costus_ sp.) that had been stuck together by a thysanopteron, _Anaphothrips_ sp. He pointed out that the oötheca would not adhere to leaves that were not stuck together but would fall to the ground.

Seín (_in_ Rehn and Hebard, 1927) in Puerto Rico found _Aglaopteryx facies_ in abandoned cocoons of _Megalopyge krugii_ (Dewitz) and in leaves webbed together by caterpillars and in abandoned spiders nests. Wolcott (1950; and _in_ Rehn and Hebard, 1927) also found _A. facies_ in the empty cocoons of _M. krugii_ and in the larval tents of _Tetralopha scabridella_ Ragonot on _Inga vera_ (coffee shade tree); and "Where there are no butterfly-nests, it lives in abandoned spider-nests on the leaves of other forest trees." Cotton (_in_ Wolcott, 1950) found the type of _Aglaopteryx absimilis_ also living in the abandoned cocoon of _M. krugii_ on bucare trees in Puerto Rico. Wolcott (1950) reported that _Plectoptera dorsalis_, _Plectoptera infulata_, and _Plectoptera rhabdota_ have been found living in trees between leaves or in "butterfly-nests" of _Tetralopha scabridella_ in leaves of _Inga vera_, or nests of _Pilocrocis secernalis_ (Möschler) in the leaves of _Petitia domingensis_ in the mountains of Puerto Rico. Seín (_in_ Rehn and Hebard, 1927) had collected _P. rhabdota_ in the nest of larvae of _T. scabridella_.

Wolcott (1950) reported that _Nyctibora lutzi_ had been found in a large rotten stump associating with "'comején' termites [_Nasutitermes costalis_ (Holmgren)], yellow wood-ants and rhinoceros beetle grubs." Rehn and Hebard (1927) found _Simblerastes jamaicanus_ in numbers in the debris of an abandoned termites' nest in Jamaica: "To what extent the species is dependent upon the protection of the termite or other structures remains to be determined."

In Virginia _Cryptocercus punctulatus_ has been found living in the same galleries with _Reticulitermes_ sp., and on the Pacific Coast it has been found occupying the same log with _Termopsis_ sp. (Cleveland et al., 1934).

Shelford (1909) found one male and one female of _Balta platysoma_ in a nest of a spider of the genus _Phryganoporus_ and assumed a symbiotic association. Chopard (1924) recorded _Mareta acutiventris_ from empty nests of spiders on Barkuda

Island, India; nothing is known of the relationships, if any, between these cockroaches and spiders.

Chopard (1924c) found _Margattea_ sp. in the nest of the ant _Acropyga acutiventris_ Roger; he also found _Margattea_ sp., _Periplaneta_ sp., _Polyphaga indica_, and _Temnopteryx obliquetruncata_ in deserted termite mounds in India. However, he believed that none of these species were more than accidental associates of the host insects; he considered them hygrophilous cockroaches which had found a retreat in the nests.

McClure (1936) obtained a large nest of _Vespula maculata_ (Linnaeus) [= _Vespa maculata_] in March in Illinois. In it were living 65 nymphs of _Parcoblatta pensylvanica_, 3 spiders (_Philodromus pernix_ Blackwall), 2 immature spiders (_Drassus_ sp.), and 6 mites. Balduf (1936) observed four individuals of _Parcoblatta pensylvanica_ in a nest of _Vespula maculata_; he suggested that they probably fed on dead bodies and organic wastes of the wasps. However, Rau (1940) has observed this cockroach devour a _Polistes_ larva in its cell. Although we do not imply that a commensal relationship exists between _Parcoblatta_ and the wasp, it is well to recall a statement by Rothschild and Clay (1957): "A commensal relationship is potentially even more dangerous than a merely social tie, for by nature it is more intimate. The closer the association, the more easily is the balance upset. One partner can then suddenly take a mean advantage of the other."

Cockroach nymphs may enter bees' nests where, according to Imamura (_in_ Sonan, 1924), they do not feed on honey or pollen but presumably feed on excreta of bees or anything scattered by bees in their nest; the bees are not disturbed by the cockroaches.

Cockroaches that have been found in the burrows of vertebrates are listed on pages 23-25.

Paulian (1950) found immature cockroaches in the nests of birds (Ploceinae) in Madagascar and Ivory Coast. All nests of _Fondia_ sp. examined in Madagascar contained many cockroaches, and Paulian believed that the blattid was a species peculiar to the nests of birds. Three nests of _Ploceus_ sp. in Ivory Coast yielded one or two cockroaches each in association with more numerous mites, Psocoptera, Heteroptera, beetles, and lepidopterous larvae (Delamare, Deboutteville and Paulian, 1952). These last cited workers also found four cockroaches in a nest of _Estrildine_ sp., and two in a nest of an undetermined bird, all in association with other arthropods. Moulton (1912) observed large numbers of _Symploce cavernicola_ and _Periplaneta australasiae_ swarming in soft bird guano on the floor of caves in Borneo. Abdulali (1942) found in India many _Periplaneta americana_ in caves containing the edible-nest swift; there was no indication of association of the cockroaches with the birds. Danforth (_in_ Wolcott, 1950) reported finding large numbers of _Aglaopteryx facies_ "in the nests of the grey kingbird, in the region of the Cartagena Lagoon [Puerto Rico], 'living among the twigs.'" In Trinidad, Kevan found a male of _Blaberus discoidalis_ in a bird's nest (Princis and Kevan, 1955).

Davis (_in_ Rehn and Hebard, 1914a) stated that "At Punta Gorda [Florida] there was a vacant house at the end of the town frequented at night by a Nanny and Billy goat, and on warm evenings many _Periplaneta australasiae_ would run about on the piazza floor and on the sides of the house. They were seen feeding on the excrement of the goats and were no doubt to a great degree dependent upon them." This is another example of a coprophagous insect that has taken advantage of a particular situation favorable to its survival. Similar associations exist in which many of the domiciliary cockroaches feed on the feces of man and domestic and other animals (Roth and Willis, 1957a).

XVI. COCKROACHES AS PREDATORS

INTERSPECIES PREDATION

Tepper (1893) made the broad statement that the majority of Australian and Polynesian cockroaches appear to be wholly carnivorous, eating other insects, eggs, and larvae. He stated that, because of their voracity and cannibalistic tendencies, the carnivorous species lead more or less solitary lives so that one rarely meets several in close proximity; they are never very numerous at any time because the stronger devour the weaker in the absence of other prey. Tepper stated that Australian species of _Ischnoptera_ hunt for their prey among the foliage of shrubs, and that Australian species of _Cutilia_ [= _Drymaplaneta_, Hebard (1943)] run about actively on the surface, or ascend shrubs and trees in quest of living insects and therefore are highly beneficial. Tepper (1894) also stated that _Geoscapheus robustus_ ate earthworms, grubs, and caterpillars. Froggatt (1906) and Marlatt (1915) attributed to Tepper the statement that cockroaches, like _Epilampra notabilis_, which are found out-of-doors in Australia, are carnivorous and feed on caterpillars and other soft-bodied insects; but Froggatt (1907) believed that this alleged behavior needed confirmation.

A number of observations have been recorded which indicate that sometimes cockroaches may be predatory. According to Ealand (1915), nymphs of the cockroach _Pseudomops cincta_ fed on the Argentine ant _Iridomyrmex humilis_. In the laboratory, _Eurycotis floridana_ has been observed to catch and devour the wasp _Anastatus floridanus_ which parasitizes the eggs of _Eurycotis_ (Roth and Willis, 1954a). _Parcoblatta pensylvanica_ was observed devouring a larva of _Polistes_ sp. in its cell in a deserted wasps' nest (Rau, 1940). Brigham (1866) saw a cockroach kill and eat a centipede four or five inches long. Annandale (1910) described the destruction in Calcutta of termites by _Periplaneta americana_. During a heavy rain storm many termites flew into the dining room and were set upon by the cockroaches which seized them with their mandibles and began to gnaw their abdomens. If disturbed, the cockroaches carried the termites away in their mandibles without using their legs to seize, hold, or carry the prey. Sometimes only the abdomen, but other times the whole body with the exception of the wings, was devoured. Perhaps this observation led Allyn (Anonymous, 1937) to theorize that, first, cockroaches could eradicate termites from houses, and then the blattids in turn could be eliminated. Falls (1938) has pointed out the unfeasibility of this idea. _Blattella vaga_

has shown some tendency to eat plant lice (Flock, 1941a). Certain small cockroaches found beneath cane leaf-sheaths, in the Philippine Islands, preyed in part upon leafhoppers (Uichanco, _in_ Williams et al., 1931).

Takahashi (1924) stated that the American cockroach will eat the eggs of the hemipteron _Cantao ocellatus_ (Thunberg). Cunliffe (1952) observed mite-infested cockroaches (_Blatta orientalis_, _Blattella germanica_, and/or _Periplaneta americana_) dislodge and eat the mite _Pimeliaphilus podapolipophagus_. Sonan (1924) reported that cockroaches (_P. americana_ and _P. australasiae_) devoured the egg clusters and first instar larvae of _Prodenia litula_ and the first instar larvae of _Attacus atlas_ which were being reared in the laboratory. Lederer (1952) stated that _Periplaneta americana_ ate reptile eggs in the aquarium at Frankfurt am Main. Pettit (1940) stated that cockroaches "are said to have destroyed a large colony of dermestids used to skeletonize carcasses at the University of Kansas."

DeFraula (1780) believed that his silent "gryllon" [obviously _Blatta orientalis_ from his drawings; see Willemet (1784)] was the enemy of the chirping species of cricket, because after the cockroach became established in his home he no longer heard crickets chirping. Gilbert White (1905 ed.), writing in England in the late 18th century, stated that "Poda says that these [_Blatta orientalis_] and house crickets will not associate together; but he is mistaken in that assertion"; however, in August 1792 White noted that "Since the _blattae_ have been so much kept under, the crickets have greatly increased in number." For several years Jolivet (1950) had observed changes in a mixed population of _Blatta orientalis_ and _Acheta domesticus_ in an old kitchen in France. He suggested that the cyclical fluctuations in the relative abundance of the cockroaches and crickets might be caused by reciprocal predatism with one species more susceptible than the other at certain stages. Mallis (1954) has stated that crickets prey on other insects as well as on one another. Lhéritier (1951) had also observed crickets becoming rare in bakeries in France, having been superseded everywhere by _B. orientalis_; however, he doubted that Jolivet's hypothesis was the correct explanation and suggested that the higher optimum temperature requirements of crickets might be the regulating factor. Lederer (1952) stated that the number of crickets decreased in the aquarium buildings at Frankfurt am Main as the population of American cockroaches increased.

Platyzosteria novae seelandiae was found under the bark of trees in New Zealand devouring bugs (Walker _in_ Shelford, 1909b).

For years it has been believed that cockroaches feed on bedbugs (_Cimex lectularius_ L.) and this statement has been repeated in many reference works and articles. Ealand (1915) stated that cockroaches devour bedbugs with avidity. Even today similar statements are to be found in the literature. "In the old sailing ship days, they [cockroaches] were often welcomed by crews because of the belief that they would eradicate a population of bedbugs. This belief was based on scientific fact, as cockroaches are known as predators of bedbugs" (Monro, 1951). Cockroaches will often "help rid a house of bedbugs by devouring all the little parasites they can capture" (Gaul, 1953). The basis for this belief may have originated with a statement by Webster (1834) who wrote that bedbugs disappeared aboard "H.M. Sloop

Chanticleer" when cockroaches made their appearance. Newman (1855) reported the observations of a friend who claimed to have seen a cockroach seize a bedbug in an infested boardinghouse in London. In 1920 Purdy reintroduced cockroaches into a house from which they had been exterminated, in order to control the bedbugs which had become established. According to a popular account by Lillingston (1934) African natives are said to ask sailors for a cockroach or two to be used to hunt bedbugs.

In Siberia, Burr (1926, 1939) found _Blattella germanica_ and bedbugs inhabiting the same room. Mellanby (1939) studied the populations of an animal house in which bedbugs and cockroaches occurred in large numbers; the bugs apparently were not attacked and their numbers increased greatly over a period of a few weeks (Johnson and Mellanby, 1939). Wille (1920) placed starved _B. germanica_ with bedbugs for 20 days, but the cockroaches failed to attack the bugs. In India, captive adults and nymphs of two species of house cockroaches would not touch living bedbugs or their eggs (Cornwall, 1916). In laboratory experiments Gulati (1930) found that _Periplaneta americana_ ate young bedbugs which had soft, blood-filled abdomens; adult bedbugs with harder exoskeletons sometimes were rejected. The maximum number of bedbugs eaten by a cockroach was 3 out of 12 during a period of 48 hours. Johnson and Mellanby (1939), also in laboratory experiments, were unable to show that bedbugs can be controlled by _Blatta orientalis_ or that bedbugs are eaten to any extent by them. The existing evidence indicates that there is little basis for the often repeated statement that cockroaches destroy bedbugs in nature. As Lorando (1929) pointed out, assassin bugs, cockroaches, and red ants can hardly be considered as practical factors in bedbug control, though he did recommend the use of spiders.

According to Martini (1952), cockroaches prey on mosquitoes and sand flies but we have been unable to find any original sources for these statements; the only reference we have found in which cockroaches and _Phlebotomus_ are mentioned together is a paper by Whittingham and Rook (1923); they fed ground-up cockroaches to larvae of _Phlebotomus papatasii_. Wharton (1951) reported that cockroaches and other predators attacked mosquitoes knocked down by insecticides and affected the number recovered.

Cockroaches will on occasion attack and bite animals other than insects. In an earlier paper (1957a) we discussed about 20 reports of cockroaches biting man. The injury is usually confined to abrasion of the callused portions of hands and feet but may result in small wounds in the softer skin of the face and neck. We failed to include the following reference in the above-mentioned paper. Sonan (1924) had his toes and breast nibbled by cockroaches on Hiyakejima Island during sleep. He had previously learned from a policeman that _Periplaneta americana_ and _P. australasiae_ nibbled people on that island, but he had hardly believed it before he experienced the biting himself.

INTRASPECIES PREDATION

Those who have reared cockroaches in the laboratory have undoubtedly seen cannibalism occur in the cultures. Cannibalism has been observed among the

common domiciliary species of cockroaches as well as laboratory colonies of _Leucophaea maderae_ (Scharrer, 1953), and _Blaberus craniifer_[12] (Saupe, 1928). Edmunds (1957)

reported that cannibalism was common in a laboratory colony of _Periplaneta brunnea_ and that egg capsules deposited by a female were often eaten by the other cockroaches.

Periplaneta americana occasionally ate other cockroaches and their oöthecae and also attacked members of their own species (Lederer, 1952). Griffiths and Tauber (1942) recorded the killing of male American cockroaches by females of the species: "One female was especially vicious and attacked each new male as he was introduced into the container. Most of such males had molted less than 2 days previously. Older males were more capable of defending themselves against attacks of these cannibalistic females." Even though adequate food may be present, females of _Periplaneta americana_ may eat their own eggs (Klein, 1933). Some females may regularly eat their oöthecae as soon as they are dropped (Griffiths and Tauber, 1942). To be completely eaten an oötheca generally must be attacked before it has hardened. If a hole is eaten in one side of the capsule, the cockroach may devour the eggs and leave a portion of the oötheca. Frequently only the keel or a part of the keel is eaten and when this occurs the eggs fail to hatch and usually do not complete development because of the rapid loss of water (Roth and Willis, 1955). When adults of _P. americana_ and _P. australasiae_ were deprived of food, both males and females ate newly deposited eggs and, finally, the females ate the males (Sonan, 1924).

Parcoblatta virginica in laboratory cultures also may eat part of its oöthecae; in this species only the soft end of the recently deposited oötheca was eaten (Roth, unpublished data, 1957).

Cros (1942) observed oöthecae-bearing females of _Blatta orientalis_ attack and kill males of the same species which were attempting to mate; these males were then eaten by the females. Cros also observed injured and recently molted nymphs of _B. orientalis_ to be eaten by others of the same species.

Pettit (1940) noted that cannibalism in his culture of _Blattella germanica_ occurred only when the insects were molting. Adult insects attacked the molting cockroaches more often than did the nymphs. However, nymphs after the fourth instar occasionally set upon other molting nymphs. First-to third-instar nymphs rarely victimized their mates. The victims were all older than third instar; the later stadia were progressively more subject to attack, and molting adults suffered the greatest mortality. No direct correlation was noted between population density and cannibalism.

German cockroaches may attack newly molted nymphs of their own kind and cause them to deflate (Gould and Deay, 1938). Lhéritier (1951) has observed the hatching nymphs of _B. germanica_ being devoured by their congeners even before they have left the oötheca.

Nauphoeta cinerea in laboratory cultures will eat newly hatched young of the same species (Roth and Willis, 1954; Willis et al., 1958). In Hawaii, in nature, _N. cinerea_ may kill and eat the cypress cockroach, _Diploptera punctata_ (Illingworth, 1942; Fullaway and Krauss, 1945).

Bunting (1956) stated that species of _Neoblattella_ are omnivorous with carnivorous and cannibalistic tendencies. An adult female _Panchlora_ sp. was killed and eaten by _Neoblattella_ sp. in captivity. A male, provisionally identified as _N. celeripes_, was killed and partly eaten by two other males of the same species.

The factors influencing the extent of cannibalism among cockroaches are not completely known. According to Wille (1920) hunger was not the cause of cannibalism in _Blattella germanica_. Wille claimed that the tendency toward cannibalism increased at high temperatures and decreased at low temperatures. Pettit (1940) also noted this effect. Gould and Deay (1938) stated that under crowded laboratory conditions, when there was a scarcity of food, cannibalism among _Periplaneta americana_ was common. The injured cockroaches and those unable to molt were often eaten. Adair (1923) made similar observations. Undoubtedly, conditions of crowding, availability of food, temperature and other factors all influence cannibalism, but practically no experimental work has been done on this subject.

It is interesting, in comparison with the above positive examples of cannibalism, that both Saupe (1928) and Roeser (1940) observed no cannibalism during extensive studies with _Pycnoscelus surinamensis_. In fact, Roeser stated that there was never a case of cannibalism in spite of long hunger periods imposed on both nymphal and adult insects.

XVII. ASSOCIATIONS AMONG COCKROACHES

Besides preying on their own species or on other blattids, cockroaches exhibit additional symbiotic relationships among themselves. These relationships are (1) the familial associations of parent and offspring, (2) gregariousness, (3) intraspecies fighting, (4) interspecies compatibility, and (5) interspecies antagonism. There are some inconsistencies between observations made on the same species by different workers, which only further observation and experimentation will explain. Some of the reported observations are unique; this is especially true for the feral species. Because of the paucity of information, it is impossible at this time to make valid generalizations about some of these interesting relationships.

FAMILIAL ASSOCIATIONS

The females of many species of cockroaches insure varying degrees of protection to the developing young in their ways of disposing of the oötheca after it has been formed. The extent of this association between the mother and her developing progeny varies from the minimum amount of time spent by oviparous females in concealing their oöthecae, to the duration of embryogenesis in the so-called

viviparous species, a period of over a month or more.

Haber (1920a) observed a female of _Periplaneta americana_ chew a groove in a piece of pasteboard into which she attempted to deposit her oötheca. The oötheca failed to adhere to the shallow hole and fell to the floor. After several futile attempts to replace the oötheca in the hole, the female finally left the egg case on the floor of the cage and coated it with an oral secretion to which she attached bits of trash. During this operation she chased other females away when they ventured near the site. Qadri (1938) described the behavior of the female of _Blatta orientalis_ in concealing her oötheca in a hole that she dug in sand; she deposited the egg case in the hole, coated it with saliva and sand, and then refilled the pit. Rau (1943) described in detail how females of _P. americana_ and _B. orientalis_ covered their oöthecae with wood dust or sand in holes they had prepared in the substrate. Both species placed a sticky oral secretion in the holes and then deposited their oöthecae therein. After coating the oöthecae with more sticky secretion, the females adjusted the oöthecae so that the keels were uppermost and then carefully concealed the oöthecae with the excavated debris. Both females spent over an hour in the act. Rau (1924) previously reported that of 90 oöthecae deposited by _B. orientalis_ in jars containing earth and trash, 36 were placed in crevices or excavated holes, and 38 were hidden by being covered with dirt stuck to them with saliva; only 16 were left uncovered.

Edmunds (1957) described oviposition by _Periplaneta brunnea_. Some females spent from 30 to 40 minutes secreting from the mouth a frothy substance that was smeared on the substrate; the egg capsule was deposited in the secretion and covered with additional froth, which hardened into a very strong cement. Some females spent as long as two hours coating the capsule after it was deposited. It was not stated whether the oötheca was otherwise concealed. The female remained with her body over the oötheca for several hours and drove away other cockroaches which approached.

Sonan (1924) observed that _Periplaneta americana_ and _Periplaneta australasiae_ spent from 40 minutes to an hour covering their oöthecae, and that if the females were frightened away from this activity, they returned again to complete it. As well as excavating holes in the substrate in which to deposit its oöthecae, _P. americana_ also avails itself of readymade crevices of appropriate size (Ehrlich, 1943). Species of _Epilampra_ in Malaya were said by Annandale (1900) to deposit their oöthecae in crevices in floating logs just above the water line. However, Shelford (1906) stated that four genera (including _Epilampra_) of the subfamily Epilamprinae are "viviparous," in which event the females would carry their oöthecae within their bodies during embryogenesis and would not place the oöthecae in crevices in logs.

The female of _Cryptocercus punctulatus_ was observed to make a groove in a piece of wood, then carry her oötheca 6 inches from where she had dropped it and place it in the groove; she covered the oötheca so that only a portion was visible (Cleveland _in_ Cleveland et al., 1934). Dr. W. L. Nutting (personal communication, 1954) collected a number of oöthecae of _C. punctulatus_ in the field and found each

one almost completely sealed off with bits of wood in a deep groove in the roof of a chamber in a log. The keel of the oötheca was visible but the rest was well camouflaged. He stated that "The adult pair usually frequents the chamber at this time, while their broods of previous years occupy neighboring galleries."

Berland (1924) observed a female of _Loboptera decipiens_ filling a hole (the abandoned nest of a hymenopteron) with earth that she carried in her mouth; he later found her oötheca behind the earthen barricade which she had erected.

In summary, the following species of oviparous cockroaches have been observed concealing their oöthecae (only those references not previously cited are given): _Blatta orientalis_; _Cryptocercus punctulatus_; _Ectobius sylvester_ (Harz, 1956, 1957); _Epilampra_ sp.; _Eurycotis floridana_ (Roth and Willis, 1954a); _Loboptera decipiens_; _Balta scripta_, _Methana curvigera_, _Methana marginalis_, and _Methana caneae_ (Pope, 1953a); _Pelmatosilpha marginalis_, _Pelmatosilpha purpurascens_, and _Nauclidas nigra_ (Bunting, 1956); _Periplaneta americana_ (Haber, 1919; Adair, 1923; Seín, 1923; Nigam, 1933; Gould and Deay, 1938; Rau, 1940a); _Periplaneta australasiae_ (Girault, 1915b; Spencer, 1943; Pope, 1953); _Periplaneta brunnea_ (Roth and Willis, unpublished data, 1958); _Periplaneta fuliginosa_ (Gould and Deay, 1940); _Periplaneta ignota_ (Pope, 1953); _Supella supellectilium_ (Flock, 1941). Undoubtedly other oviparous species that drop their oöthecae long before the eggs hatch also make some attempt to conceal the oöthecae by placing them in crevices or covering them with debris.

Sometimes the oöthecae are deposited but not concealed. Hafez and Afifi (1956) reported that in Egypt _Supella supellectilium_ attaches its oötheca to a suitable substrate with a gummy oral secretion but leaves the egg capsule otherwise exposed. We (1954) have noticed similar behavior in laboratory colonies of this species and of _Blatta orientalis_, as have Gould and Deay (1940). Cornelius (1853) stated that the female of _B. orientalis_ takes care of the safety of her offspring to the extent of usually dropping her oöthecae in places which are dry and raised above the ground, although rarely one also may find some oöthecae scattered on the ground. For lack of suitable material females of _Periplaneta americana_ sometimes did not conceal their oöthecae (Nigam, 1933). Frequently in laboratory colonies _P. americana_ merely drops the oöthecae loosely in sand or food "in contrast to _P. australasiae_, which almost always went to considerable trouble to fasten their eggs securely and to conceal them with debris" (Pope, 1953). If conditions under which _Nauclidas nigra_ is kept are not suitable, the female will drop her oötheca anywhere (Bunting, 1956). Rau (1940) stated that the female of _Parcoblatta pensylvanica_ does not conceal her oötheca. However, Gould and Deay (1940) stated that this species deposits its oöthecae loosely behind bark. _Ellipsidion affine_ and _Ellipsidion australe_ attach their oöthecae to bark or the underside of leaves but apparently make no attempt to conceal them (Pope, 1953a).

The females of most of the above species have no further familial association with their offspring. The eggs hatch with no attention from the mother who is probably not even in the vicinity at that time. The young apparently do not react to the presence of the parent, as such, after hatching. This is not unexpected, as several additional

oöthecae may have been deposited by these oviparous females before the eggs of the first oöthecae hatch. However, a different behavior is encountered among species that do not form a second oötheca until after the eggs of the first have hatched (see below) and in the so-called colonial species.

Shaw (1925) reported that in Australia both _Panesthia australis_ and _Panesthia laevicollis_ appear to live in families, and that one usually finds a pair of adults associated with from 12 to 20 nymphs in different stages of development; he continued, "it is only where the molts are very abundant that one loses sight of this familial habit." Tillyard (1926) also stated that the Australian species of _Panesthia_ live in burrows in soil in strict family communities of a pair of adults and 10 to 20 nymphs. A related colonial species, _Cryptocercus punctulatus_, lives in both sound and rotten logs in colonies consisting of a pair of adults and 15 or 20 nymphs, probably representing two or three broods (Cleveland et al., 1934; Cleveland, 1948). Chopard (1938) has cited this association as an example of gregariousness, which it may well be; however, the presence of only one pair of adults in each colony suggests a more intimate relationship.

Among species of _Blattella_ and certain other genera with similar reproductive habits the female carries her oötheca clasped in her genital cavity with the posterior portion projecting behind her. Each normal oötheca is carried for approximately the duration of embryogenesis and is not dropped until, or shortly before, hatching. We have seen (1954, fig. 65) newly hatched nymphs of _Blattella vaga_ crawl over the body of the mother who stood quietly near the dropped oötheca; this female raised her wings and some of the nymphs crawled under them onto the dorsal surface of her abdomen. The nymphs seemed to feed on the grease covering the mother's body. The association was short-lived, however, and soon the nymphs scattered. Pettit (1940) stated that when hatching of _Blattella germanica_ occurs in the open (on a table top), the nymphs may remain near the capsule only a few minutes. Ledoux (1945) found that newly hatched nymphs of _B. germanica_ remained together without shelter in a single, sparse group. If the nymphs were separated by blowing on them, the group quickly reassembled, usually in the same spot. Ledoux showed that this gregarious grouping of first-instar nymphs was not necessarily a familial association by placing nymphs from two oöthecae together. In groups of 8 to 12 nymphs there was a perfect intermingling of the offspring from the two different females.

It is among the so-called viviparous cockroaches that the greatest number of observations have been made of postparturient associations between female cockroaches and their offspring. The females of these species carry their oöthecae in brood sacs within their bodies until embryogenesis has been completed. This behavior ensures protection of the young from desiccation and attack by parasites (Roth and Willis, 1955a). (See Roth and Willis, 1958a, for an analysis of oviparity and viviparity in the Blattaria.) Shelford (1906, 1916) reported that he had captured a female of _Pseudophoraspis nebulosa_ in Borneo with numerous young nymphs clinging to the undersurface of her abdomen. He also recalled that there was in the Hope Museum (Oxford) a female of _Phlebonotus pallens_ to which the following label was attached: "'Ceylon ... carries its young beneath its wing covers. 1878.'" Pruthi

(1933) found in South India another female of _P. pallens_ which was carrying over a dozen young nymphs on her back beneath her wings. In his paper Pruthi reproduced a photograph of this specimen with the light-colored nymphs in place on the back of the female. Hanitsch (1933) reported having seen a museum specimen from Luzon, Philippine Islands, of the apterous female of _Perisphaerus glomeriformis_ with nymphs still clinging to her undersurface; he also reported having seen a museum specimen of a female of _Ellipsidion variegatum_ from Australia with four young clinging to the upper side of the apex of her tegmina and six to the oötheca which projected beyond her body. Presumably this specimen was giving birth when captured. Gurney (1954; personal communication, 1958) stated that specimens of _Perisphaerus_ sp. from Mindanao and Luzon have been found with young nymphs clinging to the middle and hind coxae. The first-instar nymph has an elongate face and specialized galeae. Karny (1925) also observed that at the slightest alarm the young of some species of Phoraspidinae creep under the dome-shaped front wings of the mother.

The newly hatched young of _Leucophaea maderae_ have also been seen congregated under the mother on several occasions. Seín (1923) stated that after being born, the nymphs of this species gather under the mother and accompany her at night in her excursions in search of food. Pessôa and Corrêa (1928) reported that "During the first days the free larvae hide under the adult cockroach which becomes restless and active in contrast to its usual slow gait." Wolcott (1950) stated that "They are not only gregarious, but the mother broods over her young, and together they sally forth at night in search for food, until they are of such a size as to mingle with their elders."

The African mountain cockroach _Aptera fusca_ has been observed during late summer and early winter in familial groups beneath loose bark, under stones, and in dead leaves (Skaife, 1954): "Each party consists of a number of black young ones, together with one, two or more adult females and perhaps a winged male or two. Later on they scatter and live more or less solitary lives." In Malaya Karny (1924) often found phoraspidine females between leaves surrounded by about 20 young nymphs. He stated that one also often found females of _Perisphaerus armadillo_ surrounded by pale, yellowish-white young; similarly he had observed that _Archiblatta hoevenii_ was found mostly in colonies made up of mothers and their young. The duration of these associations is not known.

Saupe (1928) noticed that the newly hatched nymphs of _Blaberus craniifer_ (see footnote II, p. 322) collected together under the body of their mother and stated that this is as pronounced a case of brood care as Zacher had observed with _Pycnoscelus surinamensis_. Nutting (1953) stated that "A degree of maternal solicitude is exhibited by this roach [_B. craniifer_], for many times I have observed the female to remain motionless for an hour or more with her unpigmented brood clustered around and beneath her body." We, too, have observed similar behavior in laboratory colonies of _B. craniifer_ and _Leucophaea maderae_.

Chopard (1950) noted that after hatching the young of _Gromphadorhina laevigata_ remained grouped around the female for some time; the mother stood

motionless, high on her legs, with her thorax curved up to make room for the brood which hid under her body. We (unpublished data, 1958) have seen young nymphs of _Gromphadorhina portentosa_ also stay near their mother for some time after birth; the mother at this time produced a characteristic hissing sound when she was only slightly disturbed by the movement of our hand near her and her brood. The sound is produced as air is expelled through the second abdominal spiracle. We have seen recently hatched nymphs of _Nauphoeta cinerea_ crawl beneath the mother, even under her wings, where they remained about an hour (Willis et al., 1958). Bunting (1956) observed a female of _Blaberus discoidalis_ collect a mound of debris into which she inserted the tip of her abdomen; he found young in the mound later the same day. This female showed no maternal care for the young after birth. Whole families of cockroaches may be found in bromeliads in Brazil (Ohaus, 1900). Hebard (1920) observed a colony of adults and young of _Dendroblatta sobrina_ on a tree trunk in the Panama Canal Zone.

Whether any of the above associations exemplify maternal care for the newly hatched young is questionable. The behavior of the mother, beyond placing her eggs in a suitable location, seems to be entirely passive. The first-instar nymphs are the active partners in these associations, and they may merely be seeking shelter under the nearest convenient object rather than under the mother as such. More extensive studies of some of these relationships will be needed before claims for maternal care, as suggested by Scott (1929), can be substantiated.

GREGARIOUSNESS

Casual statements that cockroaches are gregarious are often encountered in the literature. There has been some argument to the effect that large numbers of these insects seeking the same environment in a limited space would appear to be gregarious, whereas there is probably no true social tendency (Rau, 1924). Reactions of cockroaches to certain stimuli in the environment undoubtedly do result in aggregations of individuals. However, as Chopard (1938) has pointed out, it is difficult to assign the respective parts played in assembling by the attraction of the milieu and by gregarious instincts. Chopard (1938) also stated that Orthoptera with a gregarious tendency are found rarely isolated; one finds them, on the contrary nearly always collected in the same shelters, close together, as if conscious of a need for contact between themselves. He continued further that one can be tempted to attribute the assembling to taxes but that interattraction equally plays an important role; for example, if one places a large number of cockroaches in a container and offers them similar shelters composed of cardboard tubes, one finds that nearly all the individuals will assemble in one of the tubes, ignoring the others. Pettit (1940) claimed that in _Blattella germanica_ gregariousness seemed to depend on the mutual attractiveness of body secretions as well as a thigmopositive behavior and love of warmth.

Ledoux (1945) has studied experimentally gregariousness and social interattraction in _Blatta orientalis_ and _Blattella germanica_. He also found that the cockroaches tended to collect in shelters containing other cockroaches. He concluded

that group formation is not the result of chance, but is a social phenomenon, and that interattraction is mainly olfactory, conditioned by (1) positive chemotaxis to odors emitted by the cockroaches themselves, (2) positive hygrotaxis, and (3) thigmotaxis. He found also that large groups are not stable and tend to break into smaller groups.

Gregariousness in the Orthoptera varies in intensity according to the species and within a species according to the age or physiological state of the insects (Chopard, 1938). This is well exemplified by several of the blattid species discussed below.

Gregarious groupings of cockroaches have been observed most frequently among the domiciliary species. A few examples will suffice. Gal'kov (1926) observed heavy infestations of undetermined cockroaches in workers' living quarters in the Ural region: "In the corners near the stove, the cockroaches covered the walls in a dense carpet." After fumigating he collected about 135,000 dead cockroaches from one barracks and about 475,000 from another. We have reviewed a few other examples of heavy infestations in our 1957(a) paper.

Periplaneta americana was observed by Gould and Deay (1938) in an old meat-packing building in Indiana. Adult cockroaches were present in large numbers between closely placed beams, but the nymphs were more common in cracks between bricks. Clusters of several hundred cockroaches were seen on the open walls of the cold, dark hide room. Gould and Deay stated "American roaches of all sizes live together in perfect harmony. Young nymphs have been noted in clusters underneath adults and crawling over the adults as they wander about in rearing jars." In the monkey house of the Hamburg zoo, _P. americana_ spent most of the day in the cellars resting on the walls in groups of about 200 individuals (Brecher, 1929). Lederer (1952) noted that in closed, dark, heated spaces under the aquarium at Frankfurt am Main, _P. americana_ rested in groups of 20 to 30 individuals; he stated that it was remarkable that the "herd" divided itself into groups each of which usually contained insects of the same age or stage of development. Eads (1954) found _P. americana_ in 40 percent of 762 sewer manholes in Tyler, Tex.; 13 percent of 670 of these manholes were heavily infested with 100 or more cockroaches in each. Other heavy sewer infestations have been reviewed in our 1957(a) paper.

Ehrlich (1943) has stated that _Periplaneta americana_ exhibits social behavior. For instance, cockroaches of various ages inhabit a fairly large space jointly; the adults and older nymphs sense approaches with their antennae and warn and protect the young by a beating of wings and by body movements. There is complete utilization of the available living space; the imagos drive older nymphs from their resting places, and the older nymphs drive out the younger ones, until all cracks, depending on their size, are occupied by various age groups of different sizes. In his experiments Ehrlich observed that in cages with no hiding places the cockroaches would group together; when given a choice of small and large shelters, _P. americana_ hid only under the larger ones that could shelter more insects. Finally, the cockroaches ceased to bite and fight each other when they crowded together in the face of danger.

Of _Blatta orientalis_ Marlatt (1915) stated "This species is notably gregarious in habit, individuals living together in colonies in the most amicable way, the small ones being allowed by the larger ones to sit on them, run over them, and nestle beneath them without any resentment being shown." Haber (1919) also observed that this species is often noticed "huddled together, the younger ones crawling over, around, and beneath the older ones."

Wille (1920) observed that nymphs of _Blattella germanica_ remained almost constantly in groups during the first and second instars, but less so during the third instar. He believed that the aggregations of young occurred because they could occupy narrow crevices where the larger insects could not penetrate. At usual room temperatures the older nymphs and adults lived completely isolated, but at certain temperatures they gathered together in large, tightly pressed groups.

Supella supellectilium is said to be gregarious (Gould and Deay, 1940). The smaller nymphs aggregate in small groups in rearing containers, but the older ones remain separate from one another (Hafez and Afifi, 1956). _Leucophaea maderae_ is sociable and rarely found alone; in their favorite hiding places, hills of these cockroaches can be seen hanging together (Seín, 1923). Wolcott (1950) also stated that _L. maderae_ is gregarious. Annandale (1900) observed that in the "Siamese Malay States" large colonies of _Periplaneta australasiae_ conceal themselves in hollows of bamboo logs from which houses are built. Moulton (1912) stated that he was astonished at the large numbers of _P. australasiae_ and _Symploce cavernicola_ that he saw swarming on the sides of caves of Mt. Jibong, Borneo.

Rehn and Hebard (1905) stated that in Key West, Fla., _Eurycotis floridana_ fairly swarmed under the coquina boulders in the woods, in groups of a dozen containing both young and adults; _Pycnoscelus surinamensis_ was very abundant in the same type of habitat. Caudell (1905) also found the young of _E. floridana_ with the mature individuals. Hebard (1917) in his discussion of _Lattiblattella rehni_ again mentioned finding frequent colonies of _E. floridana_ in Florida. He also found many specimens of _Blaberus craniifer_ under boards on the ground at Key West. He found _Parcoblatta lata_ numerous under bark of dead pine trees in Alabama. However, Dowdy (1955), in an ecological study of oak-hickory forest in Missouri, stated that "_Parcoblatta_ [sp.] were never recorded as being gregarious, in fact they were mostly solitary. However, in some cases two were found together." Yet Blatchley (1895) stated of _Parcoblatta pensylvanica_ that in the winter in Indiana "One cannot pull the loose bark from an old log without dislodging a colony of from ten to a hundred of the nymphs of various sizes." Males of _Parcoblatta virginica_ were said to be often gregarious beneath loose bark and under chunks and rubbish (Blatchley, 1920).

Rehn and Hebard (1927) quoted observations made earlier by Hebard on _Byrsotria fumigata_ in Cuba: "I found the specimens under flat stones, sometimes in colonies of 3 or 4 mature specimens and numbers of immature individuals in all stages of development." These observers also reported that _Aspiduchus borinquen_ was found in Puerto Rico in a limestone cavern by thousands in the grass and on the walls. J. W. H. Rehn (1951a) stated that a related species, apparently _Aspiduchus

cavernicola_, was seen in great numbers on the side walls and roof of a cave in Puerto Rico, but it was not possible to collect any of these and, we infer, confirm the species. Rehn and Hebard (1927) in their account of _Simblerastes jamaicanus_ reported finding it in numbers in a termite nest. Pemberton and Williams (1938) stated that _Diploptera punctata_ is of gregarious habits in Hawaii. Saupe (1928) observed a strong "herd instinct" in all age groups of _Blaberus craniifer_. Bunting (personal communication, 1956) stated that large nymphs and adults of _Blaberus discoidalis_ "congregate in narrow cracks or on the underside of some low object. The younger nymphs keep in close communities of approximately the same age." Sonan (1924) stated that in Formosa(?) _Salganea morio_ is usually found in groups of six or seven in decayed trees. Species of the genus _Litopeltis_ may be found in small groups as they are somewhat gregarious (Rehn, 1928).

The physiological or psychological effects of gregariousness, or lack of it, are interesting aspects of the basic phenomenon. Landowski (1937) studied in _Blatta orientalis_ the effect on development and growth of the transition from life in complete isolation to life in groups. He kept nymphs in groups of 1, 2, 4, 8, and 16 in jars of identical size and shape. Landowski found that (1) mortality increased with the size of the group and with age, as each animal occupied more of the available space. [Presumably these factors are less detrimental in nature where the group is unconfined.] He further found that (2) life in complete isolation extended the time required to produce an adult insect; and (3) the mean weight of the adult insect was, generally, in inverse proportion to the number of nymphs raised together; isolated insects usually attained the greatest adult weight.

Similarly, Griffiths and Tauber (1942a) found that isolation extended the period of nymphal development in _Periplaneta americana_. As most of their isolates died before reaching maturity, these workers concluded that the American cockroach does not thrive when individually isolated and that several individuals must be together for optimum development to occur. Pettit (1940, 1940a) observed that isolated nymphs of _Blattella germanica_ take longer to mature than those reared in groups. Wallick (1954) found indications in _B. germanica_ that there is an inverse relationship between population density and individual weight; as the population decreased the weight increased. He also noted an inverse relationship between population density and life expectancy in this species.

We (Willis et al., 1958) have confirmed the above observations that _Blattella germanica_, _Blatta orientalis_, and _Periplaneta americana_ complete nymphal development in less time when reared in groups rather than individually. We (loc. cit.) also found that nymphs of the following additional species matured more quickly when reared in groups: _Eurycotis floridana_, _Periplaneta fuliginosa_, _Supella supellectilium_, _Nauphoeta cinerea_, and _Pycnoscelus surinamensis_; only a very slight decrease in the average length of the developmental period was found in grouped nymphs of _Leucophaea maderae_.

Wharton et al. (1954) observed that virgin adult males of _Periplaneta americana_ that had been individually isolated upon emergence were almost wholly unresponsive to the sexually stimulating, female odor for a test period of four weeks. Similar males

of comparable age that were kept in groups reacted strongly from the sixth day on. Removal of reactive males from the group inhibited the reaction in these isolates, but the response returned when the insects were regrouped. We (1952) had similarly observed that no isolated male of _Blattella germanica_ was ever seen to give a courting response without having received some form of external stimulation. Yet when numbers of males were kept together isolated from females, on several occasions the males became active and a few individuals gave a courting response. As the sexual stimulus is received by the male of _B. germanica_ through contact rather than odor, as in _P. americana_, presumably it was mutual contact between the grouped males that released the courting activity.

Cloudsley-Thompson (1953a), in his studies of diurnal rhythms in _Periplaneta americana_, observed a steady decline in total activity in successive 24-hour cycles: "When two cockroaches, even of different species (_P. americana_ and _P. australasiae_) were kept together, this depression did not appear to set in so readily." The associates apparently kept each other active.

Isolated females of _Periplaneta americana_ can be conditioned to run a simple maze with less time and fewer errors per trial than when paired or when a member of a group of three (Gates and Allee, 1933). There was less activity, and accordingly fewer errors per minute, among cockroaches tested as pairs and groups of three than as isolated individuals. This observation should not be contrasted with that of Cloudsley-Thompson (1953a), cited above, because the intervals during which activity was observed were quite different.

In the above account we have presumed that aggregations of some species are indications of gregariousness. However, until gregariousness has been proved experimentally for each species, we concede that reactions to environmental stimuli might be sufficient to bring about some of the observed groupings without any interaction between individuals.

In concluding this section we note that Tepper (1893) stated that carnivorous cockroaches in Australia lead more or less solitary lives, and that one rarely meets several together in close proximity. Takahashi (1940) observed that in Formosa _Blattella humbertiana_ does not have a tendency to throng together. Rau (1947) stated that the adults of _Ischnoptera deropeltiformis_ showed no tendency toward gregariousness, but in the laboratory newly hatched young lived close together under bark and remained together throughout the nymphal stages. We wonder whether this gregariousness was not imposed by the restricted quarters of the cage. As mentioned above, Dowdy (1955) did not find _Parcoblatta_ sp. to be gregarious in the field.

INTRASPECIES FIGHTING

Fighting occurs among cockroaches of the same species over food or shelter or between males. Saupe (1928) observed late-instar nymphs of _Blaberus craniifer_ attack each other and even adults. Additional records cited in the section on intraspecies predation (p. 322) imply fighting within a species. Rau (1924) saw a male of _Blatta orientalis_ attack another male in copula and bite away a large portion of

its wing. Two other males in the container had their wings badly torn overnight, presumably as a result of fighting.

Ehrlich (1943) stated that individuals of _Periplaneta americana_ that are feeding will ward off intruders by spreading their wings and pushing with their hind legs. However, the intruder will approach again and again biting the feeder in the legs and wings. Frequently the odor of approaching food was sufficient to cause the cockroaches to fight and bite each other. Biting and fighting also occurred when individuals of this species defended their daytime hiding places. A position of attack is assumed when two antagonistic individuals of _P. americana_ meet (Ehrlich, 1943, fig. 14). The insects raise their bodies slightly above the ground, by extending their legs, and they stretch their heads forward horizontally so that their mouth parts protrude; when the insects jump at each other, they may wound each other severely in the soft parts of the body. Fighting between sexually excited males resulted in injury to their legs, wings, cerci, and other parts of the body. Frequently an insect that could no longer defend itself was killed. Lederer (1952) also made similar but less extensive observations on fighting in this species.

Pettit (1940) quoted Woodruff as stating that nymphs of _Blattella germanica_, apparently healthy and perfectly normal, would do battle for no apparent cause other than a chance meeting, and that occasionally the fight was to the finish, the loser being eaten. Pettit could not substantiate such voracious attacks, although he saw nymphs engage in fights lasting about two seconds during which one would be driven off by vigorous bites on legs or cerci. Small nymphs of _B. germanica_ tended to ignore each other, but third-and later-instar nymphs would engage in "quarrels" of short duration when two met. Pettit noted that males of _B. germanica_ that were crowded together quickly set upon, but did not always kill, other cockroaches introduced into their cage. When he isolated a dozen males in a small cage, they became quarrelsome and three of the group were killed and partly eaten. After several days the surviving males had taken positions so that each was equidistant from his neighbors. Some of these males attacked other males and a female that were introduced, by biting their legs and cerci. Females under similar conditions were much less aggressive, although Pettit saw some females that roved about biting all large members of the group that were within easy reach.

We have frequently observed aggressive behavior between males of _Nauphoeta cinerea_, which resulted in torn wings. The males would wrestle with each other rolling over and over.

INTERSPECIES COMPATIBILITY

We agree in essence with Chopard (1938) who stated that it is improper to speak of associations apropos of the ecological distribution of Orthoptera. He continued that it is clearly evident that different species of Orthoptera, which are found grouped on a territory more or less narrowly limited, have no interdependence among them. Their grouping results uniquely from almost similar reactions to the different factors which characterize this limited milieu. There is neither interdependence nor interaction; the

grouping is a false biocoenose, born under the action of the environment, and does not survive a modification of this milieu.

However, as there are numerous examples of mutual toleration between different species as well as examples of incompatibility, the subject has more than academic interest even if no true ecological significance. On the other hand, further study may show that certain of these associations are definitely ecological, particularly among the feral species. As might be expected, most of the following examples pertain to domiciliary cockroaches.

Dozier (1920) occasionally found _Periplaneta americana_ with _Eurycotis floridana_ in decaying stumps, beneath loose bark of decayed trees, and beneath corded wood. Adair (1923) stated that in his house in Egypt _Periplaneta americana_, _Blatta orientalis_, and _Blattella germanica_ were found together in a cupboard. Sambon (1925) found _B. orientalis_ and _B. germanica_ side by side but not fraternizing in a home in Italy. Gould and Deay (1938) observed that apartments over stores were infested with both _B. germanica_ and _P. americana_, but did not indicate whether these occupied the same microhabitat. Gould and Deay (1940) observed that in the Purdue University greenhouse _Periplaneta fuliginosa_ was found "under benches, boxes, pots and other objects in association with the American roach." Dr. L. A. Hetrick (personal communication, 1954) wrote us that several summers before he had had a mixed infestation of cockroaches, which included _Periplaneta australasiae_, _Periplaneta fuliginosa_, and _Pycnoscelus surinamensis_, in his chicken shed.

Eads (personal communication, 1955), in response to our inquiry about the mixed populations of cockroaches that he had reported infesting sewers in Texas (Eads et al., 1954), stated that "Each of the ten colonies of _B. orientalis_ found in Tyler manholes were associated with larger colonies of _P. americana_. True breeding colonies of _B. orientalis_ appeared to be present since all the developmental stages were taken. The same situation existed with the _P. fuliginosa_ and the two species of _Parcoblatta_. Larger colonies of _P. americana_ were associated with the other species in each case. From our limited observations the two species always appeared to be perfectly compatible." Eads et al. (1954) had found _Periplaneta fuliginosa_ in three manholes, _Parcoblatta bolliana_ in one manhole and _Parcoblatta pensylvanica_ in one manhole. We assume that the groups of each species were spacially discrete so that they were recognizable as colonies. Dr. T. A. Olson (personal communication, 1958) has observed two or more species of cockroaches in a single structure but never in mixed colonies. Each species was separated physically from the others. Olson concluded that cockroaches of different species do not mingle freely unless forced to do so by some special environmental condition. Pettit (1940) found _B. germanica_ and _P. americana_ similarly separated in the same building or even in the same basement laboratory.

Perkins (1899) found _Lobopterella dimidiatipes_ generally living in company with the young of _Periplaneta australasiae_ in Hawaii. Rehn and Hebard (1914) in Florida found _P. australasiae_ abundant with _Periplaneta americana_ on a quarter-boat. They also noted that the forficulid _Marava_ [= _Prolabia_] _arachidis_ (Yersin)

appeared in numbers in a kitchen after dark accompanied by swarms of _P. americana_. These workers also found _Leurolestes pallidus_ in a fruit store in Key West "where the species was common in a pile of old burlap bags and in cracks under the stands which it shared with one fairly large colony of _Blattella germanica_, occasional specimens of _Holocompsa nitidula_, a few specimens of _Periplaneta americana_, and one specimen of _Supella supellectilium_." They also found _H. nitidula_ with _Blaberus craniifer_ "between old boards in a woodshed, where nymphs were more numerous than adults."

Rehn and Hebard (1914) stated of _Supella supellectilium_ in Florida that "The females were all taken in cupboards where _Blattella germanica_ was found in swarms." The association in human habitations of _S. supellectilium_ and _B. germanica_ has been reported also by Sein (1923), Puerto Rico; Shaw (1924), Australia; Mallis (1954): "German and brown-banded roaches were often found in the same crevice."; Anonymous (1958), Texas; and Anonymous (1958a), Georgia. Gould and Deay (1940) stated that other species of cockroaches, especially _B. germanica_, may be found with _S. supellectilium_ in the same part of a building. Yet Shaw (1925) stated that "when _Supella supellectilium_ Serv. invades places already occupied by _Blattella germanica_ L., it tends to oust the latter."

Blaberus discoidalis has been found in homes or in fruit debris in Puerto Rico in company with the more common, domiciliary species _Leucophaea maderae_, but never in abundance (Sein, 1923; Wolcott, 1950). Illingworth (1915) in Hawaii found _Symploce hospes_ associated with _Nauphoeta cinerea_, _Graptoblatta notulata_, and _Diploptera punctata_.

Hebard (1917) found _Aglaopteryx diaphana_ in a bromeliad on a forest tree in Jamaica together with _Nyctibora laevigata_ and numerous _Cariblatta insularis_. He also found numerous _Aglaopteryx gemma_ under signs on longleaf pines in Alabama with occasional specimens of _Parcoblatta lata_. In Virginia he found _Parcoblatta uhleriana_ in a decaying chestnut log with _Cryptocercus punctulalus_. In Florida he found _Latiblattella rehni_ with _Eurycotis floridana_ and, more rarely, with _Periplaneta australasiae_ under bark of pine trees. In Key West he found _Symploce hospes_ in the cupboard of a hotel with swarms of _Blattella germanica_ and a few _Supella supellectilium_.

Rehn and Hebard (1927) in their study of West Indian blattids reported finding _Neoblattella proserpina_ in epiphytic bromeliads in Jamaica in company with _Neoblattella eurydice_ and _Neoblattella dryas_. They also list most of the associations cited by Hebard (1917).

Ramme (1923) reported that he found in Germany four species of _Ectobius_ (_lapponicus_, _lucidus_, _pallidus_, and _sylvester_) living together in an area about 50 m. by 200 m. Although he had stated that his specimens of _E. lucidus_ were a distinct species in 1923, Ramme (1951) later decided that they were a form of _E. sylvester_, _E. sylvester_ f. _lucidus_.

Dow (1955) reported trapping _Blattella germanica_, _Periplaneta americana_, and _Periplaneta brunnea_ in houses and privies in south Texas. At our request Dr. Dow

(personal communication, 1958) analyzed his records to determine whether there were indications of associations between these species, with the following results:

As stated in my published note, the roaches were at first classified to genus only. The 83 _Periplaneta_ subsequently identified to species represented 28 different collections, 11 from houses and 17 from privies, all in Pharr, Texas. Tabulation of the data shows first that _P. americana_ was taken only once in a house and that _P. brunnea_ was taken only 4 times in privies. Of course this distribution greatly reduces the probability that they would be caught together, and it is not surprising that _P. americana_ was trapped alone in the single house collection. _P. brunnea_, however, was trapped with _P. americana_ 2 of the 4 times it occurred in privy collections.

To investigate the occurrence of _Periplaneta_ with and without _Blattella_, an analysis has been made of 560 trap collections taken in 40 houses and 40 associated privies in Pharr, Texas, in weekly intervals (from May 14 to June 23 [1948]). In the houses, _Periplaneta_ and _Blattella_ were caught in the same jar 26 times, _Periplaneta_ alone 12 times, _Blattella_ alone 83 times, and neither genus 159 times. In a fourfold table, the value of chi-square (14.7) is significant and indicates that the frequencies are not proportional. The number of times _Periplaneta_ and _Blattella_ actually occurred together (26) is, however, much larger than the expected number calculated from the row and column frequencies (14.8). In the privies, _Periplaneta_ and _Blattella_ were caught in the same jar 9 times, _Periplaneta_ alone 50 times, _Blattella_ alone 18 times, and neither genus 203 times. In a fourfold table, the value of chi-square (1.95) is not significant but the same type of disproportion is evident and the expected frequency of both genera in one trap is 5.7, lower than the actual frequency of 9. Both immature and adult roaches are included in this analysis.

The above evidence would be more satisfactory if based on more extensive data. There is also a possible objection in that the traps were operated for at least overnight, during which time one species could theoretically supplant another. Of course, it is doubtful that there is anything involved here like territory (in the ornithologists' sense). On the other hand, it is well to consider that _Periplaneta_ and _Blattella_ are both likely to be more abundant in the same type of favorable location and that this factor might offset in part some direct antagonism between the species.

The only known specimen of _Ischnoptera podoces_ was captured in company with the type series of _Cariblatta nebulicola_, in dead leaf litter in Jamaica (Rehn and Hebard, 1927). In Florida _Periplaneta australasiae_ was often taken in company with _Pycnoscelus surinamensis_ and _Eurycotis floridana_ (Blatchley, 1920).

INTERSPECIES ANTAGONISM

In contrast to the presumably amicable associations mentioned above, other observations in the literature seem to indicate that some species of cockroaches are incompatible when they attempt to occupy the same habitat niche. Marlatt (1915) stated "Rarely do two of the domestic species occur together in the same house. Often, also, of two neighboring districts one may be infested with one species, while

in the other a distinct species is the commoner one. The different species are thus seemingly somewhat antagonistic, and it is even supposed that they may prey upon one another, the less numerous species being often driven out." Phelps (1924) stated "Roaches of different species are rarely found together, although roaches of the same species live together on very amicable terms."

In 1859 Darwin (1887) stated that "In Russia the small Asiatic cockroach [_Blattella germanica?_] has everywhere driven before it its great congener [_Blatta orientalis?_]." Yet in France Girard (1877) suggested that the oriental cockroach be introduced into a restaurant infested with the German cockroach as the best way to expel the latter, because the more robust species drives away cockroaches of smaller size. Wille (1920) in Germany found usually only one species of cockroach in a house. Yet when he placed _B. orientalis_ and _B. germanica_ together, there were no reciprocal attacks even by hungry individuals. Wille concluded that because of their greater speed, smaller size, greater number of eggs, and faster development, the German cockroaches eat the available food and so make the environment unfavorable for the oriental. However, he noted that cases may be seen in which the opposite is also possible. Laing (1946; British Museum [Natural History], 1951) observed that in the British Isles _B. orientalis_ seems to have lost its dominant position to _B. germanica_ in recent years; it was stated that these species are not as a rule found together and that the greater rapidity of breeding and ability to climb of _B. germanica_, as well as the layout of modern buildings, are some of the factors that favor the spread of _B. germanica_. Ledoux (1945) found that first-instar nymphs of _B. germanica_ and fourth-instar nymphs of _B. orientalis_, adults of _B. germanica_ and sixth-instar nymphs of _B. orientalis_, as well as adults of both species, did not form mixed groups. However, when he combined fifth-and sixth-instar nymphs of _B. germanica_ with fourth-and fifth-instar nymphs of _B. orientalis_, which are all practically of equal size, sometimes he would find mixed groups, but generally the groups were distinct. Lucas (1912) stated that Burr had found _B. germanica_ and _B. orientalis_ swarming within a rubbish heap in England; presumably both colonies were breeding and multiplying and one species was not detrimental to the presence of the other.

Shaw (1925) claimed that _Supella supellectilium_ tended to oust _Blattella germanica_, but Pope (1953) thought it doubtful in Queensland. Wolcott (1950) stated that "The larger and more powerful domestic cockroaches, _Periplaneta americana_ (L.), _P. australasiae_ (F.) and _P. brunnea_ Burmeister have very definitely fallen behind in Puerto Rico in competition with the little German roach." Pessôa and Corêa (1928) observed that other species of cockroaches were rare in Brazil in houses that were infested with _Leucophaea maderae_. Lederer (1952) noticed that in the reptile house of the aquarium at Frankfort am Main _Blatta orientalis_ was obviously kept down by _Blattella germanica_, even before the appearance of _P. americana_. However, _B. germanica_ was not driven out of the reptile house by _P. americana_ although the populations of each fluctuated for about 22 years after the American cockroach had settled there; both species occupied separate resting places. Lederer further observed that within four years of the introduction of _P. americana_ into the crocodile house, none of the original infestation of _B. orientalis_ could be found; a

small colony of _Pycnoscelus surinamensis_ in the reptile house was apparently also driven out by _P. americana_. Chopard (1932, 1938) stated that the oriental cockroach does not exist in company with _P. americana_ which very probably destroys it. Pettit (1940) kept _B. germanica_ and _P. americana_ together in a cage for several weeks but neither species gave any indication of feeding on the other.

Froggatt (1906) stated that "It is probable that the advent of the larger and more formidable American cockroach into Australia has led to the retirement or destruction of our indigenous species" [presumably _Periplaneta australasiae_]. Tillyard (1926) noted that this statement is incorrect as neither species is native to Australia. Yet Shaw (1925) stated that in Australia "When both species live together in the same places, _australasiae_ Fabr. will probably be found gradually to displace _americana_ L." Local fluctuations in the relative abundance of these species could be a basis for such dissimilar observations. However, MacDougall (1925) observed that in the plant houses of the Royal Botanical Garden, Edinburgh, the Australian cockroach seemed to have overcome the American which had been more numerous in former years.

In conclusion, we emphasize that many of the above observations are merely tentative impressions gathered by workers who have watched many species of cockroaches in nature. Obviously, additional observations coupled with appropriate experimentation will be needed to disclose the true structure of each presumed association and to resolve apparent discrepancies. Although we are greatly indebted to the cited authors for their contributions to the known information, we anticipate that future results of cleverly designed laboratory experiments will do much to dispel the uncertainty that still surrounds our knowledge of the relations of the Blattaria to each other.

XVIII. DEFENSE OF COCKROACHES AGAINST PREDATORS

Irritating or repellent secretions provide many animals belonging to widely unrelated groups with a more or less potent means of defence....

It will be seen that this method of defence does not rest merely upon a passive unpalatable attribute, but upon an active emission of the unpalatable substance which, since it occurs when the animal is seized or threatened by an enemy, enforces its effectiveness. In its highest development we find different forms whose specialized habits and modified structure enables them to _project_ secretion at the enemy, and thus to discourage attack.

COTT (1940)

There are very few records indicating that cockroaches are unaccepted as food by other animals. Hutson (1943) found that the duck, guinea fowl, and pigeon would not normally eat _Pycnoscelus surinamensis_, and in his experiments with the chicken eye worm he had to force-feed his birds with infected cockroaches. Lederer (1952)

found that insectivorous birds in the Zoological Garden, Frankfurt am Main, either refused hardened (as opposed to teneral) American cockroaches or ate them unwillingly. Carpenter (1925) reported that a monkey (_Cercopithecus_) failed to feed on cockroaches and suggested that the insects' odor made them repugnant; however, there are a number of positive records of monkeys feeding on cockroaches (see pp. 284-286).

Cockroaches may escape capture by predators through evasive behavior, concealment, protective coloration, mimicry, or secretion of malodorous materials. Nocturnal cockroaches may avoid predators that are active during the day (Crawford, 1934), but nocturnal predators are apparently quite successful in capturing cockroaches. Some cockroaches may be protected by their swiftness, others by their resemblance to vegetation (Williams, 1928). The habit of squeezing into narrow cracks may afford cockroaches some protection.

Burrowing forms such as _Pycnoscelus_ may spend much time in underground cells (Roeser, 1940). Polyphagids rapidly burrow into sand (Fausek, 1906), where they may be protected from predators. Tepper (1893) discovered that a very large Australian cockroach, _Geoscapheus robustus_, had its fore legs, especially the tibiae, adapted for digging. He observed this species in captivity and in 1894 reported that it appeared to sink into the soil without raising any considerable amount above the surface and that it did not form an unobstructed tunnel. Another large Australian cockroach, _Macropanesthia rhinocerus_, burrows about two feet below the surface of sandy soil; it also makes nests among pine roots and the nymphs rarely appear above ground (Henson _in_ Day, 1950). Tepper (1893) observed that Australian cockroaches of the genera _Epilampra_ and _Oniscosoma_ buried themselves in loose soil and dust. Baker (_in_ Rehn, 1930) observed that _Styphon bakeri_ is found in humus and rubble in the Dutch West Indies where "It is sluggish in the open, but wedges into the humus quite quickly."

Therea nuptialis, found in India, conceals itself at the roots of fig trees, etc. The small hairs on its elytra retain sufficient dust to conceal it, or at any rate to render it inconspicuous, when not on the wing (Annandale, _in_ Chopard, 1924c). Rehn and Hebard (1914) observed that the nymphs of _Blaberus craniifer_[13] at Key West, Fla., "were usually found half buried in loose damp earth under boards, where they remained motionless, looking much like lumps of earth (with which they were usually much dusted) until disturbed." Hebard (1917) reported of _Monastria biguttata_ from Brazil that "All of the juveniles are heavily coated with foreign particles" which adhere "to a multitude of closely placed, minute and usually curved spines, which cover the dorsal surface and marginal portions of the ventral surface."

It is apparent from the numbers of predators reported herein that many animals are not deterred by the odorous secretions of cockroaches; these secretions, because they may seem repugnant to man, are often claimed to be repellent to predators. However, Cott (1940) points out that "There are many instances in which protective devices and associated warning colours are known to be ineffectual against certain enemies. But this does not necessarily imply that they are not on the whole beneficial to the species attacked." Certain cockroach secretions may well be repellent to many

predators, but as this is a purely negative aspect of the predator-prey relationship little thus far has been observed or published. Potential prey that successfully defends itself against attack is never found in a predator's stomach.

Cockroaches have a variety of glands which secrete odorous materials. Certain secretions, produced by tergal or dorsal glands in males, are involved in sexual behavior; the females feed on the secretion from these glands prior to copulating (Roth and Willis, 1954). However, other secretions which are produced by both sexes are ejected or given off when the insect is disturbed; undoubtedly these are defensive weapons that are used against predators. Very few experiments or observations are on record to show how effective these secretions may be in protecting the cockroach. Although the morphology of some of the glands has been described, relatively little is known about the chemistry of their secretions.

Many species of Australian cockroaches have been reported to emit "disgusting" odors, though the glands producing these secretions have not been described, nor is the chemistry of the compounds known. _Cosmozosteria lateralis_ exposed two orange-red spots on the abdomen while emitting a pungent odor which deterred a collector from capturing it (Shelford, 1912). Another Australian species, _Platyzosteria castanea_, when disturbed on barren ground tilts forward on the vertex and straddles out the posterior legs, supporting itself in a vertical position on the head and tarsi; in assuming this attitude it will squirt a foetid fluid as a fine spray for a distance of 6 or 7 inches (Shaw, 1914). Spencer (1892) mentions the pungent odor given off by a cockroach which had been accidentally cut in two. Rageau (1956) stated that in the New Hebrides and New Caledonia _Cutilia nitida_ emits, when disturbed, a corrosive liquid with an extremely disagreeable odor.

The adults of _Eurycotis floridana_ emit an odorous fluid when seized (Rehn and Hebard, 1905). The fluid, which may irritate sensitive skin areas, may be ejected as a spray for a distance of several inches. This secretion has been identified as 2-hexenal (Roth et al., 1956), and the ventral abdominal glands which produce it have been described (Stay, 1957). Eisner (personal communication, 1958) has found that the toad _Bufo marinus_ and the frog _Rana pipiens_ invariably spit out adults of _E. floridana_ which they have seized. The odor of 2-hexenal was strongly apparent after these attacks, and the insect was never damaged. However, the lizard _Anolis equestris_ seized and crushed _E. floridana_ before releasing its hold and dropping the insect 5 to 10 minutes later. The blue jay _Cyanocitta cristata_ readily attacked adults of _E. floridana_ and killed them but did not eat the insects until after the odor had dissipated; however, the bird carried nymphs of _E. floridana_ to its perch and ate them. Nymphs of this species do not secrete 2-hexenal (Roth et al., 1956). Recently, 2-hexenal has been tested for its antibacterial activity and has been found to be active against seven species of pathogenic bacteria (Valcurone and Baggini, 1957). _Eurycotis decipiens_ from Trinidad also ejects a fluid which may produce toxic symptoms such as vertigo and nausea (Bunting _in_ Roth and Willis, 1957a).

Large reservoirs of glands similar in appearance and position to those of _Eurycotis floridana_ are present in the adults of both sexes of _Neostylopyga rhombifolia_ and _Platyzosteria novae seelandiae_. Walker (1904) and Longstaff (_in_

Shelford, 1912) noted that the latter species had a strong odor. Roth (unpublished data, 1957) found that the secretion of _P. novae seelandiae_ when ejected is grayish or milky in color. In the reservoirs of the ventral gland of this insect the secretion is a milky liquid containing floating greenish globules. Both infrared and mass spectrographic analyses show that the secretion is a mixture containing 2-hexenal, the aldehyde that is found in _E. floridana_. Eisner (personal communication, 1958) observed that the lizard _Anolis carolinensis_ immediately released _Neostylopyga rhombifolia_ without injury, but that _Bufo marinus_, _Anolis equistris_, and _Cyanocitta cristata_ ate the insect despite the secretion; several unidentified spiders and the ant _Pogonomyrmex badius_ were not repelled by the secretion of _N. rhombifolia_.

Dorsal and ventral glands have been found in both sexes of _Blatta orientalis_ and _Periplaneta americana_ (Minchin, 1888, 1890; Kul'vets, 1898; Oettinger, 1906; Harrison, 1906; Liang, 1956). The ventral glands are found in the same general region as those of _Eurycotis_. We have also found similar ventrally located glands in both _Periplaneta australasiae_, and _P. brunnea_. The reservoirs which store the secretion of the ventral glands are smaller in _Blatta_ and _Periplaneta_ spp. than those found in _Eurycotis_, _Neostylopyga_, or _Platyzosteria_.

In _Blatta orientalis_ the dorsal glands can be everted by pressure on the abdomen; the secretion in these glands, according to Haase (1889), has the typical oriental cockroach odor. Although the dorsal glands of the oriental cockroach are usually given a defensive role (Haase, 1889, 1889a; Kul'vets, 1898; Oettinger, 1906; Konček, 1924), the functions of secretions of these nonepigamic dorsal glands and the ventral glands are still open to question. It is possible that some of the odors produced by cockroaches have functions other than defense or sex attraction. For example, Ledoux (1945) showed that the species odor is largely responsible for the gregarious behavior shown by _Blatta orientalis_ and _Blattella germanica_. The olfactory stimulus acts over a short distance only, and the source of this odor in the insect is unknown. By washing _Blattella germanica_ in warm chloroform Dusham (1918) extracted a wax which had the odor of the German cockroach. However, there is no evidence to show that cockroaches respond to the same cockroach odors that are detected by man.

Certain cockroaches have recently been found to have odorous secretions which are produced in tracheal glands. In _Diploptera punctata_ the tracheae leading to the second abdominal spiracles of nymphs and adults are modified into odoriferous glands which produce a mixture of 2-ethyl-1,4-benzoquinone; 2-methyl-1,4-benzoquinone; and _para_ benzoquinone; this material is ejected as a means of defense. The offensive odor emitted by adults and nymphs of _Leucophaea maderae_ also issues from the second abdominal spiracles (Roth and Stay, 1958).

Diploptera is capable of ejecting its quinones from either its right or left tracheal gland according to which side of the insect is attacked (pl. 36, A-B). Eisner (1958) found that the secretion repelled the ant _Pogonomyrmex badius_ (Latreille) (pl. 36, C) and the beetle _Galerita janus_ Fabricius when they attacked the cockroach. The spider _Lycosa helluo_ Walckenaer was repelled by large nymphs and adults of _D.

punctata_ but young nymphs were usually eaten promptly (Eisner, 1958).

Bordas (1901, 1908) believed that the "conglobate" gland (Miall and Denny, 1886), found in males of _Periplaneta americana_ and _Blatta orientalis_, was an odoriferous gland used for defense, but Gupta (1947) has shown that in all probability this gland (the phallic gland) secretes the outermost covering of the spermatophore.

What appears to be mimicry occurs in some species of Blattaria. The nymphs of many Panchlorini and Blaberinae vaguely resemble sow bugs (Chopard, 1938). Certain members of the Perisphaerini (e.g., _Perisphaerus glomeriformis_) from the Malayan region which resemble sow bugs (Annandale, 1900; Hanitsch, 1915) can roll themselves up into a ball thus hiding their antennae and legs (Lucas, 1862). Although these cockroaches are found among dead leaves or under stones, in places in which sow bugs are also found, the benefit to either or both forms is questionable; Annandale (1900) believed that the crustacean and the cockroach, living under similar conditions, developed the same general body shape. Rolling up into a ball is nothing more than an exaggeration of a reflex common to many young cockroaches, that is, an arched position which these insects assume when they immobilize themselves in response to certain stimuli (Chopard, 1938).

There are cockroaches that resemble various Coleoptera and Hemiptera (Belt, 1874; Shelford, 1912; Hanitsch, 1915). Some look like cerambycids, lampyrids, coccinellids, pentatomids, etc. Perhaps the most striking examples are the resemblances of cockroaches in the genus _Prosoplecta_ of the Epilamprinae to beetles of the family Coccinellidae; Shelford (1912) has figured a number of species of _Prosoplecta_ together with the species of beetles which they seem to have taken for models. Williams (1928) mentioned diurnal cockroaches which by a combination of markings, shape, posture, and active flight about vegetation suggest certain wasps.

Unfortunately, practically nothing is known about the behavior of these so-called mimics and models or their relationships with predators in the field. For the most part, the examples are based on a comparison of pinned insects from museum collections (Burr, 1899); for this reason Chopard (1938) believed that not much value should be placed on superficial resemblances of this kind. However, we believe that a lack of knowledge of cockroach mimicry is not a valid reason for rejecting the idea that mimicry, if it occurs, may be of some benefit in the survival of mimetic species. Certainly Cott's (1940) voluminous compilation of the literature on adaptive coloration should make the most skeptic hesitate to conclude dogmatically that these instances of mimicry are merely accidental and meaningless.

XIX. THE BIOLOGICAL CONTROL OF COCKROACHES

In the Navy [Japanese] a seaman who has captured 300 cockroaches will be granted one day special shore leave. They call it "shore leave for cockroaches." The purpose is to promote extermination of cockroaches in a warship because, on the one hand, any warship suffers from numerous cockroaches, and, on the other hand, any seaman likes shore leave.... The formalities for a shore leave for cockroaches are as follows. A

seaman keeps cockroaches which he captured (mainly _B. germanica_, because _P. americana_ and _P. australasiae_ are seldom found in Japan) in a bottle or in a bag until the number reaches 300. Then he brings them to the deck officer to get the confirmation that he has actually captured more than 300 cockroaches. If the deck officer confirms it, the seaman goes to a cabin where a petty officer reports that the deck officer confirmed the number of cockroaches. The petty officer signs the seaman's name, name of division, rank, and date to be on shore leave in the log book for cockroach shore leaves. The petty officer brings the log book again to the deck officer to get his approval and then goes to the commander for the final approval. In the Navy, they have another special shore leave for rats. In this system, a seaman gets one day shore leave for one rat. The formalities for the latter are the same as for the former, and there is a log book for the rat shore leave in the petty officer's quarters. The author took advantage of these systems frequently.

SONAN (1924)

Little is known of the effects of predatism and parasitism on natural populations of cockroaches. Many statements in the literature are very general; yet there are a few data on egg parasites (e.g., _Tetrastichus hagenowii_) which suggest that, in the absence of parasites, populations of domestic cockroaches might be much larger than they are in certain areas. We have summarized the literature on natural control and also that on the use by man of predators and parasites in the biological control of cockroaches. However, because of the paucity of information, we have been unable to evaluate the effectiveness of biological control in reducing the numbers of pest cockroaches. This is an area that might reward further investigation.

INVERTEBRATES

Scorpions.--In Puerto Rico, cockroaches are probably the principal food of the scorpions which live in old houses, on tree trunks, etc. (Seín, 1923). The staple diet of scorpions in Arizona is the small cockroach commonly known as the water bug (Stahnke, 1949); in the part of Arizona where he resides, Stahnke (personal communication, 1953) says that the "water-bug" is most generally _Supella supellectilium_ although _Blattella germanica_ is also found, but less abundantly.

Spiders.--Jefferys (1760) mentioned a large spider which was protected in the Antilles and especially on Guadeloupe because it hunted down and fed on cockroaches; the spider was reputed to be common in every house. Sir Hans Sloane (1725, _in_ Cowan, 1865) reported that residents of Jamaica kept spiders in their houses to destroy cockroaches. Takahashi (1924) reported that, in the Taihoku area of Formosa, human habitations contained large numbers of spiders which caught and ate cockroaches. Smith (_in_ Marlatt, 1915) reported that Brazilians encourage large house spiders because they tend to rid the house of "other insect pests." In British Guiana tarantulas were kept in a bungalow to control _Periplaneta_ and _Pycnoscelus_ (Beebe, 1925a).

Ants.--A Madam Merian noticed that ants cleared houses of cockroaches (Kirby and Spence, 1822). A small reddish-yellow ant, called Pucchuçiçi by Peruvian Indians, pursued and destroyed a cockroach called Chilicabra which was a pest in native huts (Tschudi, 1847). Schwabe (1950b) found swarms of ants attacking living _Pycnoscelus surinamensis_ and stated that ants are probably the chief enemy of this cockroach in Hawaii. Wallace (1891) stated that in Africa a band of driver ants may enter a house and clear it of cockroaches and other arthropods. In British Guiana, Beebe, (1925) found that several times a year army ants cleared the laboratory of all cockroaches and tarantulas.

Wasp egg parasites.--Matsumura (1917, _in_ Asano, 1937) proposed that parasitic wasps such as _Evania_ and _Brachygaster_ be protected in Japan as the natural enemies of cockroaches. In one area in France, 20 percent of the oöthecae of _Loboptera decipiens_ were parasitized by _Zeuxevania splendidula_ (Genieys, 1924). Edmunds (1952a) found that 12 percent of 459 oöthecae of _Parcoblatta_ collected during December through April of 1950-51 in Ohio were parasitized; evaniids accounted for about 7 percent of the parasitization. Additional collection data in 1951-52 Edmunds (1953a) showed that 8.7 percent of 320 wood-cockroach oöthecae were parasitized; 2.8 percent of these parasites were evaniids; almost 13 percent of the egg capsules collected showed evidence of previous parasite emergence. Cameron (1957) reported that oöthecae of _Periplaneta americana_ collected in Saudi Arabia were 29 percent parasitized in March and 25 percent parasitized in October by _Evania appendigaster_. Sonan (1924) found 1 of 65 oöthecae of _P. americana_ and _P. australasiae_ parasitized by _E. appendigaster_ in Formosa.

Cottam (1922) stated that the increase of _Supella supellectilium_ in Khartoum was checked by a wasp egg-parasite that was later identified as _Anastatus tenuipes_ (see p. 246) (Ferrière, 1930, 1935). In this country, this wasp seemed to be effective in decreasing the numbers of _Supella_ in certain areas in Arizona (Flock, 1941).

In Formosa, _Tetrastichus hagenowii_ was an important parasite of cockroach eggs (Maki, 1937). Sonan (1924) reported 30 percent parasitization of 65 oöthecae of _Periplaneta americana_ and _P. australasiae_ collected in Formosa. In Bangalore, India, the natural parasitization of randomly collected oöthecae of _P. americana_ varied from 21 percent (of 495 oöthecae), July 1947-June 1948, and 43 percent (of 288 oöthecae), July-December 1948, to 57 percent (of 178 oöthecae), July-October 1949 (Usman, 1949). Cameron (1955) obtained _T. hagenowii_ from oöthecae collected in Trinidad, B.W.I., and Saudi Arabia; some 15 percent of the oöthecae of _P. americana_ and _P. australasiae_ collected in October in Trinidad were parasitized; a later collection (March) was 34 percent parasitized; a small sample of _P. americana_ oöthecae was 65 percent parasitized. The oöthecae collected in Saudi Arabia in March were 20 percent parasitized. Plank (1947) found that the eggs of the American cockroach in Puerto Rico (probably in laboratory cultures) were so heavily parasitized by _T. hagenowii_ that he had to use _P. australasiae_ for experimental purposes; in 1950 Plank stated that more than 50 percent of American cockroach oöthecae were parasitized.

Fahringer (1922) stated that _Prosevania punctata_ could be used to eradicate cockroaches, but he did not test his hypothesis. Marlatt (1902) felt that the usefulness of _Evania appendigaster_ in biological control was impaired by _Tetrastichus_ acting as a hyperparasite (see footnote 6, p. 236). However, Wolcott (1951) stated that in Puerto Rico _E. appendigaster_ is quite abundant and is a factor of considerable importance in controlling cockroaches. Kadocsa (1921) stated that _Brachygaster minutus_ and _Evania appendigaster_ were not important in the biological control of cockroaches. These general statements are not supported by experimental evidence.

It is likely that the smaller wasp egg parasites are more effective than the evaniids in controlling cockroaches. Only one evaniid develops in a parasitized oötheca, but many individuals of the other wasps develop in one oötheca and the number of females that emerge is usually large. However, Cameron (1957) concluded that, with a parasitism rate of 25 to 29 percent and three to four generations a year, against one or less for the host, _Evania appendigaster_ in the areas where it is established is a valuable control agent.

The use of specific egg parasites to control cockroaches has not been attempted extensively. Cros (1942) liberated a species of _Tetrastichus_ (=_Eulophus_ sp.; see p. 254) in his home in Algeria to control the oriental cockroach; as far as we know, he did not report the parasite's effectiveness in reducing the cockroach population. According to Zimmerman (1948) _Comperia merceti_, when accidentally imported, practically wiped out _Supella supellectilium_ in parts of Hawaii; he claimed to have controlled the brown-banded cockroach in a store building with this parasite. In some parts of Honolulu, almost 100 percent of the oöthecae of this cockroach were parasitized (Zimmerman, 1944). We (1954b) ran some simulated field tests in which we liberated _Tetrastichus hagenowii_ in rooms artificially seeded with oöthecae; from 28 to 83 percent of American cockroach oöthecae and 56 percent of oriental cockroach oöthecae were parasitized during these tests.

Evania appendigaster was introduced from Hawaii into Canton Island in 1940 against _Periplaneta americana_, and it has become established (Dumbleton, 1957). This parasite was also successfully introduced into Samoa (Dumbleton, 1957).

Cockroach-hunting wasps.--An earnest attempt has been made to establish in Hawaii wasps that prey on cockroaches. Just how effective these wasps are in controlling cockroaches is still unknown. _Dolichurus stantoni_ was introduced from the Philippines in 1917 and spread to several of the Islands (Swezey, 1920, 1921; Williams, 1944). Bridwell (1920) stated that as a result of this introduction there was a great decrease in cockroaches of the genus "_Phyllodromia_." A number of _Podium haematogastrum_ from Brazil were liberated in Honolulu (Williams, 1925) but did not become established (Williams, 1928). The effectiveness of _Podium_ was questioned by Williams (1928) who observed that _Podium_ "destroyed innumerable Blattidae, which nonetheless swarmed in their neighborhood, and I must confess from my observations on the various cockroach-hunting wasps that the blattid more than holds its own alongside its enemy."

Introductions of _Ampulex_ have proved more successful. _Ampulex canaliculata_ was introduced into Hawaii from the United States (Williams, 1928a, 1929). Williams also introduced _A. compressa_ into Hawaii in 1940, and the species was reared in large numbers for distribution (Pemberton, 1942). _A. compressa_ has since become established on most of the Islands (Pemberton, 1945a, 1947; Williams, 1946; Van Zwaluwenburg, 1950). The thousand of _A. compressa_ now found in the Hawaiian Islands are all descendants of three wasps captured in Noumea, New Caledonia (Williams, 1944). According to Williams (1941), the number of cockroaches was noticeably reduced at the University of Hawaii poultry farm, where some _A. compressa_ were released. Pemberton (1953) believed that this wasp has become sufficiently abundant to be of definite value. Simmonds (1941) recommended importing _A. compressa_ into Fiji for cockroach control. _A. compressa_ was introduced from Hawaii into Guam in 1954 against _Periplaneta americana_ and into the Cook Islands in 1955 against _Periplaneta_ spp.; it is not yet known whether the parasite became established in either place (Dumbleton, 1957).

VERTEBRATES

... on conserve avec soin les crapauds dans les maisons, et que les dames les tolèrent, même sous leurs robes, en raison de leurs continuels services, car ils se promènent sans cesse à la recherche des Kakerlacs.

GIRARD (1877)

Toads.--_Bufo marinus_ was first introduced into Puerto Rico from Barbados in 1920 to reduce several major insect pests including cockroaches (Leonard, 1933). It was introduced from Puerto Rico into Hawaii by C. E. Pemberton in 1932 where it rapidly became established; it has since been distributed throughout the Pacific area. _B. marinus_ is one of the world's largest toads; it attains a body length (exclusive of the hind legs) of 7 to 9 inches (Oliver, 1949) and has been kept alive for more than 11 years in captivity (Pemberton, 1945). Alicata (1938) placed giant toads in a fenced area in Hawaii containing an infestation of _Pycnoscelus surinamensis_; after 24 hours the toads were dissected and each was found to have eaten from 11 to 25 cockroaches. Illingworth (1941) found that 40 to 90 percent of 53 stools of this toad in Hawaii contained remains of _P. surinamensis_. Alicata (1947) recommended the maintenance of _B. marinus_ in poultry yards to reduce the population of _P. surinamensis_, the vector of the chicken eye worm.

Toads have also been recommended for controlling cockroaches in houses (Meech, 1889; Sweetman, 1936). Girard (1877) cited a note in a French newspaper which stated that toads were kept in houses in Cuba to control the American cockroach.

Tree frogs.--Tree frogs enclosed in a room overnight were said to effectively clear it of cockroaches (Marlatt, 1915); on sugar plantations in Australia, these amphibians were encouraged in houses and kept as pets because they hunted and devoured large brown cockroaches (Froggatt, 1906).

Birds.--In Guadeloupe, Dutertre (1654) claimed that all the fowls of the country were fond of small cockroaches and lived on practically nothing else. In Hawaii (Zimmerman, 1948) and in the Lesser Antilles (Ballou, 1912) cockroaches are eaten by poultry whenever the birds can find them. In Puerto Rico, Wetmore (1916) stated that owls kept in houses feed extensively on cockroaches; the stomach of one owl which had been kept in a native house was filled entirely with cockroaches. In British Guiana, Beebe (1925) found that cockroaches were eaten by 27 species of birds.

Reptiles.--H. (1800) claimed that two lizards cleared his house of the "true brown cockroach" and suggested that lizards be used for cockroach control because the reptiles are docile and harmless. On Arno Atoll geckos and night-feeding skinks eat large numbers of cockroaches (Usinger and La Rivers, 1953). According to Wolcott (1924) the number of cockroaches eaten by lizards is surprisingly large considering the nocturnal habits of these insects. Beebe (1925a) kept geckos in a bungalow to help control _Periplaneta_ and _Pycnoscelus_.

Mammalia.--Cowan (1865) stated that in England hedgehogs were often kept domesticated in kitchens to destroy cockroaches. This writer also stated that a lemur was kept on board ship to destroy cockroaches.

Large numbers of the American and Australian cockroaches were eaten by the mongoose in Hawaii (Perkins, 1913).

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PLATES

[Illustration: PL. 2 _Blaberus craniifer_, nymph. (Photograph by Jack Salmon.)]

[Illustration: PL. 3 _Blaberus giganteus_, c. × 2.2. (Photograph by Jack Salmon.)]

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[Illustration: PL. 28 A-B, Gregarines (_Diplocystis_ sp.?) from _Blaberus craniifer_. A, Organisms removed from intestine, × 50. B, Organisms removed from hemocoele, × 32. C, Gregarine cysts in feces of _Leucophaea maderae_, × 12.]

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A, Undetermined mermithid that parasitizes _Ectobius pallidus_. × 9. The worm has partly emerged from the neck region of the cockroach. (Reproduced from Roth and Willis [1957].) B, Undetermined gordian worm that parasitized _Eurycotis floridana_ shown beside its host, × 1.8. (Specimen courtesy of Dr. T. Eisner.)]

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[Illustration: PL. 35 _Ampulex compressa_ attacking _Periplaneta_ sp. (presumably _americana_), about natural size.

A, The wasp finds a cockroach. B, She stings the prey in the thorax. C, She then leads the disabled cockroach (antennae clipped) to her nest. D, The wasp's egg was placed on the coxa of the cockroach's right mesothoracic leg where it hatched. E, Portion of the host's abdomen removed to show feeding larva. F, New adult wasp emerging from dead host. (Reproduced from F. X. Williams [1942] from the color paintings of the late W. Twigg-Smith, through the courtesy of F. A. Bianchi.)]

[Illustration: PL. 36 Chemical defense of _Diploptera punctata_ against predators; the spray pattern is displayed on KI-starch indicator paper. A, Spray pattern after right mesothoracic leg was pinched. B, Cumulative spray pattern after left mesothoracic leg of the same insect was pinched. C, The defensive glands of the cockroach on the left had been excised, and it is under persistent attack by ants from a laboratory colony of _Pogonomyrmex badius_ (Latreille). The intact cockroach on the right was also attacked by the ants, but it discharged a spray of quinones and repelled the attackers. (From Eisner [1958], through the courtesy of Dr. T. Eisner.)]

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[Illustration: Decoration]

FOOTNOTES:

[1] Part of the cost of publication of this monograph was borne by the Office of Naval Research, Department of the Navy (through the American Institute of Biological Sciences), and by the Quartermaster Research and Engineering Center, Department of the Army.

[2] Present address of both authors, Central Research Laboratories, United Fruit Co., Upland Road, Norwood, Mass.

[3] Names of organisms preceded by an asterisk (*) are known or suspected pathogens of vertebrates. These records were presented with annotations in our 1957a paper on the medical and veterinary importance of cockroaches. For that reason the annotations have not been repeated herein, although the records have been included to make the listing of the biotic associates of cockroaches substantially complete.

[4] The following helminths also have been stated in the literature to pass their intermediate stages in cockroaches: _Hymenolepis diminuta_ (Rudolphi, 1819) [Blanchard (1891)]; _Inermicapsifer madagascariensis_ (Davaine in Grenet, 1870) [Baer (1956)]; _Spirocerca sanguinolenta_ (Rudolphi, 1819) [Seurat (1913)]. These doubtful records are discussed in Roth and Willis (1957a).

[5] Barber (1939) has shown that the correct spelling of the name of the type genus is _Ripidius_ and not _Rhipidius_, as it is frequently written, and that, consequently, the family should be Ripiphoridae and not Rhipiphoridae.

[6] Page 11 of Marlatt (1902) has been cited erroneously so many times in support of host records for _T. hagenowii_ and _Evania appendigaster_ that we are quoting the pertinent section below. In the section preceding the quoted material Marlatt discusses the American, Australian, oriental, and German cockroaches. There is nothing in the paper to connect any of these cockroaches specifically with the parasites mentioned below:

NATURAL ENEMIES AND PARASITES

"In Europe the egg capsules of the cockroach are often parasitized by an ichneumon fly (_Evania appendigaster_). This insect has become widely distributed over the world following its host insect, and has been redescribed under a great many different names. It was found in Cuba as early as 1829, and has been several times collected in the United States. Unfortunately, its usefulness as a means of keeping the roach in check by destroying the egg capsules is greatly impaired by the occurrence of another ichneumon fly (_Entedon hagenowi_), which is parasitic upon the first. This is also a European species which has been brought over with its host parasite. If the true egg capsule parasite of the roach could have been introduced into this country without this secondary parasite, its usefulness would doubtless have been very much greater. The secondary parasite, however, seems to have been introduced early, and has been found in Cuba and Florida, and probably occurs as widely as its host and prevents the latter from multiplying very greatly."

[7] Brues et al. (1954) include this family in the Encyrtidae.

[8] archy (Marquis, 1931) was living in a dream world when he typed:

"there is always something to be thankful for you would not think that a cockroach had much ground for optimism but as the fishing season opens up i grow more and more cheerful at the thought that nobody ever got the notion of using cockroaches for bait"

[9] Classification of Amphibia and Reptilia follows Hegner (1936).

[10] Classification of Amphibia and Reptilia follows Hegner (1936).

[11] Classification of Mammalia follows Simpson (1945).

[12] The West Indian "_Blabera fusca_ Brunner" of Saupe (1928) is obviously _Blaberus craniifer_ as can readily be seen from a comparison of Saupe's figures of the pronotal shields of his species with the descriptions by Hebard (1917) of _B. craniifer_ and _Blaberus atropos_ (Stoll). The Chilean _B. fusca_ Brunner is a junior synonym of _B. atropos_ (Stoll), a South American insect (Hebard, 1917).

[13] The species was recorded by these authors as _Blaberus atropos_. (Hebard, 1917 p. 204, footnote 327.)

Transcriber's Notes:

- symbol for: female indiv. replaced by [F] - symbol for: male indiv. replaced by [M] - symbol for: 'dagger' replaced by [cross] - Different languages (English, French, German, Danish, Latin, etc.) have different spellings for example in titles of books, etc. These remain as in the original. - Several blank lines inserted to enhance readability of text(!) - Several page numbers in 'table of contents' corrected, according to real page numbers. - caret used for superscripted characters; eg.: abc^{...} - underscore used for subscripted characters; eg.: abc_{...}